

בית הספר להנדסה
School of Engineering

ד"ר יעקב רזניק

חשבון דיפרנציאלי ואינטגרלי 2

תרגילים ופתרונות

תשס"ח 2007/2008

זהויות טריגונומטריות בסיסיות

$$1) \sin^2 \alpha + \cos^2 \alpha = 1$$

$$6) 1 + \cos 2\alpha = 2\cos^2 \alpha$$

$$2) 1 + \tan^2 \alpha = \frac{1}{\cos^2 \alpha}$$

$$7) 1 - \cos 2\alpha = 2\sin^2 \alpha$$

$$3) \sin 2\alpha = 2\sin \alpha \cdot \cos \alpha$$

$$8) \sin(\alpha + \beta) = \sin \alpha \cdot \cos \beta + \sin \beta \cdot \cos \alpha$$

$$4) \cos 2\alpha = \cos^2 \alpha - \sin^2 \alpha$$

$$9) \sin(\alpha - \beta) = \sin \alpha \cdot \cos \beta - \sin \beta \cdot \cos \alpha$$

$$5) \tan 2\alpha = \frac{2\tan \alpha}{1 - \tan^2 \alpha}$$

$$10) \cos(\alpha + \beta) = \cos \alpha \cdot \cos \beta - \sin \alpha \cdot \sin \beta$$

$$11) \cos(\alpha - \beta) = \cos \alpha \cdot \cos \beta + \sin \alpha \cdot \sin \beta$$

$$11a) \tan(\alpha \pm \beta) = \frac{\tan \alpha \pm \tan \beta}{1 \mp \tan \alpha \tan \beta}$$

$$12) \sin \alpha + \sin \beta = 2\sin \frac{\alpha + \beta}{2} \cdot \cos \frac{\alpha - \beta}{2}$$

$$16) \sin \alpha \cdot \cos \beta = \frac{1}{2} [\sin(\alpha + \beta) + \sin(\alpha - \beta)]$$

$$13) \sin \alpha - \sin \beta = 2\sin \frac{\alpha - \beta}{2} \cdot \cos \frac{\alpha + \beta}{2}$$

$$17) \cos \alpha \cdot \cos \beta = \frac{1}{2} [\cos(\alpha + \beta) + \cos(\alpha - \beta)]$$

$$14) \cos \alpha + \cos \beta = 2\cos \frac{\alpha + \beta}{2} \cdot \cos \frac{\alpha - \beta}{2}$$

$$18) \sin \alpha \cdot \sin \beta = -\frac{1}{2} [\cos(\alpha + \beta) - \cos(\alpha - \beta)]$$

$$15) \cos \alpha - \cos \beta = 2\sin \frac{\alpha + \beta}{2} \cdot \sin \frac{\beta - \alpha}{2}$$

$$19) \sin 3\alpha = -\sin^3 \alpha + 3\sin \alpha \cdot \cos^2 \alpha$$

$$20) \cos 3\alpha = \cos^3 \alpha - 3\cos \alpha \cdot \sin^2 \alpha$$

נגזרות של פונקציות אלמנטריות:

$$1) (x^n)' = n x^{n-1}$$

$$6) (\sin x)' = \cos x$$

$$11) (\operatorname{ch} x)' = \operatorname{sh} x$$

$$2) (a^x)' = a^x \ln a$$

$$7) (\cos x)' = -\sin x$$

$$12) (\arcsin x)' = \frac{1}{\sqrt{1-x^2}}$$

$$3) (e^x)' = e^x$$

$$8) (\tan x)' = \frac{1}{\cos^2 x}$$

$$13) (\arccos x)' = -\frac{1}{\sqrt{1-x^2}}$$

$$4) (\log_a x)' = \frac{1}{x} \log_a e$$

$$9) (\cot x)' = -\frac{1}{\sin^2 x}$$

$$14) (\arctan x)' = \frac{1}{1+x^2}$$

$$5) (\ln x)' = \frac{1}{x}$$

$$10) (\operatorname{sh} x)' = \operatorname{ch} x$$

$$15) (\operatorname{arccot} x)' = -\frac{1}{1+x^2}$$

כללי האינטגרציה

$$1) \int x^n dx = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1)$$

$$8) \int \frac{dx}{\sin^2 x} = -\operatorname{ctg} x + C$$

$$2) \int a^x dx = \frac{a^x}{\ln a} + C$$

$$9) \int \operatorname{sh} x dx = \operatorname{ch} x + C$$

$$3) \int e^x dx = e^x + C$$

$$10) \int \operatorname{ch} x dx = \operatorname{sh} x + C$$

$$4) \int \frac{dx}{x} = \ln|x| + C$$

$$11) \int \frac{dx}{\sqrt{a^2 - x^2}} = \arcsin \frac{x}{a} + C$$

$$5) \int \sin x dx = -\cos x + C$$

$$12) \int \frac{dx}{a^2 + x^2} = \frac{1}{a} \arctan \frac{x}{a} + C$$

$$6) \int \cos x dx = \sin x + C$$

$$13) \int \frac{dx}{a^2 - x^2} = \frac{1}{2a} \ln \left| \frac{a+x}{a-x} \right| + C$$

$$7) \int \frac{dx}{\cos^2 x} = \tan x + C$$

$$13') \int \frac{dx}{x^2 - a^2} = \frac{1}{2a} \ln \left| \frac{x-a}{x+a} \right| + C$$

$$14) \int \frac{dx}{\sqrt{x^2 \pm a^2}} = \ln \left| x + \sqrt{x^2 \pm a^2} \right| + C$$

תרגיל מס' 1

גבולות ורציפות של פונקציות של משתנים אחדים

X

הוכח שהגבול הנתון לא קיים:

- $$1) \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{x-y}{x+y} \qquad 2) \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{x^2 + 2y^2 - 3xy}{x^2 + 3y^2} \qquad 3) \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} e^{\frac{x}{x^2+y^2}}$$
- $$4) \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{x^2 y}{x^4 + y^2} \qquad 5) \lim_{\substack{x \rightarrow +0 \\ y \rightarrow +0}} \frac{x^3 + 2y^3 - x}{\sqrt{y^6 + x^2}}$$
- $$6) \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{\sin(xy)}{xy} \sin \frac{1}{\sqrt{x^2 + y^2}} \qquad 7) \lim_{\substack{x \rightarrow 1 \\ y \rightarrow 0}} \frac{\ln(x+y)}{\sqrt{x^2 + y^2 - 2x + 1}}$$

חשב את הגבול הנתון:

- $$8) \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} (x^2 + y^2) \cos \frac{1}{xy} \qquad 9) \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{\sin(2xy)}{y}$$
- $$10) \lim_{\substack{x \rightarrow 2 \\ y \rightarrow \infty}} \left(1 + \frac{x}{y}\right)^y \qquad 11) \lim_{\substack{x \rightarrow \infty \\ y \rightarrow 3}} \left(1 + \frac{1}{x}\right)^{\frac{x^2}{x+y}}$$
- $$12) \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{x^2 + y^2}{\sqrt{x^2 + y^2} + 1 - 1} \qquad 13) \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \sin(x^2 + y^2)$$
- $$14) \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{x^4}{x^2 + y^2} \qquad 15) \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{\sin(xy)}{\sqrt{x^2 + y^2}}$$
- $$16) \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{\sin(x^3 + y^3)}{x^2 + y^2} \qquad 17) \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{2x^3 + 3y^4}{x^2 + y^2}$$
- $$18) \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{x^5 - x^2 y^3}{x^4 + y^4} \qquad 19) \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{x^5 - \cos(x^2 + y^2) + 1}{(x^2 + y^2)^2}$$

ב

בדוק את רציפות הפונקציה הנתונה בנקודה $A(0,0)$:

$$1) f(x,y) = \begin{cases} \frac{x^4 + y^5}{x^3 + y^3} & , (x,y) \neq (0,0) \\ 0 & , (x,y) = (0,0) \end{cases}$$

$$2) f(x,y) = \begin{cases} (x+y) \sin \frac{1}{x^2} \cos \frac{1}{y} & , (x,y) \neq (0,0) \\ 0 & , (x,y) = (0,0) \end{cases}$$

$$3) f(x,y) = \begin{cases} \frac{x^2 + y}{\sqrt{x^2 + y^2}} & , (x,y) \neq (0,0) \\ 0 & , (x,y) = (0,0) \end{cases}$$

$$4) f(x,y) = \begin{cases} \frac{xy}{x^2 + y^2} & , (x,y) \neq (0,0) \\ 0 & , (x,y) = (0,0) \end{cases}$$

$$5) f(x,y) = \begin{cases} \frac{\sqrt[3]{y} \sin(x^2)}{x^2 + y^2} & , (x,y) \neq (0,0) \\ 0 & , (x,y) = (0,0) \end{cases}$$

$$6) f(x,y) = \begin{cases} \frac{x^4 - x^3y - xy^3 + y^4}{\sqrt{x^6 + y^6}} & , (x,y) \neq (0,0) \\ 0 & , (x,y) = (0,0) \end{cases}$$

תרגיל מס' 2

נגזרות חלקיות ודיפרנציאביליות של פונקציות של משתנים אחדים

מצא את כל הנגזרות החלקיות הראשונות עבור הפונקציה הנתונה:

$$1) z = x^3 + 3x^2y - y^3$$

$$2) z = \ln(x^2 + y^2)$$

$$3) z = \frac{y}{x}$$

$$4) z = \arctan \frac{y}{x}$$

$$5) z = \frac{xy}{x-y}$$

$$6) z = \ln \left(\frac{1}{\sqrt[3]{x}} - \frac{1}{\sqrt[3]{y}} \right)$$

$$7) u = xe^{-yx}$$

$$8) u = \frac{y}{x} + \frac{z}{y} - \frac{x}{z}$$

$$9) z = \frac{2x-t}{x+2t}$$

$$10) z = \sqrt{x^2 + y^2 - 2xy \cos t}$$

$$11) z = \arcsin(t\sqrt{x})$$

$$12) z = \cos(ax - by)$$

$$13) z = \frac{x}{3y-2x}$$

$$14) u = \ln \sin(x-2t)$$

$$15) u = \arcsin \frac{y}{x}$$

$$16) u = \sin^2(x+y) - \sin^2 x - \sin^2 y$$

$$17) z = x \arctan(y)$$

$$18) z = \ln(\sqrt{x} - \sqrt{y})$$

$$19) u = 3xyz + xy + xz + yz$$

$$20) u = \sin x \cos y + \cos x \tan(yz)$$

$$\text{הוכח: } x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = \frac{1}{2}$$

$$(21) \text{ נתונה הפונקציה: } z = \ln(\sqrt{x} + \sqrt{y})$$

$$\text{הוכח: } 2x \frac{\partial u}{\partial x} + t \frac{\partial u}{\partial t} = 0$$

$$(22) \text{ נתונה הפונקציה: } u = e^{x/t^2}$$

$$\text{הוכח: } \frac{x}{y} \cdot \frac{\partial u}{\partial x} + \frac{1}{\ln x} \cdot \frac{\partial u}{\partial y} = 2u$$

$$(23) \text{ נתונה הפונקציה: } u = x^y$$

$$\text{הוכח: } \left(\frac{\partial u}{\partial x} \right)^2 + \left(\frac{\partial u}{\partial y} \right)^2 + \left(\frac{\partial u}{\partial z} \right)^2 = 1$$

$$(24) \text{ נתונה הפונקציה: } u = \sqrt{x^2 + y^2 + z^2}$$

$$\text{הוכח: } x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = \frac{z}{\ln y}$$

$$(25) \text{ נתונה הפונקציה: } z = e^{x/y} \ln y$$

$$\text{הוכח: } L \frac{\partial T}{\partial L} + g \frac{\partial T}{\partial g} = 0$$

$$(26) \text{ נתונה הפונקציה: } T = \pi \sqrt{\frac{L}{g}}$$

מצא את הנגזרות החלקיות הראשונות בנקודה $(0,0)$ עבור הפונקציה הנתונה:

$$27) f(x,y) = \begin{cases} \frac{2xy}{x^2+y^2} , & (x,y) \neq (0,0) \\ 0 , & (x,y) = (0,0) \end{cases}$$

$$28) f(x,y) = \begin{cases} (x^2+y^2) \cos \frac{\pi}{x^2+y^2} , & (x,y) \neq (0,0) \\ 0 , & (x,y) = (0,0) \end{cases}$$

$$29) f(x,y) = \begin{cases} \frac{xy}{\sqrt{x^2+y^2}} , & (x,y) \neq (0,0) \\ 0 , & (x,y) = (0,0) \end{cases}$$

$$30) f(x,y) = \begin{cases} \frac{x^3+y^3}{x^2+3y^2} , & (x,y) \neq (0,0) \\ 0 , & (x,y) = (0,0) \end{cases}$$

$$31) f(x,y) = \begin{cases} y^{4/3} \ln(y^2 + x^{4/3}) , & (x,y) \neq (0,0) \\ 0 , & (x,y) = (0,0) \end{cases}$$

(32) בדוק, האם הפונקציה:

$$f(x,y) = \begin{cases} \frac{xy}{\sqrt{x^2+y^2}} , & (x,y) \neq (0,0) \\ 0 , & (x,y) = (0,0) \end{cases}$$

דיפרנציאביליות בנקודה $(0,0)$?

(33) בדוק, האם הפונקציה:

$$f(x,y) = \begin{cases} (x^2+y^2) \sin \frac{1}{x^2+y^2} , & (x,y) \neq (0,0) \\ 0 , & (x,y) = (0,0) \end{cases}$$

דיפרנציאביליות בנקודה $(0,0)$?

(34) עבור הפונקציה: $f(x,y) = \sqrt[3]{xy}$

(a) מצא לפי ההגדרה את הנגזרות החלקיות: f'_x ו- f'_y בנקודה $(0,0)$.

(b) בדוק, האם הנגזרות f'_x , f'_y רציפות בנקודה $(0,0)$?

(c) בדוק את הדיפרנציאביליות של הפונקציה בנקודה $(0,0)$.

תרגיל מס' 3

יישומים של מושג הדיפרנציאל

א. חשב בקירוב:

- | | | |
|--------------------------------|-----------------------------|------------------------------------|
| 1) $1.02^{4.05}$ | 2) $\ln(0.09^3 + 0.99^3)$ | 3) $1.02^3 \cdot 0.97^2$ |
| 4) $\sqrt{5e^{0.02} + 2.03^2}$ | 5) $\sqrt{4.05^2 + 2.93^2}$ | 6) $\sqrt{1.04^{1.99} + \ln 1.02}$ |

ב. בחישובים בתרגילים הבאים השתמש במושג הדיפרנציאל:

1. בתיבה ריבועית אורך צלע הבסיס: $a = 6 \text{ cm}$ ואורך גובה התיבה: $h = 9 \text{ cm}$. מגדילים את הגובה ב- 1 mm ומקטינים את צלע הבסיס ב- 1 mm . מצא את שינוי נפח התיבה (באחוזים).
2. הרדיוס החיצוני (R) והרדיוס הפנימי (r) של הטבעת שווים בהתאמה: $r = 4 \text{ cm}$, $R = 5 \text{ cm}$. מגדילים את כל אחד מהרדיוסים ב- 1 mm . מצא את שינוי שטח הטבעת (באחוזים).
3. בגליל ישר רדיוס הבסיס: $R = 5 \text{ cm}$ וגובה הגליל: $h = 7 \text{ cm}$. מגדילים את רדיוס הבסיס ב- 0.8 mm ומקטינים את הגובה ב- 2 mm . מצא את שינוי נפח הגליל (באחוזים).
4. אורך אחד מאלכסוני המעויין: $a = 20 \text{ cm}$ ואורך האלכסון השני: $b = 10 \text{ cm}$. מקטינים את האלכסון הגדול ב- 4 mm ומגדילים את האלכסון הקטן ב- 2 mm . מצא את שינוי שטח המעויין (באחוזים).
5. במעויין אורך הצלע: $a = 6 \text{ cm}$ והזווית החדה: $\alpha = 45^\circ$. מגדילים את צלע המעויין ב- 1.5 mm ומקטינים את הזווית החדה ב- 1.5° . מצא את שינוי שטח המעויין (באחוזים).
6. במשולש ישר זווית אחת מהזוויות החדות: $\alpha = 30^\circ$ והניצב שלידה: $b = 10 \text{ cm}$. מגדילים את הזווית α ב- 1° ומקטינים את הניצב שלידה ב- 2.5 mm . מצא את שינוי שטח המשולש (באחוזים).

ג. חישוב שגיאות

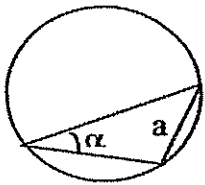
1. מודדים את אורכי צלעות המלבן: $a = 1 \text{ m} \pm 2 \text{ cm}$, $b = 0.5 \text{ m} \pm 2 \text{ cm}$. מצא את שטח המלבן.
2. נתון נפח של מנסרה: $V = (1000 \pm 20) \text{ cm}^3$ ושטח הבסיס שלה: $S = (90 \pm 1) \text{ cm}^2$. מצא את גובה המנסרה.
3. נתון שטח של מעויין: $S = (85 \pm 1) \text{ cm}^2$ ואורך אחד מהאלכסונים שלו: $a = (9.0 \pm 0.2) \text{ cm}$. מצא את אורך האלכסון השני.

4. נתונות מידות של תיבה: $a = (20.0 \pm 0.3) \text{ cm}$, $b = (14.0 \pm 0.2) \text{ cm}$, $c = (5.0 \pm 0.1) \text{ cm}$. מצא את אורך אלכסון התיבה.

5. נתון שטח של גזרה עיגולית: $S = (33 \pm 1) \text{ cm}^2$ ורדיוס הגזרה: $R = 8.20 \text{ cm} \pm 0.5 \text{ mm}$. מצא את הזווית המרכזית של הגזרה (במעלות).

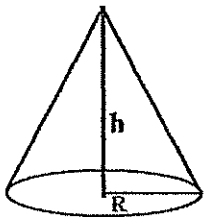
6. במעגל שרדיוסו: $R = 9 \text{ cm} \pm 2 \text{ mm}$ מודדים את אורך הקשת: $\ell = 21 \text{ cm} \pm 4 \text{ mm}$. מצא את הזווית המרכזית, הנשענת על הקשת הנמדדת (במעלות).

7. במשולש ישר זווית מודדים את היתר: $c = (75 \pm 2) \text{ m}$ ואחד מהניצבים: $a = (32 \pm 1) \text{ m}$. מצא את הזווית החדה שליוני הניצב הנמדד (במעלות).



8. במעגל בעל הקוטר: $d = (6.3 \pm 0.1) \text{ cm}$ מודדים אורך של המיתר: $a = (2.30 \pm 0.05) \text{ cm}$. מצא את הזווית ההיקפית α , הנשענת על המיתר.

9. המשולש ABC חסום במעגל. מודדים את אורך הצלע AC: $a = (13.4 \pm 0.1) \text{ cm}$ ואת הזווית $\angle B$, שמול הצלע AC: $\alpha = 52^\circ \pm 1^\circ$. מצא את שטח העיגול.



10. מודדים את המידות של החרוט הישר: $R = (7.0 \pm 0.2) \text{ cm}$, $h = (10.0 \pm 0.3) \text{ cm}$. מצא את נפח החרוט.

תשובות

א

1) 1.08, 2) -0.03, 3) 1.00, 4) 3.037, 5) 4.998, 6) 1.05

ב

1) -2%, 2) 2%, 3) 0.3%, 4) 0, 5) 2%, 6) -1%

ג

1) $(0.50 \pm 0.03) \text{ m}^2$, 2) $(11.1 \pm 0.3) \text{ cm}$, 3) $(18.9 \pm 0.6) \text{ cm}$, 4) $(25.0 \pm 0.4) \text{ cm}$, 5) $56^\circ \pm 2^\circ$, 6) $134^\circ \pm 6^\circ$, 7) $64.8^\circ \pm 1.6^\circ$, 8) $21.4^\circ \pm 0.8^\circ$, 9) $(227 \pm 10) \text{ cm}^2$, 10) $(510 \pm 40) \text{ cm}^3$

תרגיל מס' 4

נגזרות של פונקציה מורכבת, פונקציה סתומה ויישומים של נגזרות חלקיות בהנדסת המרחב

נגזרות של פונקציה מורכבת ופונקציה סתומה

(1) חשב את הנגזרות החלקיות $\frac{\partial z}{\partial y}$ ו- $\frac{\partial z}{\partial x}$ אם נתון: $z = u^2 \ln v$, $u = \frac{y}{x}$, $v = x^2 + y^2$

(2) חשב את הנגזרות החלקיות $\frac{\partial z}{\partial y}$ ו- $\frac{\partial z}{\partial x}$ אם נתון: $z = u^2 v - v^2 u$, $u = x \sin y$, $v = y \cos x$

(3) הראה כי הפונקציה $z = y \cdot \varphi(\cos(x-y))$ מהווה פתרון של המשוואה: $\frac{\partial z}{\partial x} + \frac{\partial z}{\partial y} = \frac{z}{y}$, כאשר φ היא פונקציה כלשהי.

הוכח:

(4) הפונקציה: $z = f(x, y)$, כאשר $x = r \cos \varphi$, $y = r \sin \varphi$, מקיימת את השוויון:

$$\left(\frac{\partial z}{\partial r}\right)^2 + \left(\frac{1}{r} \cdot \frac{\partial z}{\partial \varphi}\right)^2 = \left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2$$

(5) הפונקציה: $z = f(x + ay)$ מקיימת את המשוואה: $\frac{\partial z}{\partial y} = a \frac{\partial z}{\partial x}$

(6) הפונקציה: $z = y \cdot \varphi(x^2 - y^2)$ מקיימת את המשוואה: $\frac{1}{x} \cdot \frac{\partial z}{\partial x} + \frac{1}{y} \frac{\partial z}{\partial y} = \frac{z}{y^2}$

(7) הפונקציה: $p = f(x + mt, y + nt)$ מקיימת את המשוואה: $m \frac{\partial p}{\partial x} + n \frac{\partial p}{\partial y} = \frac{\partial p}{\partial t}$

(8) הפונקציה $z = f(x, y)$ מוגדרת על-ידי: $z^3 \ln\left(\frac{ex}{\pi}\right) - z(z+1)\sin y + 2\cos x = 0$

מצא את ערך הנגזרת $\frac{\partial z}{\partial x}$ בנקודה $(\pi, \pi/2)$.

(9) הפונקציה $z = f(x, y)$ מוגדרת על-ידי: $e^{\sin z} = z^2 xy$

חשב את הסכום: $\frac{\partial z}{\partial x} + \frac{\partial z}{\partial y}$ בנקודה שבה מתקיים: $x = y$, $z = \pi$

(10) הפונקציה $z = z(x, y)$ מוגדרת על-ידי: $z \cdot \ln(x+z) - \frac{xy}{z} = 0$

חשב את הנגזרות $\frac{\partial z}{\partial y}$ ו- $\frac{\partial z}{\partial x}$

(11) מצא את הדיפרנציאל dz , אם מתקיים: $yz = \arctan(xz)$

יישומים של נגזרות חלקיות בהנדסת המרחב

א. מצא את משוואת הנורמל למשטח $F(x, y, z) = 0$ בנקודה A :

- 1) $x^2 - y^2 - z^2 = 0$, $A(5, 4, 3)$ 2) $z = \sin \frac{x}{y}$, $A(\pi, 1, 0)$
 3) $(z^2 - x^2)xyz - y^5 = 5$, $A(1, 1, 2)$

ב. מצא את משוואת המישור המשיק למשטח $F(x, y, z) = 0$ בנקודה A :

- 1) $x^2 + y^2 + z^2 = 9$, $A(1, 2, 2)$ 2) $z = \frac{x^2}{4} + \frac{y^2}{9}$, $A(2, 3, 2)$
 3) $xy - z = 0$, $A(3, 4, 12)$ 4) $e^z - z + xy = 3$, $A(2, 1, 0)$
 5) $z = x^2 - y^2$, $A(1, 1/2, 3/4)$ 6) $xe^{x(y-1)} - z = 0$, $A(1, 2, e)$

תשובות

נגזרות של פונקציה מורכבת ופונקציה סתומה

- 1) $z'_x = 2u \left(\frac{u}{v}x - \frac{y}{x^2} \ln v \right)$, $z'_y = 2u \left(\frac{\ln v}{x} + y \frac{u}{v} \right)$,
 2) $z'_x = (2uv - v^2) \sin y - (u^2 - 2uv)y \sin x$, $z'_y = (2uv - v^2)x \cos y + (u^2 - 2uv) \cos x$,
 8) $-\frac{8}{7\pi}$, 9) $\frac{2\pi^2}{2+\pi}$, 10) $z'_x = \frac{yz(x+z) - z^3}{z^3 + 2xy(x+z)}$, $z'_y = \frac{xz(x+z)}{z^3 + 2xy(x+z)}$, 11) $dz = \frac{z dx - z(1+x^2z^2)dy}{y(1+x^2z^2) - x}$

יישומים של נגזרות חלקיות בהנדסת המרחב

- א**
- 1) $\frac{x-5}{5} = \frac{y-4}{-4} = \frac{z-3}{-3}$, 2) $\frac{x-\pi}{-1} = \frac{y-1}{\pi} = \frac{z}{-1}$, 3) $\frac{x-1}{2} = \frac{y-1}{1} = \frac{z-2}{11}$

- ב**
- 1) $x+2y+2z-9=0$, 2) $3x+2y-3z-6=0$, 3) $4x+3y-z-12=0$, 4) $x+2y-4=0$
 5) $8x-4y-4z-3=0$, 6) $2ex+ey-z-3e=0$

תרגיל מס' 5

נגזרת לפי כיוון (נגזרת מכוונת) וגרדיאנט של פונקציה

מצא את הנגזרת של השדה הסקלרי $u = f(x, y, z)$ בנקודה A בכיוון הווקטור $\bar{a}$:

$$1) u = xy + yz + zx, \quad A(2, 1, 3), \quad \bar{a} = 3\bar{i} + 4\bar{j} + 12\bar{k}$$

$$2) u = \ln(x^2 + y^2 + z^2), \quad A(1, 2, 1), \quad \bar{a} = 2\bar{i} + 4\bar{j} + 4\bar{k}$$

$$3) u = \ln(e^x + e^y + e^z), \quad A(0, 0, 0), \quad \bar{a} = \bar{i} + \bar{j} + \bar{k}$$

$$4) u = \frac{x^2}{4} + \frac{y^2}{9} - z^2, \quad A(2, 3, 1), \quad \bar{a} = 4\bar{i} - 3\bar{j}$$

מצא את הנגזרת של השדה הסקלרי $u = f(x, y, z)$ בנקודה A בכיוון הווקטור $\overline{AN}$:

$$5) u = x^2 - 2xz + y^2, \quad A(1, 2, -1), \quad N(2, 4, -3)$$

$$6) u = \arcsin \frac{z}{\sqrt{x^2 + y^2}}, \quad A(1, 1, 1), \quad N(3, 2, 3)$$

מצא את הגרדיאנט של השדה הסקלרי $u = f(x, y, z)$ בנקודה A :

$$7) u = xyz, \quad A(2, 1, 1)$$

$$8) u = x^2 + y^2 - z^2, \quad A(1, -1, 2)$$

$$9) u = \arctan(xyz) + \operatorname{arccot}(xyz), \quad A(x, y, z)$$

$$10) u = \sqrt{x^2 + y^2 + z^2}, \quad A(x, y, z)$$

(11) מצא את הנגזרת של השדה הסקלרי $u = \arctan xy$ בנקודה $P(1, 1)$ שנמצאת על הפרבולה $y = x^2$ בכיוון של המשיק לפרבולה בנקודה P (בכיוון העלייה של y).

(12) מצא את הנגזרת של השדה הסקלרי $u = \ln(x^2 + y^2)$ בנקודה $P(1, 2)$ שנמצאת על הפרבולה $y^2 = 4x$ בכיוון של המשיק לפרבולה בנקודה P (בכיוון העלייה של y).

(13) מצא את הנגזרת של השדה הסקלרי $u = \arctan \frac{y}{x}$ בנקודה $P(2, -2)$ שנמצאת על המעגל $x^2 + y^2 - 4x = 0$ בכיוון של המשיק למעגל בנקודה P (בכיוון העלייה של x).

(14) מצא את הנגזרת של השדה הסקלרי $u = 2xy + y^2$ בנקודה $P(\sqrt{2}, 1)$ שנמצאת על האליפסה $\frac{x^2}{4} + \frac{y^2}{2} = 1$ בכיוון של הנורמל החיצוני לאליפסה בנקודה P .

(15) תהיו $u = u(x, y, z)$ ו- $v = v(x, y, z)$ פונקציות דיפרנציאביליות בנקודה $A(x, y, z)$. הוכח שבנקודה A מתקיים:

$$a) \operatorname{grad}(\lambda u) = \lambda \operatorname{grad} u \quad (\lambda = \text{const})$$

$$b) \operatorname{grad}(u + v) = \operatorname{grad} u + \operatorname{grad} v$$

$$c) \operatorname{grad}(u \cdot v) = v \operatorname{grad} u + u \operatorname{grad} v$$

$$d) \operatorname{grad}\left(\frac{u}{v}\right) = \frac{v \operatorname{grad} u - u \operatorname{grad} v}{v^2} \quad (v \neq 0)$$

16) נתון השדה הסקלרי $u = x^2 y \sqrt{z}$. פי כמה קצב שינוי הפונקציה u בנקודה $A(2,1,1)$ בכיוון של הווקטור $\vec{a} = \vec{i} + 2\vec{j} + 2\vec{k}$ קטן מקצב השינוי המקסימלי של הפונקציה u בנקודה A ?

17) מצא את הגרדיאנט של המכפלה הסקלרית $\vec{c} \cdot \vec{r}$ כאשר $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$ ו- $\vec{c}$ הוא וקטור קבוע.

18) נתונות שתי הפונקציות: $v = x + y + 2\sqrt{xy}$, $u = \sqrt{x^2 + y^2}$. מצא את הזווית ϕ בין $\text{grad } u$ ל- $\text{grad } v$ בנקודה $A(1,1)$.

19) הוכח שעבור הווקטור $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$ מתקיים: $\text{grad}(|\vec{r}|^n) = n|\vec{r}|^{n-2} \vec{r}$.

20) חשב את הזווית ϕ בין הגרדיאנטים של השדה הסקלרי $u = \arctan \frac{x}{y}$ בנקודות $A(1,1)$ ו- $B(-1,-1)$.

21) מצא מקום גיאומטרי של נקודות שבהן עבור הגרדיאנט $\vec{g}$ של השדה הסקלרי $u = \sin(x+y)$ מתקיים: $\vec{g} = \vec{i} + \vec{j}$.

22) נתון השדה הסקלרי $u = x^2 y + 2xz$. מצא את וקטור היחידה של נורמל למשטח הרמה של השדה בנקודה $A(2,-2,3)$.

23) הנקודה $A(1,-1,2)$ נמצאת על משטח הרמה של השדה הסקלרי $u = 2xz^2 - 3xy - 4x$. מצא את משוואת המשטח ואת משוואת המישור המשיק למשטח בנקודה A .

24) מצא את הגרדיאנט של השדה הסקלרי $u = (\vec{a} \cdot \vec{r}) \cdot (\vec{b} \cdot \vec{r})$ כאשר $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$ ו- $\vec{a}, \vec{b}$ הם וקטורים קבועים.

25) מצא את הנגזרת של השדה הסקלרי $u = yze^x$ בנקודה $A(0,0,1)$ בכיוון של גרדיאנט השדה בנקודה A .

26) נתונים שני שדות סקלריים: $u = u(x,y,z)$ ו- $v = v(x,y,z)$ שעבורם מתקיים: בנקודה A הווקטורים $\text{grad } u$ ו- $\text{grad } v$ מאונכים זה לזה.

חשב את הנגזרת $\frac{\partial u}{\partial \ell}$ בנקודה A בכיוון של $\text{grad } v$.

27) מצא מקום גיאומטרי של נקודות שבהן מודול של גרדיאנט של השדה הסקלרי $u = \ln \sqrt{x^2 + y^2 + z^2}$ שווה ל-1.

תרגיל מס' 6

נגזרות חלקיות מסדר גבוה

א. נגזרות חלקיות מסדר גבוה

$$(1) \text{ מצא את הנגזרות: } \frac{\partial^2 z}{\partial x^2}, \frac{\partial^2 z}{\partial x \partial y}, \frac{\partial^2 z}{\partial y^2} \text{ כאשר } z = c \sqrt{\frac{x^2}{a^2} + \frac{y^2}{b^2}}$$

$$(2) \text{ מצא את הנגזרות: } \frac{\partial^2 z}{\partial x^2}, \frac{\partial^2 z}{\partial x \partial y}, \frac{\partial^2 z}{\partial y^2} \text{ כאשר } z = \ln(x^2 + y)$$

$$(3) \text{ מצא את הנגזרת: } \frac{\partial^2 z}{\partial x \partial y} \text{ כאשר } z = \sqrt{2xy + y^2}$$

$$(4) \text{ מצא את הנגזרת: } \frac{\partial^2 r}{\partial x^2} \text{ כאשר } r = \sqrt{x^2 + y^2 + z^2}$$

$$(5) \text{ מצא את הנגזרת: } \frac{\partial^3 z}{\partial x \partial y^2} \text{ כאשר } z = \sin(xy)$$

$$(6) \text{ הראה, כי } \frac{\partial^2 z}{\partial x \partial y} = \frac{\partial^2 z}{\partial y \partial x} \text{ כאשר } z = x^y$$

$$(7) \text{ הראה, כי הפונקציה } u = \arctan \frac{y}{x} \text{ מהווה פתרון של משוואת לפלס:}$$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

$$(8) \text{ הראה, כי הפונקציה } u = \ln \frac{1}{r} \text{ , כאשר } r = \sqrt{(x-a)^2 + (y-b)^2} \text{ מהווה פתרון של משוואת לפלס.}$$

$$(9) \text{ הראה, כי הפונקציה } u(x,t) = A \sin(a\lambda t + \phi) \cdot \sin(\lambda x) \text{ מהווה פתרון של}$$

$$\text{משוואת הגלים: } \frac{\partial^2 u}{\partial t^2} = a^2 \frac{\partial^2 u}{\partial x^2}$$

תרגיל מס' 7

אקסטרמום פונקציה של מספר משתנים

מצא את נקודות הקיצון (מקסימום ומינימום לוקלי) של פונקציה נתונה:

$$1) z = x^2 - xy + y^2 + 9x - 6y + 20$$

$$2) z = y\sqrt{x} - y^2 - x + 6y$$

$$3) z = x^3 + 8y^3 - 6xy + 1$$

$$4) z = 2xy - 4x - 2y$$

$$5) z = e^{\frac{x}{2}}(x + y^2)$$

$$6) z = xy(6 - 2x - y)$$

$$7) z = x^2 + 3xy - 8\ln x - 6\ln y$$

$$8) z = (x^2 + y^2)e^{-(x^2 + y^2)} \quad (x^2 + y^2 < 1)$$

$$9) z = e^{x-y}(x^2 - 2y^2)$$

מצא ערך מקסימאלי וערך מינימאלי של פונקציה נתונה בתחום הנתון (מקסימום ומינימום מוחלט):

$$10) z = xy - x^2y - \frac{1}{2}xy^2, \quad D = \{(x, y) \mid 0 \leq x \leq 1, 0 \leq y \leq 2\}$$

$$11) z = x^3 + 3y^2 - 3xy, \quad D = \{(x, y) \mid 0 \leq x \leq 2, 0 \leq y \leq 1\}$$

$$12) z = x^2 - 10xy^2 + 10x$$

כאשר התחום מוגבל על-ידי הקווים: $x=1, y=0, 2x+7y-14=0$

$$13) z = x^2 + y^2 + 4xy$$

כאשר התחום מוגבל על-ידי הקווים: $x^2 + y^2 = 1, y=0, y = \sqrt{3}x$

בתרגילים הבאים השתמש בשיטת כופלי לגרנז'.

מצא נקודות אקסטרמום של פונקציה עם האילוץ הנתון (אקסטרמום בתנאי):

$$14) z = x^2 - y^2 + 1, \quad 4x - y = 15$$

$$15) z = x^3 + 3x + y^3 + 3y, \quad x^2 + y^2 = 2$$

$$16) z = \frac{1}{x} + \frac{1}{y}, \quad \frac{1}{x^2} + \frac{1}{y^2} = \frac{1}{2}$$

$$17) z = xy, \quad x^2 + y^2 = 2$$

$$18) z = \frac{4}{x} + \frac{1}{y}, \quad x + y = 3$$

$$19) z = x - y, \quad \tan x - 3\tan y = 0 \quad \left(0 < x < \frac{\pi}{2}, 0 < y < \frac{\pi}{2}\right)$$

$$20) u = xy^2z^3, \quad x+y+z=12 \quad (x>0, y>0, z>0)$$

$$21) u = 2x+y-z+1, \quad x^2+y^2+2z^2=22$$

$$22) u = xyz, \quad xy+xz+yz=3 \quad (x>0, y>0, z>0)$$

$$23) u = x-2y+2z, \quad x^2+y^2+z^2=1$$

(24) בדוק, האם לפונקציה $u = x+y+z$ עם האילוץ $\frac{1}{x} + \frac{1}{y} + \frac{1}{z} = 1$ יש אקסטרמום בתנאי בנקודה $A(1,1,-1)$.

(25) האם נכונה הטענה: לפונקציה $u = xy+yz$ עם האילוצים $y+z=2$, $x^2+y^2=2$ יש מקסימום בתנאי בנקודה $A(1,1,1)$?

(26) האם נכונה הטענה: לפונקציה $u = xyz$ עם האילוצים $x+y+z=5$, $xy+yz+zx=8$ יש מקסימום בתנאי בנקודה $A(1,2,2)$?

(27) מצא נקודות אקסטרמום של הפונקציה $u = x^2+y^2+z^2$ עם האילוצים $x^2+y^2=2$, $x+y+2z=0$.

(28) מצא ערך מקסימאלי וערך מינימאלי של הפונקציה $z = 1-x^2-y^2$ בתחום $D = \{(x,y) \mid (x-1)^2 + (y-1)^2 \leq 1\}$.

(29) מצא ערך מקסימאלי וערך מינימאלי של הפונקציה $z = 4x-y$ בתחום הסגור שמוגבל על-ידי הקווים: $x^2-y^2=15$, $x=2\sqrt{6}$, $y=0$.

תרגיל מס' 8

האינטגרל הכפול

א' רשום את האינטגרל הכפול $I = \iint_D f(x,y) dS$ באמצעות האינטגרל החוזר אם תחום האינטגרציה D מוגבל על-ידי קווים הבאים (עשה שרטוט):

- 1) $y = x^2 - 2, y = -x^2 \quad (x \leq 0)$
- 2) $y = \sqrt{2-x^2}, y = x^2 \quad (x \leq 0)$
- 3) $x^2 + y^2 = 2, x = 0, y = x, \quad (y \geq 0)$
- 4) $y = \cos x, y = \sin x \quad 0 \leq x \leq \frac{\pi}{4}$
- 5) $y = -\sqrt{2-x^2}, y = x, y = 0 \quad (x \leq 0)$
- 6) $y = -\sqrt{2-x^2}, y = -\sqrt{x}, y = 0 \quad (x \geq 0)$
- 7) $y = e^{-x}, y = x^2, x = 0, x = -1$
- 8) $(y+2)^2 + x^2 = 4, x^2 + y^2 = 4, y = 0 \quad (x < 0)$
- 9) $x^2 + y^2 = 1, y = \ln x, y = 1$
- 10) $y = \arcsin x, y = -\frac{\pi}{2}x + \pi, x = 0$
- 11) $y = \ln x, y = x^2, y = 0, y = 1$

ב. חשב את האינטגרלים:

- 1) $\int_0^1 dy \int_1^2 (x^2 + y^2) dx$
- 2) $\int_3^4 dx \int_1^2 \frac{dy}{(x+y)^2}$
- 3) $\int_0^{2\pi} d\theta \int_{a \sin \theta}^a r dr$
- 4) $\int_0^a dx \int_{x/a}^x \frac{x}{x^2 + y^2} dy$
- 5) $\int_0^a dy \int_{y-a}^{2y} xy dx$
- 6) $\int_{b/2}^b dp \int_0^{\pi/2} \rho d\theta$

ג. מצא את נפח הגוף, הנמצא מתחת לגוף הפונקציה $z = f(x,y)$ בתחום D , המוגבל על-ידי הקווים הנתונים:

- 1) $z = 4 - x^2 - y^2 \quad D: x = 0, x = 1, y = 0, y = 3/2$
- 2) $z = x + y \quad D: x = 0, y = 1, y = x$
- 3) $z = x^2 + y \quad D: y = x^2, y^2 = x$
- 4) $z = x^2/y^2 \quad D: x = 2, y = x, xy = 1$
- 5) $z = \sqrt{1-x^2-y^2} \quad D: x^2 + y^2 = 1, x = 0, y = 0 \quad (x \geq 0, y \geq 0)$
- 6) $z = \frac{y}{e^{3x}} \quad D: x = 0, y = e^x, y = e^{2x} - 2$
- 7) $z = e^{x+y} \quad D: x = 1, x = \ln 2, y = x, y = \ln x$

תרגיל מס' 9

שינוי סדר האינטגרציה

החלף את סדר האינטגרציה באינטגרלים הבאים:

$$1) \int_0^1 dy \int_0^{1-y} f(x, y) dx$$

$$2) \int_{-a}^a dx \int_{-\sqrt{a^2-x^2}}^{\sqrt{a^2-x^2}} f(x, y) dy$$

$$3) \int_{-1}^1 dy \int_{-1}^y f(x, y) dx$$

$$4) \int_0^a dy \int_y^{y+2a} f(x, y) dx$$

$$5) \int_0^2 dx \int_{-\sqrt{2x-x^2}}^{\sqrt{2x-x^2}} f(x, y) dy$$

$$6) \int_0^4 dx \int_{3x^2}^{12x} f(x, y) dy$$

$$7) \int_0^1 dx \int_{2x}^{3x} f(x, y) dy$$

$$8) \int_0^a dx \int_{\frac{1}{2a}(a^2-x^2)}^{\sqrt{a^2-x^2}} f(x, y) dy$$

$$9) \int_{a/2}^a dx \int_0^{\sqrt{2ax-x^2}} f(x, y) dy$$

$$10) \int_{-6}^2 dy \int_{\frac{y^2}{4}-1}^{2-y} f(x, y) dx$$

$$11) \int_1^3 dx \int_{x^2}^{x+9} f(x, y) dy$$

$$12) \int_0^4 dy \int_y^{10-y} f(x, y) dx$$

$$13) \int_1^3 dx \int_{x/3}^{2x} f(x, y) dy$$

$$14) \int_0^3 dx \int_0^{\sqrt{25-x^2}} f(x, y) dy$$

$$15) \int_{-1}^2 dx \int_{x^2}^{x+2} f(x, y) dy$$

$$16) \int_0^{2a} dx \int_{\sqrt{2ax-x^2}}^{\sqrt{4ax}} f(x, y) dy$$

$$17) \int_0^1 dy \int_{-\sqrt{1-y^2}}^{1-y} f(x, y) dx$$

$$18) \int_0^1 dy \int_{y^2/2}^{\sqrt{3-y^2}} f(x, y) dx$$

תרגיל מס' 10

האינטגרל הכפול בקואורדינטות קוטביות

א. רשום את האינטגרל הכפול $I = \iint_D f(x,y) dx dy$ בקואורדינטות קוטביות וקבע את

גבולות האינטגרציה עבור התחום D שמוגבל על-ידי הקווים הנתונים:

- | | |
|---|--|
| 1) $x^2 + y^2 = a^2$ | 2) $x^2 + y^2 = 2Rx$ |
| 3) $x^2 + y^2 = 4$, $x^2 + y^2 = 1$, $x=0$, $y=0$
($x \geq 0$, $y \geq 0$) | 4) $x^2 + y^2 = 4x$, $y=x$, $y=-x$
($y \leq x$, $y \geq -x$) |
| 5) $x^2 + y^2 = R^2$, $y=x$, $y=0$
($y \geq 0$, $y \geq x$) | 6) $x^2 + y^2 = 2x$, $x^2 + y^2 = x$,
$y = \sqrt{3}x/3$, $y = \sqrt{3}x$ |
| 7) $y=0$, $y=1$, $x=0$, $x=1$ | 8) $x^2 + y^2 = 4$, $x+y=2$
($x \geq 0$, $y \geq 0$) |

ב. השתמש בקואורדינטות קוטביות כדי לחשב את האינטגרל הכפול הנתון בתחום D שמוגבל על-ידי הקווים הנתונים:

- 1) $\iint_D (x^2 + y^2) dx dy$, $D: x^2 + y^2 = R^2$ ($y \geq 0$)
- 2) $\iint_D e^{x^2+y^2} dx dy$, $D: x^2 + y^2 = 1$ ($x \geq 0$, $y \geq 0$)
- 3) $\iint_D \sqrt{x^2 + y^2} dx dy$, $D: x^2 + y^2 = a^2$, $x=0$, $y=0$ ($x \geq 0$, $y \geq 0$)
- 4) $\iint_D \ln(x^2 + y^2) dx dy$, $D: x^2 + y^2 = e^2$, $x^2 + y^2 = e^4$
- 5) $\iint_D \frac{\sin \sqrt{x^2 + y^2}}{\sqrt{x^2 + y^2}} dx dy$, $D: x^2 + y^2 = \frac{\pi^2}{9}$, $x^2 + y^2 = \pi^2$
- 6) $\iint_D \frac{\ln(x^2 + y^2)}{x^2 + y^2} dx dy$, $D: x^2 + y^2 = 1$, $x^2 + y^2 = e^2$
- 7) $\iint_D \frac{dx dy}{x^2 + y^2 + 1}$, $D: x^2 + y^2 = 1$, $y=0$ ($y \geq 0$)
- 8) $\iint_D \left(1 - \frac{y^2}{x^2}\right) dx dy$, $D: x^2 + y^2 = \pi^2$, $y=x$, $y=0$ ($y \geq 0$, $y < x$)
- 9) $\iint_D (x^2 + y^2) dx dy$, $D: x^2 + y^2 = 2ax$
- 10) $\iint_D (3a - x - y) dx dy$, $D: x^2 + y^2 = a^2$
- 11) $\iint_D \arctan \frac{y}{x} dx dy$, $D: x^2 + y^2 = 1$, $x^2 + y^2 = 9$, $y = \sqrt{3}x/3$, $y = \sqrt{3}x$
($y \geq \sqrt{3}x/3$, $y \leq \sqrt{3}x$)

תרגיל מס' 11

החלפת המשתנים באינטגרל כפול. מקרה כללי

כדי לחשב את האינטגרל $I = \iint_D f(x,y) dx dy$ כאשר התחום D מוגבל על-ידי קווים

נתונים, עושים את התמרת הקואורדינטות T באמצעות הנוסחאות הנתונות:

$$T: \varphi(x,y)=u, \quad \psi(x,y)=v$$

א. מצא את תמונת התחום D במישור UV .

ב. רשום את האינטגרל הכפול I באינטגרל חוזר בקואורדינטות החדשות u, v .

1) $D: y=2x-3, y=2x+5, y=-x, y=5-x, \quad T: y-2x=u, x+y=v$

2) $D: y=e^x, y=2e^x, y=2e^{-x}, y=4e^{-x}, \quad T: ye^{-x}=u, ye^x=v$

3) $D: x^2+y^2=2x, x^2+y^2=4x, y=x, y=2x, \quad T: \frac{x^2+y^2}{x}=u, \frac{y}{x}=v$

4) $D: y=x, y=-x, x=1, \quad T: x-y=u, x+y=v$

חשב את האינטגרל הנתון I כאשר התחום D מוגבל על-ידי קווים נתונים:

5) $I = \iint_D \frac{y-x}{2x+y} dx dy, \quad D: y=x+4, y=x+1, y=6-2x, y=8-2x$

6) $I = \iint_D \sqrt{36-4x^2-9y^2} dx dy, \quad D: \frac{x^2}{9} + \frac{y^2}{4} = 1$

7) $I = \iint_D \frac{x^2}{y^4} \cos\left(\frac{x^2}{y}\right) dx dy, \quad D: y=2x^2, y=4x^2, y=\sqrt{x}, y=\sqrt{2x}$

8) $I = \iint_D (x^2+2xy) dx dy, \quad D: y=\frac{1}{x^2}, y=\frac{4}{x^2}, y=x+1, y=x-1$

9) $I = \iint_D \frac{(x+y)^3}{x^2} dx dy, \quad D: y=1-x, y=2-x, y=\frac{1}{2}x, y=2x$

10) $I = \iint_D \cos \frac{x-y}{x+y} dx dy, \quad D: y=0, x=0, y=1-x$
(השתמש בהחלפת המשתנים: $x+y=v, x-y=u$)

11) $I = \iint_D e^{\frac{x-y}{x+y}} dx dy, \quad D: y=1-x, y=2-x, y=0, x=0$
(השתמש בהחלפת המשתנים: $x+y=v, x-y=u$)

12) $I = \iint_D \cos \frac{(x+y)^2}{2} dx dy, \quad D: y=1-x, y=2-x, y=2x, x=0$

(13) מצא נפח מתחת לגרף הפונקציה: $f(x,y) = x^3y + xy^3$ בתחום D , שמוגבל על-ידי שני המעגלים: $x^2 + y^2 = 9$, $x^2 + y^2 = 16$ ועל-ידי שתי ההיפרבולות: $x^2 - y^2 = 1$, $x^2 - y^2 = 4$, ברביע הראשון.

(14) מצא שטח של צורה, שמוגבלת על-ידי הישרים:
 $y = x+2$, $y = x$, $y = 5-2x$, $y = 8-2x$.

(15) מצא מסה של לוחית, שמוגבלת על-ידי הקווים:
 $y = \sqrt{x}$, $y = 2\sqrt{x}$, $y = \frac{1}{x}$, $y = \frac{2}{x}$
 כאשר הצפיפות המשטחית γ של חומר הלוחית:
 $\gamma = f(x,y) = \left(\ln y - \frac{1}{2} \ln x \right)^2$.

(16) מצא נפח מתחת לפרבולואיד: $z = 16 - x^2 - y^2$ בתחום D , שמוגבל על-ידי האליפסה:
 $D = \left\{ (x,y) \mid \frac{x^2}{16} + \frac{y^2}{4} \leq 1 \right\}$

(17) מצא שטח של צורה, שמוגבלת על-ידי הקווים:
 $x^2 + 4y^2 = 8$, $x^2 + 4y^2 = 16$, $y = x$, $y = 2x$
 ברביע הראשון.

(18) מצא מסה של לוחית, שמוגבלת על-ידי שתי האליפסות ושתי ההיפרבולות:
 $\frac{x^2}{6} + \frac{y^2}{4} = 1$, $\frac{x^2}{3} + \frac{y^2}{2} = 1$, $2x^2 - 3y^2 = 1$, $4x^2 - 6y^2 = 1$ ($x > 0, y > 0$)
 כאשר הצפיפות המשטחית γ של חומר הלוחית: $\gamma = f(x,y) = 16xy$.

(19) מצא שטח של צורה, שמוגבלת על-ידי הקווים:
 $x^2 + y^2 = 2x$, $x^2 + y^2 = 4x$, $y = -\frac{\sqrt{2}}{2}x$, $y = \frac{\sqrt{2}}{2}x$.

תרגיל מס' 12

האינטגרל המשולש

א. רשום את האינטגרל המשולש $I = \iiint_V f(x,y,z) dx dy dz$ באמצעות האינטגרל החוזר אם

תחום האינטגרציה V מוגבל על-ידי:

- (1) מישורים: $y=0, z=0, x=2, z=5, y=2x$
 - (2) מישורים: $y=0, z=0, x=3, y=4, z=3x$
 - (3) מישורים: $x=0, y=0, z=0, x+y+z=1$
 - (4) מישור: $z=0$, כדור: $x^2+y^2+z^2=1$ ($z \geq 0$)
 - (5) מישורים: $x=0, z=0, x+y+z=2$, משטח גלילי: $y=\sqrt{x}$
 - (6) מישורים: $x=0, y=0, z=0, x+y=1$, פרבולואיד: $z=x^2+y^2$
 - (7) מישורים: $x=0, z=0, y=1-x, y=x-1, z=0, x=0$ ($x \geq 0$), פרבולואיד: $z=1-x^2-y^2$
 - (8) מישור: $z=0$, פרבולואיד: $z=4-x^2-y^2$, גליל: $x^2+y^2=2x$
- (כך שהנקודה $(1,0,0)$ היא בתוך התחום)

ב. מצא את מסת הגוף, המוגבל על-ידי משטחים נתונים, כאשר $\gamma = f(x,y,z)$ היא צפיפות בכל נקודה של הגוף:

- (1) קובייה: $x=0, x=a, y=0, y=a, z=0, z=a$
 $f(x,y,z)=x+y+z$
- (2) מנסרה: $x=0, y=0, y=3, z=0, x+z=2$
 $f(x,y,z)=x$
- (3) פירמידה: $x=0, y=0, z=0, x+y+z=a$
 $f(x,y,z)=z$
- (4) חלק של כדור: $x \geq 0, y \geq 0, z \geq 0, x^2+y^2+z^2=1$
 $f(x,y,z)=xyz$
- (5) פירמידה: $x=0, y=0, z=0, x+y+z=1$
 $f(x,y,z)=\frac{1}{(x+y+z+1)^3}$
- (6) חלק של פרבולואיד היפרבולי: $x=1, y=0, y=x, z=0, z=xy$
 $f(x,y,z)=xy^2z^3$
- (7) מנסרה: $x=0, y=0, z=0, z=a, x+y=b$
 $f(x,y,z)=2x+3y-z$
- (8) קובייה: $x=1, x=2, y=1, y=2, z=1, z=2$
 $f(x,y,z)=\frac{1}{(x+y+z)^3}$

תרגיל מס' 13

האינטגרל המשולש בקואורדינטות גליליות וכדוריות

א. רשום את האינטגרל $I = \iiint_V f(x, y, z) dx dy dz$ בקואורדינטות גליליות וקבע את גבולות

האינטגרציה כאשר התחום V מוגבל על-ידי:

(1) מישורים: $x^2 + y^2 = R^2$, $z=0$, $z=1$, $y=x$, $y=\sqrt{3}x$, $(x \geq 0, y \geq 0)$ גליל:

(2) פרבולואיד: $az = x^2 + y^2$, כדור: $x^2 + y^2 + z^2 = 2a^2$

(3) חרוט: $z^2 = x^2 + y^2$, כדור: $x^2 + y^2 + z^2 = 2Rz$

(כך שהנקודה $(0,0,R)$ היא בתוך התחום)

(4) פרבולואיד: $z = x^2 + y^2$, גליל: $x^2 + y^2 = 2x$

(5) כדורים: $x^2 + y^2 + z^2 = R^2$, $x^2 + y^2 + (z-R)^2 = R^2$ (החלק המשותף של הכדורים)

רשום את האינטגרל $I = \iiint_V f(x, y, z) dx dy dz$ בקואורדינטות כדוריות וקבע את גבולות

האינטגרציה כאשר התחום V מוגבל על-ידי:

(6) כדור: $x^2 + y^2 + z^2 = R^2$, מישורים: $x=0$, $y=0$, $z=0$, $(x \geq 0, y \geq 0, z \geq 0)$

(7) כדורים: $x^2 + y^2 + z^2 = R^2$, $x^2 + y^2 + z^2 = 2R^2$ (שיכבה כדורית)

ב. חשב את האינטגרל $I = \iiint_V f(x, y, z) dx dy dz$ באמצעות מעבר לקואורדינטות גליליות

או כדוריות כאשר נתונה הפונקציה $f(x, y, z)$ והתחום V (המשטחים המגבילים)

1) $f(x, y, z) = \sqrt{x^2 + y^2 + z^2}$, $V: x^2 + y^2 + z^2 = R^2$

2) $f(x, y, z) = z$, $V: z^2 = x^2 + y^2$ ($z \geq 0$) , $z=h$

3) $f(x, y, z) = x^2 + y^2$, $V: 2z = x^2 + y^2$, $z=2$

4) $f(x, y, z) = \sqrt{1 + (x^2 + y^2 + z^2)^{3/2}}$, $V: x^2 + y^2 + z^2 = 1$

5) $f(x, y, z) = (x^2 + y^2 + z^2)^3$, $V: x^2 + y^2 = 1$, $z=0$, $z=1$

6) $f(x, y, z) = z$, $V: x^2 + y^2 + z^2 = R^2$, $x=0$, $y=0$, $z=0$ ($x \geq 0$, $y \geq 0$, $z \geq 0$)

7) $f(x, y, z) = \frac{1}{\sqrt{x^2 + y^2 + z^2}}$, $V: x^2 + y^2 + z^2 = a^2$, $x^2 + y^2 + z^2 = 4a^2$

חשב את נפח הגוף, שמוגבל על-ידי המשטחים הנתונים:

8) $az = x^2 + y^2$, $2az = a^2 - x^2 - y^2$

9) $z^2 = x^2 + y^2$, $z = 6 - x^2 - y^2$

10) $z^2 = x^2 + y^2$, $x^2 + y^2 + z^2 = a^2$ ($z \geq 0$)

תרגיל מס' 14

האינטגרל הקווי ממין ראשון והאינטגרל הקווי ממין שני

חשב את האינטגרל לפי המסלול הנתון:

- 1) $\int_L \sqrt{y} \, ds$, $L: x=a(t-\sin t), y=a(1-\cos t) \quad (0 \leq t \leq 2\pi)$
- 2) $\int_L \sqrt{x^2+y^2} \, ds$, $L: x=a(\cos t+t\sin t), y=a(\sin t-t\cos t) \quad (0 \leq t \leq 2\pi)$
- 3) $\int_{AB} x^2 \, ds$, $AB: y=\ln x, x_A=\sqrt{3}, x_B=2\sqrt{2}$
- 4) $\int_{AB} \frac{ds}{\sqrt{x^2+y^2}}$, $AB: (\text{ישר}) A(0,-2), B(4,0)$
- 5) $\int_L \frac{ds}{x^2+y^2+z^2}$, $L: x=a\cos t, y=a\sin t, z=bt \quad (0 \leq t \leq 2\pi)$
- 6) $\int_{AB} \frac{y}{\sqrt{x}} \, ds$, $AB: y^2=\frac{4}{9}x^3, x_A=3, x_B=8 \quad (y>0)$
- 7) $\int_L (2x+y) \, ds$, $L: OAB(\text{משולש}) O(0,0), A(1,0), B(0,2)$
- 8) $\int_L xy \, ds$, $L: OBCD(\text{מלבן}) O(0,0), B(4,0), C(4,2), D(0,2)$
- 9) $\int_{OB} y \, ds$, $OB: y=\sqrt{2x}, O(0,0), B(2,2)$
- 10) $\int_L (x^2+y^2)^n \, ds$, $L: x=a\cos t, y=a\sin t \quad (0 \leq t \leq 2\pi)$
- 11) $\int_L xy \, dx + y^2 \, dy$, $L: x=t^2, y=t \quad (1 \leq t \leq 2)$
- 12) $\int_L (x+y) \, dx + (x-y) \, dy$, $L: x=R\cos t, y=R\sin t \quad (0 \leq t \leq \pi/2)$
- 13) $\int_L (x+y) \, dx - x \, dy$, $L: OAB(\text{קו שבור}) O(0,0), A(2,0), B(4,2)$
- 14) $\oint_L x \, dy$, $L: OAB(\text{משולש}) O(0,0), A(2,0), B(2,2)$ (בכיוון חיובי)

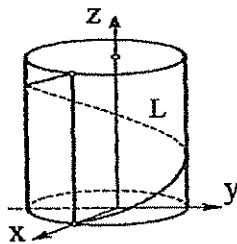
- 15) $\int_{OB} x dy - y dx$, $OB: y = x^3$, $O(0,0)$, $B(2,8)$
- 16) $\oint_L (x^2 - y) dx$, $L: (\text{מלבן}) x=0, y=0, x=1, y=2$ (בכיוון חיובי)
- 17) $\int_{OB} \sqrt{x} dx + \sqrt{y} dy$, $OB: y = x^2$, $O(0,0)$, $B(1,1)$
- 18) $\int_{OB} (x+y) dx$, $OB: y = x^2/2$, $O(0,0)$, $B(2,2)$
- 19) $\int_{OB} x^2 dx + y^2 dy$, $OB: y = \sqrt{x}$, $O(0,0)$, $B(1,1)$
- 20) $\int_{AB} \frac{dy}{x} - \frac{dx}{y}$, $AB: x = r \cos t, y = r \sin t$ ($\pi/6 \leq t \leq \pi/3$)
- 21) $\int_{AO} xy dx$, $AO: y = \sin x$, $A(\pi,0)$, $O(0,0)$
- 22) $\int_{AB} (x^2 - y^2) dx + xy dy$, $AB: (\text{ישר}) A(1,1), B(3,4)$
- 23) $\int_{OB} y dx - (y + x^2) dy$, $OB: y = 2x - x^2$, $O(0,0)$, $B(2,0)$
- 24) $\oint_L 2x dy - 3y dx$, $L: ABC (\text{משולש}) A(1,2), B(3,1), C(2,5)$ (בכיוון חיובי)
- 25) $\oint_L x^2 y dx + x^3 dy$, $L: (\text{מעגל סגור}) y^2 = x, y = x^2$ (בכיוון חיובי)
- 26) $\int_{OB} xy dx + yz dy + zx dz$, $OB: x = \cos t, y = \sin t, z = 1$ ($0 \leq t \leq \pi/2$)
- 27) $\int_{OB} x dx + y dy + z dz$, $OB: (\text{ישר}) O(0,0,0), B(1,2,2)$
- 28) $\int_{AB} (x^2 + y^2) dx + xy dy$, $AB: y = e^x$, $A(0,1), B(1,e)$
- 29) $\int_{OB} y dx + z dy + x dz$, $OB: (\text{ישר}) O(0,0,0), B(a,a,a)$
- 30) $\int_L x dy - y dx$, $L: x = a \cos^3 t, y = a \sin^3 t$ ($0 \leq t \leq 2\pi$)
- 31) $\int_L \frac{y^2 dx - x^2 dy}{x^2 + y^2}$, $L: x = a \cos t, y = a \sin t$ ($0 \leq t \leq \pi$)

(32) חשב את המסה של החוט בצורת רבע המעגל: $x^2 + y^2 = 1$ ($x \geq 0, y \geq 0$) כאשר הצפיפות הלינארית של חומר החוט: $\gamma = f(x, y) = x + y$.

(33) חשב את עבודת הכוח: $\vec{F} = -y^2 \vec{i} + x^2 \vec{j}$ לאורך החצי העליון של האליפסה: $\frac{x^2}{9} + \frac{y^2}{4} = 1$ (לפי כיוון השעון).

(34) חשב את עבודת הכוח: $\vec{F} = \arctan \frac{x}{y} \vec{j}$ לאורך קשת הפרבולה: $y = \sqrt{x}$ מהנקודה $O(0,0)$ עד הנקודה $A(1,1)$.

(35) חוט מתוח כלוא בין הנקודות: $A(0, -2)$ ו- $B(4, 0)$. מצא את מסת החוט, אם הצפיפות הלינארית של חומר החוט: $\gamma = f(x, y) = \frac{1}{x-y}$.

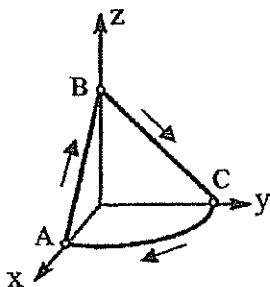


(36) המסלול L של חלקיק הוא קו הבורג: $x = a \cos t, y = a \sin t, z = bt$ ($0 \leq t \leq 2\pi$) אחד מהכוחות המופעלים על החלקיק: $\vec{F} = yz \vec{i} + xz \vec{j} + xy \vec{k}$ האם נכונה הטענה: עבודת הכוח $\vec{F}$ שווה ל-0:

(37) חשב את מסת החוט בצורת קו הבורג החרוטי: $x = t \cos t, y = t \sin t, z = t$ ($0 \leq t \leq 4\pi$) כאשר הצפיפות הלינארית של חומר החוט: $\gamma = f(x, y, z) = z$.

(38) גוף נקודתי נע בישר מהנקודה $O(0,0)$ לנקודה $B(\pi, 2\pi)$. מצא את עבודת הכוח $\vec{F} = -x \cos y \vec{i} + y \sin x \vec{j}$, שמופעל על הגוף.

(39) חשב את מסת החוט: $x = \ln(1+t^2), y = 2 \arctan t - t$ ($0 \leq t \leq 1$) כאשר הצפיפות הלינארית של חומר החוט: $\gamma = f(x, y) = e^{-x} \cdot y$.



(40) המסלול הסגור L של גוף נקודתי מורכב מהקטע AB של הישר $x+z=1$ במישור $y=0$ (ראה שרטוט), מהקטע BC של הישר $y+z=1$ במישור $x=0$ ומהחלק CA של המעגל $x^2 + y^2 = 1$ במישור $z=0$. מצא את עבודת הכוח $\vec{F} = xy \cos(xz) \vec{i} + \cos(xz) \vec{j} + xy \vec{k}$, שמופעל על הגוף.

תרגיל מס' 15

משפט גרין ואי-תלות של אינטגרל קווי במסלול האינטגרציה

א. השתמש במשפט גרין כדי לחשב אינטגרל קווי:

$$1) \oint_L (x-y)dx + x dy, \quad L: ABCD \text{ (ריבוע) } A(-a,a), B(-a,-a), C(a,-a), D(a,a), \text{ (בכיוון חיובי)}$$

$$2) \oint_L (x-y)dx + (x+y)dy, \quad L: \text{ (מעגל) } x^2 + y^2 = R^2, \text{ (בכיוון שלילי)}$$

$$3) \oint_L (x+y)dx - (x-y)dy, \quad L: \text{ (אליפסה) } \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, \text{ (בכיוון שלילי)}$$

$$4) \oint_L y dx + 2x dy, \quad L: ABCD \text{ (מעויין) } A(3,0), B(0,2), C(-3,0), D(0,-2), \text{ (בכיוון חיובי)}$$

$$5) \oint_L -x^2 y dx + xy^2 dy, \quad L: \text{ (מעגל) } x^2 + y^2 = R^2, \text{ (בכיוון שלילי)}$$

$$6) \oint_L y^2 dx + (x+y)^2 dy, \quad L: ABC \text{ (משולש) } A(a,0), B(a,a), C(0,a), \text{ (בכיוון חיובי)}$$

$$7) \oint_L \frac{dx}{y} - \frac{dy}{x}, \quad L: ABC \text{ (משולש) } A(1,1), B(2,1), C(2,2), \text{ (בכיוון חיובי)}$$

$$8) \oint_L 2(x^2 + y^2)dx + (x+y)^2 dy, \quad L: ABC \text{ (משולש) } A(1,1), B(2,2), C(1,3), \text{ (בכיוון חיובי)}$$

$$9) \oint_L \sqrt{x^2 + y^2} dx + y \left[xy + \ln(x + \sqrt{x^2 + y^2}) \right] dy, \quad L: ABCD \text{ (מלבן) } A(1,0), B(4,0), C(4,2), D(1,2), \text{ (בכיוון שלילי)}$$

$$10) \oint_L (xy + x + y)dx + (xy + x - y)dy, \quad L: \text{ (מעגל) } x^2 + y^2 = ax, \text{ (בכיוון חיובי)}$$

ב. בתרגילים הבאים השלם את מסלול האינטגרציה L למסלול סגור והשתמש במשפט גרין כדי לחשב את האינטגרל הנתון:

(1) חשב את האינטגרל: $I = \int_L (y + e^{x^2}) dx + (2x + ye^{y^2}) dy$, כאשר המסלול L הוא החצי

הימני של המעגל $x^2 + y^2 = 2y$ הכלוא בין הנקודות A(0,2) ו-O(0,0) לפי כיוון השעון.

(2) חשב את האינטגרל: $I = \int_L \vec{F} d\vec{r}$, כאשר $\vec{F} = (2xy + e^{y^2})\vec{i} + x(x + 2ye^{y^2} + 1)\vec{j}$

ומסלול האינטגרציה L הוא קשת של הפרבולה $y = 1 - x^2$, הכלואה בין הנקודות A(-1,0) ו-B(1,0) לפי כיוון השעון.

(3) חשב את האינטגרל: $I = \int_L \vec{F} d\vec{r}$, כאשר $\vec{F} = \frac{x+2}{\sqrt{(x+2)^2 + y^2}}\vec{i} + \left(\frac{y}{\sqrt{(x+2)^2 + y^2}} + 2x \right)\vec{j}$

ומסלול האינטגרציה L הוא החצי השמאלי של המעגל $x^2 + y^2 = 1$, נגד כיוון השעון.

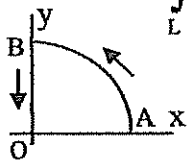
(4) חשב את האינטגרל: $I = \int_L (2y \cos^2 x + 2xye^{x^2y}) dx + x^2e^{x^2y} dy$, כאשר המסלול L

הוא חלק של גרף הפונקציה $y = \cos x$ בתחום $[-\pi/2, \pi/2]$ לפי כיוון השעון.

(5) חשב את האינטגרל: $I = \int_L \vec{F} d\vec{r}$, כאשר $\vec{F} = y^2\vec{i} + (x^2 + \sin(y^2))\vec{j}$, ומסלול

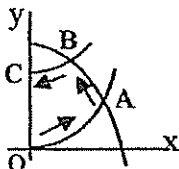
האינטגרציה L הוא החצי העליון של האליפסה $4x^2 + 9y^2 = 36$, נגד כיוון השעון.

(6) חשב את האינטגרל: $I = \int_L 2x \ln(1 + x^2 + y^2) dx + (2y + 1) \ln(1 + x^2 + y^2) dy$



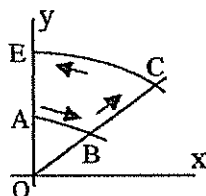
כאשר מסלול האינטגרציה L מורכב מרבע של המעגל $x^2 + y^2 = 1$ (הקשת AB) והקטע BO על ציר ה-Y (ראה שרטוט).

(7) חשב את האינטגרל: $I = \int_L (e^{\sin y} - 2x^5y) dx + (x \cos y e^{\sin y} - x^2y^2) dy$



כאשר מסלול האינטגרציה L מורכב מהקשת OA של הפרבולה $y = x^2$, מהקשת AB של הפרבולה $y = 5 - x^2$ ומהקשת BC של הפרבולה $y = x^2 + 4$ (ראה שרטוט).

(8) חשב: $I = \int_L \vec{F} d\vec{r}$, כאשר $\vec{F} = (x^2 \sin x - 2y^2)\vec{i} + \left(x^2 + \cos \frac{\pi y}{2} \right)\vec{j}$



ומסלול האינטגרציה L הוא הקשת AB של האליפסה $x^2 + 2y^2 = 2$, הקטע BC של הישר $y = x$ והקשת CE של האליפסה $x^2 + 2y^2 = 8$ (ראה שרטוט).

ג. בדוק, האם האינטגרל הנתון תלוי במסלול האינטגרציה. אם לאו, חשב אותו לפי מסלול כלשהו, שמחבר את שתי הנקודות הנתונות (כיוון האינטגרציה מוגדר על-ידי סימון המסלול. לדוגמה: המסלול AB פירושו - מהנקודה A לנקודה B).

$$1) I = \int_{AB} (y \cos(xy) - \cos y) dx + (x \cos(xy) - x \sin y) dy, \quad AB: A(-1,0), B(1,0)$$

$$2) I = \int_{OB} (x+y) dx + (x-y) dy, \quad OB: O(0,0), B(\pi, \pi)$$

$$3) I = \int_{OB} 2xy dx + x^2 dy, \quad OB: O(0,0), B(2,1)$$

$$4) I = \int_{AB} \left(\frac{1}{x+y} + \frac{x}{y} \right) dx + \left(\frac{1}{x+y} - \frac{x^2}{y^2} \right) dy, \quad x+y \neq 0, y \neq 0, \quad AB: A(2,1), B(3,5)$$

$$5) I = \int_{AB} \left(\frac{y}{x^2+y^2} - \frac{y}{x^2} \right) dx - \left(\frac{x}{x^2+y^2} + \frac{1}{x} \right) dy, \quad (x,y) \neq (0,0), x \neq 0, \\ AB: A(2,1), B(1,0)$$

$$6) I = \int_{AB} \cos 2y dx - 2x \sin 2y dy, \quad AB: A(1, \pi/6), B(2, \pi/4)$$

$$7) I = \int_{AB} \frac{x dx + y dy}{x^2 + y^2}, \quad (x,y) \neq (0,0), \quad AB: A(3,4), B(5,12)$$

$$8) I = \int_{AB} \left(\frac{x}{\sqrt{x^2+y^2}} + y \right) dx + \left(\frac{y}{\sqrt{x^2+y^2}} + x \right) dy, \quad (x,y) \neq (0,0), \\ AB: A(3,4), B(12,5)$$

ד. הוכח, שהשדה הוקטורי הנתון $\vec{F} = P(x,y)\vec{i} + Q(x,y)\vec{j}$ הוא שדה משמר, מצא

את פונקציית הפוטנציאל שלו וחשב את האינטגרל $\int_A^B \vec{F} d\vec{r}$ לאורך הקטע AB.

$$1) \vec{F} = x \cos x \vec{i} + \vec{j}, \quad A(2\pi, 1), B(\pi, -1)$$

$$2) \vec{F} = (1+xy)e^{xy} \vec{i} + x^2 e^{xy} \vec{j}, \quad A(0,1), B(2,0)$$

$$3) \vec{F} = y \ln x \vec{i} + x(\ln x - 1) \vec{j}, \quad A(1, \pi), B(e, 2\pi)$$

$$4) \vec{F} = 2xy \sin(2x^2y) \vec{i} + x^2 \sin(2x^2y) \vec{j}, \quad A(\sqrt{\pi}, 1/2), B(\pi, 0)$$

$$5) \vec{F} = \tan(x+2y) \vec{i} + 2 \tan(x+2y) \vec{j}, \quad A(0, -\pi), B(\pi/3, 0)$$

$$6) \vec{F} = \left(\frac{y}{x^2+y^2} - \frac{y}{x^2} \right) \vec{i} + \left(-\frac{x}{x^2+y^2} + \frac{1}{x} \right) \vec{j}, \quad A(2,0), B(3,3)$$

תרגיל מס' 16

האינטגרל המשטחי

חשב את האינטגרל המשטחי ממין ראשון $I = \iint_S f(x, y, z) dS$ לפי המשטח S , כאשר:

$$(1) \quad f(x, y, z) = z + 2x + \frac{4y}{3}, \quad S: \text{חלק של המישור } 6x + 4y + 3z = 12 \text{ באוקטנט הראשון} \\ (x \geq 0, y \geq 0, z \geq 0)$$

$$(2) \quad f(x, y, z) = xyz, \quad S: \text{חלק של המישור } x + y + z = 1 \text{ באוקטנט הראשון} \\ (x \geq 0, y \geq 0, z \geq 0)$$

$$(3) \quad f(x, y, z) = x^2 + y^2 + 3z^2, \quad S: \text{חלק של המשטח החרוטי } z = \sqrt{x^2 + y^2} \text{ בין המישורים:} \\ z = 1, z = 0$$

$$(4) \quad f(x, y, z) = x^2 + y^2 + z - 1/2, \quad S: \text{חלק של הפרבולואיד } 2z = 2 - x^2 - y^2, \quad z \geq 0$$

$$(5) \quad f(x, y, z) = x(y + z), \quad S: \text{חלק של המשטח הגלילי } x = \sqrt{1 - y^2} \text{ בין המישורים:} \\ z = 1, z = 0$$

$$(6) \quad f(x, y, z) = 3x^2 + 5y^2 + 3z^2 - 2, \quad S: \text{חלק של המשטח החרוטי } y = \sqrt{x^2 + z^2} \text{ בין} \\ y = 1, y = 0 \text{ המישורים}$$

$$(7) \quad f(x, y, z) = x^2 + y^2, \quad S: \text{ספירה (פני הכדור)} \quad x^2 + y^2 + z^2 = R^2$$

$$(8) \quad f(x, y, z) = x^2 + y^2 + z, \quad S: \text{החצי העליון של הספירה } z = \sqrt{R^2 - x^2 - y^2}$$

חשב את האינטגרל המשטחי ממין שני:

$$(9) \quad \iint_S \vec{F} d\vec{S}, \quad \text{כאשר } \vec{F} = (x^2 + y^2 + 3z^2)\vec{k}, \quad S: \text{הצד החיצוני של המשטח החרוטי}$$

$$z = \sqrt{x^2 + y^2} \text{ בין המישורים: } z = 2, z = 0$$

$$(10) \quad \iint_S (x^2 - 2y^2 + 6z) dx dy, \quad S: \text{הצד התחתון של המשטח הגלילי } 6z = y^2 \text{ המוגבל על-ידי}$$

$$\text{המישורים: } x = 3, x = 0, z = 6$$

$$(11) \quad \iint_S \vec{F} d\vec{S}, \quad \text{כאשר } \vec{F} = (x^2 + y^2 + z^2)\vec{k}, \quad S: \text{הצד הפנימי של החלק העליון של חצי}$$

$$\text{הספירה: } z = \sqrt{R^2 - x^2 - y^2}, \text{ הנגזר על-ידי המשטח החרוטי } z = \sqrt{x^2 + y^2}$$

$$S: \text{הצד הפנימי של המשטח הגלילי } 2y = x^2, \text{ המוגבל על-ידי} \quad \iint_S (x^2 + y + z^2) dx dz \quad (12)$$

המישורים: $y=2, z=1, z=0$.

$$S: \text{הצד החיצוני של המשטח החרוטי} \quad \bar{F} = (x^2 + y^2 + z^2) \bar{j} \quad \iint_S \bar{F} d\bar{S} \quad (13)$$

בין המישורים: $y = \sqrt{x^2 + z^2}, y=1, y=0$.

$$S: \text{הצד העליון של חלק המישור } x + y + z - 6 = 0, \text{ הנגזר} \quad \iint_S (2x + 3y + 4z) dx dy \quad (14)$$

על-ידי המשטח הגלילי $\frac{x^2}{4} + \frac{y^2}{9} = 1$.

$$S: \text{הצד החיצוני של חלק המשטח הגלילי} \quad \bar{F} = (x^2 + z^2) \bar{i} \quad \iint_S \bar{F} d\bar{S} \quad (15)$$

בין המישורים: $x = \sqrt{9 - y^2}, z=2, z=0$.

$$S: \text{הצד העליון של חלק המישור } x + z - 1 = 0, \quad \bar{F} = x \bar{i} + y \bar{j} + z \bar{k} \quad \iint_S \bar{F} d\bar{S} \quad (16)$$

המוגבל על-ידי המישורים: $y=4, z=0, y=0, x=0$.

$$S: \text{הצד העליון של חלק המישור } x + y + z = a, \quad \iint_S x dy dz + y dx dz + z dx dy \quad (17)$$

באוקטנט הראשון ($x \geq 0, y \geq 0, z \geq 0$).

$$S: \text{הצד העליון של חצי הספרה} \quad \bar{F} = x^2 y^2 z \bar{k} \quad \iint_S \bar{F} d\bar{S} \quad (18)$$

$z = \sqrt{1 - x^2 - y^2}$.

$$S: \text{הצד העליון של הפרבולואיד} \quad \iint_S x^2 dy dz + y^2 dx dz + z^2 dx dy \quad (19)$$

$2az = a^2 - x^2 - y^2$ באוקטנט השני ($x \leq 0, y \geq 0, z \geq 0$).

$$S: \text{הצד החיצוני של פני הקובייה:} \quad \oiint_S x dy dz + y dx dz + z dx dy \quad (20)$$

$z=1, z=0, y=1, y=0, x=1, x=0$.

$$S: \text{הצד החיצוני של פני הפירמידה:} \quad \oiint_S xy dy dz + yz dx dz + xz dx dy \quad (21)$$

$x + y + z = 1, z=0, y=0, x=0$.

$$S: \text{הצד החיצוני של הספרה:} \quad \bar{F} = x \bar{i} + y \bar{j} + z \bar{k} \quad \oiint_S \bar{F} d\bar{S} \quad (22)$$

$x^2 + y^2 + z^2 = R^2$.

(23) חשב את המסה של המשטח החרוטי: $\frac{z^2}{4} = x^2 + y^2$, $0 \leq z \leq 2$ כאשר הצפיפות

$$\gamma(x, y, z) = \sqrt{x^2 + y^2} \text{ של חומר:}$$

(24) חשב את השטף של השדה הוקטורי $\vec{F} = z\vec{i} + (1-z)\vec{j} + xy\vec{k}$ דרך חלק המישור:

$$x+y=1, \quad 0 \leq z \leq 1, \quad y \geq 0, \quad x \geq 0$$

כאשר וקטור הנורמל למשטח יוצר זווית חדה עם ציר ה-Y.

(25) חשב את המסה של הלוחית המשולשת: $(x \geq 0, y \geq 0, z \geq 0)$, $x+y+z=1$ כאשר

$$\gamma(x, y, z) = \frac{1}{(1+x+z)^2} \text{ הצפיפות המשטחית של החומר הלוחית:}$$

(26) חשב את השטף של השדה הוקטורי $\vec{F} = x^2\vec{i} + y^2\vec{j} + z^2\vec{k}$ דרך הצד העליון של המישור:

$$x+y+z=1 \quad (x \geq 0, y \geq 0, z \geq 0) \text{ באוקטנט הראשון}$$

(27) חשב את השטף של השדה הוקטורי $\vec{F} = y\vec{j} + z^2\vec{k}$ דרך הפרבולואיד: $y = x^2 + z^2$,

$$0 \leq y \leq 1 \text{ בכיוון של הנורמל החיצוני ביחס לתחום, שמוגבל ע"י המשטח.}$$

(28) חשב את השטף של השדה הוקטורי $\vec{F} = x\vec{i} + y\vec{j} + z\vec{k}$ דרך המשטח הגלילי:

$$0 \leq z \leq h, \quad x^2 + y^2 = R^2 \text{ בכיוון של הנורמל החיצוני למשטח.}$$

(29) יהי: $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$. נגדיר את השדה הוקטורי: $\vec{F} = |\vec{r}| \cdot \vec{r}$. חשב את השטף של $\vec{F}$

$$\text{דרך חצד החיצוני של הספרה: } x^2 + y^2 + z^2 = R^2$$

(30) חשב את השטף של השדה הוקטורי $\vec{F} = yz\vec{i} + x^2\vec{j} + yz\vec{k}$ דרך חצי הספרה:

$$y \geq 0, \quad x^2 + y^2 + z^2 = R^2 \text{ בכיוון של הנורמל החיצוני ביחס לתחום, שמוגבל ע"י המשטח.}$$

(31) חשב את המסה של חלק המשטח החרוטי: $z = \sqrt{x^2 + y^2}$ מעל חצי העיגול במישור $z=0$:

$$x^2 + y^2 = 2x, \quad y \geq 0, \text{ כאשר הצפיפות המשטחית של החומר: } \gamma(x, y, z) = xy + yz + xz$$

(32) חשב את השטף של השדה הוקטורי $\vec{F} = x^3\vec{i} + y^3\vec{j} + z^3\vec{k}$ דרך הפרבולואיד:

$$z \geq 0, \quad z = 2 - x^2 - y^2 \text{ בכיוון של הנורמל החיצוני ביחס לתחום, שמוגבל ע"י המשטח.}$$

(33) חשב את השטף של השדה הוקטורי $\vec{F} = x\vec{i} + y^2\vec{j} + z^2\vec{k}$ דרך הצד החיצוני של פני

$$\text{הגליל, שמוגבל על-ידי המשטחים: } z = -h, \quad z = h, \quad x^2 + y^2 = R^2$$

תרגיל מס' 17

משפט גאוס

בתרגילים מסומן: V - תחום תלת-מימדי, S - שפת התחום.

בעזרת משפט גאוס הצג את האינטגרל המשטחי לפי הצד החיצוני של שפת התחום S :

$$\oint_S P(x, y, z) dydz + Q(x, y, z) dx dz + R(x, y, z) dx dy$$

כאינטגרל משולש לפי התחום V :

$$1) \oint_S xy^2 dydz + yz^2 dx dz + x^2 z dx dy$$

$$2) \oint_S \frac{x dydz}{\sqrt{x^2 + y^2 + z^2}} + \frac{y dx dz}{\sqrt{x^2 + y^2 + z^2}} + \frac{z dx dy}{\sqrt{x^2 + y^2 + z^2}}$$

$$3) \oint_S \frac{\partial f}{\partial x} dydz + \frac{\partial f}{\partial y} dx dz + \frac{\partial f}{\partial z} dx dy$$

חשב את האינטגרל המשטחי לפי הצד החיצוני של שפת התחום המרחבי הנתון:

$$4) \oint_S x dydz + y dx dz + z dx dy, \quad V: (\text{כדור}) \quad x^2 + y^2 + z^2 = R^2$$

$$5) \oint_S x^2 dydz + y^2 dx dz + z^2 dx dy, \quad V: (\text{קובייה}) \quad x=0, x=1, y=0, y=1, z=0, z=1$$

$$6) \oint_S x^3 dydz + y^3 dx dz + z^3 dx dy, \quad V: (\text{כדור}) \quad x^2 + y^2 + z^2 = R^2$$

$$7) \oint_S x^2 dydz + y^2 dx dz + z^2 dx dy, \quad V: (\text{חרוט סגור}) \quad \frac{z^2}{b^2} = \frac{x^2}{a^2} + \frac{y^2}{a^2}, \quad z=0, z=b$$

8) נתון השדה הוקטורי: $\vec{F} = \vec{m} \times \vec{p}$, כאשר $\vec{p} = x\vec{i} + y^2\vec{j} + z^3\vec{k}$ ו- $\vec{m}$ הוא וקטור קבוע:

$$\vec{m} = a\vec{i} + b\vec{j} + c\vec{k}$$

חשב את השטף של $\vec{F}$ דרך הצד החיצוני של הספירה: $x^2 + y^2 + z^2 = R^2$.

9) חשב את השטף של השדה הוקטורי $\vec{F} = yz^2\vec{i} - 6yz^2\vec{j} - (z - 2z^3)\vec{k}$ דרך הצד החיצוני S

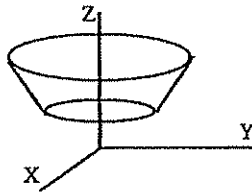
של פני המקבילון, שבנוי על הוקטורים: $\vec{OA}$, $\vec{OB}$, $\vec{OC}$, כאשר קואורדינטות הנקודות:
 $O(0,0,0)$, $A(1,-2,1)$, $B(3,2,1)$, $C(1,0,-1)$

(10) חשב את השטף של השדה הוקטורי $\vec{F} = \frac{x^3}{a^2}\vec{i} + \frac{y^3}{b^2}\vec{j} + \frac{z^3}{c^2}\vec{k}$ דרך הצד הפנימי S

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$

של שפת האליפסואיד: רמז: ניתן להשתמש בקואורדינטות כדוריות כלליות:

$$x = a \rho \sin \theta \cos \varphi, \quad y = b \rho \sin \theta \sin \varphi, \quad z = c \rho \cos \theta$$



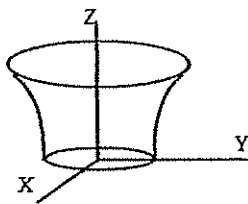
(11) מצא את השטף של השדה הוקטורי $\vec{F} = x\vec{i} + y\vec{j} + z\vec{k}$ דרך הצד החיצוני של שפת התחום V, כאשר V הוא חרוט קטום, שמוגבל על-ידי המשטחים:

$$S_1: z = \sqrt{x^2 + y^2}, \quad S_2: z = 1, \quad S_3: z = 2$$

חשב את שטף הוקטור בשתי דרכים:

א. בלי להשתמש בנוסחת גאוס.

ב. בהשתמש בנוסחת גאוס.



(12) מצא את השטף של השדה הוקטורי $\vec{F} = -xz\vec{i} - yz\vec{j} + (x^2 + y^2)\vec{k}$ דרך הצד החיצוני של שפת התחום V, כאשר V מוגבל על-ידי

חלק ההיפרבולואיד $S_1: z^2 + 1 = x^2 + y^2$ ועל-ידי שני המישורים

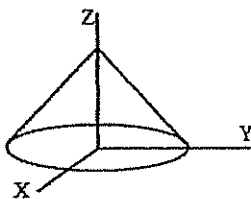
$$S_2: z = 0, \quad S_3: z = \sqrt{3}$$

חשב את שטף הוקטור בשתי דרכים:

א. בלי להשתמש בנוסחת גאוס.

ב. בהשתמש בנוסחת גאוס.

בתרגילים הבאים השתמש במשפט גאוס כדי לחשב שטף של שדה וקטורי דרך משטח לא סגור.



(13) חשב את השטף של השדה הוקטורי

$$\vec{F} = (xz^2 + y^2)\vec{i} + (yx^2 + z^2)\vec{j} + (zy^2 + x^2)\vec{k}$$

דרך המשטח הצדדי S של החרוט $z = 1 - \sqrt{x^2 + y^2}$ בכיוון של הנורמל למשטח, היוצר זווית חדה עם הכיוון החיובי של ציר ה-Z.

(14) חשב את השטף של השדה הוקטורי $\vec{F} = (x + y^2z)\vec{i} + (y + xz^2)\vec{j} + (1 - z)\vec{k}$ דרך הצד

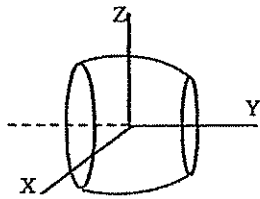
החיצוני S של מעטפת הפירמידה המלבנית OABCD, כאשר קואורדינטות קדקודי הבסיס:

$$O(0,0,0), \quad A(4,0,0), \quad B(4,3,0), \quad C(0,3,0)$$

וקדקוד הפירמידה נמצא בנקודה D(3,2,5).

(15) חשב את השטף של השדה הוקטורי $\vec{F} = (x+z)\vec{i} + (y+x)\vec{j} + z\vec{k}$ דרך הצד החיצוני S של מעטפת הפירמידה המרובעת ABCDE, כאשר קואורדינטות קדקודי הבסיס: $B(0,0,0)$, $C(1,1,0)$, $D(3,-1,0)$, $E(2,-2,0)$ וקדקוד הפירמידה נמצא בנקודה $A(4,5,3)$.

(16) נתון השדה הסקלרי: $u = x^2 + yz$. חשב את שטף הוקטור $\vec{F} = \text{grad } u$ דרך חצי הספרה: $S: x^2 + y^2 + z^2 = R^2$, $y \geq 0$ בכיוון של הנורמל למשטח, היוצר זווית חדה עם הכיוון החיובי של ציר ה-Y.



(17) חשב את השטף של השדה הוקטורי $\vec{F} = 2xy\vec{i} - y^2\vec{j} + 2z\vec{k}$ דרך הצד הפנימי S של חלק הספרה: $x^2 + y^2 + z^2 = 9$, $-1 \leq y \leq 2$.

תרגיל מס' 18

רוטור של וקטור ומשפט סטוקס

א

(1) חשב את הרוטור של השדה הווקטורי הנתון $\vec{F}$:

$$a) \vec{F} = z^2 \vec{i} + y^2 \vec{j} + x^2 \vec{k}$$

$$b) \vec{F} = (x+z) \vec{i} + (y+z) \vec{j} + (x^2+z) \vec{k}$$

$$c) \vec{F} = (x^2+y^2) \vec{i} + (y^2+z^2) \vec{j} + (z^2+x^2) \vec{k}$$

(2) נתון השדה הווקטורי: $\vec{F} = |\vec{r}| \vec{r}$, כאשר $\vec{r} = x \vec{i} + y \vec{j} + z \vec{k}$. חשב את $\text{rot} \vec{F}$.

(3) יהיו $\vec{F}$ שדה וקטורי ו- λ סקלר. הוכח: $\text{rot}(\lambda \vec{F}) = \lambda \text{rot} \vec{F}$.

(4) נתון השדה הווקטורי: $\vec{F} = \vec{m} |\vec{r}|$, כאשר $\vec{r} = x \vec{i} + y \vec{j} + z \vec{k}$ ו- $\vec{m}$ הוא וקטור קבוע. חשב את $\text{rot} \vec{F}$. $\vec{m} = a \vec{i} + b \vec{j} + c \vec{k}$.

(5) האם נכונה הטענה: $\text{rot}(\vec{u} + \vec{v}) = \text{rot} \vec{u} + \text{rot} \vec{v}$?

ב

בעזרת נוסחת סטוקס חשב את האינטגרל הנתון
(כיוון האינטגרציה נגד כיוון השעון, כאשר מסתכלים על המסלול L מלמעלה)

$$1) \oint_L (z-y)dx + (x-z)dy + (y-x)dz \quad L: \Delta ABC \quad A(a,0,0), B(0,a,0), C(0,0,a)$$

$$2) \oint_L x(z-y)dx + y(x-z)dy + z(y-x)dz \quad L: \Delta ABC \quad A(a,0,0), B(0,a,0), C(0,0,a)$$

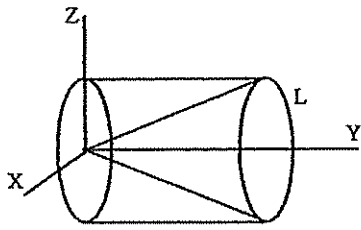
$$3) \oint_L (2z-x)dx + (x+2z)dy + 3zdz \quad L: \Delta ABC \quad A(4,0,0), B(0,1,0), C(0,0,4)$$

$$4) \oint_L y^3 dx + 3xy^2 dy + x^2 dz \quad L: \Delta ABC \quad A(2,0,0), B(0,1,0), C(0,0,3)$$

$$5) \oint_L x^2 y^3 dx + dy + z dz \quad L: (\text{מעגל}) \quad x^2 + y^2 + z^2 = a^2, z=0$$

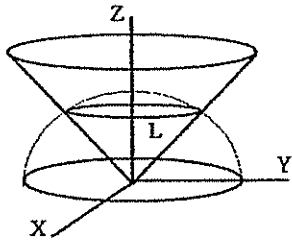
$$6) \oint_L y dx + z dy + x dz \quad L: (\text{מעגל}) \quad x^2 + y^2 + z^2 = a^2, x+y+z=0$$

השתמש בנוסחת סטוקס כדי לחשב את הצירקולציה $I = \oint_L \vec{F} d\vec{r}$ של הוקטור $\vec{F}$.



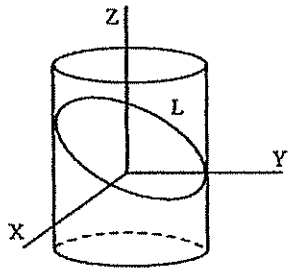
$$\vec{F} = (z^3 + z)\vec{i} + 2y^2\vec{j} + (x - x^3)\vec{k} \quad (7)$$

L הוא קו החיתוך של המשטח הגלילי $x^2 + z^2 = 4$ עם המשטח החרוטי $y = \sqrt{x^2 + z^2}$. כיוון האינטגרציה לפי כיוון השעון, כאשר מסתכלים על המסלול L מימין.



$$\vec{F} = (x^2 + y)\vec{i} + (x^2 + z)\vec{j} + (y^2 - xz)\vec{k} \quad (8)$$

L הוא קו החיתוך של הספרה $x^2 + y^2 + z^2 = 8$ עם המשטח החרוטי $z^2 = x^2 + y^2$, $z \geq 0$. כיוון האינטגרציה נגד כיוון השעון, כאשר מסתכלים על המסלול L מלמעלה.

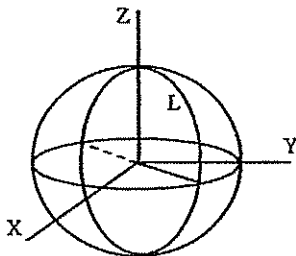


$$\vec{F} = y^2\vec{i} + z^2\vec{j} \quad (9)$$

L הוא קו החיתוך של המשטח הגלילי $x^2 + y^2 = 9$ עם המישור $3y + 4z = 5$. כיוון האינטגרציה נגד כיוון השעון, כאשר מסתכלים על המסלול L מלמעלה.

$$\vec{F} = (y - x)\vec{i} + (y + x)\vec{j} + (y + z)\vec{k} \quad (10)$$

L הוא קו החיתוך של המשטח הגלילי $x^2 + y^2 = 1$ עם המישור $x + y + z = 1$. כיוון האינטגרציה נגד כיוון השעון, כאשר מסתכלים על המסלול L מלמעלה.



$$\vec{F} = x\vec{i} - z\vec{j} + y\vec{k} \quad (11)$$

L הוא קו החיתוך של הספרה $x^2 + y^2 + z^2 = 1$ עם המישור $x = y$. כיוון האינטגרציה נגד כיוון השעון, כאשר מסתכלים על המסלול L מהקצה של הוקטור $\vec{i}$.

חשב שטף של $\text{rot } \vec{F}$ דרך המשטח הנתון $S = \iint_S \text{rot } \vec{F} d\vec{S}$ בכיוון נתון:

$$\vec{F} = (y^2 - 4y + z)\vec{i} + x^2(y - 4)\vec{j} + z^2(16 - y^2)\vec{k} \quad (12)$$

S הוא חלק של הפרבולואיד: $y = x^2 + z^2$, $0 \leq y \leq 4$. בכיוון של הנורמל החיצוני ביחס לתחום, שמוגבל על-ידי הפרבולואיד.

$$\vec{F} = -y^3\vec{i} + x^3\vec{j} + xyz\vec{k} \quad (13)$$

S הוא החצי העליון של הספרה: $x^2 + y^2 + z^2 = R^2$, $z \geq 0$. בכיוון של הנורמל החיצוני של הספרה.

$$\vec{F} = (2y + xz + \sqrt{x^2 + y^2})\vec{i} + y \left[z + \ln(x + \sqrt{x^2 + y^2}) \right] \vec{j} + (z^2 + z)\vec{k} \quad (14)$$

S הוא הצד החיצוני של מעטפת פירמידה משולשת, כאשר קדקודי בסיס הפירמידה: $A(1,1,0)$, $B(5,1,0)$, $C(1,4,0)$ וקדקוד הפירמידה נמצא בנקודה $D(2,2,6)$.



בדוק, האם $\vec{F}$ הוא שדה משמר ואם כן, מצא את פונקציית הפוטנציאל של $\vec{F}$.

$$1) \vec{F} = 2xyz \vec{i} + x^2z \vec{j} + x^2y \vec{k} \qquad 2) \vec{F} = e^x \sin y \vec{i} + e^x \sin y \vec{j} + z \vec{k}$$

$$3) \vec{F} = (y+z) \vec{i} + (x+z) \vec{j} + (x+y) \vec{k} \qquad 4) \vec{F} = x^3 \vec{i} + y^3 \vec{j} + xz^3 \vec{k}$$

$$5) \vec{F} = yz \cos(xy) \vec{i} + xz \cos(xy) \vec{j} + \sin(xy) \vec{k}$$

$$6) \vec{F} = f(x, y, z) (\vec{i} + \vec{j} + \vec{k}) \quad , \quad f(x, y, z) = \frac{1}{x+y+z}$$

$$7) \vec{F} = \left(\sqrt{z} + \frac{y}{2\sqrt{x}} \right) \vec{i} + \left(\sqrt{x} + \frac{z}{2\sqrt{y}} \right) \vec{j} + \left(\sqrt{y} + \frac{x}{2\sqrt{z}} \right) \vec{k}$$

גוף נקודתי נע מנקודה A לנקודה B. $\vec{F}$ הוא כוח, שמופעל על הגוף.
הוכח, ש- $\vec{F}$ הוא שדה משמר וחשב את עבודת הכוח.

$$8) \vec{F} = x \vec{i} - y^2 \vec{j} + z \vec{k} \quad , \quad A(1, -1, 2), B(2, 1, 3)$$

$$9) \vec{F} = yz \vec{i} + zx \vec{j} + xy \vec{k} \quad , \quad A(0, 2, 3), B(3, 2, 1)$$

$$10) \vec{F} = x(y^2 + z^2) \vec{i} + y(x^2 + z^2) \vec{j} + z(x^2 + y^2) \vec{k} \quad , \quad A(-1, 1, -1), B(1, 1, 2)$$

$$11) \vec{F} = \frac{yz}{1+x^2y^2z^2} \vec{i} + \frac{xz}{1+x^2y^2z^2} \vec{j} + \frac{xy}{1+x^2y^2z^2} \vec{k} \quad , \quad A(\sqrt{2}, 0, 0), B(1, 1, 1)$$

תרגיל מס' 19

טורים מספריים

א. רשום את שלושת האיברים הראשונים של הטור הנתון ובדוק את התכנסות הטור לפי קריטריון ההשוואה או לפי התנאי ההכרחי להתכנסות הטור.

$$\begin{array}{llll}
 1) \sum_{n=1}^{\infty} \frac{1}{\sqrt{n}} & 2) \sum_{n=1}^{\infty} \frac{1}{n5^{n-1}} & 3) \sum_{n=2}^{\infty} \frac{1}{\ln n} & 4) \sum_{n=2}^{\infty} \left(\frac{n}{n-1} \right)^n \\
 5) \sum_{n=1}^{\infty} \frac{1}{1000n-2} & 6) \sum_{n=2}^{\infty} \frac{1}{\sqrt[n]{10}} & 7) \sum_{n=2}^{\infty} \frac{1}{\sqrt{n(n-1)}} & 8) \sum_{n=1}^{\infty} n \ln \left(1 + \frac{1}{n} \right)
 \end{array}$$

ב. בדוק את התכנסות הטור לפי הקריטריון האינטגרלי.

$$\begin{array}{ll}
 9) 1 + \frac{1}{3} + \frac{1}{5} + \frac{1}{7} + \dots & 10) 1 + \frac{1}{\sqrt{4}} + \frac{1}{\sqrt{7}} + \frac{1}{\sqrt{10}} + \dots \\
 11) \frac{1}{2^3} + \frac{2}{3^3} + \frac{3}{4^3} + \frac{4}{5^3} + \dots & 12) \frac{1}{1+1^2} + \frac{1}{1+2^2} + \frac{1}{1+3^2} + \frac{1}{1+4^2} + \dots \\
 13) \frac{1}{1+1^2} + \frac{2}{1+2^2} + \frac{3}{1+3^2} + \frac{4}{1+4^2} + \dots & 14) \frac{1}{3^2-1} + \frac{1}{5^2-1} + \frac{1}{7^2-1} + \frac{1}{9^2-1} + \dots \\
 15) \frac{1}{2\ln 2} + \frac{1}{3\ln 3} + \frac{1}{4\ln 4} + \frac{1}{5\ln 5} + \dots & 16) \frac{1}{2\ln^2 2} + \frac{1}{3\ln^2 3} + \frac{1}{4\ln^2 4} + \frac{1}{5\ln^2 5} + \dots
 \end{array}$$

ג. רשום את שלושת האיברים הראשונים של הטור הנתון ובדוק את התכנסות הטור לפי קריטריון זלמבר.

$$\begin{array}{lll}
 17) \sum_{n=1}^{\infty} \frac{2n}{3^n} & 18) \sum_{n=1}^{\infty} \frac{2^{n-1}}{n!} & 19) \sum_{n=1}^{\infty} \frac{n!}{1 \cdot 3 \cdot 5 \cdot \dots \cdot (2n-1)} \\
 20) \sum_{n=1}^{\infty} \frac{3^{n-1}}{2^{n-1}(2n-1)} & 21) \sum_{n=1}^{\infty} \frac{(2n-1)!}{2 \cdot 4 \cdot 6 \cdot \dots \cdot 2n} & 22) \sum_{n=1}^{\infty} \frac{4n-3}{\sqrt{n} \cdot 3^n}
 \end{array}$$

ד. בדוק את התכנסות הטור לפי קריטריון קושי.

$$\begin{array}{llll}
 23) \sum_{n=1}^{\infty} \left(\frac{n+1}{2n-1} \right)^n & 24) \sum_{n=1}^{\infty} \left(\frac{n}{3n-1} \right)^{2n-1} & 25) \sum_{n=1}^{\infty} \frac{1}{\ln^n(n+1)} & 26) \sum_{n=1}^{\infty} \arcsin^n \frac{1}{n} \\
 27) \sum_{n=1}^{\infty} \frac{(n+1)^{n^2}}{n^{n^2} 3^n} & 28) \sum_{n=1}^{\infty} \frac{n^3}{e^n} & 29) \sum_{n=1}^{\infty} \left(\frac{n-2}{n+1} \right)^{n(n-2)}
 \end{array}$$

ה. בדוק את התכנסות הטור (לא חיובי).

$$\begin{array}{lll}
 30) \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2+1} & 31) \sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt[3]{n^2}} & 32) \sum_{n=1}^{\infty} \frac{3^n \sin n}{5^n} \\
 33) \sum_{n=2}^{\infty} \frac{(-1)^n n}{n\sqrt{n}-1} & 34) \sum_{n=2}^{\infty} (-1)^n \frac{\ln n}{n}
 \end{array}$$

תרגיל מס' 20

טורים מספריים (תרגילים שונים)

בדוק את התכנסות הטורים.

$$1) \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{\sqrt{n}}$$

$$2) \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{(2n-1)^2}$$

$$3) \sum_{n=1}^{\infty} \frac{1}{(2n-1)\sqrt{2n-1}}$$

$$4) \sum_{n=1}^{\infty} \frac{1}{100n+1}$$

$$5) \sum_{n=2}^{\infty} \frac{(-1)^n}{n \ln n}$$

$$6) \sum_{n=1}^{\infty} \frac{n}{1+n^4}$$

$$7) \sum_{n=1}^{\infty} \frac{3^n - 1}{(2^n - 1)n}$$

$$8) \sum_{n=1}^{\infty} \frac{1}{(3n-2)^2}$$

$$9) \sum_{n=1}^{\infty} \frac{2n-1}{2^n}$$

$$10) \sum_{n=1}^{\infty} \frac{\sin(\pi n/5)}{(\ln 10)^n}$$

$$11) \sum_{n=1}^{\infty} \frac{2n}{(2n-1)!}$$

$$12) \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{\sqrt{2n+1}}$$

$$13) \sum_{n=1}^{\infty} \frac{(-1)^n}{n^3}$$

$$14) \sum_{n=1}^{\infty} \frac{\cos(\pi n)}{n}$$

$$15) \sum_{n=2}^{\infty} \frac{1}{n \ln^3 n}$$

$$16) \sum_{n=10}^{\infty} \frac{1}{n \cdot \ln n \cdot \ln \ln n}$$

$$17) \sum_{n=2}^{\infty} \frac{1}{n^2 - n}$$

$$18) \sum_{n=2}^{\infty} \frac{1}{n^2 \ln n}$$

$$19) \sum_{n=1}^{\infty} \frac{(n^2 + 1)^{n^3+1}}{n^{2n^3+2}}$$

$$20) \sum_{n=1}^{\infty} \frac{1}{\sqrt{n} e^{\sqrt{n}}}$$

$$21) \sum_{n=2}^{\infty} n \ln \left(\frac{n+1}{n-1} \right)$$

$$22) \sum_{n=1}^{\infty} \frac{3^n}{(2n)!}$$

$$23) \sum_{n=1}^{\infty} \frac{n^3}{e^n}$$

$$24) \sum_{n=2}^{\infty} \frac{1}{n \sqrt[n]{e^2}}$$

$$25) \sum_{n=1}^{\infty} \frac{e^n}{\sqrt{n} \cdot 2^n}$$

$$26) \sum_{n=1}^{\infty} \frac{n^2}{(3n)!}$$

$$27) \sum_{n=1}^{\infty} \frac{n^n}{n!}$$

תשובות

(1) מתכנס בתנאי, (2) מתכנס בהחלט, (3) מתכנס, (4) מתבדר, (5) מתכנס בתנאי, (6) מתכנס, (7) מתבדר, (8) מתכנס, (9) מתכנס, (10) מתכנס בהחלט, (11) מתכנס, (12) מתכנס בתנאי, (13) מתכנס בהחלט, (14) מתכנס בתנאי, (15) מתכנס, (16) מתבדר, (17) מתכנס, (18) מתכנס, (19) מתבדר, (20) מתכנס, (21) מתבדר, (22) מתכנס, (23) מתכנס, (24) מתבדר, (25) מתבדר, (26) מתכנס, (27) מתבדר

תרגיל מס' 21

תחום ההתכנסות של טור חזקות

מצא את אינטרוול ההתכנסות של הטור.

$$1) \sum_{n=1}^{\infty} \frac{x^{n-1}}{3^{n-1}n}$$

$$2) \sum_{n=1}^{\infty} \frac{(-1)^n x^{2n-1}}{5^n \sqrt{n+1}}$$

$$3) \sum_{n=1}^{\infty} \frac{2^n x^n}{(2n+1)^2 \sqrt{3^n}}$$

$$4) \sum_{n=1}^{\infty} \frac{x^n}{n!}$$

$$5) \sum_{n=1}^{\infty} \frac{(-x)^{n-1}}{n}$$

$$6) \sum_{n=1}^{\infty} \frac{3^n x^n}{\sqrt{(3n-2)2^n}}$$

$$7) \sum_{n=1}^{\infty} \frac{(-1)^{n-1} x^{2n-2}}{3^{n-1} n \sqrt{n}}$$

$$8) \sum_{n=1}^{\infty} \frac{(10x)^n}{\sqrt{n}}$$

$$9) \sum_{n=1}^{\infty} x^{n-1}$$

$$10) \sum_{n=1}^{\infty} \frac{x^{2n-1}}{2n-1}$$

$$11) \sum_{n=1}^{\infty} \frac{n^5 x^{2n}}{2n-1}$$

$$12) \sum_{n=1}^{\infty} n! x^n$$

$$13) \sum_{n=2}^{\infty} \frac{x^{n-1}}{n 3^n \ln n}$$

$$14) \sum_{n=1}^{\infty} \frac{(x-3)^n}{n 5^n}$$

$$15) \sum_{n=1}^{\infty} \frac{(x-1)^{2n}}{n 9^n}$$

$$16) \sum_{n=1}^{\infty} \frac{(x+3)^n}{n^2}$$

$$17) \sum_{n=1}^{\infty} \frac{(x+5)^{2n-1}}{2n \cdot 4^n}$$

$$18) \sum_{n=2}^{\infty} \frac{(x+1)^{n-1}}{n \ln^2 n}$$

$$19) \sum_{n=1}^{\infty} n^n (x+3)^n$$

תשובות

- 1) $-3 \leq x < 3$, 2) $-\sqrt{5} \leq x \leq \sqrt{5}$, 3) $-\sqrt{3}/2 \leq x \leq \sqrt{3}/2$, 4) $-\infty < x < \infty$, 5) $-1 < x \leq 1$,
 6) $-\sqrt{2}/3 \leq x \leq \sqrt{2}/3$, 7) $-\sqrt{3} \leq x \leq \sqrt{3}$, 8) $-1/10 \leq x < 1/10$, 9) $-1 < x < 1$, 10) $-1 < x < 1$,
 11) $-1 < x < 1$, 12) $x=0$, 13) $-3 \leq x < 3$, 14) $-2 \leq x < 8$, 15) $-2 < x < 4$, 16) $-4 \leq x \leq -2$,
 17) $-7 < x < -3$, 18) $-2 \leq x \leq 0$, 19) $x=-3$

תרגיל מס' 22

טור טיילור

פיתוחים של פונקציות לטור מקלורן השימושיים ביותר (בכל הנוסחאות $n=0,1,2,\dots$):

$$1) e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots + \frac{x^n}{n!} + \dots \quad -\infty < x < \infty$$

$$2) \cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots + (-1)^n \frac{x^{2n}}{(2n)!} + \dots \quad -\infty < x < \infty$$

$$3) \sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots + (-1)^n \frac{x^{2n+1}}{(2n+1)!} + \dots \quad -\infty < x < \infty$$

$$4) \ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots + (-1)^n \frac{x^{n+1}}{n+1} + \dots \quad -1 < x \leq 1$$

$$5) \arctan x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \dots + (-1)^n \frac{x^{2n+1}}{2n+1} + \dots \quad -1 < x < 1$$

$$6) \frac{1}{1+x} = 1 - x + x^2 - x^3 + \dots + (-1)^n x^n + \dots \quad -1 < x < 1$$

$$7) \frac{1}{1-x} = 1 + x + x^2 + x^3 + \dots + x^n + \dots \quad -1 < x < 1$$

$$8) (1+x)^m = 1 + \frac{m}{1!}x + \frac{m(m-1)}{2!}x^2 + \frac{m(m-1)(m-2)}{3!}x^3 + \dots + \frac{m(m-1)\dots(m-n+1)}{n!}x^n + \dots \quad -1 < x < 1$$

א. השתמש בפיתוחים הידועים של פונקציות לטור מקלורן כדי לפתח את הפונקציה הנתונה לטור חזקות בסביבת הנקודה הנתונה:

$$1) f(x) = e^{-x^2}, x_0 = 0 \quad 2) f(x) = \sinh x, x_0 = 0 \quad 3) f(x) = \cosh x, x_0 = 0$$

$$4) f(x) = 2^x, x_0 = 0 \quad 5) f(x) = \cos^2 x, x_0 = 0 \quad 6) f(x) = \sin^2 x, x_0 = 0$$

$$7) f(x) = \frac{1}{2-3x}, x_0 = 0 \quad 8) f(x) = \frac{x}{9+x^2}, x_0 = 0 \quad 9) f(x) = xe^{-2x}, x_0 = 0$$

$$10) f(x) = x \cos 3x^2, x_0 = 0 \quad 11) f(x) = \ln x, x_0 = 1$$

$$12) f(x) = \frac{1}{x}, x_0 = 2 \quad 13) f(x) = \frac{1}{\sqrt{1+x}}, x_0 = 0$$

$$14) f(x) = \ln(2-3x+x^2), x_0 = 0 \quad 15) f(x) = \frac{2x+3}{x^2+3x+2}, x_0 = 0$$

ב. חשב בקירוב את ערך הפונקציה עם זיוק של 0.00001:

$$1) \sqrt[3]{1.1}$$

$$2) \ln 1.04$$

$$3) \frac{1}{\sqrt[3]{e}}$$

$$4) \cos 18^\circ$$

ג. בתרגילים הבאים השתמש בגזירה ואינטגרציה של טורי חזקות.
(1) מצא פיתוח לטור מקלורן עבור הפונקציה:

$$\begin{array}{ll} \text{a) } f(x) = \ln(1+x) & -1 < x < 1 \\ \text{b) } f(x) = \arctan x & -1 < x < 1 \\ \text{c) } f(x) = \ln \frac{1+x}{1-x} & -1 < x < 1 \\ \text{d) } f(x) = \arcsin x & -1 < x < 1 \end{array}$$

(2) מצא סכום של טור חזקות (כאן $n=1,2,3,\dots$):

$$\begin{array}{ll} \text{a) } 1+2x+3x^2+\dots+nx^{n-1}+\dots & -1 < x < 1 \\ \text{b) } x+\frac{x^2}{2}+\frac{x^3}{3}+\frac{x^4}{4}+\dots+\frac{x^n}{n}+\dots & -1 < x < 1 \\ \text{c) } -2x+4x^3-6x^5+\dots+(-1)^n 2n x^{2n-1}+\dots & -1 < x < 1 \\ \text{d) } 2x+\frac{3}{1!}x^2+\frac{4}{2!}x^3+\dots+\frac{n+1}{(n-1)!}x^n+\dots & -\infty < x < \infty \end{array}$$

ד. חשב בקירוב את האינטגרל I עם דיוק של ε :

$$\begin{array}{l} 1) I = \int_0^{0.2} \frac{\sin x}{x} dx, \quad \varepsilon = 0.0001 \\ 2) I = \int_0^{0.1} \frac{\ln(1+x)}{x} dx, \quad \varepsilon = 0.001 \\ 3) I = \int_0^{0.5} \frac{1-\cos x}{x^2} dx, \quad \varepsilon = 0.0001 \end{array}$$

ה. חשב את הנגזרת מסדר n של הפונקציה $f(x)$ בנקודה $x=0$:

$$\begin{array}{ll} 1) f(x) = \frac{x}{1-x}, \quad n=5 & 2) f(x) = \frac{1}{1+2x^2}, \quad \text{a) } n=6, \text{ b) } n=9 \\ 3) f(x) = \frac{x}{1-x^2}, \quad n=5 & 4) f(x) = x \sin x, \quad n=8 \end{array}$$

פתרונות

גבולות ורציפות של פונקציות של משתנים אחדים

(X)

$$1) \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{x-y}{x+y} = |y=kx| = \lim_{x \rightarrow 0} \frac{x(1-k)}{x(1+k)} = \frac{1-k}{1+k} \Rightarrow (\text{הגבול לא קיים})$$

$$2) \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{x^2+2y^2-3xy}{x^2+3y^2} = |y=kx| = \lim_{x \rightarrow 0} \frac{x^2(1+2k^2-3k)}{x^2(1+3k^2)} = \frac{1+2k^2-3k}{1+3k^2}$$

הגבול לא קיים

$$3) L = \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} e^{\frac{x}{x^2+y^2}} = \left| \begin{matrix} x=r \cos \varphi \\ y=r \sin \varphi \end{matrix} \right| = \lim_{r \rightarrow 0} e^{\frac{r \cos \varphi}{r^2}} = \lim_{r \rightarrow 0} e^{\frac{\cos \varphi}{r}}$$

הגבול לא קיים, $\varphi=0$ נקודת זרימה, $\varphi=\pi$ נקודת זרימה

$$L = \lim_{r \rightarrow 0} e^{\frac{\cos 0}{r}} = \lim_{r \rightarrow 0} e^{\frac{1}{r}} = |e^\infty| = \infty$$

$$L = \lim_{r \rightarrow 0} e^{\frac{\cos \pi}{r}} = \lim_{r \rightarrow 0} e^{-\frac{1}{r}} = |e^{-\infty}| = \frac{1}{e^\infty} = 0$$

הגבול לא קיים

$$4) \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{x^2 y}{x^4+y^2} = |y=kx^2| = \lim_{x \rightarrow 0} \frac{kx^4}{x^4(1+k^2)} = \frac{k}{1+k^2}$$

הגבול לא קיים

$$5) \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{x^3+2y^3-x}{\sqrt{y^6+x^2}} = |y=k^3\sqrt{x}| = \lim_{x \rightarrow 0} \frac{x^3+2k^3x-x}{\sqrt{k^6x^2+x^2}} = \lim_{x \rightarrow 0} \frac{x^3+x(2k^3-1)}{x\sqrt{k^6+1}} =$$

$$= \lim_{x \rightarrow 0} \frac{x^3}{x\sqrt{k^6+1}} + \lim_{x \rightarrow 0} \frac{x(2k^3-1)}{x\sqrt{k^6+1}} = \lim_{x \rightarrow 0} \frac{x^2}{\sqrt{k^6+1}} + \frac{2k^3-1}{\sqrt{k^6+1}} = 0 + \frac{2k^3-1}{\sqrt{k^6+1}}$$

הגבול לא קיים

$$6) \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{\sin(xy)}{xy} \sin \frac{1}{\sqrt{x^2+y^2}} = \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{\sin(xy)}{xy} \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \sin \frac{1}{\sqrt{x^2+y^2}} =$$

$$= 1 \cdot \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \sin \frac{1}{\sqrt{x^2+y^2}} = \left| \begin{matrix} x=r \cos \varphi \\ y=r \sin \varphi \end{matrix} \right| = \lim_{r \rightarrow 0} \sin \frac{1}{r} \Rightarrow (\text{הגבול לא קיים})$$

$$7) \lim_{\substack{x \rightarrow 1 \\ y \rightarrow 0}} \frac{\ln(x+y)}{\sqrt{x^2+y^2}-2x+1} = \lim_{\substack{x \rightarrow 1 \\ y \rightarrow 0}} \frac{\ln[(x-1)+1+y]}{\sqrt{(x-1)^2+y^2}} = \left| \begin{matrix} x-1=t \\ t \rightarrow 0 \end{matrix} \right| = \lim_{\substack{t \rightarrow 0 \\ y \rightarrow 0}} \frac{\ln[1+(t+y)]}{\sqrt{t^2+y^2}} =$$

$$= \left| \begin{matrix} \text{הגבול לא קיים} \\ \lim_{x \rightarrow 0} \frac{\ln(1+x)}{x} = 1 \end{matrix} \right| = \lim_{\substack{t \rightarrow 0 \\ y \rightarrow 0}} \frac{\ln(1+(t+y))}{t+y} \cdot \frac{t+y}{\sqrt{t^2+y^2}} = 1 \cdot \lim_{\substack{t \rightarrow 0 \\ y \rightarrow 0}} \frac{t+y}{\sqrt{t^2+y^2}} = |t=ky| =$$

$$= \lim_{y \rightarrow 0} \frac{y(k+1)}{y\sqrt{k^2+1}} = \frac{k+1}{\sqrt{k^2+1}}$$

הגבול לא קיים

$$8) L = \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} (x^2 + y^2) \cos \frac{1}{xy} = \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} (x^2 + y^2) \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \cos \frac{1}{xy}$$

הגורם הראשון שולל 0-δ והגורם השני $\cos \frac{1}{xy}$ מתחבט. לכן $L = 0$ (הגורם הראשון) = 0

$$9) L = \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{\sin(2xy)}{y} = \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{\sin(2xy)}{2xy} \cdot 2x = 1 \cdot \lim_{x \rightarrow 0} 2x = 0$$

$$10) L = \lim_{\substack{y \rightarrow \infty \\ x \rightarrow 2}} \left(1 + \frac{x}{y}\right)^y = \left((1+0)^\infty\right) = \lim_{\substack{y \rightarrow \infty \\ x \rightarrow 2}} \left(1 + x \cdot \frac{1}{y}\right)^y = e^2$$

$$11) L = \lim_{\substack{x \rightarrow \infty \\ y \rightarrow 3}} \left(1 + \frac{1}{x}\right)^{\frac{x^2}{x+y}} = \lim_{\substack{x \rightarrow \infty \\ y \rightarrow 3}} \left[\left(1 + \frac{1}{x}\right)^x\right]^{\frac{x}{x+y}} = \lim_{\substack{x \rightarrow \infty \\ y \rightarrow 3}} \left[\left(1 + \frac{1}{x}\right)^x\right]^{\frac{1}{1+y/x}} = e^{\frac{1}{1+0}} = e$$

$$12) L = \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{x^2 + y^2}{\sqrt{x^2 + y^2 + 1} - 1} = \begin{cases} x = r \cos \varphi \\ y = r \sin \varphi \end{cases} = \lim_{r \rightarrow 0} \frac{r^2}{\sqrt{r^2 + 1} - 1} = \lim_{r \rightarrow 0} \frac{r^2(\sqrt{r^2 + 1} + 1)}{(r^2 + 1) - 1} =$$

$$= \lim_{r \rightarrow 0} (\sqrt{r^2 + 1} + 1) = 2$$

$$13) \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \sin(x^2 + y^2) = \begin{cases} x = r \cos \varphi \\ y = r \sin \varphi \end{cases} = \lim_{r \rightarrow 0} \sin(r^2) = \sin 0 = 0$$

$$14) L = \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{x^4}{x^2 + y^2}$$

נסתכל על הצורה $(0,0)$: נראה שהגורם הראשון הוא 0

$$0 < \left| \frac{x^4}{x^2 + y^2} \right| < \left| \frac{x^4}{x^2} \right| = x^2 \xrightarrow{x \rightarrow 0} 0$$

לכן $L = 0$ (הגורם הראשון) = 0

$$15) L = \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{\sin(xy)}{\sqrt{x^2 + y^2}} = \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{\sin(xy)}{xy} \cdot \frac{xy}{\sqrt{x^2 + y^2}} = 1 \cdot \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{xy}{\sqrt{x^2 + y^2}}$$

$$0 < \left| \frac{xy}{\sqrt{x^2 + y^2}} \right| < \left| \frac{xy}{\sqrt{x^2}} \right| = \frac{|x||y|}{|x|} = |y| \xrightarrow{y \rightarrow 0} 0 \Rightarrow L = 0$$

$$16) L = \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{\sin(x^3 + y^3)}{x^2 + y^2} = \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{\sin(x^3 + y^3)}{x^3 + y^3} \cdot \frac{x^3 + y^3}{x^2 + y^2} = 1 \cdot \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{x^3 + y^3}{x^2 + y^2} = \begin{cases} x = r \cos \varphi \\ y = r \sin \varphi \end{cases}$$

$$= \lim_{r \rightarrow 0} \frac{r^3(\cos^3 \varphi + \sin^3 \varphi)}{r^2} = \lim_{r \rightarrow 0} r(\cos^3 \varphi + \sin^3 \varphi)$$

$$0 \leq |r(\cos^3 \varphi + \sin^3 \varphi)| \leq r(1+1) = 2r \xrightarrow{r \rightarrow 0} 0 \Rightarrow L = 0$$

$$17) L = \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{2x^3 + 3y^4}{x^2 + y^2} = \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{2x^3}{x^2 + y^2} + \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{3y^4}{x^2 + y^2} = L_1 + L_2$$

$$\left. \begin{aligned} 0 < \left| \frac{2x^3}{x^2 + y^2} \right| &< \left| \frac{2x^3}{x^2} \right| = 2|x| \xrightarrow{x \rightarrow 0} 0 \Rightarrow L_1 = 0 \\ 0 < \left| \frac{3y^4}{x^2 + y^2} \right| &< \left| \frac{3y^4}{y^2} \right| = 3y^2 \xrightarrow{y \rightarrow 0} 0 \Rightarrow L_2 = 0 \end{aligned} \right\} L = L_1 + L_2 = 0$$

$$18) L = \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{x^5 - x^2 y^3}{x^4 + y^4} = \left| \begin{array}{l} x = r \cos \varphi \\ y = r \sin \varphi \end{array} \right| = \lim_{r \rightarrow 0} \frac{r^5 \cos^5 \varphi - r^5 \cos^2 \varphi \sin^3 \varphi}{r^4 (\cos^4 \varphi + \sin^4 \varphi)} =$$

$$= \lim_{r \rightarrow 0} r \cdot \frac{\cos^5 \varphi - \cos^2 \varphi \sin^3 \varphi}{\cos^4 \varphi + \sin^4 \varphi}$$

$$0 \leq \left| \frac{r(\cos^5 \varphi - \cos^2 \varphi \sin^3 \varphi)}{\cos^4 \varphi + \sin^4 \varphi} \right| = \frac{r |\cos^5 \varphi + (-\cos^2 \varphi \sin^3 \varphi)|}{|\cos^4 \varphi + \sin^4 \varphi|} \leq$$

$$\leq \frac{r(|\cos^5 \varphi| + |\cos^2 \varphi \sin^3 \varphi|)}{|\cos^4 \varphi + 2\cos^2 \varphi \sin^2 \varphi + \sin^4 \varphi - 2\sin^2 \varphi \cos^2 \varphi|} \leq$$

$$\leq \frac{r(1+1)}{|(\cos^2 \varphi + \sin^2 \varphi)^2 - \frac{1}{2}(2\sin \varphi \cos \varphi)^2|} = \frac{2r}{|1^2 - \frac{1}{2}\sin^2 2\varphi|} \leq$$

$$\leq \frac{2r}{|1 - \frac{1}{2}|} = \frac{2r}{\frac{1}{2}} = 4r \xrightarrow{r \rightarrow 0} 0 \Rightarrow L = 0$$

$$19) L = \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{x^5 - \cos(x^2 + y^2) + 1}{(x^2 + y^2)^2} = \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{x^5}{(x^2 + y^2)^2} + \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{1 - \cos(x^2 + y^2)}{(x^2 + y^2)^2} = L_1 + L_2$$

$$0 < \left| \frac{x^5}{(x^2 + y^2)^2} \right| < \left| \frac{x^5}{(x^2)^2} \right| = |x| \xrightarrow{x \rightarrow 0} 0 \Rightarrow L_1 = 0$$

$$L_2 = \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{1 - \cos(x^2 + y^2)}{(x^2 + y^2)^2} = \left| \begin{array}{l} x = r \cos \varphi \\ y = r \sin \varphi \end{array} \right| = \lim_{r \rightarrow 0} \frac{1 - \cos(r^2)}{r^4} =$$

$$= \lim_{r \rightarrow 0} \frac{2\sin^2(r^2/2)}{r^4} = 2 \lim_{r \rightarrow 0} \frac{\left(\sin \frac{r^2}{2}\right)^2}{4\left(\frac{r^2}{2}\right)^2} = \frac{1}{2} \lim_{r \rightarrow 0} \left(\frac{\sin \frac{r^2}{2}}{\frac{r^2}{2}}\right)^2 = \frac{1}{2} \cdot 1^2 = \frac{1}{2}$$

$$L = 0 + \frac{1}{2} = \frac{1}{2}$$

(1)

$$1) f(x, y) = \begin{cases} \frac{x^4 + y^5}{x^3 + y^3} & , (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases}$$

$$\lim_{(x, y) \rightarrow A} f(x, y) = \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{x^4 + y^5}{x^3 + y^3} = \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{x^4}{x^3 + y^3} + \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{y^5}{x^3 + y^3} = L_1 + L_2$$

$$\left. \begin{aligned} 0 < \left| \frac{x^4}{x^3 + y^3} \right| &< \left| \frac{x^4}{x^3} \right| = |x| \xrightarrow{x \rightarrow 0} 0 \Rightarrow L_1 = 0 \\ 0 < \left| \frac{y^5}{x^3 + y^3} \right| &< \left| \frac{y^5}{y^3} \right| = y^2 \xrightarrow{y \rightarrow 0} 0 \Rightarrow L_2 = 0 \end{aligned} \right\} \Rightarrow L = L_1 + L_2 = 0$$

$$\lim_{(x, y) \rightarrow A} f(x, y) = f(A) \Rightarrow (\text{הגדרת ההמשכיות})$$

$$2) f(x, y) = \begin{cases} (x+y) \sin \frac{1}{x^2} \cos \frac{1}{y} & , (x, y) \neq (0, 0) \\ 0 & , (x, y) = (0, 0) \end{cases}$$

$$L = \lim_{(x, y) \rightarrow A} f(x, y) = \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} (x+y) \sin \frac{1}{x^2} \cos \frac{1}{y}$$

$$0 \leq |(x+y) \sin \frac{1}{x^2} \cos \frac{1}{y}| \leq |(x+y) \cdot 1 \cdot 1| = |x+y| \leq |x| + |y| \xrightarrow{\substack{x \rightarrow 0 \\ y \rightarrow 0}} 0 \Rightarrow L = 0$$

$$\lim_{(x, y) \rightarrow A} f(x, y) = f(A) \Rightarrow (\text{הגדרת ההמשכיות})$$

$$3) f(x, y) = \begin{cases} \frac{x^2 + y}{\sqrt{x^2 + y^2}} & , (x, y) \neq (0, 0) \\ 0 & , (x, y) = (0, 0) \end{cases}$$

$$L = \lim_{(x, y) \rightarrow A} f(x, y) = \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{x^2 + y}{\sqrt{x^2 + y^2}} = \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{x^2}{\sqrt{x^2 + y^2}} + \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{y}{\sqrt{x^2 + y^2}} = L_1 + L_2$$

$$0 < \left| \frac{x^2}{\sqrt{x^2 + y^2}} \right| < \left| \frac{x^2}{\sqrt{x^2}} \right| = \frac{|x|^2}{|x|} = |x| \xrightarrow{x \rightarrow 0} 0 \Rightarrow L_1 = 0$$

$$L_2 = \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{y}{\sqrt{x^2 + y^2}} = |y = kx| = \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{kx}{x\sqrt{1+k^2}} = \frac{k}{\sqrt{1+k^2}}$$

:כך נראה כי L אינו קיים. לכן L_2 אינו קיים.

$$\lim_{(x, y) \rightarrow A} f(x, y) \neq f(A) \Rightarrow (\text{הגדרת ההמשכיות})$$

$$4) f(x,y) = \begin{cases} \frac{xy}{x^2+y^2}, & (x,y) \neq 0 \\ 0, & (x,y) = 0 \end{cases}$$

$$L = \lim_{(x,y) \rightarrow A} f(x,y) = \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{xy}{x^2+y^2} = \left| \begin{matrix} x = r \cos \varphi \\ y = r \sin \varphi \end{matrix} \right| = \lim_{r \rightarrow 0} \frac{r^2 \cos \varphi \sin \varphi}{r^2} = \frac{1}{2} \sin 2\varphi$$

ה'37 108 ה'377108 $\Leftrightarrow \lim_{(x,y) \rightarrow A} f(x,y) \neq f(A)$: 108, 1017 108 1027

$$5) f(x,y) = \begin{cases} \frac{\sqrt[3]{y} \sin(x^2)}{x^2+y^2}, & (x,y) \neq (0,0) \\ 0, & (x,y) = (0,0) \end{cases}$$

$$L = \lim_{(x,y) \rightarrow A} f(x,y) = \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{\sqrt[3]{y} \sin(x^2)}{x^2+y^2} = \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{\sin(x^2)}{x^2} \cdot \frac{x^2 \sqrt[3]{y}}{x^2+y^2} = 1 \cdot \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{x^2 \sqrt[3]{y}}{x^2+y^2}$$

$$0 < \left| \frac{x^2 \sqrt[3]{y}}{x^2+y^2} \right| < \left| \frac{x^2 \sqrt[3]{y}}{x^2} \right| = |\sqrt[3]{y}| \xrightarrow{y \rightarrow 0} 0 \Rightarrow L = 0$$

$$\lim_{(x,y) \rightarrow A} f(x,y) = f(A) \Rightarrow (\text{ה'37} \text{ ה'377108})$$

$$6) f(x,y) = \begin{cases} \frac{x^4 - x^3y - xy^3 + y^4}{\sqrt{x^6 + y^6}}, & (x,y) \neq (0,0) \\ 0, & (x,y) = (0,0) \end{cases}$$

$$L = \lim_{(x,y) \rightarrow A} f(x,y) = \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{x^4 - x^3y - xy^3 + y^4}{\sqrt{x^6 + y^6}} = \left| \begin{matrix} x = r \cos \varphi \\ y = r \sin \varphi \end{matrix} \right| =$$

$$= \lim_{r \rightarrow 0} \frac{r^4 (\cos^4 \varphi - \cos^3 \varphi \sin \varphi - \cos \varphi \sin^3 \varphi + \sin^4 \varphi)}{r^3 \sqrt{\cos^6 \varphi + \sin^6 \varphi}} =$$

$$= \lim_{r \rightarrow 0} \frac{r (\cos^4 \varphi - \cos^3 \varphi \sin \varphi - \cos \varphi \sin^3 \varphi + \sin^4 \varphi)}{\sqrt{\cos^6 \varphi + \sin^6 \varphi}}$$

$$0 \leq |L| = r \cdot \frac{|\cos^4 \varphi + (-\cos^3 \varphi \sin \varphi) + (-\cos \varphi \sin^3 \varphi) + \sin^4 \varphi|}{\sqrt{(\cos^2 \varphi + \sin^2 \varphi)(\cos^4 \varphi - \cos^2 \varphi \sin^2 \varphi + \sin^4 \varphi)}} \leq$$

$$\leq r \cdot \frac{|\cos^4 \varphi| + |\cos^3 \varphi \sin \varphi| + |\cos \varphi \sin^3 \varphi| + |\sin^4 \varphi|}{\sqrt{(\cos^4 \varphi + 2 \cos^2 \varphi \sin^2 \varphi + \sin^4 \varphi) - 3 \cos^2 \varphi \sin^2 \varphi}} \leq$$

$$\leq r \cdot \frac{1+1+1+1}{\sqrt{(\cos^2 \varphi + \sin^2 \varphi)^2 - \frac{3}{4} \cdot 4 \sin^2 \varphi \cos^2 \varphi}} = \frac{4r}{\sqrt{1 - \frac{3}{4} \sin^2 2\varphi}} \leq$$

$$\leq \frac{4r}{\sqrt{1 - \frac{3}{4} \cdot 1}} = \frac{4r}{\sqrt{1/4}} = 8r \xrightarrow{r \rightarrow 0} 0 \Rightarrow L = 0$$

$$\lim_{(x,y) \rightarrow A} f(x,y) = f(A) \Rightarrow (\text{ה'37} \text{ ה'377108})$$

נגזרות חלקיות ודיפרנציאביליות של פונקציות של משתנים אחדים

$$1) z = x^3 + 3x^2y - y^3. \quad \frac{\partial z}{\partial x} = 3x^2 + 6xy = 3x(x+2y), \quad \frac{\partial z}{\partial y} = 3(x^2 - y^2)$$

$$2) z = \ln(x^2 + y^2). \quad \frac{\partial z}{\partial x} = \frac{2x}{x^2 + y^2}, \quad \frac{\partial z}{\partial y} = \frac{2y}{x^2 + y^2}$$

$$3) z = \frac{y}{x}. \quad \frac{\partial z}{\partial x} = -\frac{y}{x^2}, \quad \frac{\partial z}{\partial y} = \frac{1}{x}$$

$$4) z = \arctg \frac{y}{x}. \quad \frac{\partial z}{\partial x} = \frac{1}{1+y^2/x^2} \cdot \left(-\frac{y}{x^2}\right) = -\frac{y}{x^2+y^2}$$

$$\frac{\partial z}{\partial y} = \frac{1}{1+y^2/x^2} \cdot \frac{1}{x} = \frac{x}{x^2+y^2}$$

$$5) z = \frac{xy}{x-y}. \quad \frac{\partial z}{\partial x} = \frac{y(x-y) - xy}{(x-y)^2} = -\frac{y^2}{(x-y)^2}, \quad \frac{\partial z}{\partial y} = \frac{x^2}{(x-y)^2}$$

$$6) z = \ln\left(\frac{1}{\sqrt[3]{x}} - \frac{1}{\sqrt[3]{y}}\right). \quad \frac{\partial z}{\partial x} = \frac{-\frac{1}{3}x^{-4/3}}{\frac{1}{\sqrt[3]{x}} - \frac{1}{\sqrt[3]{y}}} = \frac{-\frac{1}{3}\sqrt[3]{x}^{-4/3}}{\frac{1}{\sqrt[3]{x}} - \frac{1}{\sqrt[3]{y}}} = \frac{\sqrt[3]{y}}{3y(\sqrt[3]{x} - \sqrt[3]{y})}$$

$$\frac{\partial z}{\partial y} = \frac{\frac{1}{3}y^{-4/3}}{\frac{1}{\sqrt[3]{x}} - \frac{1}{\sqrt[3]{y}}} = \frac{\sqrt[3]{x}}{3x(\sqrt[3]{y} - \sqrt[3]{x})}$$

$$7) u = xe^{-yx}. \quad \frac{\partial u}{\partial x} = e^{-yx} - yxe^{-yx} = e^{-yx}(1 - xy), \quad \frac{\partial u}{\partial y} = xe^{-yx}(-x) = -x^2e^{-yx}$$

$$8) u = \frac{y}{x} + \frac{z}{y} - \frac{x}{z}. \quad \frac{\partial u}{\partial x} = -\frac{y}{x^2} - \frac{1}{z}, \quad \frac{\partial u}{\partial y} = \frac{1}{x} - \frac{z}{y^2}, \quad \frac{\partial u}{\partial z} = \frac{1}{y} + \frac{x}{z^2}$$

$$9) z = \frac{2x-t}{x+2t}. \quad \frac{\partial z}{\partial x} = \frac{2(x+2t) - (2x-t) \cdot 1}{(x+2t)^2} = \frac{5t}{(x+2t)^2}$$

$$\frac{\partial z}{\partial t} = \frac{-(x+2t) - (2x-t) \cdot 2}{(x+2t)^2} = -\frac{5x}{(x+2t)^2}$$

$$10) z = \sqrt{x^2 + y^2 - 2xy \cos t}. \quad \frac{\partial z}{\partial x} = \frac{2x - 2y \cos t}{2\sqrt{x^2 + y^2 - 2xy \cos t}} = \frac{x - y \cos t}{z}$$

$$\frac{\partial z}{\partial y} = \frac{2(y - x \cos t)}{2\sqrt{x^2 + y^2 - 2xy \cos t}} = \frac{y - x \cos t}{z}, \quad \frac{\partial z}{\partial t} = \frac{xy \sin t}{z}$$

$$11) z = \arcsin(t\sqrt{x}). \quad \frac{\partial z}{\partial t} = \frac{1}{\sqrt{1-xt^2}} \sqrt{x} = \sqrt{\frac{x}{1-xt^2}}$$

$$\frac{\partial z}{\partial x} = \frac{1}{\sqrt{1-xt^2}} \cdot t \cdot \frac{1}{2\sqrt{x}} = \frac{t}{2\sqrt{x-xt^2}}$$

$$12) z = \cos(ax - by). \quad \frac{\partial z}{\partial x} = -\sin(ax - by) \cdot a = -a \sin(ax - by)$$

$$\frac{\partial z}{\partial y} = -\sin(ax - by) \cdot (-b) = b \sin(ax - by)$$

$$13) z = \frac{x}{3y-2x} \quad \frac{\partial z}{\partial x} = \frac{1 \cdot (3y-2x) - x(-2)}{(3y-2x)^2} = \frac{3y}{(3y-2x)^2}$$

$$\frac{\partial z}{\partial y} = [x(3y-2x)^{-1}]'_y = -x(3y-2x)^{-2}(3) = -\frac{3x}{(3y-2x)^2}$$

$$14) u = \ln \sin(x-2t) \quad \frac{\partial u}{\partial x} = \frac{1}{\sin(x-2t)} \cos(x-2t) \cdot 1 = \cot(x-2t)$$

$$\frac{\partial u}{\partial t} = \frac{1}{\sin(x-2t)} \cos(x-2t) (-2) = -2 \cot(x-2t)$$

$$15) u = \arcsin \frac{y}{x} \quad \frac{\partial u}{\partial x} = \frac{1}{\sqrt{1-(\frac{y}{x})^2}} \left(-\frac{y}{x^2}\right) = -\frac{y|x|}{x^2 \sqrt{x^2-y^2}}$$

$$\frac{\partial u}{\partial y} = \frac{1}{\sqrt{1-(\frac{y}{x})^2}} \cdot \frac{1}{x} = \frac{|x|}{x \sqrt{x^2-y^2}}$$

$$16) u = \sin^2(x+y) - \sin^2 x - \sin^2 y. \quad \frac{\partial u}{\partial x} = 2 \sin(x+y) \cos(x+y) \cdot 1 - 2 \sin x \cdot \cos x =$$

$$= \sin(2x+2y) - \sin 2x = 2 \sin \frac{2x+2y-2x}{2} \cos \frac{2x+2y+2x}{2} = 2 \sin y \cdot \cos(2x+y)$$

$$\frac{\partial u}{\partial y} = 2 \sin x \cdot \cos(x+2y) \quad ; \text{ស្រប ទៅ លក្ខណៈ ផ្គុំ}$$

$$17) z = x \cdot \arctg y. \quad \frac{\partial z}{\partial x} = \arctg y, \quad \frac{\partial z}{\partial y} = x \cdot \frac{1}{1+y^2} = \frac{x}{1+y^2}$$

$$18) z = \ln(\sqrt{x} - \sqrt{y}). \quad \frac{\partial z}{\partial x} = \frac{1}{\sqrt{x}-\sqrt{y}} \cdot \frac{1}{2\sqrt{x}} = \frac{1}{2(x-\sqrt{xy})}, \quad \frac{\partial z}{\partial y} = \frac{1}{2(y-\sqrt{xy})}$$

$$19) u = 3xyz + xy + xz + yz. \quad \frac{\partial u}{\partial x} = 3yz + y + z, \quad \frac{\partial u}{\partial y} = 3xz + x + z, \quad \frac{\partial u}{\partial z} = 3xy + x + y$$

$$20) u = \sin x \cdot \cos y + \cos x \cdot \tg(yz). \quad \frac{\partial u}{\partial x} = \cos x \cdot \cos y - \sin x \cdot \tg(yz)$$

$$\frac{\partial u}{\partial y} = -\sin x \sin y + \frac{z \cos x}{\cos^2(yz)}, \quad \frac{\partial u}{\partial z} = \cos x \cdot \frac{y}{\cos^2(yz)} = \frac{y \cos x}{\cos^2(yz)}$$

$$21) z = \ln(\sqrt{x} + \sqrt{y}) \Rightarrow \frac{\partial z}{\partial x} = \frac{1}{\sqrt{x} + \sqrt{y}} \cdot \frac{1}{2\sqrt{x}} = \frac{1}{2(x + \sqrt{xy})}$$

$$\frac{\partial z}{\partial y} = \frac{1}{\sqrt{x} + \sqrt{y}} \cdot \frac{1}{2\sqrt{y}} = \frac{1}{2(y + \sqrt{xy})}$$

$$x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = \frac{x}{2(x + \sqrt{xy})} + \frac{y}{2(y + \sqrt{xy})} =$$

$$= \frac{1}{2} \cdot \frac{xy + x\sqrt{xy} + xy + y\sqrt{xy}}{(x + \sqrt{xy})(y + \sqrt{xy})} = \frac{1}{2} \cdot \frac{2xy + (x+y)\sqrt{xy}}{2xy + (x+y)\sqrt{xy}} = \frac{1}{2}$$

$$22) u = e^{x/t^2} \Rightarrow \frac{\partial u}{\partial x} = e^{x/t^2} \cdot \frac{1}{t^2}, \quad \frac{\partial u}{\partial t} = e^{x/t^2} \left(-\frac{2x}{t^3}\right)$$

$$2x \frac{\partial u}{\partial x} + t \frac{\partial u}{\partial t} = 2x e^{x/t^2} \cdot \frac{1}{t^2} + t e^{x/t^2} \left(-\frac{2x}{t^3}\right) = e^{x/t^2} \left(\frac{2x}{t^2} - \frac{2x}{t^2}\right) = 0$$

$$23) u = x^y \Rightarrow \frac{\partial u}{\partial x} = y \cdot x^{y-1} = \frac{y}{x} x^y, \quad \frac{\partial u}{\partial y} = x^y \ln x$$

$$\frac{x}{y} \cdot \frac{\partial u}{\partial x} + \frac{1}{\ln x} \cdot \frac{\partial u}{\partial y} = \frac{x}{y} \cdot \frac{y}{x} \cdot x^y + \frac{1}{\ln x} \cdot x^y \ln x = 2x^y = 2u$$

$$24) u = \sqrt{x^2 + y^2 + z^2} \Rightarrow \frac{\partial u}{\partial x} = \frac{2x}{2\sqrt{x^2 + y^2 + z^2}} = \frac{x}{\sqrt{x^2 + y^2 + z^2}},$$

$$\frac{\partial u}{\partial y} = \frac{y}{\sqrt{x^2 + y^2 + z^2}}, \quad \frac{\partial u}{\partial z} = \frac{z}{\sqrt{x^2 + y^2 + z^2}}$$

$$\left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial u}{\partial y}\right)^2 + \left(\frac{\partial u}{\partial z}\right)^2 = \frac{x^2}{x^2 + y^2 + z^2} + \frac{y^2}{x^2 + y^2 + z^2} + \frac{z^2}{x^2 + y^2 + z^2} = \frac{x^2 + y^2 + z^2}{x^2 + y^2 + z^2} = 1$$

$$25) z = e^{x/y} \ln y \Rightarrow \frac{\partial z}{\partial x} = e^{x/y} \cdot \frac{1}{y} \ln y$$

$$\frac{\partial z}{\partial y} = e^{x/y} \left(-\frac{x}{y^2}\right) \ln y + e^{x/y} \frac{1}{y} = \frac{e^{x/y}}{y} \left(1 - \frac{x \ln y}{y}\right)$$

$$x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = e^{x/y} \cdot \frac{x}{y} \ln y + e^{x/y} - e^{x/y} \cdot \frac{x \ln y}{y} = e^{x/y} = \frac{e^{x/y} \ln y}{\ln y} = \frac{z}{\ln y}$$

$$26) T = \pi \sqrt{\frac{e}{g}} \Rightarrow \frac{\partial T}{\partial e} = \frac{\pi}{\sqrt{g}} \cdot \frac{1}{2\sqrt{e}}, \quad \frac{\partial T}{\partial g} = \pi \sqrt{e} \left(-\frac{1}{2} g^{-3/2}\right) = -\pi \sqrt{\frac{e}{g}} \cdot \frac{1}{2g}$$

$$e \frac{\partial T}{\partial e} + g \frac{\partial T}{\partial g} = e \cdot \frac{\pi}{\sqrt{g}} \cdot \frac{1}{2\sqrt{e}} - g \pi \sqrt{\frac{e}{g}} \cdot \frac{1}{2g} = \frac{1}{2} \pi \sqrt{\frac{e}{g}} - \frac{1}{2} \pi \sqrt{\frac{e}{g}} = 0$$

$$27) f(x, y) = \begin{cases} \frac{2xy}{x^2+y^2}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$$

$$f'_x(0, 0) = \lim_{\Delta x \rightarrow 0} \frac{\frac{2\Delta x \cdot 0}{\Delta x^2 + 0^2} - 0}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{0}{\Delta x} = 0, \quad f'_y(0, 0) = \lim_{\Delta y \rightarrow 0} \frac{\frac{2 \cdot 0 \cdot \Delta y}{0^2 + \Delta y^2} - 0}{\Delta y} = 0$$

$$28) f(x, y) = \begin{cases} (x^2+y^2) \cos \frac{\pi}{x^2+y^2}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$$

$$f'_x(0, 0) = \lim_{\Delta x \rightarrow 0} \frac{\Delta x^2 \cos \frac{\pi}{\Delta x^2} - 0}{\Delta x} = \lim_{\Delta x \rightarrow 0} \Delta x \cdot \cos \frac{\pi}{\Delta x^2} = \left| 0 \cdot \left(\sin \frac{\pi}{\Delta x^2} \right) \right| = 0$$

$$f'_y(0, 0) = \lim_{\Delta y \rightarrow 0} \frac{\Delta y^2 \cos \frac{\pi}{\Delta y^2} - 0}{\Delta y} = \lim_{\Delta y \rightarrow 0} \Delta y \cdot \cos \frac{\pi}{\Delta y^2} = 0$$

$$29) f(x, y) = \begin{cases} \frac{xy}{\sqrt{x^2+y^2}}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$$

$$f'_x(0, 0) = \lim_{\Delta x \rightarrow 0} \frac{\frac{\Delta x \cdot 0}{\sqrt{\Delta x^2 + 0}} - 0}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{0}{\Delta x} = 0, \quad f'_y(0, 0) = \lim_{\Delta y \rightarrow 0} \frac{\frac{0 \cdot \Delta y}{\sqrt{0 + \Delta y^2}} - 0}{\Delta y} = 0$$

$$30) f(x, y) = \begin{cases} \frac{x^3+y^3}{x^2+3y^2}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$$

$$f'_x(0, 0) = \lim_{\Delta x \rightarrow 0} \frac{\frac{\Delta x^3 + 0}{\Delta x^2 + 0} - 0}{\Delta x} = \lim_{\Delta x \rightarrow 0} 1 = 1, \quad f'_y(0, 0) = \lim_{\Delta y \rightarrow 0} \frac{\frac{0 + \Delta y^3}{0 + 3\Delta y^2} - 0}{\Delta y} = \lim_{\Delta y \rightarrow 0} \frac{1}{3} = \frac{1}{3}$$

$$31) f(x, y) = \begin{cases} y^{4/3} \ln(y^2 + x^{4/3}), & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$$

$$f'_x(0, 0) = \lim_{\Delta x \rightarrow 0} \frac{0 \cdot \ln(\Delta x^{4/3}) - 0}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{0}{\Delta x} = 0$$

$$f'_y(0, 0) = \lim_{\Delta y \rightarrow 0} \frac{\Delta y^{4/3} \cdot \ln(\Delta y^2) - 0}{\Delta y} = \lim_{\Delta y \rightarrow 0} \sqrt[3]{\Delta y} \cdot \ln(\Delta y^2) = \left| 0 \cdot \infty \right| = \lim_{\Delta y \rightarrow 0} \frac{\ln(\Delta y^2)}{\frac{1}{\Delta y^{1/3}}} = \left| \frac{\infty}{\infty} \right| =$$

$$= \lim_{\Delta y \rightarrow 0} \frac{\frac{2\Delta y}{\Delta y^2}}{-\frac{1}{3}\Delta y^{-4/3}} = -6 \lim_{\Delta y \rightarrow 0} \frac{\Delta y^{4/3}}{\Delta y} = -6 \lim_{\Delta y \rightarrow 0} \sqrt[3]{\Delta y} = 0$$

$$32) f(x,y) = \begin{cases} \frac{xy}{\sqrt{x^2+y^2}} & , (x,y) \neq (0,0) \\ 0 & , (x,y) = (0,0) \end{cases}$$

נבדוק את רציפות הפונקציה בנקודה $(0,0)$, כי רק פונקציה רציפה בנקודה יכולה להיות דיפרנציאביליות בנקודה הזאת.

$$L = \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} f(x,y) = \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{xy}{\sqrt{x^2+y^2}} \Rightarrow$$

$$0 < \left| \frac{xy}{\sqrt{x^2+y^2}} \right| < \left| \frac{xy}{\sqrt{x^2}} \right| < \frac{|x||y|}{|x|} = |y| \xrightarrow{y \rightarrow 0} 0 \Rightarrow L = 0 \Rightarrow \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} f(x,y) = f(0,0)$$

הפונקציה $f(x,y)$ רציפה בנקודה $(0,0)$.

נמצא את הנגזרות החלקיות בנקודה $(0,0)$.

$$f'_x(0,0) = \lim_{\Delta x \rightarrow 0} \frac{\frac{\Delta x \cdot 0}{\sqrt{\Delta x^2 + 0}} - 0}{\Delta x} = 0, \quad f'_y(0,0) = \lim_{\Delta y \rightarrow 0} \frac{\frac{0 \cdot \Delta y}{\sqrt{0 + \Delta y^2}} - 0}{\Delta y} = 0$$

לפי ההגדרה, הפונקציה $f(x,y)$ דיפרנציאביליות בנקודה $(0,0)$ אם מתקיים:

$$\Delta f(0,0) = f'_x(0,0) \cdot \Delta x + f'_y(0,0) \cdot \Delta y + \varepsilon \cdot \sqrt{\Delta x^2 + \Delta y^2}, \quad \lim_{\substack{\Delta x \rightarrow 0 \\ \Delta y \rightarrow 0}} \varepsilon = 0$$

כאשר x, y מקבלים בנקודה $(0,0)$ את התוספות $\Delta x, \Delta y$ אז הפונקציה מקבלת

$$\Delta f(0,0) = \frac{\Delta x \cdot \Delta y}{\sqrt{\Delta x^2 + \Delta y^2}} \quad \text{את התוספת השלמה}$$

לכן, אם $f(x,y)$ דיפרנציאביליות בנקודה $(0,0)$ אז צריך להתקיים:

$$\frac{\Delta x \cdot \Delta y}{\sqrt{\Delta x^2 + \Delta y^2}} = 0 \cdot \Delta x + 0 \cdot \Delta y + \varepsilon \cdot \sqrt{\Delta x^2 + \Delta y^2}$$

$$\varepsilon = \frac{\Delta x \cdot \Delta y}{\left(\sqrt{\Delta x^2 + \Delta y^2}\right)^2} = \frac{\Delta x \cdot \Delta y}{\Delta x^2 + \Delta y^2} \quad \text{מכאן נובע:}$$

נבדוק עכשיו - האם ε שואף לאפס כאשר $\Delta x, \Delta y$ שואפים לאפס.

$$\lim_{\substack{\Delta x \rightarrow 0 \\ \Delta y \rightarrow 0}} \varepsilon = \lim_{\substack{\Delta x \rightarrow 0 \\ \Delta y \rightarrow 0}} \frac{\Delta x \cdot \Delta y}{\Delta x^2 + \Delta y^2} = \lim_{\Delta x \rightarrow 0} \frac{\Delta x^2 \cdot k}{\Delta x^2 (1+k^2)} = \frac{k}{1+k^2} \neq 0$$

הפונקציה לא דיפרנציאביליות בנקודה $(0,0)$.

$$33) f(x,y) = \begin{cases} (x^2+y^2) \sin \frac{1}{x^2+y^2} & , (x,y) \neq (0,0) \\ 0 & , (x,y) = (0,0) \end{cases}$$

נבדוק את רציפות הפונקציה בנקודה $(0,0)$.

$$\begin{aligned} L = \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} f(x,y) &= \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} (x^2+y^2) \sin \frac{1}{x^2+y^2} = \lim_{r \rightarrow 0} \begin{vmatrix} x = r \cos \varphi \\ y = r \sin \varphi \end{vmatrix} (r^2) \sin \frac{1}{r^2} = \\ &= |0 \cdot (\text{פונקציה חסומה})| = 0 \Rightarrow L = 0 \Rightarrow \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} f(x,y) = f(0,0) \end{aligned}$$

הפונקציה $f(x,y)$ רציפה בנקודה $(0,0)$. נמצא את הנגזרות החלקיות בנקודה $(0,0)$.

$$\begin{aligned} f'_x(0,0) &= \lim_{\Delta x \rightarrow 0} \frac{(\Delta x^2 + 0) \sin \frac{1}{\Delta x^2 + 0} - 0}{\Delta x} = \lim_{\Delta x \rightarrow 0} \Delta x \cdot \sin \frac{1}{\Delta x^2} = 0 \\ &\text{באותה דרך נוכיח: } f'_y(0,0) = 0 \end{aligned}$$

לפי ההגדרה, הפונקציה $f(x,y)$ דיפרנציאביליות בנקודה $(0,0)$ אם מתקיים:

$$\Delta f(0,0) = f'_x(0,0) \cdot \Delta x + f'_y(0,0) \cdot \Delta y + \varepsilon \cdot \sqrt{\Delta x^2 + \Delta y^2}, \quad \lim_{\substack{\Delta x \rightarrow 0 \\ \Delta y \rightarrow 0}} \varepsilon = 0$$

כאשר x, y מקבלים בנקודה $(0,0)$ את התוספות $\Delta x, \Delta y$ אז הפונקציה מקבלת את התוספת השלמה: $\Delta f(0,0) = (\Delta x^2 + \Delta y^2) \sin \frac{1}{\Delta x^2 + \Delta y^2}$. לכן צריך להתקיים:

$$(\Delta x^2 + \Delta y^2) \sin \frac{1}{\Delta x^2 + \Delta y^2} = 0 \cdot \Delta x + 0 \cdot \Delta y + \varepsilon \cdot \sqrt{\Delta x^2 + \Delta y^2}$$

מכאן נובע:

$$\varepsilon = \frac{(\Delta x^2 + \Delta y^2)}{\sqrt{\Delta x^2 + \Delta y^2}} \sin \frac{1}{\Delta x^2 + \Delta y^2} = \sqrt{\Delta x^2 + \Delta y^2} \sin \frac{1}{\Delta x^2 + \Delta y^2}$$

נבדוק עכשיו – האם ε שואף לאפס כאשר $\Delta x, \Delta y$ שואפים לאפס.

$$\lim_{\substack{\Delta x \rightarrow 0 \\ \Delta y \rightarrow 0}} \varepsilon = \lim_{\substack{\Delta x \rightarrow 0 \\ \Delta y \rightarrow 0}} \sqrt{\Delta x^2 + \Delta y^2} \sin \frac{1}{\Delta x^2 + \Delta y^2} = \lim_{r \rightarrow 0} \begin{vmatrix} \Delta x = r \cos \varphi \\ \Delta y = r \sin \varphi \end{vmatrix} r \cdot \sin \frac{1}{r^2} = 0$$

הפונקציה דיפרנציאביליות בנקודה $(0,0)$.

$$34) f(x,y) = \sqrt[3]{xy}$$

(a) ערך של הפונקציה בנקודה $(0,0)$ שווה ל-0. לכן:

$$f'_x(0,0) = \lim_{\Delta x \rightarrow 0} \frac{\sqrt[3]{x \cdot 0} - 0}{\Delta x} = 0, \quad f'_y(0,0) = \lim_{\Delta y \rightarrow 0} \frac{\sqrt[3]{0 \cdot y} - 0}{\Delta y} = 0$$

(b) בכל נקודה בסביבת הנקודה $(0,0)$ חוץ מ- $(0,0)$ עצמה:

$$f'_x(x,y) = \frac{\partial}{\partial x}(\sqrt[3]{xy}) = \sqrt[3]{y} \cdot \frac{\partial}{\partial x}(\sqrt[3]{x}) = \sqrt[3]{y} \cdot \frac{1}{3} x^{-2/3} = \frac{1}{3} \sqrt[3]{\frac{y}{x^2}}$$

מטעמי סימטריה: $f'_y(x,y) = \frac{1}{3} \sqrt[3]{\frac{x}{y^2}}$. נבדוק – האם $f'_x(x,y)$ רציפה בנקודה $(0,0)$.

$$\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} f'_x(x,y) = \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{1}{3} \sqrt[3]{\frac{y}{x^2}} = \left| y = k \cdot x^2 \right| = \lim_{x \rightarrow 0} \frac{1}{3} \sqrt[3]{\frac{k \cdot x^2}{x^2}} = \frac{1}{3} \sqrt[3]{k}$$

הגבול לא קיים. ז"א $\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} f'_x(x,y) \neq f'_x(0,0)$ ומסיקים שהנגזרת f'_x לא

רציפה בנקודה $(0,0)$. באותה דרך נגיע לאותה המסקנה עבור הנגזרת f'_y .

(c) רציפות של הנגזרות f'_x , f'_y בנקודה – הוא התנאי המספיק לדיפרנציאביליות של הפונקציה בנקודה. לכן, אם הנגזרות לא רציפות – זה עוד לא אומר שהפונקציה לא דיפרנציאבילית. לכן נבדוק את הדיפרנציאביליות לפי ההגדרה.

הפונקציה הנתונה רציפה בנקודה $(0,0)$ (קומפוזיציה של פונקציות רציפות). לפי ההגדרה, הפונקציה $f(x,y)$ דיפרנציאבילית בנקודה $(0,0)$ אם מתקיים:

$$\Delta f(0,0) = f'_x(0,0) \cdot \Delta x + f'_y(0,0) \cdot \Delta y + \varepsilon \cdot \sqrt{\Delta x^2 + \Delta y^2}, \quad \lim_{\substack{\Delta x \rightarrow 0 \\ \Delta y \rightarrow 0}} \varepsilon = 0$$

כאשר x, y מקבלים בנקודה $(0,0)$ תוספות $\Delta x, \Delta y$ אז הפונקציה מקבלת את התוספת השלמה: $\Delta f(0,0) = \sqrt[3]{\Delta x \cdot \Delta y}$. לכן צריך להתקיים:

$$\sqrt[3]{\Delta x \cdot \Delta y} = 0 \cdot \Delta x + 0 \cdot \Delta y + \varepsilon \cdot \sqrt{\Delta x^2 + \Delta y^2}$$

$$\varepsilon = \frac{\sqrt[3]{\Delta x \cdot \Delta y}}{\sqrt{\Delta x^2 + \Delta y^2}} \quad \text{מכאן נובע:}$$

נבדוק עכשיו – האם ε שואף לאפס כאשר $\Delta x, \Delta y$ שואפים לאפס.

$$\lim_{\substack{\Delta x \rightarrow 0 \\ \Delta y \rightarrow 0}} \varepsilon = \lim_{\substack{\Delta x \rightarrow 0 \\ \Delta y \rightarrow 0}} \frac{\sqrt[3]{\Delta x \cdot \Delta y}}{\sqrt{\Delta x^2 + \Delta y^2}} = \left| \begin{array}{l} \Delta x = r \cos \varphi \\ \Delta y = r \sin \varphi \end{array} \right| = \lim_{r \rightarrow 0} \frac{\sqrt[3]{r^2 \cos \varphi \sin \varphi}}{r} =$$

$$= \lim_{r \rightarrow 0} \sqrt[3]{\frac{r^2 \cdot \frac{1}{2} \cdot 2 \cos \varphi \sin \varphi}{r^3}} = \lim_{r \rightarrow 0} \sqrt[3]{\frac{\sin 2\varphi}{2r}} \neq 0$$

הפונקציה לא דיפרנציאבילית בנקודה $(0,0)$.

יישומים של מושג הדיפרנציאל

(N)

$$1) A = 1.02^{4.05} \Rightarrow z = x^y, x=1, \Delta x=0.02, y=4, \Delta y=0.05$$

$$z(x+\Delta x, y+\Delta y) \approx z(x, y) + \frac{\partial z}{\partial x} \Delta x + \frac{\partial z}{\partial y} \Delta y = x^y + yx^{y-1} \Delta x + x^y \ln x \Delta y$$

$$A \approx 1^4 + 4 \cdot 1^3 \cdot 0.02 + 1^4 \cdot \ln 1 \cdot 0.05 = 1 + 0.08 + 0 = 1.08$$

$$2) A = \ln(0.09^3 + 0.99^3) \Rightarrow z = \ln(x^3 + y^3), x=0, \Delta x=0.09, y=1, \Delta y=-0.01$$

$$z(x+\Delta x, y+\Delta y) \approx z(x, y) + \frac{\partial z}{\partial x} \Delta x + \frac{\partial z}{\partial y} \Delta y =$$

$$= \ln(x^3 + y^3) + \frac{3x^2}{x^3 + y^3} \Delta x + \frac{3y^2}{x^3 + y^3} \Delta y = \ln(x^3 + y^3) + 3 \frac{x^2 \Delta x + y^2 \Delta y}{x^3 + y^3}$$

$$A \approx \ln(0+1) + 3 \frac{0 \cdot 0.09 - 1 \cdot 0.01}{0+1} = 0 - 3 \cdot 0.01 = -0.03$$

$$3) A = (1.02)^3 (0.97)^2 \Rightarrow z = x^3 y^2, x=1, \Delta x=0.02, y=1, \Delta y=-0.03$$

$$z(x+\Delta x, y+\Delta y) \approx z(x, y) + \frac{\partial z}{\partial x} \Delta x + \frac{\partial z}{\partial y} \Delta y = x^3 y^2 + 3x^2 y^2 \Delta x + 2x^3 y \Delta y =$$

$$= x^3 y^2 (1 + 3 \frac{\Delta x}{x} + 2 \frac{\Delta y}{y}) \Rightarrow A \approx 1 \cdot 1 \cdot (1 + 3 \frac{0.02}{1} + 2 \frac{-0.03}{1}) = 1.00$$

$$4) A = \sqrt{5e^{0.02} + 2.03^2} \Rightarrow z = \sqrt{5e^x + y^2}, x=0, \Delta x=0.02, y=2, \Delta y=0.03$$

$$z(x+\Delta x, y+\Delta y) \approx z(x, y) + \frac{\partial z}{\partial x} \Delta x + \frac{\partial z}{\partial y} \Delta y =$$

$$= \sqrt{5e^x + y^2} + \frac{5e^x \Delta x}{2\sqrt{5e^x + y^2}} + \frac{2y \Delta y}{2\sqrt{5e^x + y^2}} = \sqrt{5e^x + y^2} + \frac{2.5e^x \Delta x + y \Delta y}{\sqrt{5e^x + y^2}}$$

$$A \approx \sqrt{5 \cdot 1 + 4} + \frac{1}{\sqrt{5+4}} (2.5 \cdot 1 \cdot 0.02 + 2 \cdot 0.03) = 3 + \frac{1}{3} (0.05 + 0.06) = 3.037$$

$$5) A = \sqrt{4.05^2 + 2.93^2} \Rightarrow z = \sqrt{x^2 + y^2}, x=4, \Delta x=0.05, y=3, \Delta y=-0.07$$

$$z(x+\Delta x, y+\Delta y) \approx z(x, y) + \frac{\partial z}{\partial x} \Delta x + \frac{\partial z}{\partial y} \Delta y = \sqrt{x^2 + y^2} + \frac{x \Delta x}{\sqrt{x^2 + y^2}} + \frac{y \Delta y}{\sqrt{x^2 + y^2}}$$

$$A \approx \sqrt{4^2 + 3^2} + \frac{1}{\sqrt{16+9}} (4 \cdot 0.05 - 3 \cdot 0.07) = 5 + \frac{1}{5} (0.2 - 0.21) = 4.998$$

$$6) A = \sqrt{1.04^{1.99} + \ln 1.02} \Rightarrow u = \sqrt{x^y + \ln z}, x=1, \Delta x=0.04, y=2, \Delta y=-0.01, z=1, \Delta z=0.02$$

$$u(x+\Delta x, y+\Delta y, z+\Delta z) \approx \sqrt{x^y + \ln z} + \frac{1}{2\sqrt{x^y + \ln z}} (yx^{y-1} \Delta x + x^y \ln x \Delta y + \frac{\Delta z}{z})$$

$$A \approx \sqrt{1 + \ln 1} + \frac{1}{2\sqrt{1+0}} (2 \cdot 1 \cdot 0.04 + 1 \cdot \ln 1 \cdot \Delta y + 1 \cdot 0.02) = 1 + \frac{1}{2} \cdot 0.1 = 1.05$$

(2)

1) $\Delta h = 1 \text{ mm}$, $h = 90 \text{ mm}$, $\Delta R = -1 \text{ mm}$, $a = 60 \text{ mm}$: | $\Delta V = a^2 h$: ΔV ၁၀၀% ကို

$$\Delta V \approx dV = \frac{\partial V}{\partial a} \Delta a + \frac{\partial V}{\partial h} \Delta h = 2ah \Delta a + a^2 \Delta h = a^2 h \left(2 \frac{\Delta a}{a} + \frac{\Delta h}{h} \right).$$

$$\Delta V = V \left(2 \frac{\Delta a}{a} + \frac{\Delta h}{h} \right) \Rightarrow \Delta V \% = \frac{\Delta V}{V} 100 \% = \left(2 \frac{\Delta a}{a} + \frac{\Delta h}{h} \right) 100 \% =$$

$$= \left(2 \cdot \frac{-1}{60} + \frac{1}{90} \right) \cdot 100 \% = -2.22 \% \Rightarrow \Delta V \% = -2 \%$$

. 2% - ကို ၁၀၀% ကို

2) : | $S' = \pi(R^2 - r^2)$: ΔS ၁၀၀% ကို

$$R = 50 \text{ mm}, \Delta R = 1 \text{ mm}, r = 40 \text{ mm}, \Delta r = 1 \text{ mm}$$

$$\Delta S \approx dS = \frac{\partial S}{\partial R} \Delta R + \frac{\partial S}{\partial r} \Delta r = 2\pi R \Delta R - 2\pi r \Delta r = 2\pi(R \Delta R - r \Delta r)$$

$$\Delta S \% = \frac{\Delta S}{S} 100 \% = 2 \cdot \frac{R \Delta R - r \Delta r}{R^2 - r^2} 100 \% = 2 \cdot \frac{50 \cdot 1 - 40 \cdot 1}{50^2 - 40^2} 100 \% = 2.22 \%$$

. 2% - ကို $\Delta S \% = 2 \%$

3) : | $V = \pi R^2 h$: ΔV ၁၀၀% ကို

$$R = 50 \text{ mm}, \Delta R = 0.8 \text{ mm}, h = 70 \text{ mm}, \Delta h = -2 \text{ mm}$$

$$\Delta V \approx dV = \frac{\partial V}{\partial R} \Delta R + \frac{\partial V}{\partial h} \Delta h = \pi 2R h \Delta R + \pi R^2 \Delta h = \pi R^2 h \left(2 \frac{\Delta R}{R} + \frac{\Delta h}{h} \right)$$

$$\Delta V \% = \frac{\Delta V}{V} 100 \% = \left(2 \frac{\Delta R}{R} + \frac{\Delta h}{h} \right) 100 \% = \left(2 \cdot \frac{0.8}{50} + \frac{-2}{70} \right) 100 \% = 0.34 \%$$

. 0.3% - ကို $\Delta V \% = 0.3 \%$

4) : | $S = \frac{1}{2} ab$: ΔS ၁၀၀% ကို . ΔS ၁၀၀% ကို

$$a = 200 \text{ mm}, \Delta a = -4 \text{ mm}, b = 100 \text{ mm}, \Delta b = 2 \text{ mm}$$

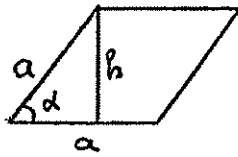
$$\Delta S \approx dS = \frac{\partial S}{\partial a} \Delta a + \frac{\partial S}{\partial b} \Delta b = \frac{1}{2} b \Delta a + \frac{1}{2} a \Delta b = \frac{1}{2} ab \left(\frac{\Delta a}{a} + \frac{\Delta b}{b} \right)$$

$$\Delta S = S \left(\frac{\Delta a}{a} + \frac{\Delta b}{b} \right) \Rightarrow \Delta S \% = \frac{\Delta S}{S} 100 \% = \left(\frac{\Delta a}{a} + \frac{\Delta b}{b} \right) 100 \% =$$

$$= \left(\frac{-4}{200} + \frac{2}{100} \right) 100 \% = 0$$

. ΔS ၁၀၀% ကို

5)



ဦးနှစ်က ဂရု . $h = a \sin \alpha$: ဦးနှစ်က အခါ

$$S = a \cdot h = a \cdot a \sin \alpha \Rightarrow S = a^2 \sin \alpha$$

$$a = 60 \text{ mm}, \Delta a = 1.5 \text{ mm}, \alpha = 45^\circ, \Delta \alpha = -1.5^\circ$$

$$\Delta S \approx dS = \frac{\partial S}{\partial a} \Delta a + \frac{\partial S}{\partial \alpha} \Delta \alpha = 2a \sin \alpha \Delta a + a^2 \cos \alpha \Delta \alpha =$$

$$= a^2 \sin \alpha \left(2 \frac{\Delta a}{a} + \frac{\Delta \alpha}{\tan \alpha} \right) = S \left(2 \frac{\Delta a}{a} + \frac{\Delta \alpha}{\tan \alpha} \right).$$

$$\Delta S \% = \frac{\Delta S}{S} 100 \% = \left(2 \frac{\Delta a}{a} + \frac{\Delta \alpha}{\tan \alpha} \right) = \left(2 \cdot \frac{1.5}{60} + \frac{-1.5^\circ \cdot \pi}{180^\circ \cdot \tan 45^\circ} \right) 100 \% = 2.4 \%$$

$$. 2 \% - \text{အ } \delta_3 \text{ ဂရု } \Delta S \% = 2 \%$$

6)

$$b = 100 \text{ mm}, \Delta b = -2.5 \text{ mm}, \alpha = 30^\circ, \Delta \alpha = 1^\circ : \text{ဦးနှစ်}$$



. $S = \frac{1}{2} b \cdot b \tan \alpha$: ဦးနှစ်က ဂရု

$$S = \frac{1}{2} b^2 \tan \alpha$$

$$\Delta S = \frac{\partial S}{\partial b} \Delta b + \frac{\partial S}{\partial \alpha} \Delta \alpha = b \tan \alpha \cdot \Delta b + \frac{1}{2} b^2 \frac{1}{\cos^2 \alpha} \Delta \alpha =$$

$$= \frac{1}{2} b^2 \tan \alpha \left(2 \frac{\Delta b}{b} + \frac{\Delta \alpha}{\cos^2 \alpha \tan \alpha} \right) = S \left(\frac{2 \Delta b}{b} + \frac{\Delta \alpha}{\sin \alpha \cos \alpha} \right) =$$

$$= S \left(\frac{2 \Delta b}{b} + \frac{2 \Delta \alpha}{\sin 2 \alpha} \right) = 2 S \left(\frac{\Delta b}{b} + \frac{\Delta \alpha}{\sin 2 \alpha} \right).$$

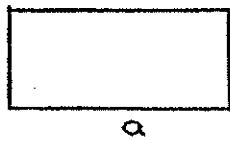
$$\Delta S \% = \frac{\Delta S}{S} 100 \% = 2 \left(\frac{\Delta b}{b} + \frac{\Delta \alpha}{\sin 2 \alpha} \right) 100 \% = 2 \left(\frac{-2.5}{100} + \frac{1^\circ \cdot \pi}{\sin 60^\circ \cdot 180^\circ} \right) 100 \% =$$

$$= -0.97 \% \Rightarrow \Delta S \% = -1 \%$$

$$1 \% - \text{အ } \delta_3 \text{ ဂရု}$$

ג. חישוב שגיאות

1)



$$S = a \cdot b \Rightarrow \Delta S = \left| \frac{\partial S}{\partial a} \right| \Delta a + \left| \frac{\partial S}{\partial b} \right| \Delta b = b \Delta a + a \Delta b =$$

$$S = 1.05 = 0.5 \text{ m}^2 \quad = ab \left(\frac{\Delta a}{a} + \frac{\Delta b}{b} \right) = S \left(\frac{\Delta a}{a} + \frac{\Delta b}{b} \right)$$

$$\Delta S = 0.5 \left(\frac{0.02}{1} + \frac{0.02}{0.5} \right) = 0.03 \text{ m}^2 \Rightarrow S = (0.50 \pm 0.03) \text{ m}^2$$

2) S - פס - h - רוחב - h - רוחב - S - שטח, $S = V/h$, $V = S \cdot h$

$$V = S \cdot h \Rightarrow h = \frac{V}{S} = \frac{1000}{90} = 11.111 \text{ cm}$$

$$\Delta h = \left| \frac{\partial h}{\partial V} \right| \Delta V + \left| \frac{\partial h}{\partial S} \right| \Delta S = \left| \frac{1}{S} \right| \Delta V + \left| -\frac{V}{S^2} \right| \Delta S = \frac{1}{S} \Delta V - \frac{V}{S^2} \Delta S =$$

$$= \frac{V}{S} \left(\frac{\Delta V}{V} + \frac{\Delta S}{S} \right) = h \left(\frac{\Delta V}{V} + \frac{\Delta S}{S} \right) = 11.111 \left(\frac{20}{1000} + \frac{1}{90} \right) = 0.345$$

$$\Delta h = 0.3 \text{ cm} \Rightarrow h = (11.1 \pm 0.3) \text{ cm}$$

3) $S = \frac{1}{2}ab$ - שטח, S - שטח, b - רוחב, a - אורך, $b = \frac{2S}{a}$

$$b = \frac{2S}{a} = \frac{2 \cdot 85}{9} = 18.888 \text{ cm}$$

$$\Delta b = \left| \frac{\partial b}{\partial S} \right| \Delta S + \left| \frac{\partial b}{\partial a} \right| \Delta a = \frac{2}{a} \Delta S + \left| -\frac{2S}{a^2} \right| \Delta a = \frac{2S}{a} \left(\frac{\Delta S}{S} + \frac{\Delta a}{a} \right) =$$

$$= b \left(\frac{\Delta S}{S} + \frac{\Delta a}{a} \right) = 18.888 \left(\frac{1}{85} + \frac{0.2}{9} \right) = 0.64 \Rightarrow \Delta b = 0.6 \text{ cm}$$

$$b = (18.9 \pm 0.6) \text{ cm}$$

4) $m = \sqrt{a^2 + b^2 + c^2}$ - אורך, a, b, c - צדדים, m - אורך

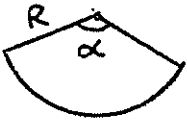
$$m = \sqrt{20^2 + 14^2 + 5^2} = 24.919 \quad \Leftarrow m = \sqrt{a^2 + b^2 + c^2} \quad ; \text{ אורך}$$

$$\Delta m = \left| \frac{\partial m}{\partial a} \right| \Delta a + \left| \frac{\partial m}{\partial b} \right| \Delta b + \left| \frac{\partial m}{\partial c} \right| \Delta c =$$

$$= \frac{1}{2\sqrt{a^2 + b^2 + c^2}} \cdot 2a \cdot \Delta a + \frac{1}{2\sqrt{a^2 + b^2 + c^2}} \cdot 2b \cdot \Delta b + \frac{1}{2\sqrt{a^2 + b^2 + c^2}} \cdot 2c \cdot \Delta c = \frac{a\Delta a + b\Delta b + c\Delta c}{\sqrt{a^2 + b^2 + c^2}}$$

$$= \frac{1}{24.919} \cdot (20 \cdot 0.3 + 14 \cdot 0.2 + 5 \cdot 0.1) = 0.373 \Rightarrow \Delta m = 0.4 \text{ cm}$$

$$m = (25.0 \pm 0.4) \text{ cm}$$

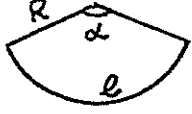
5)  $(\text{p'j'le'37} \alpha - \alpha) S = \frac{1}{2} R^2 \alpha$; $\alpha' \delta / \alpha' \gamma \text{ } \alpha' \gamma \delta \text{ } \alpha' \gamma \delta \text{ } \alpha' \gamma \delta$

$$\alpha = \frac{2S}{R^2} = \frac{2 \cdot 33}{8,2^2} = 0,9815(\text{rad}) = 0,9815 \frac{180^\circ}{\pi} = 56,239^\circ$$

$$\Delta \alpha = \left| \frac{\partial \alpha}{\partial S} \right| \Delta S + \left| \frac{\partial \alpha}{\partial R} \right| \Delta R = 2 \left(\frac{1}{R^2} \Delta S + \frac{2S}{R^3} \Delta R \right) = \frac{2S}{R^2} \left(\frac{\Delta S}{S} + 2 \frac{\Delta R}{R} \right) =$$

$$= \alpha \left(\frac{\Delta S}{S} + 2 \cdot \frac{\Delta R}{R} \right) = 56,239 \left(\frac{1}{33} + 2 \cdot \frac{0,05}{8,2} \right) = 2,39^\circ \Rightarrow \Delta \alpha = 2^\circ$$

$$\alpha = 56^\circ \pm 2^\circ$$

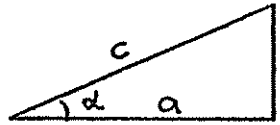
6)  $l = \alpha(\text{rad}) \cdot R \Rightarrow \alpha(\text{rad}) = \frac{l}{R} \Rightarrow \alpha^\circ = \frac{180^\circ}{\pi} \frac{l}{R}$

$$\alpha^\circ = \frac{180^\circ}{\pi} \cdot \frac{21}{9} = 133,69^\circ$$

$$\Delta \alpha = \left| \frac{\partial \alpha}{\partial l} \right| \Delta l + \left| \frac{\partial \alpha}{\partial R} \right| \Delta R = \frac{180^\circ}{\pi} \left[\frac{1}{R} \Delta l + \left| -\frac{l}{R^2} \right| \Delta R \right] =$$

$$= \frac{180^\circ l}{\pi R} \left(\frac{\Delta l}{l} + \frac{\Delta R}{R} \right) = \alpha \left(\frac{\Delta l}{l} + \frac{\Delta R}{R} \right) = 133,69 \left(\frac{4}{210} + \frac{2}{90} \right) = 5,517^\circ$$

$$\Delta \alpha = 6^\circ \Rightarrow \alpha = 134^\circ \pm 6^\circ$$

7)  $\alpha = \arccos \frac{a}{c} = \arccos \frac{32}{75} = 64,744^\circ$

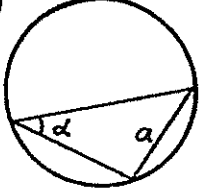
$$\Delta \alpha = \left| \frac{\partial \alpha}{\partial a} \right| \Delta a + \left| \frac{\partial \alpha}{\partial c} \right| \Delta c =$$

$$= \left| \frac{-1}{\sqrt{1 - \left(\frac{a}{c}\right)^2}} \cdot \frac{1}{c} \right| \Delta a + \left| \frac{-1}{\sqrt{1 - \left(\frac{a}{c}\right)^2}} \left(-\frac{a}{c^2}\right) \right| \Delta c = \frac{c}{\sqrt{c^2 - a^2}} \left(\frac{\Delta a}{c} + \frac{a}{c^2} \Delta c \right) =$$

$$= \frac{a}{\sqrt{c^2 - a^2}} \left(\frac{\Delta a}{a} + \frac{\Delta c}{c} \right) = \frac{32}{\sqrt{75^2 - 32^2}} \left(\frac{1}{32} + \frac{2}{75} \right) = 0,0273 \text{ rad} =$$

$$= 0,0273 \cdot \frac{180^\circ}{\pi} = 1,565^\circ \Rightarrow \Delta \alpha = 1,6^\circ$$

$$\alpha = 64,8^\circ \pm 1,6^\circ$$

8)  $d = (6,3 \pm 0,1) \text{ cm}$, $a = (2,30 \pm 0,05) \text{ cm}$

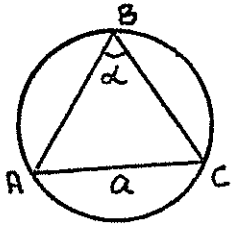
$$\frac{a}{\sin \alpha} = 2R \Rightarrow \alpha = \arcsin \frac{a}{d} = \arcsin \frac{2,3}{6,3} = 21,41^\circ$$

$$\Delta \alpha = \left| \frac{\partial \alpha}{\partial a} \right| \Delta a + \left| \frac{\partial \alpha}{\partial d} \right| \Delta d = \frac{1}{\sqrt{1 - \left(\frac{a}{d}\right)^2}} \cdot \frac{\Delta a}{d} + \frac{1}{\sqrt{1 - \left(\frac{a}{d}\right)^2}} \frac{a \Delta d}{d^2}$$

$$\Delta \alpha = \frac{1}{\sqrt{d^2 - a^2}} \left(\Delta a + \frac{a}{d} \Delta d \right) = \frac{1}{\sqrt{6,3^2 - 2,3^2}} \left(0,05 + \frac{2,3}{6,3} \cdot 0,1 \right) = 0,0147(\text{rad}) = 0,84^\circ$$

$$\Delta \alpha = 0,8^\circ \Rightarrow \alpha = 21,4^\circ \pm 0,8^\circ$$

9)



$$2R \cdot \sin \alpha = a \quad ; \quad \text{ပိုမိုသော အချက်များ}$$

$$S = \pi R^2 = \frac{\pi}{4} \cdot \frac{a^2}{\sin^2 \alpha} = \frac{\pi}{4} \cdot \frac{13.4^2}{\sin^2 52^\circ} = 227.11$$

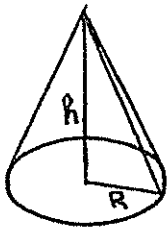
$$\Delta S = \left| \frac{\partial S}{\partial a} \right| \Delta a + \left| \frac{\partial S}{\partial \alpha} \right| \Delta \alpha =$$

$$= \frac{\pi}{4} \left[\frac{2a}{\sin^2 \alpha} \Delta a + \frac{2a^2}{\sin^3 \alpha} \cos \alpha \Delta \alpha \right] = \frac{\pi a^2}{4 \sin^2 \alpha} \cdot 2 \left(\frac{\Delta a}{a} + \frac{\Delta \alpha}{\tan \alpha} \right) =$$

$$= 2S \left(\frac{\Delta a}{a} + \frac{\Delta \alpha}{\tan \alpha} \right) = 2 \cdot 227.11 \left(\frac{0.1}{13.4} + \frac{10 \cdot \pi}{180^\circ \tan 52^\circ} \right) = 9.58$$

$$\Delta S = 10 \text{ cm}^2 \Rightarrow S = (227 \pm 10) \text{ cm}^2$$

10)



$$V = \frac{1}{3} \pi R^2 h = \frac{1}{3} \pi 7^2 \cdot 10 = 513.13 \text{ cm}^3$$

အချက်များကို ပိုမိုသော အချက်များ

$$\Delta V = \left| \frac{\partial V}{\partial R} \right| \Delta R + \left| \frac{\partial V}{\partial h} \right| \Delta h = \frac{1}{3} \pi 2R h \Delta R + \frac{1}{3} \pi R^2 \Delta h =$$

$$= \frac{1}{3} \pi R^2 h \left(2 \frac{\Delta R}{R} + \frac{\Delta h}{h} \right) = 513.13 \left(\frac{2 \cdot 0.2}{7} + \frac{0.3}{10} \right) = 44.71$$

$$\Delta V = 40 \text{ cm}^3 \Rightarrow V = (510 \pm 40) \text{ cm}^3$$

נגזרות של פונקציה מורכבת ופונקציה סתומה.

1) $z = u^2 \ln v$, $u = \frac{y}{x}$, $v = x^2 + y^2 \Rightarrow$

$$\frac{\partial z}{\partial x} = \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial x} = 2u \ln v \left(-\frac{y}{x^2}\right) + \frac{u^2}{v} \cdot 2x =$$

$$= 2u \left(\frac{y}{v} x - \ln v \cdot \frac{y}{x^2} \right)$$

$$\frac{\partial z}{\partial y} = \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial y} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial y} = 2u \ln v \cdot \frac{1}{x} + \frac{u^2}{v} \cdot 2y = 2u \left(\frac{\ln v}{x} + y \frac{u}{v} \right)$$

2) $z = u^2 v - v^2 u$, $u = x \sin y$, $v = y \cos x \Rightarrow$

$$\frac{\partial z}{\partial x} = \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial x} = (2uv - v^2) \sin y - (u^2 - 2uv) y \sin x$$

$$\frac{\partial z}{\partial y} = \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial y} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial y} = (2uv - v^2) x \cos y + (u^2 - 2uv) \cos x$$

3) $z = y \cdot \varphi(u)$, $u = \cos(x-y)$

$$\frac{\partial z}{\partial x} = y \cdot \frac{\partial \varphi}{\partial u} \cdot \frac{\partial u}{\partial x} = y \cdot \varphi'_u (-\sin(x-y)) = -y \sin(x-y) \cdot \varphi'_u$$

$$\frac{\partial z}{\partial y} = \varphi(u) + y \frac{\partial \varphi}{\partial u} \cdot \frac{\partial u}{\partial y} = \varphi(u) + y \varphi'_u (-\sin(x-y))(-1) = \varphi(u) + y \sin(x-y) \cdot \varphi'_u$$

$$\frac{\partial z}{\partial x} + \frac{\partial z}{\partial y} = -y \sin(x-y) \varphi'_u + \varphi(u) + y \sin(x-y) \cdot \varphi'_u = \varphi(u) = \frac{y \cdot \varphi(u)}{y} = \frac{z}{y} \quad (\delta, e, n)$$

4) $z = f(x, y)$, $x = r \cos \varphi$, $y = r \sin \varphi$

$$\frac{\partial z}{\partial r} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial r} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial r} = z'_x \cos \varphi + z'_y \sin \varphi$$

$$\frac{\partial z}{\partial \varphi} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial \varphi} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial \varphi} = z'_x (-r \sin \varphi) + z'_y r \cos \varphi$$

$$\left(\frac{\partial z}{\partial r} \right)^2 + \left(\frac{1}{r} \frac{\partial z}{\partial \varphi} \right)^2 = (z'_x \cos \varphi + z'_y \sin \varphi)^2 + (z'_y \cos \varphi - z'_x \sin \varphi)^2 =$$

$$= z'^2_x \cos^2 \varphi + 2 z'_x z'_y \sin \varphi \cos \varphi + z'^2_y \sin^2 \varphi +$$

$$+ z'^2_y \cos^2 \varphi - 2 z'_y z'_x \sin \varphi \cos \varphi + z'^2_x \sin^2 \varphi =$$

$$= z'^2_x (\cos^2 \varphi + \sin^2 \varphi) + z'^2_y (\sin^2 \varphi + \cos^2 \varphi) = z'^2_x + z'^2_y \Rightarrow (\delta, e, n)$$

5) $z = f(x+ay) \Rightarrow x+ay=u \Rightarrow z=f(u)$

$$\frac{\partial z}{\partial x} = \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial x} = z'_u \cdot 1 = z'_u, \Rightarrow \frac{\partial z}{\partial y} = \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial y} = z'_u \cdot a$$

$$\frac{\partial z}{\partial y} = z'_u \cdot a = a \cdot z'_u = a \frac{\partial z}{\partial x} \Rightarrow (\delta, e, n)$$

$$6) z = y \cdot \varphi(x^2 - y^2) \Rightarrow x^2 - y^2 = u \Rightarrow z = y \cdot \varphi(u)$$

$$\frac{\partial z}{\partial x} = \frac{\partial}{\partial x}(y \varphi(u)) = y \cdot \frac{\partial \varphi}{\partial x} = y \frac{\partial \varphi}{\partial u} \cdot \frac{\partial u}{\partial x} = y \cdot \frac{\partial \varphi}{\partial u} \cdot 2x = 2xy \frac{\partial \varphi}{\partial u}$$

$$\frac{\partial z}{\partial y} = \frac{\partial}{\partial y}(y \varphi(u)) = 1 \cdot \varphi(u) + y \frac{\partial \varphi}{\partial y} = \varphi(u) + y \frac{\partial \varphi}{\partial u} \cdot \frac{\partial u}{\partial y} =$$

$$= \varphi(u) + y \frac{\partial \varphi}{\partial u} (-2y) = \varphi(u) - 2y^2 \frac{\partial \varphi}{\partial u}$$

$$\frac{1}{x} \cdot \frac{\partial z}{\partial x} + \frac{1}{y} \frac{\partial z}{\partial y} = \frac{1}{x} \cdot 2xy \varphi'_u + \frac{1}{y} [\varphi(u) - 2y^2 \varphi'_u] =$$

$$= 2y \varphi'_u + \frac{1}{y} \varphi(u) - 2y \varphi'_u = \frac{1}{y} \varphi(u) = \frac{y \cdot \varphi(u)}{y^2} = \frac{z}{y^2} \Rightarrow (\delta, e, n).$$

$$7) p = f(x+mt, y+nt) \Rightarrow u = x+mt, v = y+nt, \Rightarrow p = f(u, v).$$

$$\frac{\partial p}{\partial t} = \frac{\partial f}{\partial u} \cdot \frac{\partial u}{\partial t} + \frac{\partial f}{\partial v} \cdot \frac{\partial v}{\partial t} = f'_u \cdot m + f'_v \cdot n$$

$$\frac{\partial p}{\partial x} = \frac{\partial f}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial f}{\partial v} \cdot \frac{\partial v}{\partial x} = f'_u \cdot 1 + f'_v \cdot 0 = f'_u$$

$$\frac{\partial p}{\partial y} = \frac{\partial f}{\partial u} \cdot \frac{\partial u}{\partial y} + \frac{\partial f}{\partial v} \cdot \frac{\partial v}{\partial y} = f'_u \cdot 0 + f'_v \cdot 1 = f'_v$$

$$m \frac{\partial p}{\partial x} + n \frac{\partial p}{\partial y} = m f'_u + n f'_v = \frac{\partial p}{\partial t} \Rightarrow (\delta, e, n).$$

$$8) z^3 \ln\left(\frac{e^x}{\pi}\right) - z(z+1) \sin y + 2 \cos x = 0$$

$$\frac{\partial F}{\partial x} = \frac{z^3}{x} - 2 \sin x, \quad \frac{\partial F}{\partial z} = 3z^2 \ln\left(\frac{e^x}{\pi}\right) - (2z+1) \sin y, \quad \frac{\partial z}{\partial x} = - \frac{\frac{\partial F}{\partial x}}{\frac{\partial F}{\partial z}}$$

$$\frac{\partial z}{\partial x} = - \frac{\frac{z^3}{x} - 2 \sin x}{3z^2 \ln\left(\frac{e^x}{\pi}\right) - (2z+1) \sin y}$$

$$x = \pi, y = \pi/2 \Rightarrow z^3 \ln\left(\frac{e^\pi}{\pi}\right) - z(z+1) \sin \frac{\pi}{2} + 2 \cos \pi = 0 \Rightarrow$$

$$z^3 - z^2 - z - 2 = 0 \quad z = 2 \rightarrow \text{OK} \Rightarrow \begin{array}{l} z^3 - z^2 - z - 2 \\ \underline{z^3 - z^2 - 2z} \\ -z - 2 \\ \underline{-z - 2} \\ 0 \end{array} \quad \begin{array}{l} z^2 + z + 1 \\ \Delta < 0 \\ (p' \in]e|e|'e) \end{array}$$

$$\frac{\partial z}{\partial x} \Big|_{(\pi, \pi/2, 2)} = \frac{-\frac{8}{\pi} - 0}{3 \cdot 4 - 1 \cdot (2 \cdot 2 + 1) \cdot 1} = -\frac{8}{7\pi}$$

$$9) F(x, y, z) = e^{\sin z} - xy z^2 = 0 \quad \frac{\partial F}{\partial x} = -yz^2, \quad \frac{\partial F}{\partial y} = -xz^2, \quad \frac{\partial F}{\partial z} = e^{\sin z} \cos z - 2xy z$$

$$\frac{\partial z}{\partial x} + \frac{\partial z}{\partial y} = - \frac{\frac{\partial F}{\partial x}}{\frac{\partial F}{\partial z}} - \frac{\frac{\partial F}{\partial y}}{\frac{\partial F}{\partial z}} = - \frac{\frac{\partial F}{\partial x} + \frac{\partial F}{\partial y}}{\frac{\partial F}{\partial z}} = \frac{z^2(x+y)}{e^{\sin z} \cos z - 2xy z}$$

$$x = y, z = \pi \Rightarrow e^{\sin \pi} - x^2 \pi^2 = 0 \Rightarrow 1 - \pi^2 x^2 = 0 \Rightarrow x = y = \pm 1/\pi.$$

$$x = y = 1/\pi, z = \pi \Rightarrow \frac{\partial z}{\partial x} + \frac{\partial z}{\partial y} = \frac{\pi^2(\frac{1}{\pi} + \frac{1}{\pi})}{e^0 \cos \pi - 2 \cdot \frac{1}{\pi}} = \frac{2\pi}{-1 - 2/\pi} = -\frac{2\pi^2}{2 + \pi}$$

$$(-\frac{1}{\pi}, -\frac{1}{\pi}, \pi) \Rightarrow \frac{\partial z}{\partial x} + \frac{\partial z}{\partial y} = \frac{2\pi^2}{2 + \pi}$$

$$10) z \cdot \ln(x+z) - \frac{xy}{z} = 0 \Rightarrow F(x, y, z) = 0. \quad \frac{\partial z}{\partial x} = -\frac{\partial F / \partial x}{\partial F / \partial z}, \quad \frac{\partial z}{\partial y} = -\frac{\partial F / \partial y}{\partial F / \partial z}$$

$$\frac{\partial F}{\partial x} = \frac{z}{x+z} - \frac{y}{z} = \frac{z^2 - y(x+z)}{z(x+z)}, \quad \frac{\partial F}{\partial y} = -\frac{x}{z} \Rightarrow$$

$$\frac{\partial F}{\partial z} = \ln(x+z) + \frac{z}{x+z} + \frac{xy}{z^2} = \ln(x+z) + \frac{xy}{z^2} + \frac{z}{x+z} + \frac{xy}{z^2} = \frac{z^3 + 2xy(x+z)}{z^2(x+z)}$$

$$\frac{\partial z}{\partial x} = -\frac{\frac{z^2 - y(x+z)}{z(x+z)}}{\frac{z^3 + 2xy(x+z)}{z^2(x+z)}} = \frac{yz(x+z) - z^3}{z^3 + 2xy(x+z)} \Rightarrow \frac{\partial z}{\partial y} = -\frac{-\frac{x}{z}}{\frac{z^3 + 2xy(x+z)}{z^2(x+z)}} = \frac{xz(x+z)}{z^3 + 2xy(x+z)}$$

$$11) yz = \arctg(xz) \Rightarrow yz - \arctg(xz) = 0 \Rightarrow F(x, y, z) = 0. \quad dz = z'_x dx + z'_y dy$$

$$\frac{\partial F}{\partial x} = -\frac{z}{1+x^2 z^2}, \quad \frac{\partial F}{\partial y} = z, \quad \frac{\partial F}{\partial z} = y - \frac{x}{1+x^2 z^2} = \frac{y(1+x^2 z^2) - x}{1+x^2 z^2}$$

$$dz = \frac{\frac{z}{1+x^2 z^2}}{\frac{y(1+x^2 z^2) - x}{1+x^2 z^2}} dx - \frac{z}{\frac{y(1+x^2 z^2) - x}{1+x^2 z^2}} dy = \frac{z dx - z(1+x^2 z^2) dy}{y(1+x^2 z^2) - x}$$

יישומים של נגזרות חלקיות בהנדסת המרחב

(14)

$$1) x^2 - y^2 - z^2 = 0, \quad A(5, 4, 3)$$

$$\frac{\partial F}{\partial x} = 2x, \quad \frac{\partial F}{\partial y} = -2y, \quad \frac{\partial F}{\partial z} = -2z \Rightarrow \frac{\partial F}{\partial x}|_A = 10, \quad \frac{\partial F}{\partial y}|_A = -8, \quad \frac{\partial F}{\partial z}|_A = -6$$

$$\frac{x-5}{5} = \frac{y-4}{-4} = \frac{z-3}{-3}$$

$$2) z = \sin \frac{x}{y}, \quad A(\pi, 1, 0) \Rightarrow \sin \frac{x}{y} - z = 0$$

$$\frac{\partial F}{\partial x} = \frac{1}{y} \cos \frac{x}{y}, \quad \frac{\partial F}{\partial y} = -\frac{x}{y^2} \cos \frac{x}{y}, \quad \frac{\partial F}{\partial z} = -1.$$

$$\frac{\partial F}{\partial x}|_A = \frac{1}{1} \cos \pi = -1, \quad \frac{\partial F}{\partial y}|_A = -\pi \cos \pi = \pi, \quad \frac{\partial F}{\partial z}|_A = -1 \Rightarrow \frac{x-\pi}{-1} = \frac{y-1}{\pi} = \frac{z-0}{-1}$$

$$3) (z^2 - x^2)xy - y^5 = 5, \quad A(1, 1, 2)$$

$$\frac{\partial F}{\partial x} = -2x \cdot xyz + (z^2 - x^2)y = yz(z^2 - 3x^2), \quad \frac{\partial F}{\partial y} = xz(z^2 - y^2) - 5y^4$$

$$\frac{\partial F}{\partial z} = 2z \cdot xyz + (z^2 - x^2)x = 3xyz^2 - x^3 \Rightarrow$$

$$\frac{\partial F}{\partial x}|_A = 2(4-3) = 2, \quad \frac{\partial F}{\partial y}|_A = 2(4-1) - 5 = 1, \quad \frac{\partial F}{\partial z}|_A = 3 \cdot 1 \cdot 4 - 1 = 11$$

$$\frac{x-1}{2} = \frac{y-1}{1} = \frac{z-2}{11}$$

①

1) $x^2 + y^2 + z^2 = 9$, $A(1, 2, 2)$

$$\frac{\partial F}{\partial x} = 2x, \frac{\partial F}{\partial y} = 2y, \frac{\partial F}{\partial z} = 2z \Rightarrow \frac{\partial F}{\partial x}|_A = 2, \frac{\partial F}{\partial y}|_A = \frac{\partial F}{\partial z}|_A = 4$$

$$2(x-1) + 4(y-2) + 4(z-2) = 0 \Rightarrow x + 2y + 2z - 9 = 0$$

2) $z = \frac{x^2}{4} + \frac{y^2}{9}$, $A(2, 3, 2) \Rightarrow 9x^2 + 4y^2 - 36z = 0$

$$\frac{\partial F}{\partial x} = 18x, \frac{\partial F}{\partial y} = 8y, \frac{\partial F}{\partial z} = -36 \Rightarrow \frac{\partial F}{\partial x}|_A = 36, \frac{\partial F}{\partial y}|_A = 24, \frac{\partial F}{\partial z}|_A = -36$$

$$36(x-2) + 24(y-3) - 36(z-2) = 0 \Rightarrow 3x + 2y - 3z - 6 = 0$$

3) $xy - z = 0$, $A(3, 4, 12)$

$$\frac{\partial F}{\partial x} = y, \frac{\partial F}{\partial y} = x, \frac{\partial F}{\partial z} = -1 \Rightarrow \frac{\partial F}{\partial x}|_A = 4, \frac{\partial F}{\partial y}|_A = 3, \frac{\partial F}{\partial z}|_A = -1$$

$$4(x-3) + 3(y-4) - (z-12) = 0 \Rightarrow 4x + 3y - z - 12 = 0$$

4) $e^z - z + xy = 3$, $A(2, 1, 0)$

$$\frac{\partial F}{\partial x} = y, \frac{\partial F}{\partial y} = x, \frac{\partial F}{\partial z} = e^z - 1 \Rightarrow \frac{\partial F}{\partial x}|_A = 1, \frac{\partial F}{\partial y}|_A = 2, \frac{\partial F}{\partial z}|_A = 0$$

$$1 \cdot (x-2) + 2(y-1) + 0(z-0) = 0 \Rightarrow x + 2y - 4 = 0$$

5) $z = x^2 - y^2$, $F(x, y, z) = x^2 - y^2 - z = 0$, $A(1, 1/2, 3/4)$.

$$\frac{\partial F}{\partial x} = 2x, \frac{\partial F}{\partial y} = -2y, \frac{\partial F}{\partial z} = -1. \quad \frac{\partial F}{\partial x}|_A = 2, \frac{\partial F}{\partial y}|_A = -1, \frac{\partial F}{\partial z}|_A = -1$$

$$\vec{n} = (2, -1, -1) \Rightarrow 2(x-1) - 1(y - \frac{1}{2}) - (z - \frac{3}{4}) = 0 \Rightarrow 8x - 4y - 4z - 3 = 0$$

6) $F(x, y, z) = x e^{x(y-1)} - z = 0$, $A(1, 2, e)$.

$$\frac{\partial F}{\partial x} = e^{x(y-1)} + x e^{x(y-1)}(y-1) = e^{x(y-1)}(1 + xy - x)$$

$$\frac{\partial F}{\partial y} = x e^{x(y-1)} \cdot x = x^2 e^{x(y-1)}, \quad \frac{\partial F}{\partial z} = -1.$$

$$\frac{\partial F}{\partial x}|_A = 2e, \frac{\partial F}{\partial y}|_A = e, \frac{\partial F}{\partial z}|_A = -1. \Rightarrow \vec{n} = (2e, e, -1) \Rightarrow$$

$$2e(x-1) + e(y-2) - (z-e) = 0 \Rightarrow 2ex + ey - z - 3e = 0$$

נגזרת לפי כיוון (נגזרת מכוונת) וגרדיאנט של פונקציה

$$1) u = xy + yz + zx, A(2, 1, 3), \vec{a} = 3\vec{i} + 4\vec{j} + 12\vec{k}$$

$$\frac{\partial u}{\partial x} = y + z, \frac{\partial u}{\partial y} = x + z, \frac{\partial u}{\partial z} = y + x \Rightarrow \frac{\partial u}{\partial x}|_A = 1 + 3 = 4, \frac{\partial u}{\partial y}|_A = 2 + 3 = 5,$$

$$\frac{\partial u}{\partial z} = 1 + 2 = 3. \Rightarrow |\vec{a}| = \sqrt{3^2 + 4^2 + 12^2} = 13,$$

$$\cos \alpha = 3/13, \cos \beta = 4/13, \cos \gamma = 12/13 \Rightarrow$$

$$\frac{\partial u}{\partial \ell} = \frac{\partial u}{\partial x} \cos \alpha + \frac{\partial u}{\partial y} \cos \beta + \frac{\partial u}{\partial z} \cos \gamma = 4 \cdot \frac{3}{13} + 5 \cdot \frac{4}{13} + 3 \cdot \frac{12}{13} = 68/13$$

$$2) u = \ln(x^2 + y^2 + z^2), A(1, 2, 1), \vec{a} = 2\vec{i} + 4\vec{j} + 4\vec{k}$$

$$\frac{\partial u}{\partial x} = \frac{2x}{x^2 + y^2 + z^2}, \frac{\partial u}{\partial y} = \frac{2y}{x^2 + y^2 + z^2}, \frac{\partial u}{\partial z} = \frac{2z}{x^2 + y^2 + z^2}$$

$$\frac{\partial u}{\partial x}|_A = \frac{2 \cdot 1}{1^2 + 2^2 + 1^2} = \frac{2}{6} = \frac{1}{3}, \Rightarrow \frac{\partial u}{\partial y}|_A = \frac{2 \cdot 2}{6} = \frac{2}{3}, \frac{\partial u}{\partial z}|_A = \frac{2 \cdot 1}{6} = \frac{1}{3}$$

$$|\vec{a}| = \sqrt{4 + 16 + 16} = 6 \Rightarrow \cos \alpha = \frac{2}{6} = \frac{1}{3}, \cos \beta = \frac{4}{6} = \frac{2}{3}, \cos \gamma = \frac{4}{6} = \frac{2}{3}$$

$$\frac{\partial u}{\partial \ell} = \frac{\partial u}{\partial x} \cos \alpha + \frac{\partial u}{\partial y} \cos \beta + \frac{\partial u}{\partial z} \cos \gamma = \frac{1}{3} \cdot \frac{1}{3} + \frac{2}{3} \cdot \frac{2}{3} + \frac{1}{3} \cdot \frac{2}{3} = 7/9$$

$$3) u = \ln(e^x + e^y + e^z), A(0, 0, 0), \vec{a} = \vec{i} + \vec{j} + \vec{k}$$

$$\frac{\partial u}{\partial x} = \frac{e^x}{e^x + e^y + e^z}, \frac{\partial u}{\partial y} = \frac{e^y}{e^x + e^y + e^z}, \frac{\partial u}{\partial z} = \frac{e^z}{e^x + e^y + e^z}$$

$$\frac{\partial u}{\partial x}|_A = \frac{\partial u}{\partial y}|_A = \frac{\partial u}{\partial z}|_A = \frac{e^0}{e^0 + e^0 + e^0} = \frac{1}{3} \Rightarrow |\vec{a}| = \sqrt{1^2 + 1^2 + 1^2} = \sqrt{3}$$

$$\cos \alpha = \cos \beta = \cos \gamma = \frac{1}{\sqrt{3}}$$

$$\frac{\partial u}{\partial \ell} = \frac{\partial u}{\partial x} \cos \alpha + \frac{\partial u}{\partial y} \cos \beta + \frac{\partial u}{\partial z} \cos \gamma = 3 \cdot \frac{1}{3} \cdot \frac{1}{\sqrt{3}} = \sqrt{3}/3$$

$$4) u = \frac{x^2}{4} + \frac{y^2}{9} - z^2, A(2, 3, 1), \vec{a} = 4\vec{i} - 3\vec{j}$$

$$\frac{\partial u}{\partial x} = \frac{1}{2}x, \frac{\partial u}{\partial y} = \frac{2}{9}y, \frac{\partial u}{\partial z} = -2z, \Rightarrow$$

$$\frac{\partial u}{\partial x}|_A = \frac{1}{2} \cdot 2 = 1, \frac{\partial u}{\partial y}|_A = \frac{2}{9} \cdot 3 = \frac{2}{3}, \frac{\partial u}{\partial z}|_A = -2 \cdot 1 = -2, \Rightarrow |\vec{a}| = \sqrt{16 + 9} = 5$$

$$\cos \alpha = 4/5, \cos \beta = -3/5, \cos \gamma = 0$$

$$\frac{\partial u}{\partial \ell} = \frac{\partial u}{\partial x} \cos \alpha + \frac{\partial u}{\partial y} \cos \beta + \frac{\partial u}{\partial z} \cos \gamma = 1 \cdot \frac{4}{5} + \frac{2}{3} \cdot \left(-\frac{3}{5}\right) + (-2) \cdot 0 = 2/5$$

$$5) u = x^2 - 2xz + y^2, A(1, 2, -1), N(2, 4, -3)$$

$$\frac{\partial u}{\partial x} = 2x - 2z, \frac{\partial u}{\partial y} = 2y, \frac{\partial u}{\partial z} = -2x, \Rightarrow \frac{\partial u}{\partial x}|_A = 4, \frac{\partial u}{\partial y}|_A = 4, \frac{\partial u}{\partial z}|_A = -2$$

$$\vec{AN} = (2-1, 4-2, -3+1) = (1, 2, -2) \Rightarrow |\vec{AN}| = \sqrt{1+4+4} = 3$$

$$\cos \alpha = 1/3, \cos \beta = 2/3, \cos \gamma = -2/3$$

$$\frac{\partial u}{\partial \rho} = \frac{\partial u}{\partial x} \cos \alpha + \frac{\partial u}{\partial y} \cos \beta + \frac{\partial u}{\partial z} \cos \gamma = 4 \cdot \frac{1}{3} + 4 \cdot \frac{2}{3} - 2 \cdot \left(-\frac{2}{3}\right) = 16/3$$

$$6) u = \arcsin \frac{z}{\sqrt{x^2+y^2}}, A(1, 1, 1), N(3, 2, 3)$$

$$\frac{\partial u}{\partial x} = \frac{1}{\sqrt{1-\frac{z^2}{x^2+y^2}}} \cdot \left(-\frac{1}{2}\right) \frac{z \cdot 2x}{\sqrt{(x^2+y^2)^3}} = -\frac{xz}{\sqrt{x^2+y^2-2z^2} \cdot (x^2+y^2)}$$

$$\frac{\partial u}{\partial y} = -\frac{yz}{(x^2+y^2)\sqrt{x^2+y^2-2z^2}}, \frac{\partial u}{\partial z} = \frac{1}{\sqrt{1-\frac{z^2}{x^2+y^2}}} \cdot \frac{1}{\sqrt{x^2+y^2}} = \frac{1}{\sqrt{x^2+y^2-2z^2}}$$

$$\frac{\partial u}{\partial x}|_A = \frac{\partial u}{\partial y}|_A = -\frac{1 \cdot 1}{(1+1)\sqrt{1+1-1}} = -\frac{1}{2} \Rightarrow \frac{\partial u}{\partial z}|_A = \frac{1}{\sqrt{1+1-1}} = 1.$$

$$\vec{AN} = (3-1, 2-1, 3-1) = (2, 1, 2), \rightarrow |\vec{AN}| = \sqrt{4+1+4} = 3$$

$$\cos \alpha = 2/3, \cos \beta = 1/3, \cos \gamma = 2/3$$

$$\frac{\partial u}{\partial \rho} = \frac{\partial u}{\partial x} \cos \alpha + \frac{\partial u}{\partial y} \cos \beta + \frac{\partial u}{\partial z} \cos \gamma = -\frac{1}{2} \cdot \frac{2}{3} - \frac{1}{2} \cdot \frac{1}{3} + 1 \cdot \frac{2}{3} = 1/6$$

$$7) u = xyz, A(2, 1, 1) \Rightarrow \text{grad } u = \frac{\partial u}{\partial x} \vec{i} + \frac{\partial u}{\partial y} \vec{j} + \frac{\partial u}{\partial z} \vec{k}$$

$$\frac{\partial u}{\partial x} = yz, \frac{\partial u}{\partial y} = xz, \frac{\partial u}{\partial z} = xy, \Rightarrow \frac{\partial u}{\partial x}|_A = 1, \frac{\partial u}{\partial y}|_A = 2, \frac{\partial u}{\partial z}|_A = 2$$

$$\text{grad } u = \vec{i} + 2\vec{j} + 2\vec{k}$$

$$8) u = x^2 + y^2 - z^2, A(1, -1, 2) \Rightarrow \text{grad } u = \frac{\partial u}{\partial x} \vec{i} + \frac{\partial u}{\partial y} \vec{j} + \frac{\partial u}{\partial z} \vec{k}$$

$$\frac{\partial u}{\partial x} = 2x, \frac{\partial u}{\partial y} = 2y, \frac{\partial u}{\partial z} = -2z \Rightarrow \frac{\partial u}{\partial x}|_A = 2, \frac{\partial u}{\partial y}|_A = -2, \frac{\partial u}{\partial z}|_A = -4$$

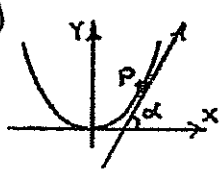
$$\text{grad } u = 2\vec{i} - 2\vec{j} - 4\vec{k}$$

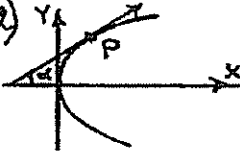
$$9) u = \arctg(xy/z) + \arctg(xyz), A(x, y, z) \Rightarrow \text{grad } u = \frac{\partial u}{\partial x} \vec{i} + \frac{\partial u}{\partial y} \vec{j} + \frac{\partial u}{\partial z} \vec{k}$$

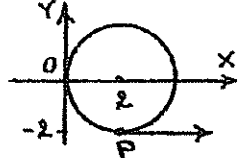
$$\frac{\partial u}{\partial x} = \frac{yz}{1+x^2y^2z^2} - \frac{yz}{1+x^2y^2z^2} = 0, \frac{\partial u}{\partial y} = 0, \frac{\partial u}{\partial z} = 0 \Rightarrow \text{grad } u = 0.$$

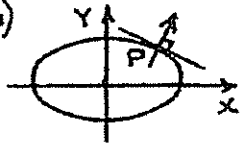
$$10) u = r, r = \sqrt{x^2+y^2+z^2}, A(x, y, z). \Rightarrow \text{grad } u = \frac{\partial u}{\partial x} \vec{i} + \frac{\partial u}{\partial y} \vec{j} + \frac{\partial u}{\partial z} \vec{k}$$

$$\frac{\partial u}{\partial x} = \frac{x}{r}, \frac{\partial u}{\partial y} = \frac{y}{r}, \frac{\partial u}{\partial z} = \frac{z}{r} \Rightarrow \text{grad } u = \frac{1}{r}(x\vec{i} + y\vec{j} + z\vec{k}).$$

11)  $y^2 = 2x \Rightarrow y'(P) = \tan \alpha = 2 \Rightarrow \cos \alpha = \frac{1}{\sqrt{1+\tan^2 \alpha}} \Rightarrow$
 $\cos \alpha = \frac{1}{\sqrt{1+4}} = \frac{1}{\sqrt{5}} \Rightarrow \cos \beta = \cos(90^\circ - \alpha) = \sin \alpha$
 $\sin \alpha = \sqrt{1 - \cos^2 \alpha} = \sqrt{1 - \frac{1}{5}} = \frac{2}{\sqrt{5}} \Rightarrow \frac{\partial u}{\partial x} = \frac{y}{1+x^2+y^2}, \frac{\partial u}{\partial y} = \frac{x}{1+x^2+y^2}$
 $\frac{\partial u}{\partial x}|_P = \frac{1}{1+1} = \frac{1}{2}, \frac{\partial u}{\partial y}|_P = \frac{1}{1+1} = \frac{1}{2} \Rightarrow \frac{\partial u}{\partial \rho} = \frac{1}{2} \cdot \frac{1}{\sqrt{5}} + \frac{1}{2} \cdot \frac{2}{\sqrt{5}} = \frac{3}{2\sqrt{5}}$

12)  $y^2 = 4x \Rightarrow 2yy' = 4 \Rightarrow y' = \frac{2}{y} \Rightarrow P(1,2) \Rightarrow$
 $y'(P) = \frac{2}{2} = 1 \Rightarrow \alpha = 45^\circ \Rightarrow \beta = 45^\circ \Rightarrow$
 $\cos \alpha = \cos \beta = \frac{\sqrt{2}}{2} \Rightarrow \frac{\partial u}{\partial x} = \frac{2x}{x^2+y^2}, \frac{\partial u}{\partial y} = \frac{2y}{x^2+y^2}$
 $\frac{\partial u}{\partial x}|_P = \frac{2 \cdot 1}{1^2+2^2} = \frac{2}{5}, \frac{\partial u}{\partial y}|_P = \frac{2 \cdot 2}{1^2+2^2} = \frac{4}{5}$
 $\frac{\partial u}{\partial \rho} = \frac{2}{5} \cdot \frac{\sqrt{2}}{2} + \frac{4}{5} \cdot \frac{\sqrt{2}}{2} = \frac{6\sqrt{2}}{5} \Rightarrow \frac{\partial u}{\partial \rho} = \frac{3\sqrt{2}}{5}$

13)  $x^2 + y^2 - 4x = 0 \Rightarrow (x^2 - 2 \cdot 2x + 4) + y^2 = 4 \Rightarrow$
 $(x-2)^2 + y^2 = 2^2$
 הריא שמהלך נגזרת נקודה P(2,2) ונגזרת נקודה P(2,2) : $x^2 + y^2 - 4x = 0$: נגזרת נקודה P(2,2) : $2x + 2yy' - 4 = 0 \Rightarrow y' = \frac{2-x}{y} \Rightarrow y'|_P = y'|_{(2,2)} = \frac{2-2}{2} = 0 \Rightarrow$
 $\cos \alpha = 1, \cos \beta = 0 \Rightarrow \frac{\partial u}{\partial x} = \frac{1}{1+(y/x)^2} \left(-\frac{y}{x^2}\right) = -\frac{y}{x^2+y^2}$
 $\frac{\partial u}{\partial y} = \frac{1}{1+(y/x)^2} \cdot \frac{1}{x} = \frac{x}{x^2+y^2} \Rightarrow \frac{\partial u}{\partial x}|_P = \frac{\partial u}{\partial y}|_P = \frac{1}{4} \Rightarrow \frac{\partial u}{\partial \rho} = \frac{1}{4} \cdot 1 + \frac{1}{4} \cdot 0 = \frac{1}{4}$

14)  $\frac{x^2}{4} + \frac{y^2}{2} = 1 \Rightarrow \frac{x}{2} + yy' = 0 \Rightarrow y' = -\frac{x}{2y} \Rightarrow y'(P) = -\frac{\sqrt{2}}{2}$
 :m נגזרת נקודה P(2,2) : $m = \tan \alpha = -\frac{1}{y'(P)} = -\frac{1 \cdot 2}{-\sqrt{2}} = \sqrt{2} \Rightarrow \cos \alpha = \frac{1}{\sqrt{1+\tan^2 \alpha}}$
 $\cos \alpha = \frac{1}{\sqrt{1+2}} = \frac{1}{\sqrt{3}} \Rightarrow \cos \beta = \sin \alpha = \sqrt{1 - \cos^2 \alpha} = \frac{\sqrt{2}}{\sqrt{3}}$
 $\frac{\partial u}{\partial x} = 2y, \frac{\partial u}{\partial y} = 2x + 2y \Rightarrow \frac{\partial u}{\partial x}|_P = 2, \frac{\partial u}{\partial y}|_P = 2 + 2\sqrt{2}$
 $\frac{\partial u}{\partial \rho} = 2 \cdot \frac{1}{\sqrt{3}} + (2 + 2\sqrt{2}) \cdot \frac{\sqrt{2}}{\sqrt{3}} = \frac{1}{\sqrt{3}} (6 + 2\sqrt{2}) = \frac{2(3 + \sqrt{2})\sqrt{3}}{3} = \frac{2\sqrt{3}}{3} (3 + \sqrt{2})$

$$15) \textcircled{a} \operatorname{grad}(\lambda u) = \frac{\partial(\lambda u)}{\partial x} \vec{i} + \frac{\partial(\lambda u)}{\partial y} \vec{j} + \frac{\partial(\lambda u)}{\partial z} \vec{k} =$$

$$= \lambda \frac{\partial u}{\partial x} \vec{i} + \lambda \frac{\partial u}{\partial y} \vec{j} + \lambda \frac{\partial u}{\partial z} \vec{k} = \lambda \left(\frac{\partial u}{\partial x} \vec{i} + \frac{\partial u}{\partial y} \vec{j} + \frac{\partial u}{\partial z} \vec{k} \right) = \lambda \cdot \operatorname{grad} u$$

$$\textcircled{b} \operatorname{grad}(u+v) = \frac{\partial}{\partial x}(u+v) \vec{i} + \frac{\partial}{\partial y}(u+v) \vec{j} + \frac{\partial}{\partial z}(u+v) \vec{k} =$$

$$= \frac{\partial u}{\partial x} \vec{i} + \frac{\partial v}{\partial x} \vec{i} + \frac{\partial u}{\partial y} \vec{j} + \frac{\partial v}{\partial y} \vec{j} + \frac{\partial u}{\partial z} \vec{k} + \frac{\partial v}{\partial z} \vec{k} =$$

$$= \left(\frac{\partial u}{\partial x} \vec{i} + \frac{\partial u}{\partial y} \vec{j} + \frac{\partial u}{\partial z} \vec{k} \right) + \left(\frac{\partial v}{\partial x} \vec{i} + \frac{\partial v}{\partial y} \vec{j} + \frac{\partial v}{\partial z} \vec{k} \right) = \operatorname{grad} u + \operatorname{grad} v$$

$$\textcircled{c} \operatorname{grad}(uv) = \frac{\partial(uv)}{\partial x} \vec{i} + \frac{\partial(uv)}{\partial y} \vec{j} + \frac{\partial(uv)}{\partial z} \vec{k} =$$

$$= \left(v \frac{\partial u}{\partial x} + u \frac{\partial v}{\partial x} \right) \vec{i} + \left(v \frac{\partial u}{\partial y} + u \frac{\partial v}{\partial y} \right) \vec{j} + \left(v \frac{\partial u}{\partial z} + u \frac{\partial v}{\partial z} \right) \vec{k} =$$

$$= v \left(\frac{\partial u}{\partial x} \vec{i} + \frac{\partial u}{\partial y} \vec{j} + \frac{\partial u}{\partial z} \vec{k} \right) + u \left(\frac{\partial v}{\partial x} \vec{i} + \frac{\partial v}{\partial y} \vec{j} + \frac{\partial v}{\partial z} \vec{k} \right) = v \cdot \operatorname{grad} u + u \cdot \operatorname{grad} v$$

$$\textcircled{d} \operatorname{grad} \left(\frac{u}{v} \right) = \frac{\partial \left(\frac{u}{v} \right)}{\partial x} \vec{i} + \frac{\partial \left(\frac{u}{v} \right)}{\partial y} \vec{j} + \frac{\partial \left(\frac{u}{v} \right)}{\partial z} \vec{k} =$$

$$= \frac{\frac{\partial u}{\partial x} \cdot v - u \frac{\partial v}{\partial x}}{v^2} \vec{i} + \frac{\frac{\partial u}{\partial y} \cdot v - u \frac{\partial v}{\partial y}}{v^2} \vec{j} + \frac{\frac{\partial u}{\partial z} \cdot v - u \frac{\partial v}{\partial z}}{v^2} \vec{k}$$

$$= \frac{1}{v^2} \left[\left(\frac{\partial u}{\partial x} v \vec{i} + \frac{\partial u}{\partial y} v \vec{j} + \frac{\partial u}{\partial z} v \vec{k} \right) - \left(u \frac{\partial v}{\partial x} \vec{i} + u \frac{\partial v}{\partial y} \vec{j} + u \frac{\partial v}{\partial z} \vec{k} \right) \right] =$$

$$= \frac{1}{v^2} \left[v \left(\frac{\partial u}{\partial x} \vec{i} + \frac{\partial u}{\partial y} \vec{j} + \frac{\partial u}{\partial z} \vec{k} \right) - u \left(\frac{\partial v}{\partial x} \vec{i} + \frac{\partial v}{\partial y} \vec{j} + \frac{\partial v}{\partial z} \vec{k} \right) \right] =$$

$$= \frac{1}{v^2} (v \operatorname{grad} u - u \operatorname{grad} v) = \frac{v \cdot \operatorname{grad} u - u \cdot \operatorname{grad} v}{v^2}$$

16) A ગણતરના ઇન્કોર્ડના બિંદુએ સ્કેલર, $\vec{g} = \operatorname{grad} u|_A$ નો
 $\frac{\partial u}{\partial \rho}|_A$ નો આ સેન્ટર ગણતરના સેન્ટરના $|\vec{g}|$ નો

$$\frac{\partial u}{\partial x} = 2xy\sqrt{z}, \quad \frac{\partial u}{\partial y} = x^2\sqrt{z}, \quad \frac{\partial u}{\partial z} = \frac{x^2y}{2\sqrt{z}}$$

$$\frac{\partial u}{\partial x}|_A = 2 \cdot 2 \cdot 1 = 4, \quad \frac{\partial u}{\partial y}|_A = 2^2 \sqrt{1} = 4, \quad \frac{\partial u}{\partial z}|_A = \frac{2^2 \cdot 1}{2 \cdot 1} = 2 \Rightarrow$$

$$\vec{g} = \operatorname{grad} u|_A = 4\vec{i} + 4\vec{j} + 2\vec{k} \Rightarrow |\vec{g}| = \sqrt{4^2 + 4^2 + 2^2} = 6$$

$$\vec{a}^0 = \frac{\vec{a}}{|\vec{a}|} = \frac{1}{\sqrt{1^2 + 2^2 + 2^2}} (\vec{i} + 2\vec{j} + 2\vec{k}) = \frac{1}{3} \vec{i} + \frac{2}{3} \vec{j} + \frac{2}{3} \vec{k}$$

$$\frac{\partial u}{\partial \rho}|_A = \operatorname{grad} u|_A \cdot \vec{a}^0 = \vec{g} \cdot \vec{a}^0 = 4 \cdot \frac{1}{3} + 4 \cdot \frac{2}{3} + 2 \cdot \frac{2}{3} = \frac{16}{3} \Rightarrow \frac{|\vec{g}|}{\frac{\partial u}{\partial \rho}|_A} = \frac{6 \cdot 3}{16} = 1.125$$

બિંદુએ સ્કેલર ~ 1.125 છે, આ સેન્ટર ગણતરના સેન્ટરના ઇન્કોર્ડ

$$17) \vec{r} = x\vec{i} + y\vec{j} + z\vec{k}, \quad \vec{c} = \text{const} \Rightarrow \vec{c} = c_1\vec{i} + c_2\vec{j} + c_3\vec{k}, \quad (c_1, c_2, c_3 \in \mathbb{R})$$

$$\vec{c} \cdot \vec{r} = c_1 x + c_2 y + c_3 z \Rightarrow$$

$$\text{grad}(\vec{c} \cdot \vec{r}) = \vec{i} \cdot \frac{\partial}{\partial x} (c_1 x + c_2 y + c_3 z) + \vec{j} \cdot \frac{\partial}{\partial y} (c_1 x + c_2 y + c_3 z) + \vec{k} \cdot \frac{\partial}{\partial z} (c_1 x + c_2 y + c_3 z) =$$

$$= \vec{i} \cdot c_1 + \vec{j} \cdot c_2 + \vec{k} \cdot c_3 = c_1\vec{i} + c_2\vec{j} + c_3\vec{k} = \vec{c}$$

$$18) u = \sqrt{x^2 + y^2} \Rightarrow \text{grad } u = \frac{\partial u}{\partial x} \vec{i} + \frac{\partial u}{\partial y} \vec{j} = \frac{x}{\sqrt{x^2 + y^2}} \vec{i} + \frac{y}{\sqrt{x^2 + y^2}} \vec{j}$$

$$v = x + y + 2\sqrt{xy} \Rightarrow \text{grad } v = (1 + \frac{y}{\sqrt{xy}}) \vec{i} + (1 + \frac{x}{\sqrt{xy}}) \vec{j}$$

$$A(1,1) \Rightarrow \vec{g}_1 = \text{grad } u|_A = \frac{1}{\sqrt{2}} \vec{i} + \frac{1}{\sqrt{2}} \vec{j}, \quad \vec{g}_2 = \text{grad } v|_A = 2\vec{i} + 2\vec{j}$$

$$\vec{g}_2 = 2\sqrt{2} \vec{g}_1 \Rightarrow \vec{g}_2 \parallel \vec{g}_1 \Rightarrow \varphi = \widehat{\vec{g}_1 \vec{g}_2} = 0$$

$$19) \vec{r} = x\vec{i} + y\vec{j} + z\vec{k} \Rightarrow |\vec{r}| = \sqrt{x^2 + y^2 + z^2}$$

$$\text{grad } |\vec{r}|^n = \text{grad} (\sqrt{x^2 + y^2 + z^2})^n = \text{grad} (x^2 + y^2 + z^2)^{n/2} =$$

$$= \vec{i} \frac{\partial}{\partial x} (x^2 + y^2 + z^2)^{n/2} + \vec{j} \frac{\partial}{\partial y} (x^2 + y^2 + z^2)^{n/2} + \vec{k} \frac{\partial}{\partial z} (x^2 + y^2 + z^2)^{n/2} =$$

$$= \vec{i} \cdot \frac{n}{2} (x^2 + y^2 + z^2)^{\frac{n}{2}-1} \cdot 2x + \vec{j} \cdot \frac{n}{2} (x^2 + y^2 + z^2)^{\frac{n}{2}-1} \cdot 2y + \vec{k} \cdot \frac{n}{2} (x^2 + y^2 + z^2)^{\frac{n}{2}-1} \cdot 2z =$$

$$= \vec{i} n \left[(\sqrt{x^2 + y^2 + z^2})^2 \right]^{\frac{n}{2}-1} \cdot x + \vec{j} n \left[(\sqrt{x^2 + y^2 + z^2})^2 \right]^{\frac{n}{2}-1} \cdot y + \vec{k} n \left[(\sqrt{x^2 + y^2 + z^2})^2 \right]^{\frac{n}{2}-1} \cdot z =$$

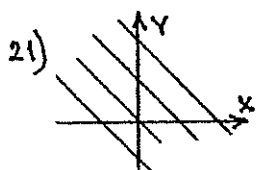
$$= \vec{i} n |\vec{r}|^{n-2} \cdot x + \vec{j} n |\vec{r}|^{n-2} y + \vec{k} n |\vec{r}|^{n-2} z =$$

$$= n |\vec{r}|^{n-2} (x\vec{i} + y\vec{j} + z\vec{k}) = n |\vec{r}|^{n-2} \cdot \vec{r}$$

$$20) u = \arctan \frac{y}{x} \Rightarrow \frac{\partial u}{\partial x} = \frac{y}{y^2 + x^2}, \quad \frac{\partial u}{\partial y} = -\frac{x}{x^2 + y^2}$$

$$\vec{g}_1 = \text{grad } u|_{A(1,1)} = \frac{1}{2} \vec{i} - \frac{1}{2} \vec{j}, \quad \vec{g}_2 = \text{grad } u|_{B(-1,-1)} = -\frac{1}{2} \vec{i} + \frac{1}{2} \vec{j}$$

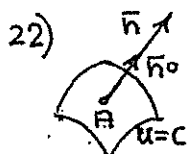
$$\cos \varphi = \frac{\vec{g}_1 \cdot \vec{g}_2}{|\vec{g}_1| |\vec{g}_2|} = \frac{\frac{1}{2}(-\frac{1}{2}) + (-\frac{1}{2})\frac{1}{2}}{\sqrt{\frac{1}{4} + \frac{1}{4}} \cdot \sqrt{\frac{1}{4} + \frac{1}{4}}} = -1 \Rightarrow \varphi = \arccos(-1) = \pi$$



$$u = \sin(x+y) \Rightarrow \text{grad } u = \cos(x+y) \vec{i} + \cos(x+y) \vec{j} \Rightarrow$$

$$\text{grad } u = \vec{i} + \vec{j} \Rightarrow \cos(x+y) = 1 \Rightarrow x+y = 2\pi k$$

$$y = -x + 2\pi k, \quad k \in \mathbb{Z}$$



$$\vec{n} = \text{grad } u = (2xy + 2z) \vec{i} + x^2 \vec{j} + 2x \vec{k}$$

$$\vec{n}(A) = \text{grad } u|_{A(2,2,3)} = -2\vec{i} + 4\vec{j} + 4\vec{k}$$

$$\vec{n}_0 = \frac{\vec{n}}{|\vec{n}|} = \frac{1}{\sqrt{4+16+16}} (-2\vec{i} + 4\vec{j} + 4\vec{k}) = -\frac{1}{3}\vec{i} + \frac{2}{3}\vec{j} + \frac{2}{3}\vec{k}$$

23) . $2xz^2 - 3xy - 4x = C \Leftarrow u=C$: זה נראה כמו שם של פוטנציאל
:כאן, הפוטנציאל של $A(1,-1,2)$ הוא 7
 $2 \cdot 1 \cdot 4 - 3 \cdot 1 \cdot (-1) - 4 \cdot 1 = C \Rightarrow C=7 \Rightarrow 2xz^2 - 3xy - 4x = 7 \rightarrow$ (פוטנציאל של פוטנציאל)
 $\vec{n} = \text{grad } u = (2z^2 - 3y - 4)\vec{i} - 3x\vec{j} - 4x\vec{k} \Rightarrow \vec{n}(A) = 7\vec{i} - 3\vec{j} + 8\vec{k}$
:כאן, $\vec{n}(\vec{r} - \vec{r}_0) = 0$; ג'ומה של המישור מולד
 $7(x-1) - 3(y+1) + 8(z-2) = 0 \Rightarrow 7x - 3y + 8z - 26 = 0$

$$24) \vec{r} = x\vec{i} + y\vec{j} + z\vec{k}, \quad \vec{a} = a_1\vec{i} + a_2\vec{j} + a_3\vec{k}, \quad \vec{b} = b_1\vec{i} + b_2\vec{j} + b_3\vec{k}$$

$$\begin{aligned} \text{grad } u &= \text{grad}[(\vec{a} \cdot \vec{r})(\vec{b} \cdot \vec{r})] = (\vec{b} \cdot \vec{r}) \text{grad}(\vec{a} \cdot \vec{r}) + (\vec{a} \cdot \vec{r}) \text{grad}(\vec{b} \cdot \vec{r}) = \\ &= (b_1x + b_2y + b_3z) \text{grad}(a_1x + a_2y + a_3z) + (a_1x + a_2y + a_3z) \text{grad}(b_1x + b_2y + b_3z) = \\ &= (b_1x + b_2y + b_3z)(a_1\vec{i} + a_2\vec{j} + a_3\vec{k}) + (a_1x + a_2y + a_3z)(b_1\vec{i} + b_2\vec{j} + b_3\vec{k}) = \\ &= (\vec{b} \cdot \vec{r})\vec{a} + (\vec{a} \cdot \vec{r})\vec{b} \end{aligned}$$

25) $\frac{\partial u}{\partial \ell}|_A = |\text{grad } u|_A| \Rightarrow \text{grad } u = yze^x \vec{i} + ze^x \vec{j} + ye^x \vec{k}$
 $\text{grad } u|_{A(0,0,1)} = 0\vec{i} + \vec{j} + 0\vec{k} = \vec{j} \Rightarrow \frac{\partial u}{\partial \ell}|_A = |\text{grad } u|_A| = 1.$

26) $\frac{\partial u}{\partial e}$ - נגזרת של u ביחס ל e , ∇u - וקטור הגרדיאנט של u בנקודה e .

$$\bar{g}_0 = \frac{\text{grad } v}{|\text{grad } v|}$$

$$\frac{\partial u}{\partial e} = \text{grad } u \cdot \bar{g}_0 = \frac{\text{grad } u \cdot \text{grad } v}{|\text{grad } v|} = \frac{1}{|\text{grad } v|} (\text{grad } u \cdot \text{grad } v)$$

$$-\frac{\partial u}{\partial e} = 0 \quad \text{כלומר, } \text{grad } u \cdot \text{grad } v = 0 \text{ כלומר, } \text{grad } u \perp \text{grad } v \text{ כלומר, } \nabla u \perp \nabla v$$

$$\begin{aligned} 2) \quad u &= \ln \sqrt{x^2 + y^2 + z^2} = \frac{1}{2} \ln(x^2 + y^2 + z^2) \\ \text{grad } u &= \frac{x}{x^2 + y^2 + z^2} \vec{i} + \frac{y}{x^2 + y^2 + z^2} \vec{j} + \frac{z}{x^2 + y^2 + z^2} \vec{k} \\ |\text{grad } u| &= \sqrt{\frac{x^2}{(x^2 + y^2 + z^2)^2} + \frac{y^2}{(x^2 + y^2 + z^2)^2} + \frac{z^2}{(x^2 + y^2 + z^2)^2}} = \\ &= \sqrt{\frac{x^2 + y^2 + z^2}{(x^2 + y^2 + z^2)^2}} = \frac{1}{\sqrt{x^2 + y^2 + z^2}} \Rightarrow |\text{grad } u| = 1 \Rightarrow \frac{1}{\sqrt{x^2 + y^2 + z^2}} = 1 \\ &\quad x^2 + y^2 + z^2 = 1. \end{aligned}$$

נגזרות חלקיות מסדר גבוה

$$1) z = c \sqrt{\frac{x^2}{a^2} + \frac{y^2}{b^2}}. \quad \frac{\partial z}{\partial x} = c \cdot \frac{2x/a^2}{2\sqrt{\frac{x^2}{a^2} + \frac{y^2}{b^2}}} = \frac{cx}{a^2 \sqrt{\frac{x^2}{a^2} + \frac{y^2}{b^2}}}$$

$$\frac{\partial^2 z}{\partial x^2} = \frac{c}{a^2} \cdot \left(\frac{\sqrt{\frac{x^2}{a^2} + \frac{y^2}{b^2}} - x \cdot \frac{\frac{2x}{a^2}}{2\sqrt{\frac{x^2}{a^2} + \frac{y^2}{b^2}}}}{\frac{x^2}{a^2} + \frac{y^2}{b^2}} \right) = \frac{c}{a^2} \cdot \frac{\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{x^2}{a^2}}{\left(\frac{x^2}{a^2} + \frac{y^2}{b^2}\right)^{3/2}} = \frac{c}{a^2} \cdot \frac{y^2(b^2)^3}{(b^2x^2 + a^2y^2)^{3/2}}$$

$$\frac{\partial^2 z}{\partial x^2} = \frac{abc y^2}{(b^2x^2 + a^2y^2)^{3/2}} \xrightarrow{(\text{ג'רנטו לנטר})} \frac{\partial^2 z}{\partial y^2} = \frac{abc \cdot x^2}{(b^2x^2 + a^2y^2)^{3/2}}$$

$$\frac{\partial^2 z}{\partial x \partial y} = \frac{cx}{a^2} \left(-\frac{1}{2}\right) \left(\frac{x^2}{a^2} + \frac{y^2}{b^2}\right)^{-\frac{3}{2}} \cdot \frac{2y}{b^2} = \frac{-cx a^2 (a^2 b^2)^{3/2} \cdot b^2 y}{(b^2x^2 + a^2y^2)^{3/2}} = -\frac{abc \cdot xy}{(b^2x^2 + a^2y^2)^{3/2}}$$

$$2) z = \ln(x^2 + y^2). \quad \frac{\partial z}{\partial x} = \frac{2x}{x^2 + y^2}, \quad \frac{\partial z}{\partial y} = \frac{2y}{x^2 + y^2}, \quad \frac{\partial^2 z}{\partial y^2} = -\frac{2}{(x^2 + y^2)^2} = -\frac{1}{(x^2 + y^2)^2}$$

$$\frac{\partial^2 z}{\partial x^2} = 2 \cdot \frac{(x^2 + y^2) - x \cdot 2x}{(x^2 + y^2)^2} = \frac{2(y^2 - x^2)}{(x^2 + y^2)^2}, \quad \frac{\partial^2 z}{\partial x \partial y} = 2x(-1)(x^2 + y^2)^{-2} = -\frac{2x}{(x^2 + y^2)^2}$$

$$3) z = \sqrt{2xy + y^2}. \quad \frac{\partial z}{\partial x} = \frac{y}{2\sqrt{2xy + y^2}} = \frac{y}{\sqrt{2xy + y^2}}$$

$$\frac{\partial^2 z}{\partial x \partial y} = \frac{\sqrt{2xy + y^2} - y \cdot \frac{x+y}{\sqrt{2xy + y^2}}}{2xy + y^2} = \frac{2xy + y^2 - xy - y^2}{(2xy + y^2)^{3/2}} = \frac{xy}{(2xy + y^2)^{3/2}}$$

$$4) r = \sqrt{x^2 + y^2 + z^2}. \quad \frac{\partial r}{\partial x} = \frac{2x}{2\sqrt{x^2 + y^2 + z^2}} = \frac{x}{r}$$

$$\frac{\partial^2 r}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{x}{r} \right) = \frac{\frac{\partial x}{\partial x} \cdot r - x \cdot \frac{\partial r}{\partial x}}{r^2} = \frac{1 \cdot r - x \cdot \frac{x}{r}}{r^2} = \frac{r^2 - x^2}{r^3}$$

$$5) z = \sin(xy). \quad \frac{\partial z}{\partial x} = y \cdot \cos(xy), \quad \frac{\partial^2 z}{\partial x \partial y} = \cos(xy) - xy \cdot \sin(xy)$$

$$\frac{\partial^3 z}{\partial x \partial y^2} = -x \cdot \sin(xy) - x [\sin(xy) + xy \cdot \cos(xy)] = -2x \sin(xy) - x^2 y \cos(xy).$$

$$6) z = x^y. \quad \frac{\partial z}{\partial x} = y \cdot x^{y-1}, \quad \frac{\partial^2 z}{\partial x \partial y} = x^{y-1} + y \cdot x^{y-1} \ln x$$

$$\frac{\partial^2 z}{\partial y^2} = x^y \ln x, \quad \frac{\partial^2 z}{\partial y \partial x} = y \cdot x^{y-1} \ln x + x^{y-1} \cdot \frac{1}{x} = x^{y-1} + y \cdot x^{y-1} \ln x \Rightarrow$$

$$\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial^2 z}{\partial y \partial x} \quad \checkmark$$

$$7) u = \arctan \frac{y}{x}. \quad \frac{\partial u}{\partial x} = \frac{-y/x^2}{1 + (\frac{y}{x})^2} = -\frac{y}{x^2 + y^2}, \quad \frac{\partial u}{\partial y} = \frac{1/x}{1 + (\frac{y}{x})^2} = \frac{x}{x^2 + y^2}$$

$$\frac{\partial^2 u}{\partial x^2} = -y \left[-(x^2 + y^2)^{-2} \cdot 2x \right] = \frac{2xy}{(x^2 + y^2)^2}, \quad \frac{\partial^2 u}{\partial y^2} = x \left[-\frac{2y}{(x^2 + y^2)^2} \right] = -\frac{2xy}{(x^2 + y^2)^2}$$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = \frac{2xy}{(x^2 + y^2)^2} - \frac{2xy}{(x^2 + y^2)^2} = 0 \Rightarrow (\text{S.E.N})$$

$$8) u = \ln \frac{1}{r}, \quad r = \sqrt{(x-a)^2 + (y-b)^2} \Rightarrow u = -\frac{1}{2} \ln [(x-a)^2 + (y-b)^2]$$

$$\cdot \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \quad \text{: (c) o d o d s a/c/e n}$$

$$\frac{\partial u}{\partial x} = -\frac{1}{2} \cdot \frac{2(x-a)}{(x-a)^2 + (y-b)^2} = -\frac{x-a}{(x-a)^2 + (y-b)^2}$$

$$\frac{\partial^2 u}{\partial x^2} = -\frac{(x-a)^2 + (y-b)^2 - (x-a) \cdot 2(x-a)}{[(x-a)^2 + (y-b)^2]^2} = \frac{(x-a)^2 - (y-b)^2}{[(x-a)^2 + (y-b)^2]^2}$$

$$\text{: } \uparrow \text{ } \downarrow \text{ } \cdot \frac{(y-b)^2 - (x-a)^2}{[(x-a)^2 + (y-b)^2]^2} \quad \text{: o h o n' o n r o n}$$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = \frac{(x-a)^2 - (y-b)^2}{[(x-a)^2 + (y-b)^2]^2} + \frac{(y-b)^2 - (x-a)^2}{[(x-a)^2 + (y-b)^2]^2} =$$

$$= \frac{\cancel{(x-a)^2} - \cancel{(y-b)^2} + (y-b)^2 - \cancel{(x-a)^2}}{[(x-a)^2 + (y-b)^2]^2} = 0 \Rightarrow (\text{S.E.N})$$

$$9) u = A \sin(a\lambda t + \varphi) \sin \lambda x \Rightarrow \frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} \stackrel{?}{=} 0$$

$$\frac{\partial u}{\partial t} = A \lambda a \cos(a\lambda t + \varphi) \sin \lambda x, \quad \frac{\partial u}{\partial x} = A \cdot \lambda \sin(a\lambda t + \varphi) \cos \lambda x$$

$$\frac{\partial^2 u}{\partial t^2} = -A \lambda^2 a^2 \sin(a\lambda t + \varphi) \sin \lambda x, \quad \frac{\partial^2 u}{\partial x^2} = -A \lambda^2 \sin(a\lambda t + \varphi) \sin \lambda x$$

$$\frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = -A \lambda^2 a^2 \sin(a\lambda t + \varphi) \sin \lambda x + a^2 A \lambda^2 \sin(a\lambda t + \varphi) \sin \lambda x = 0$$

(S.E.N).

אקסטרים פונקציה של מספר משתנים

1) $z = x^2 - xy + y^2 + 9x - 6y + 20$

$$\begin{aligned} z'_x &= 2x - y + 9 \\ z'_y &= -x + 2y - 6 \end{aligned} \Rightarrow \begin{cases} 2x - y + 9 = 0 \\ -x + 2y - 6 = 0 \end{cases} \Rightarrow \begin{matrix} x = -4 \\ y = 1 \end{matrix} \Rightarrow (-4, 1)$$

$$z''_{xx} = 2, \quad z''_{xy} = -1, \quad z''_{yy} = 2 \Rightarrow \Delta = \begin{vmatrix} 2 & -1 \\ -1 & 2 \end{vmatrix} = 4 - 1 > 0 \Rightarrow \left(\begin{vmatrix} z'_x & z'_y \end{vmatrix} \right)$$

$$z''_{xx} = 2 > 0 \rightarrow (\min) \Rightarrow z_{\min} = 16 + 4 + 1 - 36 - 6 + 20 = -1.$$

2) $z = y\sqrt{x} - y^2 - x + 6y$

$$z'_x = \frac{y}{2\sqrt{x}} - 1, \quad z'_y = \sqrt{x} - 2y + 6 \Rightarrow \begin{cases} \frac{y}{2\sqrt{x}} - 1 = 0 \\ \sqrt{x} - 2y + 6 = 0 \end{cases} \rightarrow \sqrt{x} = \frac{y}{2} \downarrow$$

$$\frac{y}{2} - 2y + 6 = 0 \Rightarrow y = 4, x = 4 \Rightarrow (4, 4)$$

$$z''_{xx} = -\frac{y}{4x\sqrt{x}}, \quad z''_{xy} = \frac{1}{2\sqrt{x}}, \quad z''_{yy} = -2 \Rightarrow z''_{xx}(4, 4) = -\frac{1}{8}, \quad z''_{xy}(4, 4) = \frac{1}{4}$$

$$\Delta = \begin{vmatrix} -\frac{1}{8} & \frac{1}{4} \\ \frac{1}{4} & -2 \end{vmatrix} = \frac{1}{4} - \frac{1}{16} > 0, \quad z''_{xx} = -\frac{1}{8} < 0 \rightarrow (\max) \Rightarrow z_{\max} = 12$$

3) $z = x^3 + 8y^3 - 6xy + 1$

$$z'_x = 3x^2 - 6y, \quad z'_y = 24y^2 - 6x \Rightarrow \begin{cases} 3x^2 - 6y = 0 \\ 24y^2 - 6x = 0 \end{cases} \rightarrow y = \frac{1}{2}x^2 \downarrow$$

$$24 \cdot \frac{1}{4}x^4 - 6x = 0 \Rightarrow 6x(x^3 - 1) = 0 \Rightarrow \begin{matrix} x = 0 \rightarrow y = 0 \\ x = 1 \rightarrow y = \frac{1}{2} \end{matrix} \Rightarrow \begin{matrix} A(0, 0) \\ B(1, \frac{1}{2}) \end{matrix}$$

$$z''_{xx} = 6x, \quad z''_{xy} = -6, \quad z''_{yy} = 48y$$

$$z''_{xx}(A) = 0, \quad z''_{yy}(A) = 0 \Rightarrow \Delta(A) = \begin{vmatrix} 0 & -6 \\ -6 & 0 \end{vmatrix} = -36 < 0 \rightarrow \left(\begin{vmatrix} z'_x & z'_y \end{vmatrix} \right)$$

$$z''_{xx}(B) = 6, \quad z''_{yy}(B) = 24 \Rightarrow \Delta(B) = \begin{vmatrix} 6 & -6 \\ -6 & 24 \end{vmatrix} = 6 \cdot 24 - 36 > 0 \rightarrow \left(\begin{vmatrix} z'_x & z'_y \end{vmatrix} \right)$$

$$z''_{xx} = 6 > 0 \rightarrow (\min) \Rightarrow z_{\min} = 1 + 8 \cdot \frac{1}{8} - 6 \cdot 1 \cdot \frac{1}{2} + 1 = 0$$

4) $z = 2xy - 4x - 2y$

$$z'_x = 2y - 4, \quad z'_y = 2x - 2 \Rightarrow \begin{cases} 2y - 4 = 0 \\ 2x - 2 = 0 \end{cases} \Rightarrow x = 1, y = 2 \Rightarrow (1, 2)$$

$$z''_{xx} = 0, \quad z''_{xy} = 2, \quad z''_{yy} = 0 \Rightarrow \Delta = \begin{vmatrix} 0 & 2 \\ 2 & 0 \end{vmatrix} = -4 < 0 \rightarrow$$

$$\left(\begin{vmatrix} z'_x & z'_y \end{vmatrix} \right)$$

$$5) z = e^{x/2} (x+y^2)$$

$$z'_x = \frac{1}{2} e^{x/2} (x+y^2) + e^{x/2} \cdot 1 = e^{x/2} (\frac{1}{2}x + \frac{1}{2}y^2 + 1) = \frac{1}{2} e^{x/2} (x+y^2+2)$$

$$z'_y = e^{x/2} \cdot 2y \Rightarrow \begin{cases} x+y^2+2=0 \\ y=0 \end{cases} \Rightarrow x+2=0 \Rightarrow (-2, 0)$$

$$z''_{xx} = \frac{1}{2} \left[\frac{1}{2} e^{x/2} (x+y^2+2) + e^{x/2} \right] = \frac{1}{4} e^{x/2} (x+y^2+4)$$

$$z''_{xy} = \frac{1}{2} e^{x/2} \cdot 2y = y e^{x/2}, \quad z''_{yy} = 2 e^{x/2}$$

$$z''_{xx}(-2, 0) = \frac{1}{4} e^{-1} (-2+4) = \frac{1}{2} e, \quad z''_{xy}(-2, 0) = 0, \quad z''_{yy}(-2, 0) = 2/e$$

$$\Delta = \begin{vmatrix} \frac{1}{2}e & 0 \\ 0 & 2/e \end{vmatrix} = \frac{1}{e^2} > 0, \quad z''_{xx} = \frac{1}{2}e > 0 \rightarrow (\min) \Rightarrow z_{\min} = -\frac{2}{e}$$

$$6) z = xy(6-2x-y) \Rightarrow z = 6xy - 2x^2y - xy^2$$

$$z'_x = 6y - 4xy - y^2 = y(6-4x-y), \quad z'_y = 6x - 2x^2 - 2xy = 2x(3-x-y)$$

$$\begin{cases} y(6-4x-y)=0 \\ x(3-x-y)=0 \end{cases} \Rightarrow \begin{aligned} &y=0 \rightarrow x(3-x)=0 \Rightarrow x=0, x=3 \Rightarrow \underline{A(0,0)}, \underline{B(3,0)} \\ &6-4x-y=0 \Rightarrow y=6-4x \rightarrow 3-x-(6-4x)=0 \\ &3x=3 \rightarrow x=1 \Rightarrow y=2 \Rightarrow \underline{C(1,2)} \end{aligned}$$

$$x=0 \rightarrow 6-y=0 \Rightarrow y=6 \Rightarrow \underline{D(0,6)}$$

$$3-x-y=0 \Rightarrow y=3-x \Rightarrow 6-4x-(3-x)=0 \Rightarrow x=3 \Rightarrow \underline{B(3,0)}$$

$$z''_{xx} = -4y, \quad z''_{xy} = 6-4x-2y, \quad z''_{yy} = -2x \Rightarrow \Delta = \begin{vmatrix} -4y & 6-4x-2y \\ 6-4x-2y & -2x \end{vmatrix}$$

$$\Delta(A) = \Delta(0,0) = \begin{vmatrix} 0 & 6 \\ 6 & 0 \end{vmatrix} = -36 < 0, \quad \Delta(B) = \begin{vmatrix} 0 & -6 \\ -6 & -6 \end{vmatrix} = -36 < 0$$

$$\Delta(C) = \Delta(1,2) = \begin{vmatrix} -8 & -2 \\ -2 & -2 \end{vmatrix} = 16-4 > 0, \quad z''_{xx} = -8 < 0 \rightarrow (\max) \Rightarrow z_{\max} = 4$$

$$\Delta(D) = \Delta(0,6) = \begin{vmatrix} -24 & -6 \\ -6 & 0 \end{vmatrix} = -36 < 0$$

$$z_{\max} = z(1,2) = 4$$

$$7) z = x^2 + 3xy - 8 \ln x - 6 \ln y \Rightarrow x > 0, y > 0$$

$$z'_x = 2x + 3y - \frac{8}{x}, \quad z'_y = 3x - \frac{6}{y}$$

$$\begin{cases} 2x + 3y - \frac{8}{x} = 0 \\ 3x - \frac{6}{y} = 0 \end{cases} \Rightarrow \begin{aligned} &2x + 6x - \frac{8}{x} = 0 \rightarrow x=1 \rightarrow y=2 \rightarrow (1,2) \\ &\nearrow \end{aligned}$$

$$z''_{xx} = 2 + \frac{8}{x^2}, \quad z''_{xy} = 3, \quad z''_{yy} = \frac{6}{y^2} \Rightarrow z''_{xx}(1,2) = 10, \quad z''_{yy}(1,2) = \frac{3}{2}$$

$$\Delta = \begin{vmatrix} 10 & 3 \\ 3 & 3/2 \end{vmatrix} = 15-9 > 0, \quad z''_{xx} = 10 > 0 \Rightarrow (\min) \Rightarrow z_{\min} = z(1,2) = 7-6 \ln 2$$

8) $z = (x^2 + y^2) e^{-(x^2 + y^2)}$, $D = \{(x, y) \mid x^2 + y^2 < 1\}$

$$z'_x = 2x e^{-(x^2 + y^2)} - 2x(x^2 + y^2) e^{-(x^2 + y^2)} = 2x e^{-(x^2 + y^2)} (1 - x^2 - y^2)$$

$$z'_y = 2y e^{-(x^2 + y^2)} (1 - x^2 - y^2)$$

$$\begin{cases} x(1 - x^2 - y^2) = 0 \\ y(1 - x^2 - y^2) = 0 \end{cases} \Rightarrow x = 0, y = 0 \Rightarrow (0, 0)$$

אין הנקודה $(0, 0)$ יש נקודה (x, y) כזו שבה $x^2 + y^2 = 1$: נבדוק $1 - x^2 - y^2 = 0$: נקודה
 $x^2 + y^2 < 1$: נקודה (x, y) כזו שבה $x^2 + y^2 < 1$: נקודה

$$z''_{xx} = [2(x - x^3 - xy^2) e^{-(x^2 + y^2)}]'_x = 2[(1 - 3x^2 - y^2) e^{-(x^2 + y^2)} - 2x(x - x^3 - xy^2) e^{-(x^2 + y^2)}]$$

$$z''_{xy} = 2[(-2xy) e^{-(x^2 + y^2)} - 2y(x - x^3 - xy^2) e^{-(x^2 + y^2)}]$$

$$z''_{yy} = [2(y - x^2y - y^3) e^{-(x^2 + y^2)}]'_y = 2[(1 - x^2 - 3y^2) e^{-(x^2 + y^2)} - 2y(y - x^2y - y^3) e^{-(x^2 + y^2)}]$$

$$z''_{xx}(0, 0) = 2(1 - 0) = 2, z''_{xy}(0, 0) = 0, z''_{yy}(0, 0) = 2 \Rightarrow \Delta = \begin{vmatrix} 2 & 0 \\ 0 & 2 \end{vmatrix} = 4 > 0$$

$$z''_{xx} = 2 > 0 \Rightarrow (\min) \Rightarrow z_{\min} = 0$$

9) $z = e^{x-y}(x^2 - 2y^2)$ $\frac{\partial z}{\partial x} = e^{x-y}(x^2 - 2y^2) + 2xe^{x-y} = e^{x-y}(x^2 - 2y^2 + 2x)$

$$\frac{\partial z}{\partial y} = -e^{x-y}(x^2 - 2y^2) - e^{x-y} \cdot 4y = -e^{x-y}(x^2 - 2y^2 + 4y)$$

$$\begin{cases} e^{x-y}(x^2 - 2y^2 + 2x) = 0 \\ -e^{x-y}(x^2 - 2y^2 + 4y) = 0 \end{cases} \Rightarrow \begin{cases} x^2 - 2y^2 + 2x = 0 \\ x^2 - 2y^2 + 4y = 0 \end{cases} \xrightarrow{(\text{החסר})} \begin{cases} 2x - 4y = 0 \\ x = 2y \end{cases} \quad (\text{נציב}) \rightarrow$$

$$4y^2 - 2y^2 + 4y = 0 \quad y(y + 2) = 0 \quad \begin{matrix} 1) y = 0 \rightarrow x = 0 \rightarrow (0, 0) \\ 2) y = -2 \rightarrow x = -4 \rightarrow (-4, -2) \end{matrix}$$

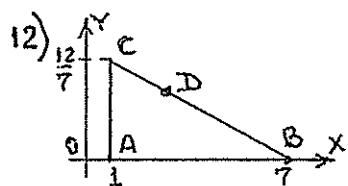
$$\frac{\partial^2 z}{\partial x^2} = e^{x-y}(x^2 - 2y^2 + 2x) + e^{x-y}(2x + 2) = e^{x-y}(x^2 - 2y^2 + 4x + 2)$$

$$\frac{\partial^2 z}{\partial y^2} = e^{x-y}(x^2 - 2y^2 + 4y) + e^{x-y}(4y - 4) = e^{x-y}(x^2 - 2y^2 + 8y - 4)$$

$$\frac{\partial^2 z}{\partial x \partial y} = -e^{x-y}(x^2 - 2y^2 + 2x) - e^{x-y} \cdot 4y = -e^{x-y}(x^2 - 2y^2 + 2x + 4y)$$

$$(0, 0) \Rightarrow \Delta = \begin{vmatrix} 2 & 0 \\ 0 & -4 \end{vmatrix} = -8 < 0 \quad (\text{נקודה קיצונית})$$

$$(-4, -2) \Rightarrow \Delta = \begin{vmatrix} -6/e^2 & 8/e^2 \\ 8/e^2 & -12/e^2 \end{vmatrix} = \frac{8}{e^4} > 0, \quad \frac{\partial^2 z}{\partial x^2} = -\frac{6}{e^2} < 0 \quad (\max) \quad \underline{z_{\max} = \frac{8}{e^2}}$$



$$z = x^2 - 10xy^2 + 10x, \quad x=1, y=0, \quad 2x+7y-14=0$$

: הפונקציה של המ"מ הנ"ל

$$z'_x = 2x - 10y^2 + 10, \quad z'_y = -20xy$$

$$\begin{cases} 2x - 10y^2 + 10 = 0 \\ -20xy = 0 \end{cases} \Rightarrow \begin{cases} x=0 \rightarrow (\text{פונקציה של } y) \\ y=0 \rightarrow (\text{פונקציה של } x) \end{cases} \Rightarrow \begin{pmatrix} \text{מציאת נקודות קיצון} \\ \text{פונקציה של } x \end{pmatrix}$$

AB: $y=0, 1 < x < 7 \Rightarrow z = x^2 + 10x \Rightarrow z' = 2x + 10 \Rightarrow 2x + 10 = 0$
 $x = -5 \Rightarrow (\text{פונקציה של } x)$

AC: $x=1, 0 < y < \frac{12}{7} \Rightarrow z = 11 - 10y^2 \Rightarrow z' = -20y \Rightarrow -20y = 0$
 $y = 0 \Rightarrow (\text{פונקציה של } y)$

CB: $y = 2 - \frac{2}{7}x, 1 < x < 7 \Rightarrow z = x^2 - 10x(2 - \frac{2}{7}x)^2 + 10x$

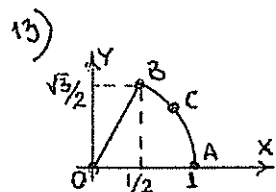
$$z = -\frac{40}{49}x^3 + \frac{87}{7}x^2 - 30x \Rightarrow z' = -\frac{120}{49}x^2 + \frac{174}{7}x - 30 \Rightarrow z' = 0 \Rightarrow$$

$$120x^2 - 1218x + 1470 = 0 \rightarrow x = 8.75 \rightarrow (\text{פונקציה של } x), \quad x = 1.4 \rightarrow y = 1.6$$

$D(1.4, 1.6)$

A(1,0)	B(7,0)	C(1, 12/7)	D(1.4, 1.6)
11	119	-18.39	-19.88

$$z_{\max} = z(7,0) = 119, \quad z_{\min} = z(1.4, 1.6) = -19.88$$



$$z = x^2 + y^2 + 4xy, \quad x^2 + y^2 = 1, \quad y=0, \quad y=\sqrt{3}x. \quad \begin{cases} x \geq 0 \\ y \geq 0 \end{cases}$$

: הפונקציה של המ"מ הנ"ל

$$z'_x = 2x + 4y, \quad z'_y = 2y + 4x \Rightarrow \begin{cases} x + 2y = 0 \\ 2x + y = 0 \end{cases} \Rightarrow (0,0) \rightarrow (\text{פונקציה של } x)$$

OA: $y=0, 0 < x < 1 \Rightarrow z = x^2, z' = 2x \Rightarrow z' = 0 \rightarrow x = 0 \rightarrow (\text{פונקציה של } x)$

: B נקודת קיצון של הפונקציה

$$\begin{cases} x^2 + y^2 = 1 \\ y = \sqrt{3}x \end{cases} \Rightarrow x^2 + 3x^2 = 1 \Rightarrow x = \frac{1}{2}, y = \frac{\sqrt{3}}{2} \Rightarrow B(\frac{1}{2}, \frac{\sqrt{3}}{2})$$

OB: $y = \sqrt{3}x, 0 < x < \frac{1}{2} \Rightarrow z = (4 + 4\sqrt{3})x^2, z' = 2(4 + 4\sqrt{3})x \Rightarrow z' = 0 \rightarrow$
 $x = 0 \rightarrow (\text{פונקציה של } x)$

BA: $y = \sqrt{1-x^2}, \frac{1}{2} < x < 1 \Rightarrow z = x^2 + 1 - x^2 + 4x\sqrt{1-x^2} \Rightarrow z = 1 + 4\sqrt{x^2 - x^4}$

$$z' = 4 \cdot \frac{x - 2x^3}{\sqrt{x^2 - x^4}} \Rightarrow x(1 - 2x^2) = 0, \quad x \neq 0 \rightarrow 1 - 2x^2 = 0 \rightarrow x = \frac{1}{\sqrt{2}} \rightarrow C(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$$

O(0,0)	A(1,0)	B(1/2, sqrt(3)/2)	C(1/sqrt(2), 1/sqrt(2))
0	1	2.73	3

$$z_{\max} = z(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}) = 3, \quad z_{\min} = z(0,0) = 0$$

14) $z = x^2 - y^2 + 1, \quad 4x - y = 15$

$$L = x^2 - y^2 + 1 + \lambda(4x - y - 15) \Rightarrow L'_x = 2x + 4\lambda, \quad L'_y = -2y - \lambda \Rightarrow$$

$$\begin{cases} 2x + 4\lambda = 0 \\ -2y - \lambda = 0 \\ 4x - y - 15 = 0 \end{cases} \Rightarrow \begin{cases} x + 2\lambda = 0 \\ 2y + \lambda = 0 \\ 4x - y = 15 \end{cases} \Rightarrow x = 4y \downarrow \quad 16y - y = 15 \Rightarrow A(4, 1), \lambda = -2$$

$$L''_{xx} = 2, \quad L''_{yy} = -2, \quad L''_{xy} = 0 \Rightarrow d^2L = 2dx^2 + 2 \cdot 0 dx dy - 2dy^2 \Rightarrow$$

$$\begin{cases} d^2L = 2dx^2 - 2dy^2 \\ 4dx - dy = 0 \rightarrow dy = 4dx \uparrow \end{cases} \quad d^2L = 2dx^2 - 32dx^2 = -32dx^2 < 0$$

:A' תנאי מספיקים, $d^2L < 0$: p' תנאי מספיקים

$$z_{\max} = z(4, 1) = 16$$

15) $z = x^3 + 3x + y^3 + 3y, \quad x^2 + y^2 = 2$

$$L = x^3 + 3x + y^3 + 3y + \lambda(x^2 + y^2 - 2) \Rightarrow$$

$$L'_x = 3x^2 + 3 + 2\lambda x, \quad L'_y = 3y^2 + 3 + 2\lambda y$$

$$(*) \begin{cases} 3x^2 + 2\lambda x + 3 = 0 \\ 3y^2 + 2\lambda y + 3 = 0 \\ x^2 + y^2 = 2 \end{cases} \Leftrightarrow \begin{cases} 3(x^2 - y^2) + 2\lambda(x - y) = 0 \\ (**) (x - y)[3(x + y) + 2\lambda] = 0 \Rightarrow x - y = 0 \Rightarrow \end{cases}$$

$$x = y \uparrow \quad x^2 + x^2 = 2 \Rightarrow x = y = 1, \lambda = -3, \quad x = y = -1, \lambda = 3$$

$$A(1, 1), \lambda = -3, \quad B(-1, -1), \lambda = 3$$

אם נבדוק את תנאי מספיקים, $3(x + y) + 2\lambda = 0$: p' תנאי מספיקים

:p' תנאי מספיקים, $x^2 + y^2 = 2$: אם נבדוק, $3(x^2 + y^2) + 2\lambda(x + y) + 6 = 0$: תנאי מספיקים

$$3 \cdot 2 + 2\lambda(x + y) + 6 = 0 \Rightarrow \lambda(x + y) = -6$$

$$\begin{cases} 3(x + y) + 2\lambda = 0 \\ \lambda(x + y) = -6 \end{cases} \Rightarrow x + y = -\frac{6}{\lambda} \uparrow \quad -\frac{18}{\lambda} + 2\lambda = 0 \Rightarrow \lambda^2 = 9 \Rightarrow \lambda = \pm 3$$

$$A(1, 1), \lambda = -3, \quad B(-1, -1), \lambda = 3$$

$$\{(x, y) | x^2 + y^2 = 2\}$$

הפונקציה $z = x^3 + 3x + y^3 + 3y$ היא פונקציה רגולרית

ב-A ו-B. הפונקציה z היא פונקציה רגולרית

$$z(A) = z(1, 1) = 8, \quad z(B) = z(-1, -1) = -8 \Rightarrow$$

$$z_{\max} = z(1, 1) = 8, \quad z_{\min} = z(-1, -1) = -8$$

18) $z = \frac{4}{x} + \frac{1}{y}, \quad x+y=3$

$L = \frac{4}{x} + \frac{1}{y} + \lambda(x+y-3) \Rightarrow L'_x = -\frac{4}{x^2} + \lambda, \quad L'_y = -\frac{1}{y^2} + \lambda$

$\begin{cases} -\frac{4}{x^2} + \lambda = 0 \\ -\frac{1}{y^2} + \lambda = 0 \\ x+y=3 \end{cases} \Rightarrow \frac{4}{x^2} - \frac{1}{y^2} = 0 \Rightarrow 4y^2 - x^2 = 0 \Rightarrow (2y-x)(2y+x) = 0$

$\begin{cases} 2y-x=0 \\ x+y=3 \end{cases} \Rightarrow x=2, y=1, \lambda=1 \quad , \quad \begin{cases} 2y+x=0 \\ x+y=3 \end{cases} \Rightarrow x=6, y=-3, \lambda=\frac{1}{9} \Rightarrow A(2,1), \lambda=1$
 $B(6,-3), \lambda=\frac{1}{9}$

$L''_{xx} = \frac{8}{x^3}, \quad L''_{yy} = \frac{2}{y^3}, \quad L''_{xy} = 0 \Rightarrow d^2L = \frac{8}{x^3} dx^2 + \frac{2}{y^3} dy^2,$

$dx+dy=0 \Rightarrow dy=-dx \Rightarrow d^2L = \frac{8}{x^3} dx^2 + \frac{2}{y^3} dx^2 = \left(\frac{8}{x^3} + \frac{2}{y^3}\right) dx^2$

$A(2,1) \Rightarrow d^2L = \left(\frac{8}{8} + \frac{2}{1}\right) dx^2 = 3 dx^2 > 0 \Rightarrow (\min) \Rightarrow z_{\min} = z(2,1) = 3$

$B(6,-3) \Rightarrow d^2L = -\frac{1}{27} dx^2 < 0 \Rightarrow (\max) \Rightarrow z_{\max} = z(6,-3) = \frac{1}{3}$

19) $z = x-y, \quad \tan x - 3 \tan y = 0, \quad 0 < x < \frac{\pi}{2}, \quad 0 < y < \frac{\pi}{2}$

$L = x-y + \lambda(\tan x - 3 \tan y) \Rightarrow L'_x = 1 + \frac{\lambda}{\cos^2 x}, \quad L'_y = -1 - \frac{3\lambda}{\cos^2 y}$

$\begin{cases} 1 + \frac{\lambda}{\cos^2 x} = 0 \Rightarrow \frac{1}{\cos^2 x} = -\frac{1}{\lambda} \Rightarrow 1 + \tan^2 x = -\frac{1}{\lambda} \Rightarrow \tan^2 x = -1 - \frac{1}{\lambda} \\ 1 + \frac{3\lambda}{\cos^2 y} = 0 \Rightarrow \frac{1}{\cos^2 y} = -\frac{1}{3\lambda} \Rightarrow 1 + \tan^2 y = -\frac{1}{3\lambda} \Rightarrow \tan^2 y = -1 - \frac{1}{3\lambda} \\ \tan x = 3 \tan y \end{cases}$

$\tan^2 x = 9 \tan^2 y \Rightarrow -1 - \frac{1}{\lambda} = 9 \left(-1 - \frac{1}{3\lambda}\right) \Rightarrow \frac{3}{\lambda} - \frac{1}{\lambda} = -8 \Rightarrow \lambda = -\frac{1}{4}$

$\frac{1}{\cos^2 x} = -\frac{1}{\lambda} \Rightarrow \cos^2 x = \frac{1}{4} \Rightarrow \cos x = \pm \frac{1}{2}$

$0 < x < \frac{\pi}{2} \Rightarrow \cos x = \frac{1}{2} \Rightarrow x = \frac{\pi}{3}$

$\frac{1}{\cos^2 y} = -\frac{1}{3\lambda} \Rightarrow \cos^2 y = \frac{3}{4} \Rightarrow \cos y = \pm \frac{\sqrt{3}}{2} \Rightarrow y = \frac{\pi}{6}$

$A\left(\frac{\pi}{3}, \frac{\pi}{6}\right), \quad \lambda = -\frac{1}{4}$

$L''_{xx} = \frac{2\lambda \sin x}{\cos^3 x} = 2\lambda \cdot \frac{\tan x}{\cos^2 x}, \quad L''_{yy} = \frac{6\lambda \sin y}{\cos^3 y} = 6\lambda \frac{\tan y}{\cos^2 y}, \quad L''_{xy} = 0$

$A\left(\frac{\pi}{3}, \frac{\pi}{6}\right), \lambda = -\frac{1}{4} \Rightarrow L''_{xx} = -\frac{2}{4} \frac{\sqrt{3}}{1/4} = -2\sqrt{3}, \quad L''_{yy} = -\frac{6}{4} \frac{\sqrt{3}/3}{3/4} = -\frac{2}{3}\sqrt{3}$

$d^2L = -2\sqrt{3} dx^2 - \frac{2}{3}\sqrt{3} dy^2 < 0 \Rightarrow (\max)$

$z_{\max} = z\left(\frac{\pi}{3}, \frac{\pi}{6}\right) = \frac{\pi}{6}$

20) $u = xy^2z^3$, $x+y+z=12$ ($x>0, y>0, z>0$)

$$L = xy^2z^3 + \lambda(x+y+z-12)$$

$$L'_x = y^2z^3 + \lambda, \quad L'_y = 2xy^2z^3 + \lambda, \quad L'_z = 3xy^2z^2 + \lambda$$

$$\begin{cases} y^2z^3 + \lambda = 0 \\ 2xy^2z^3 + \lambda = 0 \\ 3xy^2z^2 + \lambda = 0 \\ x+y+z=12 \end{cases} \Rightarrow \begin{cases} y^2z^3 = 2xy^2z^3 \\ y^2z^3 = 3xy^2z^2 \end{cases} \Rightarrow \begin{cases} yz^3(y-2x)=0 \\ y^2z^2(z-3x)=0 \end{cases} \Rightarrow \begin{cases} y-2x=0 \\ z-3x=0 \\ x+y+z=12 \end{cases}$$

$$y=2x, z=3x \Rightarrow x+2x+3x=12 \Rightarrow x=2, y=4, z=6 \Rightarrow A(2,4,6), \lambda = -4^2 \cdot 6^3$$

$$L''_{xx}=0, \quad L''_{xy}=2yz^3, \quad L''_{xz}=3y^2z^2, \quad L''_{yy}=2xz^3, \quad L''_{yz}=6xy^2z^2, \quad L''_{zz}=6xy^2z$$

$$A(2,4,6) \Rightarrow L''_{xy} = 2 \cdot 4 \cdot 6^3 = (4 \cdot 6^2) \cdot 12, \quad L''_{xz} = 3 \cdot 4^2 \cdot 6^2 = (4 \cdot 6^2) \cdot 12$$

$$L''_{yy} = 2 \cdot 2 \cdot 6^3 = (4 \cdot 6^2) \cdot 6, \quad L''_{yz} = 6 \cdot 2 \cdot 4 \cdot 6^2 = (4 \cdot 6^2) \cdot 12, \quad L''_{zz} = (4 \cdot 6^2) \cdot 8$$

$$d^2L = 4 \cdot 6^2 [0 \cdot dx^2 + 6dy^2 + 8dz^2 + 2 \cdot 12 dx dy + 2 \cdot 12 dx dz + 2 \cdot 12 dy dz]$$

$$d^2L = 8 \cdot 6^2 (3 dy^2 + 4 dz^2 + 12 dx dy + 12 dx dz + 12 dy dz)$$

$$x+y+z=12 \Rightarrow dx+dy+dz=0 \Rightarrow dz = -(dx+dy) \Rightarrow$$

$$d^2L = 288 [3 dy^2 + 4(dx+dy)^2 + 12 dx dy - 12 dx(dx+dy) - 12 dy(dx+dy)] =$$

$$= 288 (3 dy^2 + 4 dx^2 + 8 dx dy + 4 dy^2 + 12 dx dy - 12 dx^2 - 12 dx dy - 12 dx dy - 12 dy^2) = 288 (-8 dx^2 - 4 dx dy - 5 dy^2)$$

$$: -8 \cdot dx^2 - 2 \cdot 2 dx dy - 5 dy^2 : \text{מ'כיוון ש} \Delta < 0 \text{ ו} \Delta_{xx} < 0 \text{ אז } A \text{ היא נקודת מקסימום}$$

$$\Delta = \begin{vmatrix} -8 & -2 \\ -2 & -5 \end{vmatrix} = 40 - 4 > 0, \quad -8 < 0 \Rightarrow (\max) \Rightarrow u_{\max} = 2 \cdot 4^2 \cdot 6^3$$

21) $u = 2x + y - z + 1$, $x^2 + y^2 + 2z^2 = 22$

$$L = 2x + y - z + 1 + \lambda(x^2 + y^2 + 2z^2 - 22) \Rightarrow L'_x = 2 + 2\lambda x, \quad L'_y = 1 + 2\lambda y, \quad L'_z = -1 + 4\lambda z$$

$$\begin{cases} 1 + \lambda x = 0 \rightarrow x = -1/\lambda \\ 1 + 2\lambda y = 0 \rightarrow y = -1/2\lambda \\ -1 + 4\lambda z = 0 \rightarrow z = 1/4\lambda \\ x^2 + y^2 + 2z^2 = 22 \end{cases} \Rightarrow \frac{1}{\lambda^2} + \frac{1}{4\lambda^2} + \frac{2}{16\lambda^2} = 22$$

$$\lambda^2 = \frac{1}{16} \Rightarrow \lambda = \pm \frac{1}{4}$$

$$A(-4, -2, 1), \lambda = \frac{1}{4}, \quad B(4, 2, -1), \lambda = -\frac{1}{4}$$

נראה כי $x^2 + y^2 + 2z^2 = 22$ היא אליפסואיד סגור. נגדור את המרחב $\{(x, y, z) | x^2 + y^2 + 2z^2 = 22\}$ ונראה כי A ו- B הן הנקודות היחידות בהן u מקבלת ערכים קיצוניים.

$$u = 2x + y - z + 1$$

נראה כי A ו- B הן הנקודות היחידות בהן u מקבלת ערכים קיצוניים.

$$u(A) = u(-4, -2, 1) = -10, \quad u(B) = u(4, 2, -1) = 12$$

$$u_{\max} = u(4, 2, -1) = 12, \quad u_{\min} = u(-4, -2, 1) = -10$$

$$\begin{cases} 1 - \frac{\lambda}{x^2} = 0 \\ 1 - \frac{\lambda}{y^2} = 0 \\ 1 - \frac{\lambda}{z^2} = 0 \\ \frac{1}{x} + \frac{1}{y} + \frac{1}{z} = 1 \end{cases} \quad A(1,1,1) \Rightarrow \begin{cases} 1 - \lambda = 0 \\ 1 - \lambda = 0 \\ 1 - \lambda = 0 \\ 1 + 1 + 1 = 1 \end{cases} \Rightarrow \lambda = 1$$

$$L''_{xx} = \frac{2\lambda}{x^3}, L''_{yy} = \frac{2\lambda}{y^3}, L''_{zz} = \frac{2\lambda}{z^3}, L''_{xy} = L''_{xz} = L''_{yz} = 0$$

$$d^2L = 2dx^2 + 2dy^2 - 2dz^2 = 2(dx^2 + dy^2 - dz^2)$$

$$d\left(\frac{1}{x} + \frac{1}{y} + \frac{1}{z} - 1\right) = 0 \Rightarrow -\frac{1}{x^2}dx - \frac{1}{y^2}dy - \frac{1}{z^2}dz = 0$$

$$A(1,1,-1) \Rightarrow dx+dy+dz=0 \Rightarrow dz=-(dx+dy)$$

$$d^2L = 2[dx^2 + dy^2 - (dx + dy)^2] = 2[dx^2 + dy^2 - dx^2 - 2dxdy - dy^2] = -4dxdy$$

25) $u = xy + yz$, $x^2 + y^2 = 2$, $y + z = 2$, $A(1, 1, 1)$

$$L = xy + yz + \lambda(x^2 + y^2 - 2) + \mu(y + z - 2) \Rightarrow L'_x = y + 2\lambda x, \quad L'_y = x + z + 2\lambda y + \mu, \quad L'_z = y + \mu$$

$$\begin{cases} y+2\lambda x=0 \\ x+z+2\lambda y+\mu=0 \\ y+\mu=0 \\ x^2+y^2=2 \\ y+z=2 \end{cases} \Rightarrow R(1,1,1) \Rightarrow \begin{cases} 1+2\lambda=0 \\ 2+2\lambda+\mu=0 \\ 1+\mu=0 \\ 1+1 \equiv 2 \\ 1+1 \equiv 2 \end{cases} \Rightarrow \lambda=-\frac{1}{2}, \mu=-1$$

קואורדינטות הנקודה $(1, 1, 1)$, $\lambda = -\frac{1}{2}$, $\mu = -1$ מק"מ נ"מ אות המערכת, $S'' = 10$
 התנאי ההכרחי שקיים אינסוף מק"מ נ"מ.

$$L''_{xx} = 2\lambda, L''_{yy} = 2\lambda, L''_{zz} = 0, L''_{xy} = 1, L''_{xz} = 0, L''_{yz} = 1$$

$$A(1,1,1), \lambda = -\frac{1}{2} \Rightarrow L''_{xx} = -1, L''_{yy} = -1 \Rightarrow d^2L = -dx^2 - dy^2 + 2dx dy + 2dy dz$$

$$d(x^2+y^2-2)=0 \rightarrow 2xdx+2ydy=0 \quad d(y+z-2)=0 \Rightarrow dy+dz=0$$

$$A(1,1,1) \Rightarrow 2dx + 2dy = 0 \Rightarrow dx + dy = 0$$

$$\begin{cases} d^2L = -dx^2 - dy^2 + 2dx dy + 2dy dz & d^2L = -dx^2 - dx^2 - 2dx^2 - 2dx^2 = -6dx^2 \\ dx + dy = 0 \rightarrow dy = -dx \\ dy + dz = 0 \rightarrow dz = -dy \end{cases} \quad \begin{matrix} \searrow \\ \nearrow \end{matrix} \quad \begin{matrix} dz = dx \\ dz = dx \end{matrix}$$

$A(1,1,1)$ એક/જા ρ/n 'ઠી' e $d^2L < 0$
 . $d^2L < 0$

26) $u = xyz$, $x+y+z=5$, $xy+yz+zx=8$, $A(1,2,2)$

$$L = xyz + \lambda(x+y+z-5) + \mu(xy+yz+zx-8)$$

$$L'_x = yz + \lambda + \mu(y+z), L'_y = xz + \lambda + \mu(x+z), L'_z = xy + \lambda + \mu(y+x)$$

$$\begin{cases} yz + \lambda + \mu(y+z) = 0 \\ xz + \lambda + \mu(x+z) = 0 \\ xy + \lambda + \mu(y+x) = 0 \\ x+y+z=5 \\ xy+yz+zx=8 \end{cases} \Rightarrow \begin{cases} 4 + \lambda + 4\mu = 0 \\ 2 + \lambda + 3\mu = 0 \\ 2 + \lambda + 3\mu = 0 \\ 1+2+2=5 \\ 2+4+2=8 \end{cases} \Rightarrow \begin{cases} \lambda + 4\mu = -4 \\ \lambda + 3\mu = -2 \end{cases} \Rightarrow \begin{cases} \lambda = 4 \\ \mu = -2 \end{cases}$$

$\mu = -2, \lambda = 4, (1,2,2)$ נקודה קיצונית

$$L''_{xx} = 0, L''_{yy} = 0, L''_{zz} = 0, L''_{xy} = z + \mu, L''_{xz} = y + \mu, L''_{yz} = x + \mu$$

$$A(1,2,2), \mu = -2 \Rightarrow L''_{xy} = 0, L''_{xz} = 0, L''_{yz} = -1 \Rightarrow d^2L = -2dydz$$

$$d(x+y+z-5)=0 \Rightarrow dx+dy+dz=0, d(xy+yz+zx-8)=0 \Rightarrow$$

$$(y+z)dx + (x+z)dy + (y+x)dz = 0 \Rightarrow A(1,2,2) \Rightarrow 4dx + 3dy + 3dz = 0$$

$$\begin{cases} dx+dy+dz=0 \\ 4dx+3dy+3dz=0 \end{cases} \otimes 4 \Rightarrow \begin{cases} 4dx+4dy+4dz=0 \\ 4dx+3dy+3dz=0 \end{cases} \ominus \Rightarrow dy+dz=0 \Rightarrow dz=-dy$$

$$\begin{cases} d^2L = -2dydz \\ dz = -dy \end{cases} \Rightarrow d^2L = -2dy(-dy) = 2dy^2 > 0 \rightarrow (min)$$

נקודה קיצונית. נקודה קיצונית $A(1,2,2)$ קיצונית

27) $u = x^2 + y^2 + z^2$, $x^2 + y^2 = 2$, $x + y + 2z = 0$

$$L = x^2 + y^2 + z^2 + \lambda(x^2 + y^2 - 2) + \mu(x + y + 2z)$$

$$L'_x = 2x + 2\lambda x + \mu, L'_y = 2y + 2\lambda y + \mu, L'_z = 2z + 2\mu$$

$$\begin{cases} 2x + 2\lambda x + \mu = 0 \\ 2y + 2\lambda y + \mu = 0 \\ z + \mu = 0 \\ x^2 + y^2 = 2 \\ x + y + 2z = 0 \end{cases} \ominus \Rightarrow \begin{cases} 2(x-y) + 2\lambda(x-y) = 0 \\ 2(x-y)(1+\lambda) = 0 \end{cases} \Rightarrow \begin{cases} x-y=0 \\ x^2+y^2=2 \\ x+y+2z=0 \end{cases} \Rightarrow \begin{cases} y=x=\pm 1 \\ z=\mp 1 \end{cases}$$

$$A(1,1,-1), \lambda = -3/2, \mu = 1$$

$$B(-1,-1,1), \lambda = 3/2, \mu = -1$$

$$\begin{cases} 1+\lambda=0 \\ 2x+2\lambda x+\mu=0 \end{cases} \Rightarrow \mu=0 \Rightarrow z=0 \Rightarrow \begin{cases} x^2+y^2=2 \\ x+y=0 \end{cases} \Rightarrow x=\pm 1, y=\mp 1, z=0$$

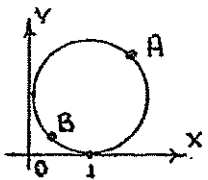
$$C(1,-1,0), \lambda = -1, \mu = 0, D(-1,1,0), \lambda = -1, \mu = 0$$

נקודה קיצונית נקודה קיצונית נקודה קיצונית נקודה קיצונית
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 $\{(x,y,z) | x^2+y^2=2, x+y+2z=0\}$: נקודה קיצונית נקודה קיצונית
 : נקודה קיצונית נקודה קיצונית נקודה קיצונית נקודה קיצונית

A(1,1,-1)	B(-1,-1,1)	C(1,-1,0)	D(-1,1,0)
3	3	2	2

$$u_{max} = u(1,1,-1) = u(-1,-1,1) = 3, u_{min} = u(1,-1,0) = u(-1,1,0) = 2$$

28)

$z = 1 - x^2 - y^2$, $D = \{(x, y) | (x-1)^2 + (y-1)^2 \leq 1\}$

 : $\delta'x$ ו $\delta'y$ של z על D מתקבלים מהצורה $z'x = -2x$, $z'y = -2y$ $\begin{cases} z'_x = 0 \\ z'_y = 0 \end{cases} \Rightarrow 0(0,0) \rightarrow$ (נקודה קריטית)

: $\delta'x$ ו $\delta'y$ של z על D מתקבלים מהצורה $z'x = -2x$, $z'y = -2y$ $\begin{cases} z'_x = 0 \\ z'_y = 0 \end{cases} \Rightarrow 0(0,0) \rightarrow$ (נקודה קריטית)

$$L = 1 - x^2 - y^2 + \lambda [(x-1)^2 + (y-1)^2 - 1]$$

$$L'_x = -2x + 2\lambda(x-1) = 2[x(\lambda-1) - \lambda], \quad L'_y = 2[y(\lambda-1) - \lambda]$$

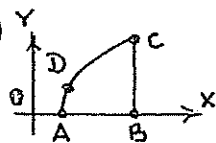
$$\begin{cases} x(\lambda-1) - \lambda = 0 \\ y(\lambda-1) - \lambda = 0 \end{cases} \Leftrightarrow (x-y)(\lambda-1) = 0 \Rightarrow x=y$$

$$(x-1)^2 + (y-1)^2 = 1 \Rightarrow 2(x-1)^2 = 1 \Rightarrow x = 1 \pm \frac{1}{\sqrt{2}}$$

$$A(1 + \frac{1}{\sqrt{2}}, 1 + \frac{1}{\sqrt{2}}), \quad B(1 - \frac{1}{\sqrt{2}}, 1 - \frac{1}{\sqrt{2}})$$

$$z(A) = 1 - 2(1 + \frac{1}{\sqrt{2}})^2 = -2(\sqrt{2} + 1) = z_{\min}, \quad z(B) = 1 - 2(1 - \frac{1}{\sqrt{2}})^2 = 2(\sqrt{2} - 1) = z_{\max}$$

29)



$$z = 4x - y, \quad y = 0, \quad x = 2\sqrt{6}, \quad x^2 - y^2 = 15$$

: $\delta'x$ ו $\delta'y$ של z על D מתקבלים מהצורה $z'x = 4$, $z'y = -1$
 $A(\sqrt{15}, 0), \quad B(2\sqrt{6}, 0)$

$$C: \begin{cases} x^2 - y^2 = 15 \\ x = 2\sqrt{6} \end{cases} \Rightarrow C(2\sqrt{6}, 3)$$

: $\delta'x$ ו $\delta'y$ של z על D מתקבלים מהצורה $z'x = 4$, $z'y = -1$

$$AB: y = 0, \quad \sqrt{15} < x < 2\sqrt{6} \Rightarrow z = 4x \Rightarrow$$
 (נקודה קריטית)

$$BC: x = 2\sqrt{6}, \quad 0 < y < 3 \Rightarrow z = 8\sqrt{6} - y \Rightarrow$$
 (נקודה קריטית)

$$AC: z = 4x - y, \quad x^2 - y^2 = 15, \quad \sqrt{15} < x < 2\sqrt{6}$$

: $\delta'x$ ו $\delta'y$ של z על D מתקבלים מהצורה $z'x = 4$, $z'y = -1$

$$L = 4x - y + \lambda(x^2 - y^2 - 15) \Rightarrow L'_x = 4 + 2\lambda x, \quad L'_y = -1 - 2\lambda y$$

$$\begin{cases} 4 + 2\lambda x = 0 \Rightarrow x = -2/\lambda \\ -1 - 2\lambda y = 0 \Rightarrow y = -1/(2\lambda) \end{cases} \Rightarrow \frac{4}{\lambda^2} - \frac{1}{4\lambda^2} = 15 \Rightarrow \frac{15}{4\lambda^2} = 15 \Rightarrow \lambda = \pm \frac{1}{2}$$

$$\lambda = \frac{1}{2} \Rightarrow x = -4 \rightarrow$$
 (נקודה קריטית), $\lambda = -\frac{1}{2} \Rightarrow x = 4 \Rightarrow D(4, 1)$

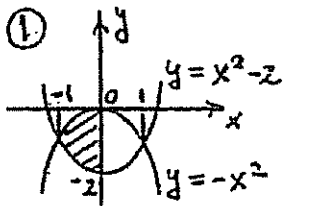
: D, C, B, A של z על D מתקבלים מהצורה $z'x = 4$, $z'y = -1$

$A(\sqrt{15}, 0)$	$B(2\sqrt{6}, 0)$	$C(2\sqrt{6}, 3)$	$D(4, 1)$
15.49	19.6	16.6	15

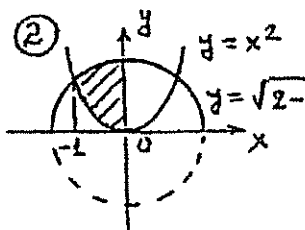
$$z_{\max} = z(2\sqrt{6}, 0) = 19.6, \quad z_{\min} = z(4, 1) = 15$$

האינטגרל הכפול

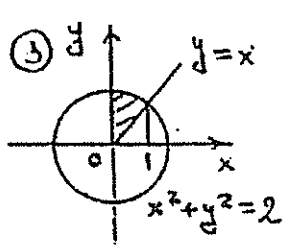
(X)

① 
$$\begin{cases} y = x^2 - 2 \\ y = -x^2 \end{cases} \quad \begin{aligned} -x^2 &= x^2 - 2 \\ 2x^2 &= 2 \\ x &= \pm 1 \end{aligned} \quad (x \leq 0)$$

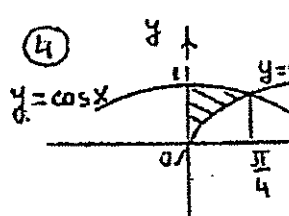
$$I = \int_{-1}^0 dx \int_{x^2-2}^{-x^2} f(x,y) dy$$

② 
$$\begin{cases} y = \sqrt{2-x^2} \\ y = x^2 \end{cases} \quad \begin{aligned} x^4 + x^2 - 2 &= 0 \\ x^2 &= t \\ t^2 + t - 2 &= 0 \\ t &= 1 \rightarrow x = \pm 1 \\ t &= -2 \text{ (לא רלוונטי)} \end{aligned} \quad (x \leq 0)$$

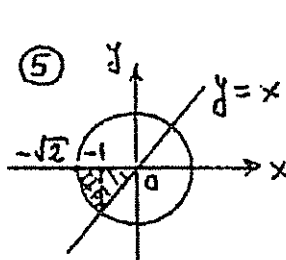
$$I = \int_{-1}^0 dx \int_{x^2}^{\sqrt{2-x^2}} f(x,y) dy$$

③ 
$$\begin{cases} x^2 + y^2 = 2 \\ y = x \end{cases} \quad \begin{aligned} x^2 + x^2 &= 2 \\ x^2 &= 1 \\ x &= \pm 1 \end{aligned} \quad (y \geq 0)$$

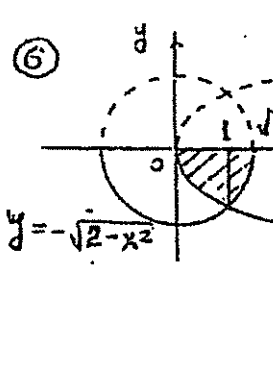
$$I = \int_0^1 dx \int_x^{\sqrt{2-x^2}} f(x,y) dy$$

④ 
$$\begin{cases} y = \cos x \\ y = \sin x \end{cases} \quad \begin{aligned} \sin x &= \cos x \\ \tan x &= 1 \\ x &= \frac{\pi}{4} + \pi k \\ x &= \frac{\pi}{4} \end{aligned} \quad 0 \leq x \leq \frac{\pi}{4}$$

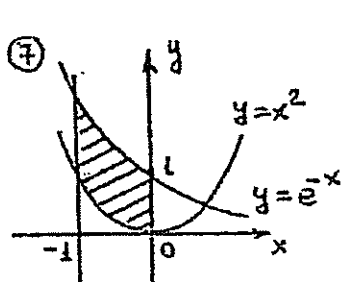
$$I = \int_0^{\frac{\pi}{4}} dx \int_{\sin x}^{\cos x} f(x,y) dy$$

⑤ 
$$\begin{cases} y = x \\ y = -\sqrt{2-x^2} \end{cases} \quad \begin{aligned} 2-x^2 &= x^2 \\ 2x^2 &= 2 \\ x &= \pm 1 \end{aligned} \quad (x \leq 0)$$

$$I = \int_{-\sqrt{2}}^{-1} dx \int_{-\sqrt{2-x^2}}^{-1} f(x,y) dy + \int_{-1}^0 dx \int_x^0 f(x,y) dy$$

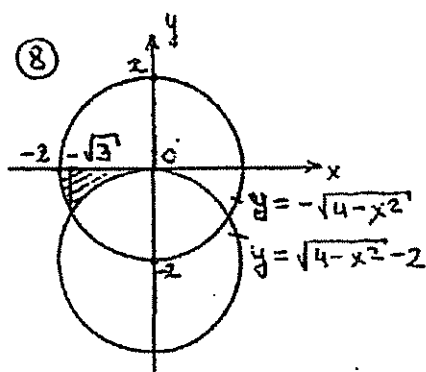
⑥ 
$$\begin{cases} y = -\sqrt{2-x^2} \\ y = -\sqrt{x} \\ y = 0 \end{cases} \quad \begin{aligned} x &= 2-x^2 \\ x^2 + x - 2 &= 0 \\ x &= 1 \\ x &= -2 \text{ (לא רלוונטי)} \end{aligned} \quad (x \geq 0)$$

$$I = \int_0^1 dx \int_{-\sqrt{x}}^0 f(x,y) dy + \int_1^{\sqrt{2}} dx \int_{-\sqrt{2-x^2}}^0 f(x,y) dy$$



$$\begin{cases} y = e^{-x} \\ y = x^2 \\ x = 0 \\ x = -1 \end{cases}$$

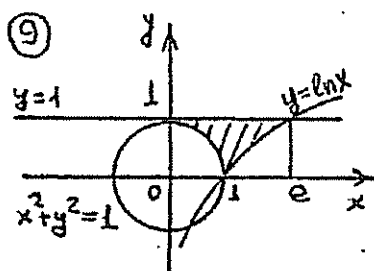
$$I = \int_{-1}^0 dx \int_{x^2}^{e^{-x}} f(x, y) dy$$



$$\begin{cases} (y+2)^2 + x^2 = 4 \\ x^2 + y^2 = 4 \\ y = 0 \\ (x < 0) \end{cases}$$

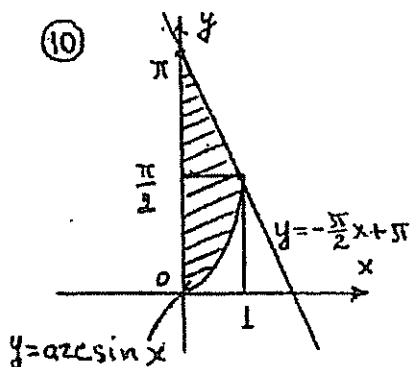
$$\begin{aligned} (y+2)^2 + x^2 &= 4 \\ y^2 + 4y + 4 + x^2 &= 4 \\ y^2 + 4y + x^2 &= 0 \\ y = -1 &\Rightarrow x^2 + (-1)^2 = 4 \\ x &= \pm\sqrt{3} \end{aligned}$$

$$I = \int_{-2}^{-\sqrt{3}} dx \int_{-\sqrt{4-x^2}}^0 f(x, y) dy + \int_{-\sqrt{3}}^0 dx \int_{\sqrt{4-x^2}-2}^0 f(x, y) dy$$



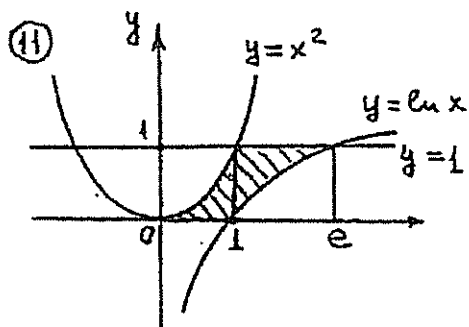
$$\begin{cases} x^2 + y^2 = 1 \\ y = \ln x \\ y = 1 \end{cases} \quad \begin{aligned} 1 &= \ln x \\ x &= e^1 \\ x &= e \end{aligned}$$

$$I = \int_0^1 dx \int_{\sqrt{1-x^2}}^1 f(x, y) dy + \int_1^e dx \int_{\ln x}^1 f(x, y) dy$$



$$\begin{cases} y = \arcsin x \\ y = -\frac{\pi}{2}x + \pi \\ x = 0 \end{cases}$$

$$I = \int_0^1 dx \int_{\arcsin x}^{-\frac{\pi}{2}x + \pi} f(x, y) dy$$



$$\begin{cases} y = \ln x \\ y = x^2 \\ y = 0 \\ y = 1 \end{cases}$$

$$\begin{cases} y = x^2 \\ y = 1 \end{cases} \Rightarrow x = \pm 1$$

$$\begin{cases} y = \ln x \\ y = 1 \end{cases} \Rightarrow x = e$$

$$I = \int_0^1 dx \int_0^{x^2} f(x, y) dy + \int_1^e dx \int_{\ln x}^1 f(x, y) dy$$

(1)

$$1) \int_0^1 dy \int_1^2 (x^2 + y^2) dx = \int_0^1 dy \left(\frac{x^3}{3} + y^2 x \right) \Big|_{x=1}^{x=2} = \int_0^1 \left(\frac{8}{3} + 2y^2 - \frac{1}{3} - y^2 \right) dy =$$

$$= \int_0^1 \left(\frac{7}{3} + y^2 \right) dy = \left(\frac{7}{3} y + \frac{y^3}{3} \right) \Big|_0^1 = \frac{1}{3} (7+1) = \frac{8}{3}$$

$$2) \int_3^4 dx \int_1^2 \frac{dy}{(x+y)^2} = \int_3^4 dx \int_1^2 \frac{d(x+y)}{(x+y)^2} = - \int_3^4 dx \left(\frac{1}{x+y} \right) \Big|_{y=1}^{y=2} = - \int_3^4 \left(\frac{1}{x+2} - \frac{1}{x+1} \right) dx =$$

$$= \int_3^4 \frac{dx}{x+1} - \int_3^4 \frac{dx}{x+2} = \left[\ln(x+1) - \ln(x+2) \right] \Big|_3^4 = \ln \frac{x+1}{x+2} \Big|_3^4 = \ln \frac{5}{6} - \ln \frac{4}{5} = \ln \frac{25}{24}$$

$$3) \int_0^{2\pi} d\theta \int_{a \sin \theta}^a r dr = \frac{1}{2} \int_0^{2\pi} d\theta \left(r^2 \Big|_{r=a \sin \theta}^{r=a} \right) = \frac{1}{2} \int_0^{2\pi} (a^2 - a^2 \sin^2 \theta) d\theta = \frac{a^2}{2} \int_0^{2\pi} \cos^2 \theta d\theta =$$

$$= \frac{a^2}{4} \int_0^{2\pi} (1 + \cos 2\theta) d\theta = \frac{a^2}{4} \left(\theta + \frac{1}{2} \sin 2\theta \right) \Big|_0^{2\pi} = \frac{a^2}{4} \cdot 2\pi = \frac{\pi a^2}{2}$$

$$4) \int_0^a dx \int_{x/a}^x \frac{x}{x^2 + y^2} dy = \int_0^a x dx \int_{x/a}^x \frac{dy}{x^2 + y^2} = \int_0^a dx \cdot x \cdot \left(\frac{1}{x} \arctan \frac{y}{x} \Big|_{y=x/a}^{y=x} \right) =$$

$$= \int_0^a x \cdot \frac{1}{x} (\arctan \frac{x}{x} - \arctan \frac{x}{ax}) dx = \int_0^a (\arctan 1 - \arctan \frac{1}{a}) dx =$$

$$= \left(\frac{\pi}{4} - \arctan \frac{1}{a} \right) x \Big|_0^a = a \left(\frac{\pi}{4} - \arctan \frac{1}{a} \right)$$

$$5) \int_0^a dy \int_{y-a}^{2y} xy dx = \int_0^a y dy \int_{y-a}^{2y} x dx = \int_0^a y dy \left(\frac{1}{2} x^2 \Big|_{y-a}^{2y} \right) = \frac{1}{2} \int_0^a y [4y^2 - (y-a)^2] dy =$$

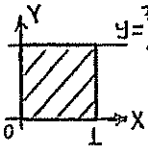
$$= \frac{1}{2} \int_0^a (3y^3 + 2y^2 a - a^2 y) dy = \frac{1}{2} \left(\frac{3}{4} y^4 + \frac{2}{3} y^3 a - \frac{1}{2} a^2 y^2 \right) \Big|_0^a =$$

$$= \frac{1}{2} \left(\frac{3}{4} a^4 + \frac{2}{3} a^4 - \frac{1}{2} a^4 \right) = \frac{11}{24} a^4$$

$$6) \int_{b/2}^b d\varphi \int_0^{\pi/2} \rho d\theta = \int_{b/2}^b \rho d\varphi \int_0^{\pi/2} d\theta = \int_{b/2}^b \rho d\varphi \left(\theta \Big|_0^{\pi/2} \right) = \frac{\pi}{2} \int_{b/2}^b \rho d\varphi = \frac{\pi}{4} \rho^2 \Big|_{b/2}^b =$$

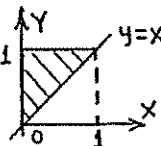
$$= \frac{\pi}{4} \left(b^2 - \frac{b^2}{4} \right) = \frac{3\pi b^2}{16}$$

(λ)

1)  $V = \iint_D (4-x^2-y^2) dx dy$, $D: x=0, x=1, y=0, y=3/2$

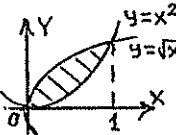
$$V = \int_0^1 dx \int_0^{3/2} (4-x^2-y^2) dy = \int_0^1 dx \left(4y - x^2y - \frac{y^3}{3} \right) \Big|_{y=0}^{y=3/2} =$$

$$= \int_0^1 \left(6 - \frac{3}{2}x^2 - \frac{9}{8} \right) dx = \frac{1}{2} \int_0^1 \left(\frac{39}{4} - 3x^2 \right) dx = \frac{1}{2} \left(\frac{39}{4}x - x^3 \right) \Big|_0^1 = \frac{1}{2} \left(\frac{39}{4} - 1 \right) = \frac{35}{8}$$

2)  $V = \iint_D (x+y) dx dy$, $D: x=0, y=1, y=x$

$$V = \int_0^1 dx \int_x^1 (x+y) dy = \int_0^1 dx \left(xy + \frac{y^2}{2} \right) \Big|_{y=x}^{y=1} = \int_0^1 \left(x + \frac{1}{2} - x^2 - \frac{x^2}{2} \right) dx =$$

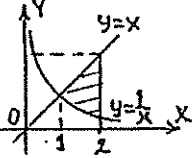
$$= \int_0^1 \left(x + \frac{1}{2} - \frac{3}{2}x^2 \right) dx = \left(\frac{x^2}{2} + \frac{1}{2}x - \frac{x^3}{2} \right) \Big|_0^1 = \frac{1}{2} + \frac{1}{2} - \frac{1}{2} = \frac{1}{2}$$

3)  $V = \iint_D (x^2+y) dx dy$, $D: y=x^2, y^2=x$

$$\begin{cases} y^2=x^2 \\ y^2=x \end{cases} \Rightarrow x=x^4 \Rightarrow x(x^3-1)=0 \Rightarrow x=0, x=1$$

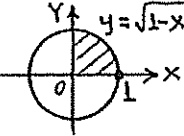
$$V = \int_0^1 dx \int_{x^2}^{\sqrt{x}} (x^2+y) dy = \int_0^1 dx \left(x^2y + \frac{y^2}{2} \right) \Big|_{x^2}^{\sqrt{x}} = \int_0^1 dx \left(x^2\sqrt{x} + \frac{x}{2} - x^4 - \frac{x^4}{2} \right) =$$

$$= \int_0^1 \left(x^{5/2} + \frac{x}{2} - \frac{3}{2}x^4 \right) dx = \left(\frac{2}{7}x^{7/2} + \frac{1}{4}x^2 - \frac{3}{10}x^5 \right) \Big|_0^1 = \frac{2}{7} + \frac{1}{4} - \frac{3}{10} = \frac{33}{140}$$

4)  $V = \iint_D \frac{x^2}{y^2} dx dy$, $D: x=2, y=x, xy=1$

$$V = \int_1^2 dx \int_{1/x}^x \frac{x^2}{y^2} dy = \int_1^2 x^2 dx \int_{1/x}^x \frac{dy}{y^2} = - \int_1^2 x^2 dx \left(\frac{1}{y} \Big|_{1/x}^x \right) = - \int_1^2 x^2 \left(\frac{1}{x} - x \right) dx =$$

$$= \int_1^2 (x^3 - x) dx = \left(\frac{x^4}{4} - \frac{x^2}{2} \right) \Big|_1^2 = \frac{1}{2} \left(\frac{16}{2} - 4 - \frac{1}{2} + 1 \right) = \frac{9}{4}$$

5)  $V = \iint_D \sqrt{1-x^2-y^2} dx dy$, $D: x=0, y=0, x^2+y^2=1, x>=0, y>=0$

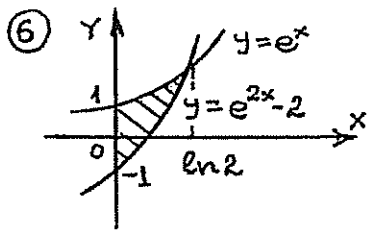
$$V = \int_0^1 dx \int_0^{\sqrt{1-x^2}} \sqrt{1-x^2-y^2} dy = \int_0^1 dx \int_0^{\sqrt{1-x^2}} \sqrt{(1-x^2)-y^2} dy =$$

$$= \left| \begin{aligned} y &= \sqrt{1-x^2} \sin t \rightarrow dy = \sqrt{1-x^2} \cos t dt, \\ y=0 &\rightarrow 0 = \sqrt{1-x^2} \sin t \Rightarrow \sin t=0 \rightarrow t=0, y=\sqrt{1-x^2} \Rightarrow \sin t=1 \Rightarrow t=\pi/2 \end{aligned} \right| =$$

$$= \int_0^1 dx \int_0^{\pi/2} \sqrt{(1-x^2)- (1-x^2) \sin^2 t} \cdot \sqrt{1-x^2} \cos t dt = \int_0^1 dx \int_0^{\pi/2} (1-x^2) \cos^2 t dt =$$

$$= \frac{1}{2} \int_0^1 (1-x^2) dx \int_0^{\pi/2} 2 \cos^2 t dt = \frac{1}{2} \left(x - \frac{x^3}{3} \right) \Big|_0^1 \int_0^{\pi/2} (1 + \cos 2t) dt = \frac{1}{2} \left(1 - \frac{1}{3} \right) \left(t + \frac{1}{2} \sin 2t \right) \Big|_0^{\pi/2} =$$

$$= \frac{1}{2} \cdot \frac{2}{3} \left(\frac{\pi}{2} + 0 \right) = \frac{\pi}{6}$$



$$e^x = e^{2x} - 2 \Rightarrow (e^x = t) \Rightarrow t^2 - t - 2 = 0$$

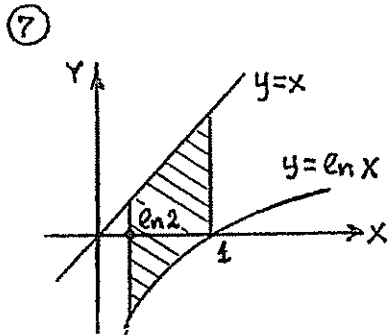
$$t = 2 \Rightarrow e^x = 2 \Rightarrow x = \ln 2 \Rightarrow (1, 1) \text{ and } (2, 2)$$

$$V = \int_0^{\ln 2} \frac{1}{e^{3x}} dx \int_{e^{2x}-2}^{e^x} y dy = \frac{1}{2} \int_0^{\ln 2} e^{-3x} dx (y^2 \Big|_{e^{2x}-2}^{e^x}) =$$

$$= \frac{1}{2} \int_0^{\ln 2} e^{-3x} [e^{2x} - (e^{2x} - 2)^2] dx = \frac{1}{2} \int_0^{\ln 2} e^{-3x} (5e^{2x} - e^{4x} - 4) dx =$$

$$= \frac{1}{2} \int_0^{\ln 2} (5e^{-x} - e^x - 4e^{-3x}) dx = \frac{1}{2} (-5e^{-x} - e^x + \frac{4}{3}e^{-3x}) \Big|_0^{\ln 2} =$$

$$= \frac{1}{2} \left[(-5e^{-\ln 2} - e^{\ln 2} + \frac{4}{3}e^{-3\ln 2}) - (-5 - 1 + \frac{4}{3}) \right] = \frac{1}{2} \left(-\frac{5}{2} - 2 + \frac{4}{3} \cdot \frac{1}{8} + 6 - \frac{4}{3} \right) = \frac{1}{6}$$



$$V = \int_{\ln 2}^1 dx \int_{\ln x}^x e^{x+y} dy = \int_{\ln 2}^1 e^x dx \int_{\ln x}^x e^y dy =$$

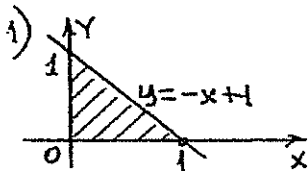
$$= \int_{\ln 2}^1 e^x dx (e^y \Big|_{\ln x}^x) = \int_{\ln 2}^1 (e^{2x} - x e^x) dx =$$

$$= \frac{1}{2} e^{2x} \Big|_{\ln 2}^1 - \int_{\ln 2}^1 x e^x dx = \frac{e^2}{2} - 2 - \int_{\ln 2}^1 x e^x dx = \left| \begin{array}{l} x=u \\ e^x dx = dv \\ v=e^x \end{array} \right| =$$

$$= \frac{e^2}{2} - 2 - [x e^x \Big|_{\ln 2}^1 - \int_{\ln 2}^1 e^x dx] = \frac{e^2}{2} - 2 - [e - 2 \ln 2 - e^x \Big|_{\ln 2}^1] =$$

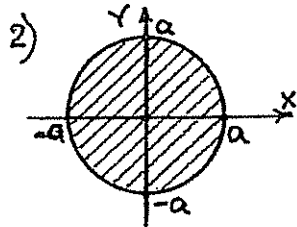
$$= \frac{e^2}{2} - 2 - e + 2 \ln 2 + e - 2 = \frac{1}{2} e^2 + 2(\ln 2 - 2) = 1.081$$

שינוי סדר האינטגרציה



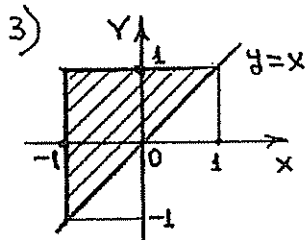
$$x = 1 - y \Rightarrow y = -x + 1 \quad (y = 1 - x)$$

$$\int_0^1 dy \int_0^{1-y} f(x, y) dx = \int_0^1 dx \int_0^{1-x} f(x, y) dy$$



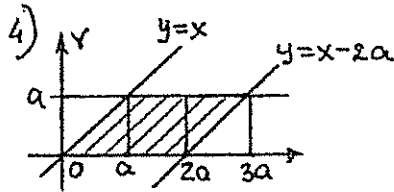
$$\left. \begin{aligned} y = -\sqrt{a^2 - x^2} &\Rightarrow x^2 + y^2 = a^2 \quad (y \leq 0) \\ y = \sqrt{a^2 - x^2} &\Rightarrow x^2 + y^2 = a^2 \quad (y \geq 0) \end{aligned} \right\} \Rightarrow x = \pm \sqrt{a^2 - y^2}$$

$$\int_{-a}^a dx \int_{-\sqrt{a^2 - x^2}}^{\sqrt{a^2 - x^2}} f(x, y) dy = \int_{-a}^a dy \int_{-\sqrt{a^2 - y^2}}^{\sqrt{a^2 - y^2}} f(x, y) dx$$



$$x = y \Rightarrow y = x$$

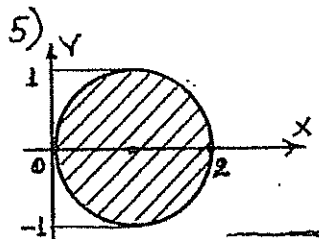
$$\int_{-1}^1 dy \int_{-1}^y f(x, y) dx = \int_{-1}^1 dx \int_x^1 f(x, y) dy$$



$$x = y \Rightarrow y = x$$

$$x = y + 2a \Rightarrow y = x - 2a$$

$$\int_0^a dy \int_y^{y+2a} f(x, y) dx = \int_0^a dx \int_0^x f(x, y) dy + \int_a^{2a} dx \int_0^a f(x, y) dy + \int_{2a}^{3a} dx \int_{x-2a}^a f(x, y) dy$$



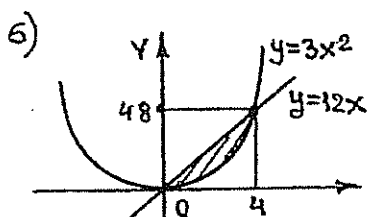
$$y = -\sqrt{2x - x^2} \Rightarrow x^2 - 2x + y^2 = 0 \rightarrow (x^2 - 2x + 1) - 1 + y^2 = 0$$

$$(x - 1)^2 + y^2 = 1^2 \quad (y \leq 0)$$

$$y = \sqrt{2x - x^2} \Rightarrow (x - 1)^2 + y^2 = 1^2 \quad (y \geq 0)$$

$$\left. \begin{aligned} & \\ & \end{aligned} \right\} \Rightarrow x = 1 \pm \sqrt{1 - y^2}$$

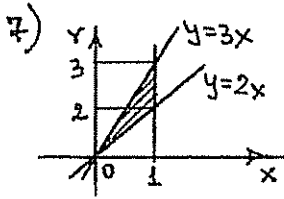
$$\int_0^2 dx \int_{-\sqrt{2x - x^2}}^{\sqrt{2x - x^2}} f(x, y) dy = \int_{-1}^1 dy \int_{1 - \sqrt{1 - y^2}}^{1 + \sqrt{1 - y^2}} f(x, y) dx$$



$$y = 3x^2 \Rightarrow x = \pm \sqrt{y/3}$$

$$y = 12x \Rightarrow x = y/12$$

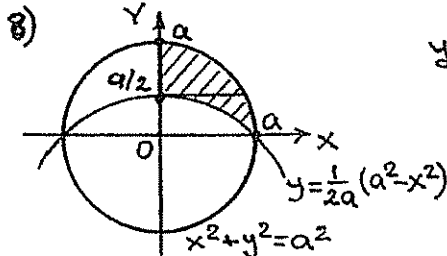
$$\int_0^4 dx \int_{3x^2}^{12x} f(x, y) dy = \int_0^{48} dy \int_{y/12}^{\sqrt{y/3}} f(x, y) dx$$



$$y=2x \Rightarrow x=y/2$$

$$y=3x \Rightarrow x=y/3$$

$$\int_0^1 dx \int_{2x}^{3x} f(x,y) dy = \int_0^2 dy \int_{y/3}^{y/2} f(x,y) dx + \int_2^3 dy \int_{y/3}^1 f(x,y) dx$$

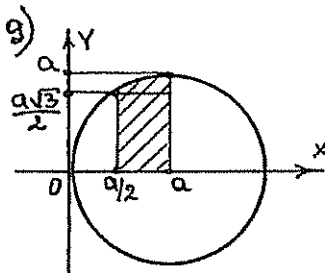


$$y = \frac{1}{2a}(a^2 - x^2) \Rightarrow x = \pm \sqrt{a^2 - 2ay}$$

$$y = \sqrt{a^2 - x^2} \Rightarrow x^2 + y^2 = a^2 \Rightarrow x = \pm \sqrt{a^2 - y^2}$$

$$\int_0^a dx \int_{\frac{1}{2a}(a^2 - x^2)}^{\sqrt{a^2 - x^2}} f(x,y) dy =$$

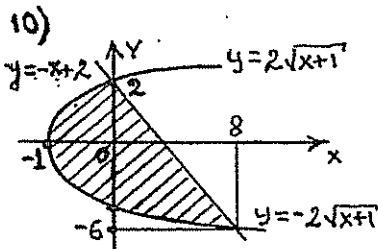
$$= \int_0^{a/2} dy \int_{\sqrt{a^2 - 2ay}}^{\sqrt{a^2 - y^2}} f(x,y) dx + \int_{a/2}^a dy \int_0^{\sqrt{a^2 - y^2}} f(x,y) dx$$



$$y = \sqrt{2ax - x^2} \Rightarrow x^2 - 2ax + y^2 = 0 \Rightarrow$$

$$(x-a)^2 + y^2 = a^2 \Rightarrow x = a \pm \sqrt{a^2 - y^2}$$

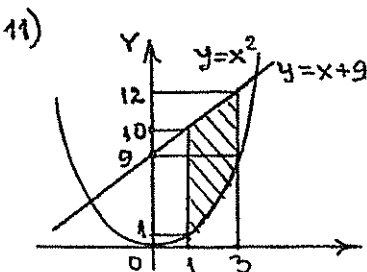
$$\int_{a/2}^a dx \int_0^{\sqrt{2ax - x^2}} f(x,y) dy = \int_0^{a\sqrt{3}/2} dy \int_{a/2}^a f(x,y) dx + \int_{a\sqrt{3}/2}^a dy \int_{a - \sqrt{a^2 - y^2}}^a f(x,y) dx$$



$$x = \frac{1}{4}y^2 - 1 \Rightarrow y = \pm 2\sqrt{x+1}$$

$$x = 2 - y \Rightarrow y = -x + 2 \quad (y = 2 - x)$$

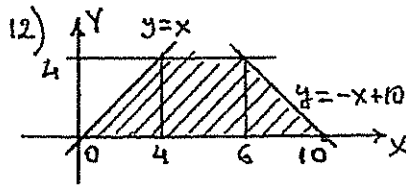
$$\int_{-6}^2 dy \int_{\frac{1}{4}y^2 - 1}^{2-y} f(x,y) dx = \int_{-1}^0 dx \int_{-2\sqrt{x+1}}^{2\sqrt{x+1}} f(x,y) dy + \int_0^8 dx \int_{-2\sqrt{x+1}}^{2-x} f(x,y) dy$$



$$y = x^2 \Rightarrow x = \pm \sqrt{y}, \quad y = x + 9 \Rightarrow x = y - 9$$

$$\int_1^3 dx \int_{x^2}^{x+9} f(x,y) dy =$$

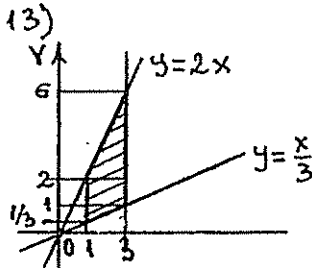
$$= \int_1^9 dy \int_1^{\sqrt{y}} f(x,y) dx + \int_9^{10} dy \int_{y-9}^3 f(x,y) dx + \int_{10}^{12} dy \int_{y-9}^3 f(x,y) dx$$



$$x=y \Rightarrow y=x, \quad x=10-y \Rightarrow y=10-x$$

$$\int_0^4 dy \int_y^{10-y} f(x,y) dx =$$

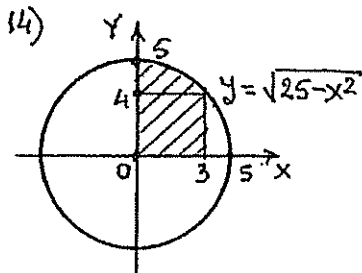
$$= \int_0^4 dx \int_0^x f(x,y) dy + \int_4^6 dx \int_0^4 f(x,y) dy + \int_6^{10} dx \int_0^{10-x} f(x,y) dy$$



$$y=\frac{x}{3} \Rightarrow x=3y, \quad y=2x \Rightarrow x=\frac{y}{2}$$

$$\int_1^3 dx \int_{x/3}^{2x} f(x,y) dy =$$

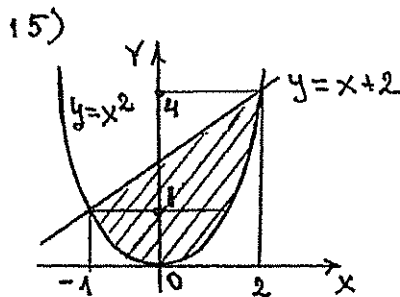
$$= \int_{1/3}^1 dy \int_1^{3y} f(x,y) dx + \int_1^2 dy \int_1^3 f(x,y) dx + \int_2^6 dy \int_{y/2}^3 f(x,y) dx$$



$$y=\sqrt{25-x^2} \Rightarrow x=\sqrt{25-y^2}$$

$$\int_0^3 dx \int_0^{\sqrt{25-x^2}} f(x,y) dy =$$

$$= \int_0^4 dy \int_0^3 f(x,y) dx + \int_4^5 dy \int_0^{\sqrt{25-y^2}} f(x,y) dx$$

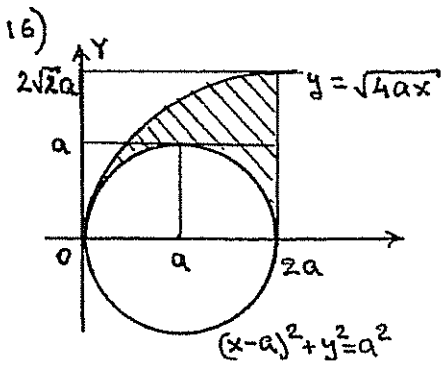


$$y=x^2 \Rightarrow x=\pm\sqrt{y}$$

$$y=x+2 \Rightarrow x=y-2$$

$$\int_{-1}^2 dx \int_{x^2}^{x+2} f(x,y) dy =$$

$$= \int_0^1 dy \int_{-\sqrt{y}}^{\sqrt{y}} f(x,y) dx + \int_1^4 dy \int_{y-2}^{\sqrt{y}} f(x,y) dx$$

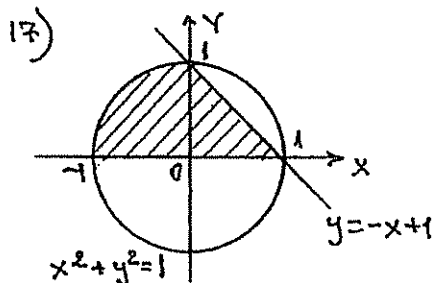


$$y = \sqrt{2ax - x^2} \Rightarrow x^2 - 2ax + y^2 = 0 \Rightarrow (x-a)^2 + y^2 = a^2 \Rightarrow x = a \pm \sqrt{a^2 - y^2}$$

$$y = \sqrt{4ax} \Rightarrow x = \frac{1}{4a} y^2$$

$$\int_0^{2a} dx \int_{\sqrt{2ax-x^2}}^{\sqrt{4ax}} f(x,y) dy =$$

$$= \int_0^a dy \int_{y^2/4a}^{a-\sqrt{a^2-y^2}} f(x,y) dx + \int_0^a dy \int_{a+\sqrt{a^2-y^2}}^{2a} f(x,y) dx + \int_a^{2\sqrt{2}a} dy \int_{y^2/4a}^{2a} f(x,y) dx$$

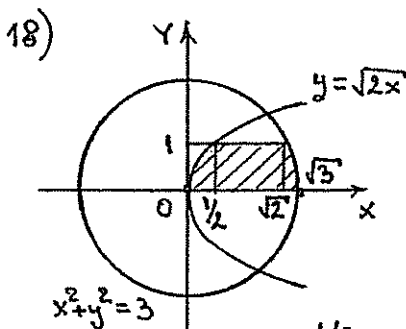


$$x = -\sqrt{1-y^2} \Rightarrow x^2 + y^2 = 1 \Rightarrow y = \pm \sqrt{1-x^2}$$

$$x = 1-y \Rightarrow y = -x+1 \quad (y=1-x)$$

$$\int_0^1 dy \int_{-\sqrt{1-y^2}}^{1-y} f(x,y) dx =$$

$$= \int_{-1}^0 dx \int_0^{\sqrt{1-x^2}} f(x,y) dy + \int_0^1 dx \int_0^{1-x} f(x,y) dy$$



$$x = \frac{1}{2} y^2 \Rightarrow y = \pm \sqrt{2x}$$

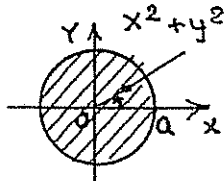
$$x = \sqrt{3-y^2} \Rightarrow x^2 + y^2 = 3 \Rightarrow y = \pm \sqrt{3-x^2}$$

$$\int_0^1 dy \int_{y^2/2}^{\sqrt{3-y^2}} f(x,y) dx =$$

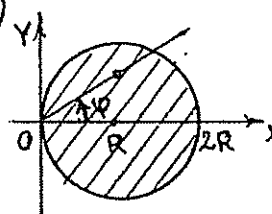
$$= \int_0^{1/2} dx \int_0^{\sqrt{2x}} f(x,y) dy + \int_{1/2}^{\sqrt{2}} dx \int_0^1 f(x,y) dy + \int_{\sqrt{2}}^{\sqrt{3}} dx \int_0^{\sqrt{3-x^2}} f(x,y) dy$$

האינטגרל הכפול בקואורדינטות קוטביות

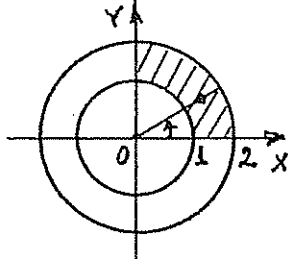
(7)

1)  $x^2 + y^2 = a^2$
 $x = \rho \cos \varphi, y = \rho \sin \varphi \Rightarrow \rho^2 (\cos^2 \varphi + \sin^2 \varphi) = a^2$
 $\rho = a \Rightarrow (\text{||} \vec{r} \text{||} = a) \Rightarrow 0 \leq \rho \leq a, 0 \leq \varphi \leq 2\pi$

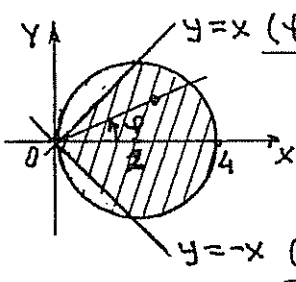
$$I = \int_0^{2\pi} d\varphi \int_0^a f(\rho \cos \varphi, \rho \sin \varphi) \rho d\rho$$

2)  $x^2 + y^2 = 2Rx \Rightarrow (x^2 - 2Rx + R^2) - R^2 + y^2 = 0 \Rightarrow (x - R)^2 + y^2 = R^2 \Rightarrow (\delta \Sigma \gamma \mathcal{N})$
 $x^2 + y^2 = 2Rx \Rightarrow \rho^2 (\cos^2 \varphi + \sin^2 \varphi) = 2R\rho \cos \varphi$
 $\rho^2 = 2R\rho \cos \varphi \Rightarrow \rho = 2R \cos \varphi \rightarrow (\text{||} \vec{r} \text{||} = 2R \cos \varphi)$
 $-\frac{\pi}{2} \leq \varphi \leq \frac{\pi}{2}, 0 \leq \rho \leq 2R \cos \varphi$

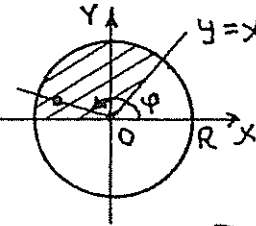
$$I = \int_{-\pi/2}^{\pi/2} d\varphi \int_0^{2R \cos \varphi} f(\rho \cos \varphi, \rho \sin \varphi) \rho d\rho$$

3)  $x^2 + y^2 = 1 \Rightarrow \rho = 1 \rightarrow (\text{||} \vec{r} \text{||} = 1)$
 $x^2 + y^2 = 4 \Rightarrow \rho = 2 \rightarrow (\text{||} \vec{r} \text{||} = 2)$
 $0 \leq \varphi \leq \pi/2, 1 \leq \rho \leq 2$

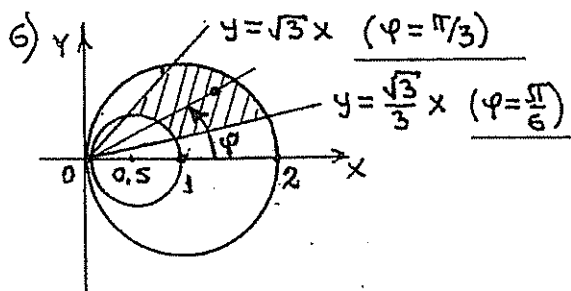
$$I = \int_0^{\pi/2} d\varphi \int_1^2 f(\rho \cos \varphi, \rho \sin \varphi) \rho d\rho$$

4)  $y = x (\varphi = \pi/4)$
 $x^2 + y^2 = 4x \Rightarrow (x^2 - 2 \cdot 2x + 4) - 4 + y^2 = 0$
 $(x - 2)^2 + y^2 = 2^2 \Rightarrow (\delta \Sigma \gamma \mathcal{N})$
 $x^2 + y^2 = 4x \Rightarrow \rho^2 = 4\rho \cos \varphi \Rightarrow \rho = 4 \cos \varphi \rightarrow (\text{||} \vec{r} \text{||} = 4 \cos \varphi)$
 $y = -x (\varphi = -\pi/4)$
 $-\frac{\pi}{4} \leq \varphi \leq \frac{\pi}{4}, 0 \leq \rho \leq 4 \cos \varphi \Rightarrow$

$$I = \int_{-\pi/4}^{\pi/4} d\varphi \int_0^{4 \cos \varphi} f(\rho \cos \varphi, \rho \sin \varphi) \rho d\rho$$

5)  $y = x (\varphi = \pi/4)$
 $x^2 + y^2 = R^2 \Rightarrow \rho = R (\text{||} \vec{r} \text{||} = R)$
 $\frac{\pi}{4} \leq \varphi \leq \pi, 0 \leq \rho \leq R$

$$I = \int_{\pi/4}^{\pi} d\varphi \int_0^R f(\rho \cos \varphi, \rho \sin \varphi) \rho d\rho$$



$$x^2 + y^2 = 2x \Rightarrow$$

$$(x-1)^2 + y^2 = 1^2 \Rightarrow (\text{сфера}).$$

$$x^2 + y^2 = x \Rightarrow$$

$$(x^2 - 2 \cdot \frac{1}{2}x + \frac{1}{4}) - \frac{1}{4} + y^2 = 0 \Rightarrow$$

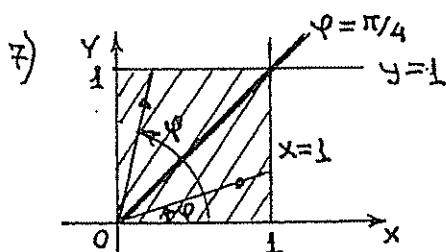
$$(x - 1/2)^2 + y^2 = (1/2)^2 \Rightarrow (\text{сфера})$$

$$x^2 + y^2 = 2x \Rightarrow \rho^2 = 2\rho \cos \varphi \Rightarrow \rho = 2 \cos \varphi \rightarrow (\text{||} \text{ } \varphi \text{ ||} \text{ } \text{||} \text{ } \varphi \text{ ||})$$

$$x^2 + y^2 = x \Rightarrow \rho^2 = \rho \cos \varphi \Rightarrow \rho = \cos \varphi \rightarrow (\text{||} \text{ } \varphi \text{ ||} \text{ } \text{||} \text{ } \varphi \text{ ||})$$

$$\frac{\pi}{6} \leq \varphi \leq \frac{\pi}{3}, \quad \cos \varphi \leq \rho \leq 2 \cos \varphi$$

$$I = \int_{\pi/6}^{\pi/3} d\varphi \int_{\cos \varphi}^{2 \cos \varphi} f(\rho \cos \varphi, \rho \sin \varphi) \rho d\rho$$



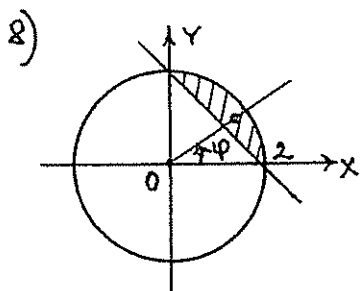
$$x=1 \Rightarrow \rho \cos \varphi = 1 \Rightarrow$$

$$\rho = \frac{1}{\cos \varphi} \rightarrow (\text{||} \text{ } \varphi \text{ ||} \text{ } \text{||} \text{ } \varphi \text{ ||})$$

$$y=1 \Rightarrow \rho \sin \varphi = 1 \Rightarrow$$

$$\rho = \frac{1}{\sin \varphi} \rightarrow (\text{||} \text{ } \varphi \text{ ||} \text{ } \text{||} \text{ } \varphi \text{ ||})$$

$$I = \int_0^{\pi/4} d\varphi \int_{\frac{1}{\cos \varphi}}^{\frac{1}{\sin \varphi}} f(\rho \cos \varphi, \rho \sin \varphi) \rho d\rho + \int_{\pi/4}^{\pi/2} d\varphi \int_0^{\frac{1}{\sin \varphi}} f(\rho \cos \varphi, \rho \sin \varphi) \rho d\rho$$



$$x^2 + y^2 = 4 \Rightarrow \rho = 2 \rightarrow (\text{||} \text{ } \varphi \text{ ||} \text{ } \text{||} \text{ } \varphi \text{ ||})$$

$$x+y=2 \Rightarrow \rho \cos \varphi + \rho \sin \varphi = 2$$

$$\rho = \frac{2}{\cos \varphi + \sin \varphi} = \frac{\sqrt{2}}{\frac{\sqrt{2}}{2} \cos \varphi + \frac{\sqrt{2}}{2} \sin \varphi} =$$

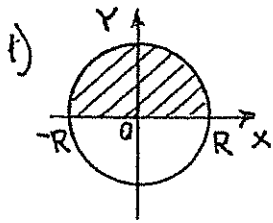
$$= \frac{\sqrt{2}}{\cos \varphi \cdot \cos \frac{\pi}{4} + \sin \varphi \sin \frac{\pi}{4}} = \sqrt{2} \frac{1}{\cos(\varphi - \pi/4)} = \sqrt{2} \sec(\varphi - \pi/4)$$

$$\rho = \sqrt{2} \sec(\varphi - \pi/4) \rightarrow (\text{||} \text{ } \varphi \text{ ||} \text{ } \text{||} \text{ } \varphi \text{ ||})$$

$$0 \leq \varphi \leq \frac{\pi}{2}, \quad \sqrt{2} \sec(\varphi - \pi/4) \leq \rho \leq 2$$

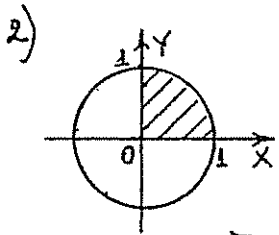
$$I = \int_0^{\pi/2} d\varphi \int_{\sqrt{2} \sec(\varphi - \pi/4)}^2 f(\rho \cos \varphi, \rho \sin \varphi) \rho d\rho$$

①



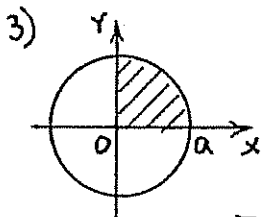
$$I = \iint_D (x^2 + y^2) dx dy \Rightarrow f(x, y) = x^2 + y^2 = \\ = \rho^2 (\cos^2 \varphi + \sin^2 \varphi) = \rho^2 \Rightarrow (\text{н'б } G \text{ н'б})$$

$$I = 2 \int_0^{\pi/2} d\varphi \int_0^R \rho^2 \cdot \rho d\rho = \frac{2}{4} \int_0^{\pi/2} d\varphi (\rho^4 \Big|_0^R) = \\ = \frac{1}{2} R^4 \int_0^{\pi/2} d\varphi = \frac{1}{2} R^4 \varphi \Big|_0^{\pi/2} = \pi R^4 / 4$$



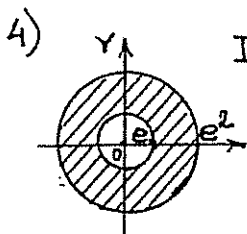
$$I = \iint_D e^{x^2 + y^2} dx dy \Rightarrow f(x, y) = e^{x^2 + y^2} = \\ = e^{\rho^2 (\cos^2 \varphi + \sin^2 \varphi)} = e^{\rho^2}$$

$$I = \int_0^{\pi/2} d\varphi \int_0^1 e^{\rho^2} \rho d\rho = \frac{1}{2} \int_0^{\pi/2} d\varphi \int_0^1 e^{\rho^2} d(\rho^2) = \frac{1}{2} \int_0^{\pi/2} d\varphi (e^{\rho^2} \Big|_0^1) \\ = \frac{1}{2} (e - 1) \int_0^{\pi/2} d\varphi = \frac{e - 1}{2} \varphi \Big|_0^{\pi/2} = \frac{\pi}{4} (e - 1)$$



$$I = \iint_D \sqrt{x^2 + y^2} dx dy \Rightarrow f(x, y) = \sqrt{x^2 + y^2} = \\ = \sqrt{\rho^2 (\cos^2 \varphi + \sin^2 \varphi)} = \rho$$

$$I = \int_0^{\pi/2} d\varphi \int_0^a \rho \cdot \rho d\rho = \frac{1}{3} \int_0^{\pi/2} d\varphi (\rho^3 \Big|_0^a) = \frac{a^3}{3} \int_0^{\pi/2} d\varphi = \\ = \frac{a^3}{3} \varphi \Big|_0^{\pi/2} = \pi a^3 / 6$$

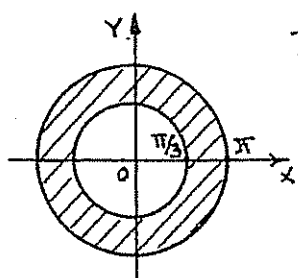


$$I = \iint_D \ln(x^2 + y^2) dx dy \Rightarrow f(x, y) = \ln \rho^2 (\cos^2 \varphi + \sin^2 \varphi) = \\ = \ln \rho^2 = 2 \ln \rho \rightarrow (\text{н'б } G \text{ н'б})$$

$$I = 4 \int_0^{\pi/2} d\varphi \int_e^{e^2} 2 \ln \rho \cdot \rho d\rho = 8 \int_0^{\pi/2} d\varphi \int_e^{e^2} \rho \ln \rho d\rho =$$

$$\left| \int \rho \ln \rho d\rho = \left| \begin{matrix} \ln \rho = u & du = \frac{1}{\rho} d\rho \\ \rho d\rho = dv & v = \rho^2 / 2 \end{matrix} \right| = \frac{1}{2} \rho^2 \ln \rho - \frac{1}{2} \int \rho d\rho = \right| = \\ = \frac{1}{2} \rho^2 \ln \rho - \frac{1}{4} \rho^2 \\ = 4 \int_0^{\pi/2} d\varphi \left[\rho^2 \left(\ln \rho - \frac{1}{2} \right) \Big|_e^{e^2} \right] = 4 \cdot \frac{1}{2} e^2 (3e^2 - 1) \int_0^{\pi/2} d\varphi = 2e^2 (3e^2 - 1) \varphi \Big|_0^{\pi/2} = \\ = \pi e^2 (3e^2 - 1)$$

5)



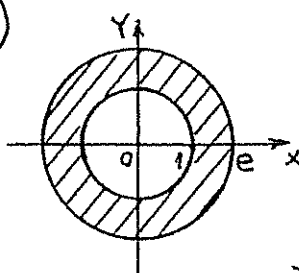
$$I = \iint_D \frac{\sin \sqrt{x^2+y^2}}{\sqrt{x^2+y^2}} dx dy, \quad f(x,y) = \frac{\sin \sqrt{x^2+y^2}}{\sqrt{x^2+y^2}} =$$

$$= \frac{\sin \rho \sqrt{\cos^2 \varphi + \sin^2 \varphi}}{\rho \sqrt{\cos^2 \varphi + \sin^2 \varphi}} = \frac{\sin \rho}{\rho} \rightarrow (\text{н'о Гр'о})$$

$$I = 4 \int_0^{\pi/2} d\varphi \int_{\pi/3}^{\pi} \frac{\sin \rho}{\rho} \rho d\rho = -4 \int_0^{\pi/2} d\varphi (\cos \rho \big|_{\pi/3}^{\pi}) =$$

$$= 4 \left(1 + \frac{1}{2}\right) \int_0^{\pi/2} d\varphi = 6 \varphi \big|_0^{\pi/2} = 3\pi$$

6)



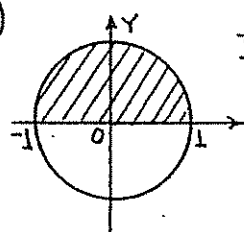
$$I = \iint_D \frac{\ln(x^2+y^2)}{x^2+y^2} dx dy \Rightarrow f(x,y) = \frac{\ln(x^2+y^2)}{x^2+y^2} =$$

$$= \frac{\ln \rho^2 (\cos^2 \varphi + \sin^2 \varphi)}{\rho^2 (\cos^2 \varphi + \sin^2 \varphi)} = \frac{2 \ln \rho}{\rho^2} \rightarrow (\text{н'о Гр'о})$$

$$I = 4 \int_0^{\pi/2} d\varphi \int_1^e \frac{2 \ln \rho}{\rho^2} \rho d\rho = 8 \int_0^{\pi/2} d\varphi \int_1^e \ln \rho d(\ln \rho) =$$

$$= \frac{8}{2} \int_0^{\pi/2} d\varphi (\ln^2 \rho \big|_1^e) = 4(e^2 - 0^2) \int_0^{\pi/2} d\varphi = 4\varphi \big|_0^{\pi/2} = 2\pi$$

7)

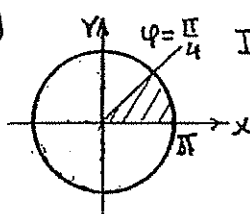


$$I = \iint_D \frac{dx dy}{x^2+y^2+1} \Rightarrow f(x,y) = \frac{1}{\rho^2 (\cos^2 \varphi + \sin^2 \varphi) + 1} = \frac{1}{\rho^2 + 1}$$

$$I = \int_0^{\pi} d\varphi \int_0^1 \frac{\rho d\rho}{\rho^2 + 1} = \frac{1}{2} \int_0^{\pi} d\varphi \int_0^1 \frac{d(\rho^2 + 1)}{\rho^2 + 1} = \frac{1}{2} \int_0^{\pi} d\varphi (\ln(\rho^2 + 1) \big|_0^1) =$$

$$= \frac{1}{2} (\ln 2 - 0) \int_0^{\pi} d\varphi = \frac{\ln 2}{2} \varphi \big|_0^{\pi} = \frac{\pi}{2} \ln 2$$

8)

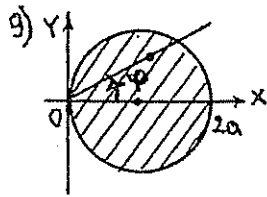


$$I = \iint_D \left(1 - \frac{y^2}{x^2}\right) dx dy \Rightarrow f(x,y) = 1 - \frac{y^2}{x^2} = 1 - \frac{\rho^2 \sin^2 \varphi}{\rho^2 \cos^2 \varphi} = 1 - \tan^2 \varphi$$

$$I = \int_0^{\pi/4} d\varphi \int_0^{\pi} (1 - \tan^2 \varphi) \rho d\rho = \int_0^{\pi/4} (1 - \tan^2 \varphi) d\varphi \int_0^{\pi} \rho d\rho =$$

$$= \frac{1}{2} \int_0^{\pi/4} (1 - \tan^2 \varphi) d\varphi (\rho^2 \big|_0^{\pi}) = \frac{\pi^2}{2} \int_0^{\pi/4} \left(1 - \frac{\sin^2 \varphi}{\cos^2 \varphi}\right) d\varphi = \frac{\pi^2}{2} \int_0^{\pi/4} \left(1 - \frac{1 - \cos^2 \varphi}{\cos^2 \varphi}\right) d\varphi =$$

$$= \frac{\pi^2}{2} \int_0^{\pi/4} \left(2 - \frac{1}{\cos^2 \varphi}\right) d\varphi = \frac{\pi^2}{2} (2\varphi - \tan \varphi) \big|_0^{\pi/4} = \frac{\pi^2}{2} \left(\frac{\pi}{2} - \tan \frac{\pi}{4}\right) = \frac{\pi^2}{4} (\pi - 2)$$



$$I = \iint_D (x^2 + y^2) dx dy \Rightarrow f(x, y) = (x^2 + y^2) = \rho^2$$

$$D: x^2 + y^2 = 2ax \Rightarrow (x-a)^2 + y^2 = a^2 \rightarrow (\text{circle})$$

$$\rho^2 (\cos^2 \varphi + \sin^2 \varphi) = 2a \rho \cos \varphi \Rightarrow \rho = 2a \cos \varphi$$

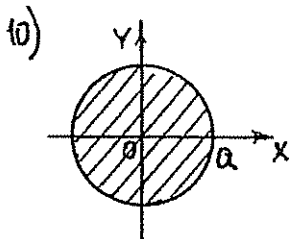
$$I = \int_{-\pi/2}^{\pi/2} d\varphi \int_0^{2a \cos \varphi} \rho^2 \cdot \rho d\rho = \frac{1}{4} \int_{-\pi/2}^{\pi/2} d\varphi (\rho^4 \Big|_0^{2a \cos \varphi}) = 4a^4 \int_{-\pi/2}^{\pi/2} \cos^4 \varphi d\varphi =$$

$$= a^4 \int_{-\pi/2}^{\pi/2} (2 \cos^2 \varphi)^2 d\varphi = a^4 \int_{-\pi/2}^{\pi/2} (1 + \cos 2\varphi)^2 d\varphi$$

$$= a^4 \int_{-\pi/2}^{\pi/2} (1 + 2 \cos 2\varphi + \cos^2 2\varphi) d\varphi = a^4 \left[\left(\varphi + \sin 2\varphi \right) \Big|_{-\pi/2}^{\pi/2} + \frac{1}{2} \int_{-\pi/2}^{\pi/2} 2 \cos^2 2\varphi d\varphi \right] =$$

$$= a^4 \left[\pi + \frac{1}{2} \int_{-\pi/2}^{\pi/2} (1 + \cos 4\varphi) d\varphi \right] = a^4 \left[\pi + \frac{1}{2} \left(\varphi + \frac{1}{4} \sin 4\varphi \right) \Big|_{-\pi/2}^{\pi/2} \right] =$$

$$= a^4 \left[\pi + \frac{1}{2} \pi \right] = \frac{3}{2} \pi a^4$$



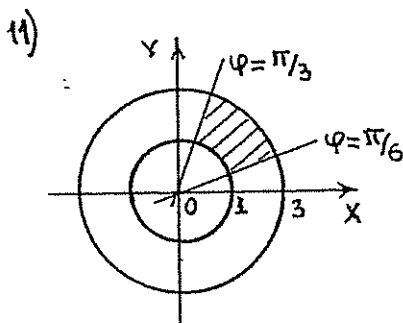
$$I = \iint_D (3a - x - y) dx dy \Rightarrow f(x, y) = 3a - x - y =$$

$$= 3a - \rho \cos \varphi - \rho \sin \varphi$$

$$I = \int_0^{2\pi} d\varphi \int_0^a (3a - \rho \cos \varphi - \rho \sin \varphi) \rho d\rho =$$

$$= \int_0^{2\pi} d\varphi \int_0^a (3a\rho - \rho^2 \cos \varphi - \rho^2 \sin \varphi) d\rho = \int_0^{2\pi} d\varphi \left[\left(\frac{3}{2} a \rho^2 - \frac{\rho^3}{3} \cos \varphi - \frac{\rho^3}{3} \sin \varphi \right) \Big|_0^a \right] =$$

$$= \int_0^{2\pi} d\varphi \left(\frac{3}{2} a^3 - \frac{a^3}{3} \cos \varphi - \frac{a^3}{3} \sin \varphi \right) = \left(\frac{3}{2} a^3 \varphi - \frac{a^3}{3} \sin \varphi + \frac{a^3}{3} \cos \varphi \right) \Big|_0^{2\pi} = 3\pi a^3$$



$$I = \iint_D \arctg \frac{y}{x} dx dy \Rightarrow f(x, y) = \arctg \frac{y}{x} =$$

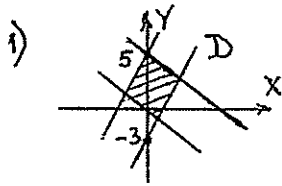
$$= \arctg \frac{\rho \sin \varphi}{\rho \cos \varphi} = \arctg (\tan \varphi) = \varphi$$

$$D: x^2 + y^2 = 1 \Rightarrow \rho = 1, \quad x^2 + y^2 = 9 \Rightarrow \rho = 3$$

$$y = \sqrt{3}x \Rightarrow \varphi = \frac{\pi}{3}, \quad y = \frac{\sqrt{3}}{3}x \Rightarrow \varphi = \frac{\pi}{6}$$

$$I = \int_{\pi/6}^{\pi/3} d\varphi \int_1^3 \varphi \rho d\rho = \int_{\pi/6}^{\pi/3} \varphi d\varphi \int_1^3 \rho d\rho = \int_{\pi/6}^{\pi/3} \varphi d\varphi \left(\frac{\rho^2}{2} \Big|_1^3 \right) = 4 \cdot \frac{\varphi^2}{2} \Big|_{\pi/6}^{\pi/3} = \pi^2/6$$

החלפת המשתנים באינטגרל כפול. מקרה כללי



$$I = \iint_D f(x, y) dx dy, \quad y = 2x - 3, y = 2x + 5, y = -x, y = 5 - x$$

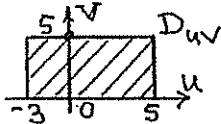
$$T: y - 2x = u, \quad x + y = v$$

$$y = 2x - 3 \rightarrow y - 2x = -3 \rightarrow u = -3 \Rightarrow -3 \leq u \leq 5$$

$$y = 2x + 5 \rightarrow y - 2x = 5 \rightarrow u = 5$$

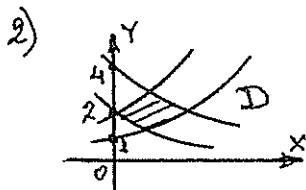
$$y = -x \rightarrow y + x = 0 \rightarrow v = 0 \Rightarrow 0 \leq v \leq 5$$

$$y = 5 - x \rightarrow y + x = 5 \rightarrow v = 5$$



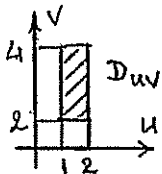
$$\begin{cases} y - 2x = u \\ x + y = v \end{cases} \Rightarrow \begin{cases} x = \frac{1}{3}v - \frac{1}{3}u \\ y = \frac{2}{3}v + \frac{1}{3}u \end{cases} \Rightarrow J = \frac{D(x, y)}{D(u, v)} = \begin{vmatrix} x'_u & y'_u \\ x'_v & y'_v \end{vmatrix} = \begin{vmatrix} -\frac{1}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{2}{3} \end{vmatrix} = -\frac{1}{3}$$

$$|J| = \frac{1}{3} \Rightarrow I = \iint_{D_{uv}} f\left(\frac{v-u}{3}, \frac{2v+u}{3}\right) |J| du dv = \frac{1}{3} \int_{-3}^5 du \int_0^5 f\left(\frac{v-u}{3}, \frac{2v+u}{3}\right) dv$$



$$I = \iint_D f(x, y) dx dy, \quad y = e^x, y = 2e^x, y = 2e^{-x}, y = 4e^{-x}$$

$$T: y e^{-x} = u, \quad y e^x = v$$



$$y = e^x \rightarrow y \cdot e^{-x} = 1 \Rightarrow u = 1 \Rightarrow 1 \leq u \leq 2$$

$$y = 2e^x \rightarrow y \cdot e^{-x} = 2 \Rightarrow u = 2$$

$$y = 2e^{-x} \rightarrow y e^x = 2 \Rightarrow v = 2 \Rightarrow 2 \leq v \leq 4$$

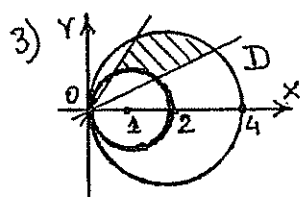
$$y = 4e^{-x} \Rightarrow y e^x = 4 \Rightarrow v = 4$$

$$\begin{cases} y e^{-x} = u \\ y e^x = v \end{cases} \Rightarrow \begin{cases} \div \rightarrow e^{2x} = \frac{v}{u} \rightarrow 2x = \ln \frac{v}{u} \Rightarrow x = \frac{1}{2}(\ln v - \ln u) \\ \otimes \rightarrow y^2 = uv \rightarrow y = \sqrt{uv} \end{cases}$$

$$J = \begin{vmatrix} x'_u & y'_u \\ x'_v & y'_v \end{vmatrix} = \begin{vmatrix} -\frac{1}{2u} & \frac{1}{2}\sqrt{\frac{v}{u}} \\ \frac{1}{2v} & \frac{1}{2}\sqrt{\frac{u}{v}} \end{vmatrix} = -\frac{1}{4} \cdot \frac{1}{\sqrt{uv}} - \frac{1}{4} \cdot \frac{1}{\sqrt{uv}} = -\frac{1}{2\sqrt{uv}}$$

$$|J| = \frac{1}{2\sqrt{uv}} \Rightarrow I = \iint_{D_{uv}} f\left(\frac{\ln v - \ln u}{2}, \sqrt{uv}\right) |J| du dv \Rightarrow$$

$$I = \frac{1}{2} \int_1^2 du \int_2^4 f\left(\frac{\ln v - \ln u}{2}, \sqrt{uv}\right) \frac{1}{\sqrt{uv}} dv$$

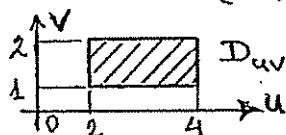


$$I = \iint_D f(x,y) dx dy, \quad x^2+y^2=2x, \quad x^2+y^2=4x$$

$$y=x, \quad y=2x$$

$$T: \frac{x^2+y^2}{x} = u, \quad \frac{y}{x} = v$$

האזור D קובע על ידי המעגלים $(x-1)^2+y^2=1$ ו- $(x-2)^2+y^2=4$



$$x^2+y^2=2x \Rightarrow \frac{x^2+y^2}{x} = 2 \Rightarrow u=2 \Rightarrow 2 \leq u \leq 4$$

$$x^2+y^2=4x \Rightarrow \frac{x^2+y^2}{x} = 4 \Rightarrow u=4$$

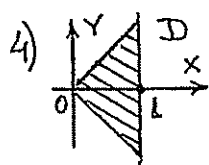
$$y=x \Rightarrow \frac{y}{x} = 1 \Rightarrow v=1 \Rightarrow 1 \leq v \leq 2$$

$$y=2x \Rightarrow \frac{y}{x} = 2 \Rightarrow v=2$$

$$\begin{cases} \frac{x^2+y^2}{x} = u \\ \frac{y}{x} = v \end{cases} \rightarrow y = xv \rightarrow \frac{x^2+x^2v^2}{x} = u \rightarrow x = \frac{u}{1+v^2} \rightarrow y = \frac{uv}{1+v^2}$$

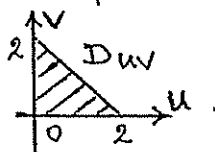
$$\begin{cases} x = \frac{u}{1+v^2} \\ y = \frac{uv}{1+v^2} \end{cases} \Rightarrow J = \begin{vmatrix} x'_u & y'_u \\ x'_v & y'_v \end{vmatrix} = \begin{vmatrix} \frac{1}{1+v^2} & \frac{v}{1+v^2} \\ -\frac{2uv}{(1+v^2)^2} & \frac{u(1-v^2)}{(1+v^2)^2} \end{vmatrix} = \frac{u}{(1+v^2)^2}$$

$$I = \iint_{D_{uv}} f\left(\frac{u}{1+v^2}, \frac{uv}{1+v^2}\right) |J| du dv = \int_2^4 du \int_1^2 f\left(\frac{u}{1+v^2}, \frac{uv}{1+v^2}\right) \frac{u}{(1+v^2)^2} dv$$



$$I = \iint_D f(x,y) dx dy, \quad y=x, \quad y=-x, \quad x=1$$

$$T: x-y=u, \quad x+y=v$$



$$y=x \Rightarrow x-y=0 \Rightarrow u=0$$

$$y=-x \Rightarrow x+y=0 \Rightarrow v=0$$

$$x=1 \Rightarrow \begin{cases} 1-y=u \\ 1+y=v \end{cases} \oplus \Rightarrow u+v=2 \Rightarrow$$

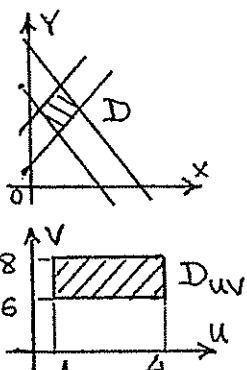
$v=2-u, v=0, u=0$: קווי הגבול של D_{uv} ו- D הם אותו דבר.

$$\begin{cases} x-y=u \\ x+y=v \end{cases} \Rightarrow \begin{cases} x = \frac{1}{2}(v+u) \\ y = \frac{1}{2}(v-u) \end{cases} \Rightarrow J = \begin{vmatrix} x'_u & y'_u \\ x'_v & y'_v \end{vmatrix} = \begin{vmatrix} 1/2 & 1/2 \\ 1/2 & -1/2 \end{vmatrix} = -\frac{1}{2}$$

$$I = \iint_{D_{uv}} f\left(\frac{v+u}{2}, \frac{v-u}{2}\right) |J| du dv = \frac{1}{2} \int_0^2 du \int_0^{2-u} f\left(\frac{v+u}{2}, \frac{v-u}{2}\right) dv$$

האזור D קובע על ידי המעגלים $(x-1)^2+y^2=1$ ו- $(x-2)^2+y^2=4$

$$\frac{1}{J} = \frac{D(u,v)}{D(x,y)} = \begin{vmatrix} u'_x & v'_x \\ u'_y & v'_y \end{vmatrix} = \begin{vmatrix} 1 & 1 \\ -1 & 1 \end{vmatrix} = 2 \Rightarrow J = \frac{1}{2}$$

5) 

$$I = \iint_D \frac{y-x}{2x+y} dx dy \quad D: y=x+1, y=x+4, y=6-2x, y=8-2x$$

$$y=x+1 \Rightarrow y-x=1 \Rightarrow y-x=u \Rightarrow 1 \leq u \leq 4$$

$$y=x+4 \Rightarrow y-x=4$$

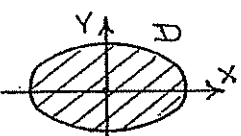
$$y=6-2x \Rightarrow y+2x=6 \Rightarrow y+2x=v \Rightarrow 6 \leq v \leq 8$$

$$y=8-2x \Rightarrow y+2x=8$$

$$\frac{1}{J} = \frac{D(u,v)}{D(x,y)} = \begin{vmatrix} u'_x & v'_x \\ u'_y & v'_y \end{vmatrix} = \begin{vmatrix} -1 & 2 \\ 1 & 1 \end{vmatrix} = -3 \Rightarrow |J| = \frac{1}{3}$$

$$f(x,y) = \frac{y-x}{2x+y} = \frac{u}{v} \Rightarrow I = \iint_{D_{uv}} \frac{u}{v} \cdot \frac{1}{3} du dv \Rightarrow$$

$$I = \frac{1}{3} \int_1^4 u du \int_6^8 \frac{dv}{v} = \frac{1}{3} \left(\frac{u^2}{2} \Big|_1^4 \right) \left(\ln v \Big|_6^8 \right) = \frac{1}{3} \left(8 - \frac{1}{2} \right) (\ln 8 - \ln 6) = \frac{5}{2} \ln \frac{4}{3}$$

6) 

$$I = \iint_D \sqrt{36-4x^2-9y^2} dx dy, \quad D: \frac{x^2}{9} + \frac{y^2}{4} = 1$$

נשתמש בקואורדינטות אליפטיות. $x=3\rho\cos\varphi, y=2\rho\sin\varphi$
 נבדוק את ההתאמה: $\frac{9\rho^2\cos^2\varphi}{9} + \frac{4\rho^2\sin^2\varphi}{4} = 1 \Rightarrow \rho^2(\cos^2\varphi + \sin^2\varphi) = 1 \Rightarrow \rho = 1$

$$\frac{9\rho^2\cos^2\varphi}{9} + \frac{4\rho^2\sin^2\varphi}{4} = 1 \Rightarrow \rho^2(\cos^2\varphi + \sin^2\varphi) = 1 \Rightarrow \rho = 1$$

$$D_{\rho\varphi} = \{(\rho, \varphi) \mid 0 \leq \varphi \leq 2\pi, 0 \leq \rho \leq 1\} : D_{\rho\varphi} \text{ התחום}$$

$$f(x,y) = \sqrt{36-4x^2-9y^2} = 6\sqrt{1-\frac{x^2}{9}-\frac{y^2}{4}} = 6\sqrt{1-\frac{9\rho^2\cos^2\varphi}{9}-\frac{4\rho^2\sin^2\varphi}{4}} =$$

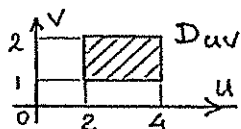
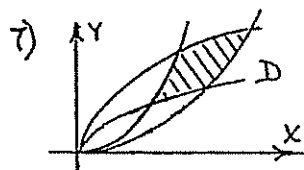
$$= 6\sqrt{1-\rho^2(\cos^2\varphi + \sin^2\varphi)} = 6\sqrt{1-\rho^2}$$

$$J = \begin{vmatrix} x'_\rho & y'_\rho \\ x'_\varphi & y'_\varphi \end{vmatrix} = \begin{vmatrix} 3\cos\varphi & 2\sin\varphi \\ -3\rho\sin\varphi & 2\rho\cos\varphi \end{vmatrix} = 6\rho(\cos^2\varphi + \sin^2\varphi) = 6\rho$$

$$I = \iint_{D_{\rho\varphi}} 6\sqrt{1-\rho^2} \cdot 6\rho d\varphi d\rho = 36 \iint_{D_{\rho\varphi}} \sqrt{1-\rho^2} \rho d\varphi d\rho =$$

$$= 36 \int_0^{2\pi} d\varphi \int_0^1 \sqrt{1-\rho^2} \rho d\rho = 36(\varphi \Big|_0^{2\pi}) \int_0^1 \sqrt{1-\rho^2} \rho d\rho =$$

$$= 36 \cdot 2\pi \left(-\frac{1}{2} \right) \int_0^1 (1-\rho^2)^{1/2} d(1-\rho^2) = -36\pi \cdot \frac{2}{3} \left[(1-\rho^2)^{3/2} \right]_0^1 = 24\pi$$



$$I = \iint_D \frac{x^2}{y^4} \cos\left(\frac{x^2}{y}\right) dx dy \quad D: y=2x^2, y=4x^2, y=\sqrt{x}, y=\sqrt{2x}$$

$$y=2x^2 \Rightarrow \frac{y}{x^2}=2, \quad y=4x^2 \Rightarrow \frac{y}{x^2}=4 \Rightarrow \frac{y}{x^2}=u \Rightarrow 2 \leq u \leq 4$$

$$y=\sqrt{x} \Rightarrow y^2=x \Rightarrow \frac{y^2}{x}=1 \Rightarrow \frac{y^2}{x}=v \Rightarrow 1 \leq v \leq 2$$

$$y=\sqrt{2x} \Rightarrow y^2=2x \Rightarrow \frac{y^2}{x}=2$$

$$\frac{1}{J} = \begin{vmatrix} u'_x & v'_x \\ u'_y & v'_y \end{vmatrix} = \begin{vmatrix} -\frac{2y}{x^3} & -\frac{y^2}{x^2} \\ \frac{1}{x^2} & \frac{2y}{x} \end{vmatrix} = -\frac{3y^2}{x^4} \Rightarrow |J| = \frac{1}{3} \frac{x^4}{y^2} = \frac{1}{3} \frac{1}{\left(\frac{y}{x^2}\right)^2} = \frac{1}{3u^2}$$

$$I = \iint_D \frac{1}{\left(\frac{y^2}{x}\right)^2} \cos \frac{1}{y/x^2} dx dy = \iint_{D_{uv}} \frac{1}{v^2} \cos \frac{1}{u} \cdot \frac{1}{3u^2} du dv =$$

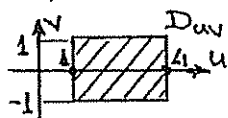
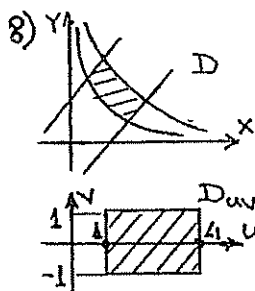
$$= \frac{1}{3} \int_1^2 \frac{dv}{v^2} \int_2^4 \frac{1}{u^2} \cos \frac{1}{u} du = \frac{1}{3} \left(-\frac{1}{v}\right) \Big|_1^2 \int_2^4 \left(-\cos \frac{1}{u}\right) d\left(\frac{1}{u}\right) = \frac{1}{3} \cdot \frac{1}{2} \left(-\sin \frac{1}{u}\right) \Big|_2^4 =$$

$$= \frac{1}{6} \left(\sin \frac{1}{2} - \sin \frac{1}{4}\right)$$

הערה: ניתן גם לחשב את האינטגרל באמצעות שיטת ההחלפה

$$\begin{cases} u = y/x^2 \rightarrow y = ux^2 \\ v = y^2/x \end{cases} \Rightarrow v = u^2 x^3 \rightarrow x = \sqrt[3]{v/u^2} \nearrow y = \sqrt[3]{v^2/u}$$

$$J = \begin{vmatrix} x'_u & y'_u \\ x'_v & y'_v \end{vmatrix} = \begin{vmatrix} -\frac{2}{3} u^{-5/3} v^{1/3} & -\frac{1}{3} u^{-4/3} v^{2/3} \\ \frac{1}{3} u^{-2/3} v^{-2/3} & \frac{2}{3} u^{-1/3} v^{-1/3} \end{vmatrix} = -\frac{1}{3} u^{-2} \Rightarrow |J| = \frac{1}{3u^2}$$



$$I = \iint_D (x^2 + 2xy) dx dy, \quad D: y=\frac{1}{x^2}, y=\frac{4}{x^2}, y=x+1, y=x-1$$

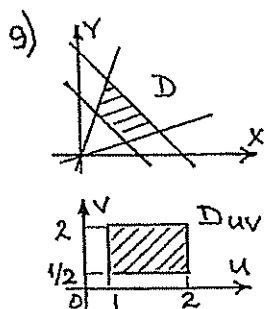
$$y=\frac{1}{x^2}, y=\frac{4}{x^2} \Rightarrow x^2 y=1, x^2 y=4 \Rightarrow x^2 y=u \Rightarrow 1 \leq u \leq 4$$

$$y=x+1, y=x-1 \Rightarrow y-x=1, y-x=-1 \Rightarrow y-x=v \Rightarrow -1 \leq v \leq 1$$

$$\frac{1}{J} = \begin{vmatrix} u'_x & v'_x \\ u'_y & v'_y \end{vmatrix} = \begin{vmatrix} 2xy & -1 \\ x^2 & 1 \end{vmatrix} = 2xy + x^2 \Rightarrow J = \frac{1}{x^2 + 2xy} > 0$$

$$f(u,v) = (x(u,v))^2 + 2x(u,v)y(u,v), \quad J(u,v) = \frac{1}{(x(u,v))^2 + 2x(u,v)y(u,v)}$$

$$I = \iint_{D_{uv}} \frac{(x^2 + 2xy)}{x^2 + 2xy} du dv = \iint_{D_{uv}} du dv = \int_{-1}^1 \int_0^4 du dv = (4-0)(1-(-1)) = 6$$



$$I = \iint_D \frac{(x+y)^3}{x^2} dx dy, D: y=1-x, y=2-x, y=\frac{1}{2}x, y=2x$$

$$y=1-x, y=2-x \Rightarrow x+y=1, x+y=2 \Rightarrow x+y=u \Rightarrow 1 \leq u \leq 2$$

$$y=\frac{1}{2}x, y=2x \Rightarrow \frac{y}{x}=\frac{1}{2}, \frac{y}{x}=2 \Rightarrow \frac{y}{x}=v \Rightarrow \frac{1}{2} \leq v \leq 2$$

$$\frac{1}{J} = \begin{vmatrix} u'_x & v'_x \\ u'_y & v'_y \end{vmatrix} = \begin{vmatrix} 1 & -y/x^2 \\ 1 & 1/x \end{vmatrix} = \frac{1}{x} + \frac{y}{x^2} = \frac{x+y}{x^2} \Rightarrow J = \frac{x^2}{x+y}$$

$$f(u,v) = \frac{(x(u,v)+y(u,v))^3}{(x(u,v))^2}, J(u,v) = \frac{(x(u,v))^2}{x(u,v)+y(u,v)} \Rightarrow$$

$$I = \iint_{D_{uv}} \frac{(x+y)^3}{x^2} \cdot \frac{x^2}{x+y} du dv = \iint_{D_{uv}} (x+y)^2 du dv = \iint_{D_{uv}} u^2 du dv =$$

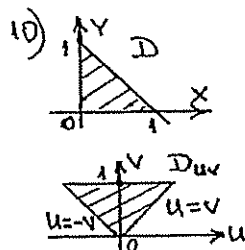
$$= \int_{1/2}^2 dv \int_1^2 u^2 du = (v|_{1/2}^2) (\frac{1}{3} u^3|_1^2) = \frac{7}{2} = 3.5$$

אנחנו גם נבדוק את זה

$$\begin{cases} u=x+y \\ v=y/x \Rightarrow y=vx \end{cases} \Rightarrow u=x(v+1) \Rightarrow x=\frac{u}{v+1} \text{ ו } y=\frac{uv}{v+1}$$

$$J = \begin{vmatrix} x'_u & y'_u \\ x'_v & y'_v \end{vmatrix} = \begin{vmatrix} \frac{1}{v+1} & \frac{v}{v+1} \\ -\frac{u}{(v+1)^2} & \frac{u}{(v+1)^2} \end{vmatrix} = \frac{u}{(v+1)^2} \Rightarrow J(x,y) = \frac{(x+y)^3}{x^2} = \frac{u^3}{(\frac{u}{v+1})^2} = u(v+1)^2$$

$$I = \iint_{D_{uv}} u(v+1)^2 \cdot \frac{u}{(v+1)^2} du dv = \iint_{D_{uv}} u^2 du dv = 3.5$$



$$I = \iint_D \cos \frac{x-y}{x+y} dx dy, D: y=0, x=0, y=1-x$$

$$\begin{cases} x-y=u \\ x+y=v \end{cases} \Rightarrow \frac{1}{J} = \begin{vmatrix} u'_x & v'_x \\ u'_y & v'_y \end{vmatrix} = \begin{vmatrix} 1 & 1 \\ -1 & 1 \end{vmatrix} = 2 \Rightarrow J = \frac{1}{2}$$

$$f(x,y) = \cos \frac{x-y}{x+y} = \cos \frac{u}{v}$$

$$\begin{cases} x-y=u \\ x+y=v \end{cases} \Rightarrow x=0 \Rightarrow \begin{cases} -y=u \\ y=v \end{cases} \Rightarrow u+v=0, y=0 \Rightarrow \begin{cases} x=u \\ x=v \end{cases} \Rightarrow u-v=0$$

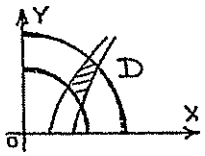
אנחנו רוצים את כל הנקודות שבהן $u-v=0$ ו- $u+v=0$ פרט ל- $y=0$ ו- $x=0$ פרט ל- $y=1-x$.
 אז כל הנקודות הן $v=1-x$ ו- $u=1-x$ כלומר $u=v$ ו- $u+v=1$: פרט ל- $y=1-x$

$$I = \iint_{D_{uv}} \cos \frac{u}{v} \cdot \frac{1}{2} du dv = \frac{1}{2} \int_0^1 dv \int_{-v}^v \cos \frac{u}{v} du = \frac{1}{2} \int_0^1 dv \cdot v \int_{-v}^v \cos \frac{u}{v} \cdot \frac{1}{v} du =$$

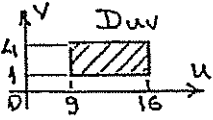
$$= \frac{1}{2} \int_0^1 v dv \int_{-v}^v \cos \frac{u}{v} d(\frac{u}{v}) = \frac{1}{2} \int_0^1 v dv (\sin \frac{u}{v} |_{-v}^v) =$$

$$= \frac{1}{2} \int_0^1 v dv (\sin 1 - \sin(-1)) = \frac{1}{2} \cdot 2 \sin 1 \int_0^1 v dv = \sin 1 \cdot \frac{1}{2} (v^2|_0^1) = \frac{1}{2} \sin 1.$$

13)



$$f(x, y) = x^3 y + x y^3, \quad D: \begin{cases} x^2 + y^2 = 9, & x^2 + y^2 = 16 \\ x^2 - y^2 = 1, & x^2 - y^2 = 4 \end{cases}$$



$$W = \iint_D f(x, y) dx dy$$

$$x^2 + y^2 = 9, x^2 + y^2 = 16 \Rightarrow x^2 + y^2 = u \Rightarrow 9 \leq u \leq 16$$

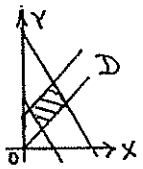
$$x^2 - y^2 = 1, x^2 - y^2 = 4 \Rightarrow x^2 - y^2 = v \Rightarrow 1 \leq v \leq 4$$

$$\frac{1}{J} = \begin{vmatrix} u'_x & v'_x \\ u'_y & v'_y \end{vmatrix} = \begin{vmatrix} 2x & 2x \\ 2y & -2y \end{vmatrix} = -8xy \Rightarrow |J| = \frac{1}{8xy} = \frac{1}{8x(u,v) \cdot y(u,v)}$$

$$W = \iint_{D_{uv}} (x^3 y - x y^3) \frac{1}{8xy} du dv = \iint_{D_{uv}} \frac{xy(x^2 + y^2)}{8xy} du dv = \frac{1}{8} \iint_{D_{uv}} (x^2 + y^2) du dv =$$

$$= \frac{1}{8} \iint_{D_{uv}} u du dv = \frac{1}{8} \int_1^4 dv \int_9^{16} u du = \frac{1}{8} (v|_1^4) \left(\frac{u^2}{2} \Big|_9^{16} \right) = \frac{1}{8} \cdot 3 \cdot \frac{1}{2} (256 - 81) = \frac{525}{16}$$

14)

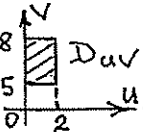


$$y = x + 2, y = x, y = 5 - 2x, y = 8 - 2x$$

$$S = \iint_D dx dy$$

$$y = x, y = x + 2 \Rightarrow y - x = 0, y - x = 2 \Rightarrow y - x = u \Rightarrow 0 \leq u \leq 2$$

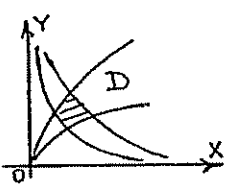
$$y = 5 - 2x, y = 8 - 2x \Rightarrow 2x + y = 5, 2x + y = 8 \Rightarrow 2x + y = v \Rightarrow 5 \leq v \leq 8$$



$$\frac{1}{J} = \begin{vmatrix} u'_x & v'_x \\ u'_y & v'_y \end{vmatrix} = \begin{vmatrix} -1 & 2 \\ 1 & 1 \end{vmatrix} = -3 \Rightarrow |J| = \frac{1}{3} \Rightarrow$$

$$S = \iint_{D_{uv}} \frac{1}{3} du dv = \frac{1}{3} \iint_{D_{uv}} du dv = \frac{1}{3} S_{D_{uv}} = \frac{1}{3} (2-0)(8-5) = 2.$$

15)

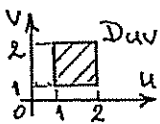


$$y = \sqrt{x}, y = 2\sqrt{x}, y = 1/x, y = 2/x, \quad f(x, y) = (\ln y - \frac{1}{2} \ln x)^2$$

$$m = \iint_D f(x, y) dx dy$$

$$y = \sqrt{x}, y = 2\sqrt{x} \Rightarrow y/\sqrt{x} = 1, y/\sqrt{x} = 2 \Rightarrow \frac{y}{\sqrt{x}} = u \Rightarrow 1 \leq u \leq 2$$

$$y = \frac{1}{x}, y = \frac{2}{x} \Rightarrow xy = 1, xy = 2 \Rightarrow xy = v \Rightarrow 1 \leq v \leq 2$$



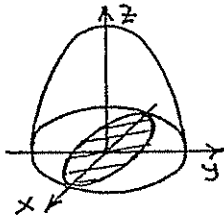
$$\frac{1}{J} = \begin{vmatrix} u'_x & v'_x \\ u'_y & v'_y \end{vmatrix} = \begin{vmatrix} -\frac{1}{2} x^{3/2} y & y \\ x^{-1/2} & x \end{vmatrix} = -\frac{y}{2\sqrt{x}} - \frac{y}{\sqrt{x}} = -\frac{3}{2} \cdot \frac{y}{\sqrt{x}}$$

$$|J| = \frac{2}{3} \cdot \frac{\sqrt{x}}{y} = \frac{2}{3} \cdot \frac{1}{y/\sqrt{x}} = \frac{2}{3u}, \quad f(x, y) = (\ln y - \frac{1}{2} \ln x)^2 = \left(\ln \frac{y}{\sqrt{x}} \right)^2 = (\ln u)^2 = \ln^2 u$$

$$m = \iint_{D_{uv}} \ln^2 u \cdot \frac{2}{3u} du dv = \frac{2}{3} \int_1^2 dv \int_1^2 \ln^2 u \cdot \frac{1}{u} du = \frac{2}{3} (v|_1^2) \int_1^2 \ln^2 u d(\ln u) =$$

$$= \frac{2}{3} \cdot \frac{1}{3} \ln^3 u \Big|_1^2 = \frac{2}{9} \ln^3 2$$

16)



$$z = f(x, y) = 16 - x^2 - y^2, \quad D = \{(x, y) \mid \frac{x^2}{16} + \frac{y^2}{4} \leq 1\}$$

$$V = \iint_D f(x, y) dx dy$$

לשם הריכוז, נשתמש במערכת הקואורדינטות הצינוריות:

$$x = 4\rho \cos \varphi, \quad y = 2\rho \sin \varphi$$

העברת המערכת למערכת הקואורדינטות הצינוריות:

$$\frac{16\rho^2 \cos^2 \varphi}{16} + \frac{4\rho^2 \sin^2 \varphi}{4} = 1 \Rightarrow \rho^2 = 1 \Rightarrow \rho = 1$$

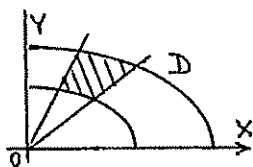
$$D_{\rho\varphi} = \{(\rho, \varphi) \mid 0 \leq \varphi < 2\pi, 0 \leq \rho \leq 1\} : \text{הקטע הריבועי}$$

$$J = \begin{vmatrix} x'_\rho & y'_\rho \\ x'_\varphi & y'_\varphi \end{vmatrix} = \begin{vmatrix} 4\cos\varphi & 2\sin\varphi \\ -4\rho\sin\varphi & 2\rho\cos\varphi \end{vmatrix} = 8\rho\cos^2\varphi + 8\rho\sin^2\varphi = 8\rho$$

$$\begin{aligned} f(x, y) &= 16 - x^2 - y^2 = 16 - 16\rho^2 \cos^2 \varphi - 4\rho^2 \sin^2 \varphi = 16 - \rho^2 [8(2\cos^2 \varphi + 2(2\sin^2 \varphi))] = \\ &= 16 - \rho^2 [8(1 + \cos 2\varphi) + 2(1 - \cos 2\varphi)] = 16 - 10\rho^2 - 6\rho^2 \cos 2\varphi. \end{aligned}$$

$$\begin{aligned} V &= \iint_{D_{\rho\varphi}} (16 - 10\rho^2 - 6\rho^2 \cos 2\varphi) \cdot 8\rho d\varphi d\rho = 8 \int_0^{2\pi} d\varphi \int_0^1 (16 - 10\rho^2 - 6\rho^2 \cos 2\varphi) \rho d\rho = \\ &= \left| \frac{8\rho^2}{2} - \frac{10\rho^4}{4} - \frac{6\rho^4 \cos 2\varphi}{4} \right|_0^1 = 8 \int_0^{2\pi} (4 - \frac{5}{2}\rho^2 - \frac{3}{2}\rho^2 \cos 2\varphi) d\varphi = \\ &= 32 \int_0^{2\pi} (4 - \frac{5}{2}\rho^2 - \frac{3}{2}\rho^2 \cos 2\varphi) d\varphi = 16 \int_0^{2\pi} (11 - 3\cos 2\varphi) d\varphi = \\ &= 16 (11\varphi - \frac{3}{2} \sin 2\varphi) \Big|_0^{2\pi} = 16 (\frac{11}{2} \cdot 2\pi - \frac{3}{2} \sin 4\pi) = 88\pi. \end{aligned}$$

17)

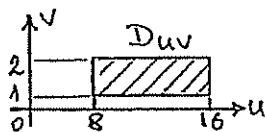


$$x^2 + 4y^2 = 8, \quad x^2 + 4y^2 = 16, \quad y = x, \quad y = 2x$$

$$S = \iint_D dx dy$$

$$x^2 + 4y^2 = 8, \quad x^2 + 4y^2 = 16 \Rightarrow x^2 + 4y^2 = u \Rightarrow 8 \leq u \leq 16$$

$$y = x, \quad y = 2x \Rightarrow \frac{y}{x} = 1, \quad \frac{y}{x} = 2 \Rightarrow \frac{y}{x} = v \Rightarrow 1 \leq v \leq 2$$



$$\frac{1}{J} = \begin{vmatrix} u'_x & v'_x \\ u'_y & v'_y \end{vmatrix} = \begin{vmatrix} 2x & -y/x^2 \\ 8y & 1/x \end{vmatrix} = 2 + 8 \frac{y^2}{x^2} = 2 + 8v^2$$

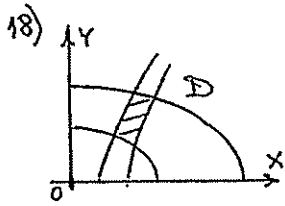
$$J = \frac{1}{2 + 8v^2} \Rightarrow S = \iint_{D_{uv}} \frac{1}{2 + 8v^2} du dv = \frac{1}{8} \int_8^{16} du \int_1^2 \frac{dv}{(\frac{1}{2})^2 + v^2} =$$

$$= \frac{1}{8} (u \Big|_8^{16}) 2 \arctan(2v) \Big|_1^2 = 2(\arctan 4 - \arctan 2)$$

השטח הנדרש הוא הפרש שטחי האליפסה החיצונה והפנימית:

$$\begin{cases} u = x^2 + 4y^2 \\ v = y/x \end{cases} \rightarrow y = vx \rightarrow x = \sqrt{\frac{u}{1+4v^2}}, \quad y = \sqrt{\frac{uv}{1+4v^2}}$$

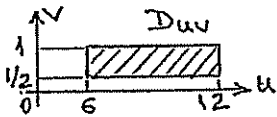
$$J = \begin{vmatrix} x'_u & y'_u \\ x'_v & y'_v \end{vmatrix} = \begin{vmatrix} \frac{1}{\sqrt{1+4v^2}} \cdot \frac{1}{2\sqrt{u}} & \frac{1}{\sqrt{1+4v^2}} \cdot \frac{1}{2\sqrt{u}} \\ -\frac{1}{2} \frac{8v\sqrt{u}}{(1+4v^2)^{3/2}} & \frac{\sqrt{u}}{(1+4v^2)^{3/2}} \end{vmatrix} = \frac{1}{2(1+4v^2)}$$



$$\frac{x^2}{3} + \frac{y^2}{2} = 1, \quad \frac{x^2}{6} + \frac{y^2}{4} = 1, \quad 2x^2 - 3y^2 = 1, \quad 4x^2 - 6y^2 = 1$$

$$\delta = \delta(x, y) = 16xy$$

$$m = \iint_D \delta \, dx \, dy$$



$$\frac{x^2}{3} + \frac{y^2}{2} = 1, \quad \frac{x^2}{6} + \frac{y^2}{4} = 1 \Rightarrow 2x^2 + 3y^2 = 6, \quad 2x^2 + 3y^2 = 12$$

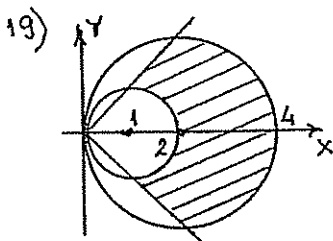
$$2x^2 + 3y^2 = u \Rightarrow 6 \leq u \leq 12$$

$$2x^2 - 3y^2 = 1, \quad 4x^2 - 6y^2 = 1 \Rightarrow 2x^2 - 3y^2 = 1, \quad 2x^2 - 3y^2 = \frac{1}{2}$$

$$2x^2 - 3y^2 = v \Rightarrow \frac{1}{2} \leq v \leq 1$$

$$\frac{1}{J} = \left| \frac{u'_x \, v'_x}{u'_y \, v'_y} \right| = \left| \frac{4x \, 4x}{6y \, -6y} \right| = -48xy \Rightarrow |J| = \frac{1}{48xy} = \frac{1}{48x(u,v)y(u,v)}$$

$$m = \iint_{D_{uv}} 16xy \cdot \frac{1}{48xy} \, du \, dv = \frac{1}{3} \iint_{D_{uv}} du \, dv = \frac{1}{3} S_{D_{uv}} = \frac{1}{3} (12-6)(1-\frac{1}{2}) = 1$$



$$x^2 + y^2 = 2x, \quad x^2 + y^2 = 4x, \quad y = \frac{\sqrt{2}}{2}x, \quad y = -\frac{\sqrt{2}}{2}x$$

$$S = \iint_D dx \, dy$$

$$\left(\begin{array}{l} \text{הגדרת המרחב} \\ (x-1)^2 + y^2 = 1, \quad (x-2)^2 + y^2 = 4 \end{array} \right)$$

$$J = \rho, \quad y = \rho \sin \varphi, \quad x = \rho \cos \varphi : \text{המרחב} \text{ מוגדר על ידי}$$

$$x^2 + y^2 = 2x \Rightarrow \rho^2 = 2\rho \cos \varphi \Rightarrow \rho = 2 \cos \varphi$$

$$x^2 + y^2 = 4x \Rightarrow \rho^2 = 4\rho \cos \varphi \Rightarrow \rho = 4 \cos \varphi$$

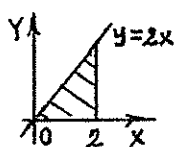
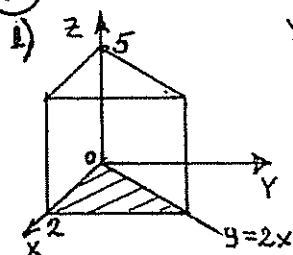
$$S = \iint_D \rho \, d\varphi \, d\rho = \int_{-\pi/4}^{\pi/4} d\varphi \int_{2 \cos \varphi}^{4 \cos \varphi} \rho \, d\rho = \left| \text{המרחב} \right| = 2 \int_0^{\pi/4} d\varphi \int_{2 \cos \varphi}^{4 \cos \varphi} \rho \, d\rho$$

$$= \int_0^{\pi/4} d\varphi \left(\rho^2 \Big|_{2 \cos \varphi}^{4 \cos \varphi} \right) = 12 \int_0^{\pi/4} \cos^2 \varphi \, d\varphi = 6 \int_0^{\pi/4} (1 + \cos 2\varphi) \, d\varphi =$$

$$= 6 \left(\varphi + \frac{1}{2} \sin 2\varphi \right) \Big|_0^{\pi/4} = 6 \left(\frac{\pi}{4} + \frac{1}{2} \sin \frac{\pi}{2} \right) = 3 \left(\frac{\pi}{2} + 1 \right)$$

האינטגרל המשולש

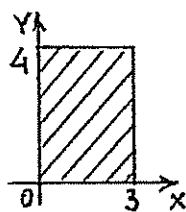
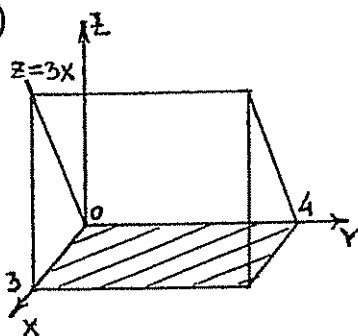
2)



$$y=0, z=0, z=5, x=2, y=2x$$

$$I = \int_0^2 dx \int_0^{2x} dy \int_0^5 f(x,y,z) dz$$

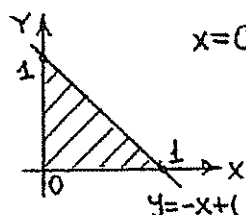
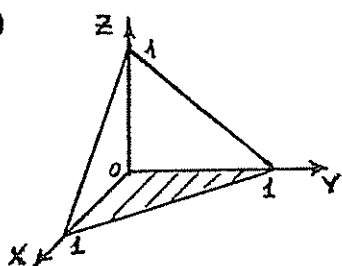
2)



$$z=0, y=0, y=4, x=3, z=3x$$

$$I = \int_0^3 dx \int_0^4 dy \int_0^{3x} f(x,y,z) dz$$

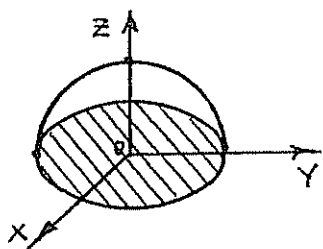
3)



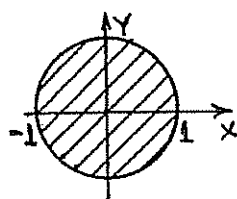
$$x=0, y=0, z=0, x+y+z=1$$

$$I = \int_0^1 dx \int_0^{1-x} dy \int_0^{1-x-y} f(x,y,z) dz$$

4)



$$z=0, x^2+y^2+z^2=1 \quad (z \geq 0)$$

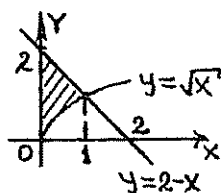
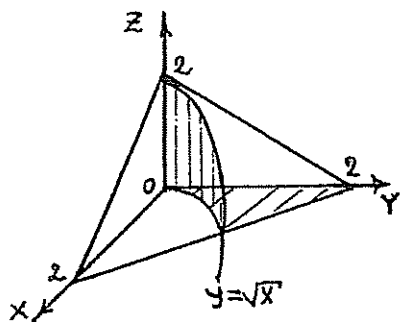


$$z = \sqrt{1-x^2-y^2}$$

$$\begin{cases} x^2+y^2+z^2=1 \\ z=0 \end{cases} \Rightarrow \begin{cases} x^2+y^2=1 \\ y=\pm\sqrt{1-x^2} \end{cases}$$

$$I = \int_{-1}^1 dx \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} dy \int_0^{\sqrt{1-x^2-y^2}} f(x,y,z) dz$$

5)



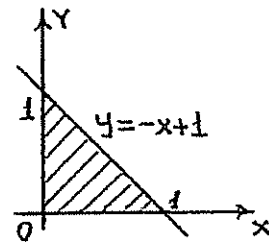
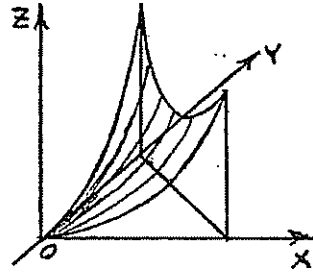
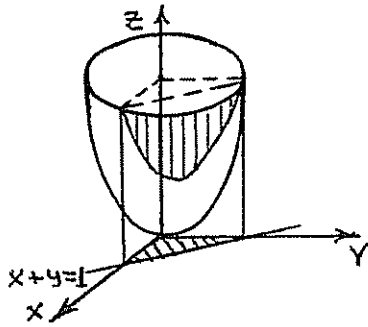
$$x=0, z=0, x+y+z=2, y=\sqrt{x}$$

$$\begin{cases} x+y+z=2 \\ z=0 \end{cases} \Rightarrow y=2-x$$

$$\begin{cases} y=2-x \\ y=\sqrt{x} \end{cases} \Rightarrow x=1$$

$$I = \int_0^1 dx \int_{\sqrt{x}}^{2-x} dy \int_0^{2-x-y} f(x,y,z) dz$$

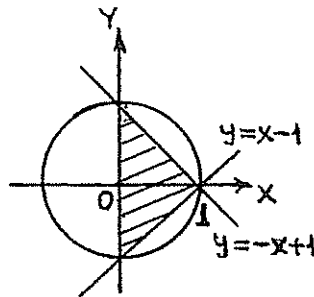
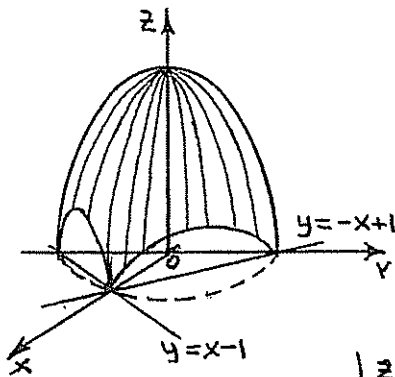
6)



$$x=0, y=0, z=0, x+y=1, z=x^2+y^2$$

$$I = \int_0^1 dx \int_0^{1-x} dy \int_0^{x^2+y^2} f(x,y,z) dz$$

7)



$$x=0 \quad (x \geq 0)$$

$$z=0$$

$$y=x-1$$

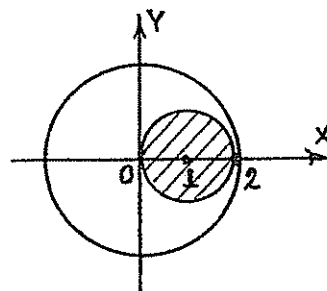
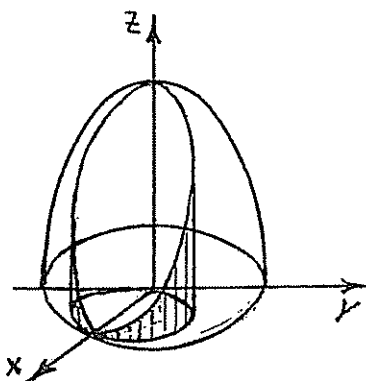
$$y=-x+1$$

$$z=1-x^2-y^2$$

$$\begin{cases} z=1-x^2-y^2 \\ z=0 \end{cases} \Rightarrow x^2+y^2=1$$

$$I = \int_0^1 dx \int_{x-1}^{1-x} dy \int_0^{1-x^2-y^2} f(x,y,z) dz$$

8)



$$z=0$$

$$z=4-x^2-y^2$$

$$x^2+y^2=2x$$

$$(x^2-2x+1)-1+y^2=0$$

$$(x-1)^2+y^2=1^2$$

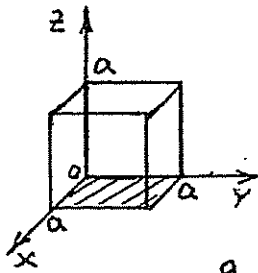
$$\begin{cases} z=4-x^2-y^2 \\ z=0 \end{cases} \Rightarrow x^2+y^2=2^2$$

$$x^2+y^2=2x \Rightarrow y = \pm \sqrt{2x-x^2}$$

$$I = \int_0^2 dx \int_{-\sqrt{2x-x^2}}^{\sqrt{2x-x^2}} dy \int_0^{4-x^2-y^2} f(x,y,z) dz$$

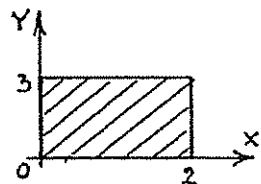
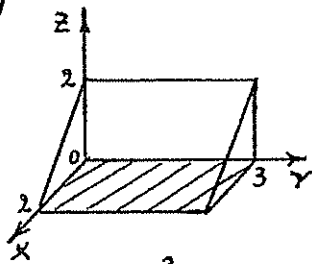
(2)

1)



$$\begin{aligned}
 M &= \iiint (x+y+z) dx dy dz = \\
 &= \int_0^a dx \int_0^a dy \int_0^a (x+y+z) dz = \int_0^a dx \int_0^a dy \left[(x+y)z + \frac{z^2}{2} \right]_0^a = \\
 &= \int_0^a dx \int_0^a dy \left[(x+y)a + \frac{a^2}{2} \right] = \frac{a^2}{2} \int_0^a dx \int_0^a dy + a \int_0^a dx \int_0^a (x+y) dy = \\
 &= \frac{a^2}{2} \int_0^a dx (y|_0^a) + a \int_0^a dx (xy + \frac{y^2}{2})|_0^a = \frac{a^3}{2} \int_0^a dx + a \int_0^a (xa + \frac{a^2}{2}) dx = \\
 &= \frac{a^3}{2} \cdot x|_0^a + \frac{a^3}{2} \int_0^a dx + a^2 \int_0^a x dx = \frac{a^4}{2} + \frac{a^4}{2} + \frac{a^4}{2} = \frac{3}{2} a^4
 \end{aligned}$$

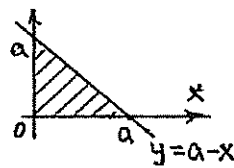
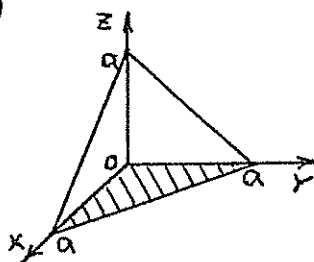
2)



$$\begin{aligned}
 M &= \iiint x dx dy dz = \\
 &= \int_0^2 x dx \int_0^{2-x} dy \int_0^{2-x} dz =
 \end{aligned}$$

$$\begin{aligned}
 &\int_0^2 x dx \int_0^{2-x} dy (z|_0^{2-x}) = \int_0^2 x(2-x) dx \int_0^{2-x} dy = \int_0^2 (2x-x^2) (y|_0^{2-x}) dx = \\
 &= 3 \left(x^2 - \frac{x^3}{3} \right) |_0^2 = 3 \left(4 - \frac{8}{3} \right) = 4
 \end{aligned}$$

3)

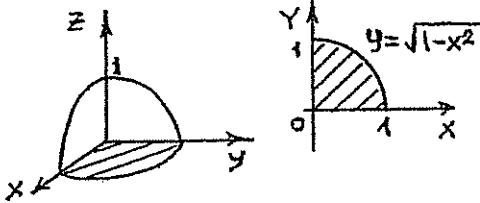


$$\begin{cases} x+y+z=a \\ z=0 \end{cases} \Rightarrow y = -x+a$$

$$M = \iiint z dx dy dz =$$

$$\begin{aligned}
 &= \int_0^a dx \int_0^{a-x} dy \int_0^{a-x-y} z dz = \frac{1}{2} \int_0^a dx \int_0^{a-x} dy (z^2|_0^{a-x-y}) = \\
 &= \frac{1}{2} \int_0^a dx \int_0^{a-x} (a-x-y)^2 dy = \frac{1}{2} \int_0^a dx \int_0^{a-x} [(a-x)^2 - 2(a-x)y + y^2] dy = \\
 &= \frac{1}{2} \int_0^a dx \left[(a-x)^2 y - (a-x)y^2 + \frac{1}{3} y^3 \right]_0^{a-x} = \frac{1}{2} \int_0^a \left[(a-x)^3 - (a-x)^3 + \frac{(a-x)^3}{3} \right] dx = \\
 &= -\frac{1}{6} \int_0^a (a-x)^3 d(a-x) = -\frac{1}{24} (a-x)^4 |_0^a = -0 + \frac{1}{24} a^4 = \frac{a^4}{24}
 \end{aligned}$$

4)



$$\begin{cases} x^2 + y^2 + z^2 = 1 \\ z = 0 \end{cases} \Rightarrow x^2 + y^2 = 1$$

$$y = \sqrt{1-x^2}$$

$$M = \iiint_V xyz \, dx \, dy \, dz =$$

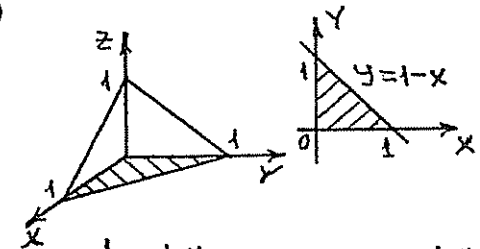
$$= \int_0^1 x \, dx \int_0^{\sqrt{1-x^2}} y \, dy \int_0^{\sqrt{1-x^2-y^2}} z \, dz = \frac{1}{2} \int_0^1 x \, dx \int_0^{\sqrt{1-x^2}} y \, dy \cdot (1-x^2-y^2) =$$

$$= \frac{1}{2} \int_0^1 x \, dx \int_0^{\sqrt{1-x^2}} (y - x^2 y - y^3) \, dy = \frac{1}{4} \int_0^1 x \, dx \left[y^2 (1-x^2 - \frac{y^2}{2}) \right]_0^{\sqrt{1-x^2}} =$$

$$= \frac{1}{4} \int_0^1 x \, dx (1-x^2) \left((1-x^2) - \frac{1-x^2}{2} \right) = \frac{1}{8} \int_0^1 (1-x^2)^2 x \, dx = -\frac{1}{16} \int_0^1 (1-x^2)^2 d(1-x^2) =$$

$$= -\frac{1}{16} \cdot \frac{1}{3} (1-x^2)^3 \Big|_0^1 = \frac{1}{48}$$

5)



$$M = \iiint_V \frac{dx \, dy \, dz}{(x+y+z+1)^3} =$$

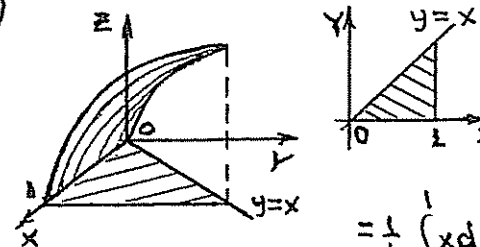
$$= \int_0^1 dx \int_0^{1-x} dy \int_0^{1-x-y} \frac{dz}{(x+y+z+1)^3} =$$

$$= -\frac{1}{2} \int_0^1 dx \int_0^{1-x} dy \left[\frac{1}{(x+y+z+1)^2} \right]_0^{1-x-y} = -\frac{1}{2} \int_0^1 dx \int_0^{1-x} dy \left[\frac{1}{2^2} - \frac{1}{(x+y+1)^2} \right] =$$

$$= -\frac{1}{2} \int_0^1 dx \left[\frac{1}{4} y + \frac{1}{(1+x)+y} \right]_0^{1-x} = -\frac{1}{2} \int_0^1 dx \left[\frac{1}{4} (1-x) + \frac{1}{2} - \frac{1}{1+x} \right] =$$

$$= -\frac{1}{2} \int_0^1 \left(\frac{3}{4} - \frac{1}{4} x - \frac{1}{x+1} \right) dx = -\frac{1}{2} \left(\frac{3}{4} x - \frac{1}{8} x^2 - \ln(x+1) \right) \Big|_0^1 = \frac{1}{2} \ln 2 - \frac{5}{16}$$

6)



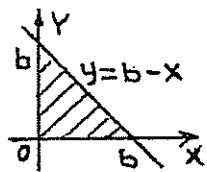
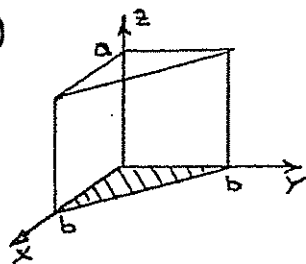
$$M = \iiint_V xy^2 z^3 \, dx \, dy \, dz =$$

$$= \int_0^1 x \, dx \int_0^x y^2 \, dy \int_0^{xy} z^3 \, dz =$$

$$= \frac{1}{4} \int_0^1 x \, dx \int_0^x y^2 \, dy (z^4 \Big|_0^{xy}) = \frac{1}{4} \int_0^1 x^5 \, dx \int_0^x y^6 \, dy =$$

$$= \frac{1}{4} \cdot \frac{1}{7} \int_0^1 x^5 (y^7 \Big|_0^x) \, dx = \frac{1}{28} \int_0^1 x^{12} \, dx = \frac{1}{28} \cdot \frac{1}{13} x^{13} \Big|_0^1 = \frac{1}{364}$$

7)



$$M = \iiint_V (2x+3y-z) dx dy dz =$$

$$= \int_0^b dx \int_0^{b-x} dy \int_0^a (2x+3y-z) dz =$$

$$= \int_0^b dx \int_0^{b-x} dy \left[(2x+3y)z - \frac{z^2}{2} \right]_0^a = a \int_0^b dx \int_0^{b-x} (2x+3y - \frac{a}{2}) dy =$$

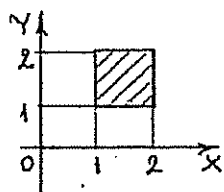
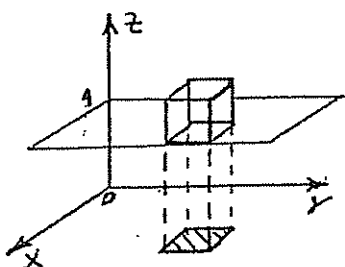
$$= a \int_0^b dx \left(2xy + \frac{3}{2}y^2 - \frac{a}{2}y \right) \Big|_0^{b-x} = a \int_0^b dx \left[y(2x + \frac{3}{2}y - \frac{a}{2}) \right] \Big|_0^{b-x} =$$

$$= a \int_0^b dx (b-x)(2x + \frac{3}{2}b - \frac{3}{2}x - \frac{a}{2}) = \frac{a}{2} \int_0^b (b-x)(x+3b-a) dx =$$

$$= \frac{a}{2} \int_0^b (3b^2 - ab - x^2 - 2bx + ax) dx = \frac{a}{2} \left(3b^2x - abx - \frac{x^3}{3} - bx^2 + \frac{ax^2}{2} \right) \Big|_0^b =$$

$$= \frac{a}{2} (3b^3 - ab^2 - \frac{1}{3}b^3 - b^3 + \frac{1}{2}ab^2) = \frac{a}{2} (\frac{5}{3}b^3 - \frac{1}{2}ab^2) = \frac{ab^2}{12} (10b - 3a)$$

8)



$$M = \iiint_V \frac{dx dy dz}{(x+y+z)^3} =$$

$$= \int_1^2 dx \int_1^2 dy \int_1^2 \frac{dz}{(x+y+z)^3} =$$

$$= -\frac{1}{2} \int_1^2 dx \int_1^2 dy \left[\frac{1}{(x+y+z)^2} \right]_1^2 =$$

$$= -\frac{1}{2} \int_1^2 dx \int_1^2 dy \left[\frac{1}{(x+y+2)^2} - \frac{1}{(x+y+1)^2} \right] = -\frac{1}{2} \int_1^2 dx \int_1^2 dy \frac{(x+y+1+x+y+2)(x+y+1-x-y-2)}{(x+y+2)^2(x+y+1)^2}$$

$$= \frac{1}{2} \int_1^2 dx \int_1^2 \frac{2x+2y+3}{(2xy+y^2+3y+x^2+3x+2)^2} dy = \frac{1}{2} \int_1^2 dx \int_1^2 \frac{d(2xy+y^2+3y+x^2+3x+2)}{(2xy+y^2+3y+x^2+3x+2)^2}$$

$$= -\frac{1}{2} \int_1^2 dx \left(\frac{1}{2xy+y^2+3y+x^2+3x+2} \right) \Big|_1^2 = -\frac{1}{2} \int_1^2 \left[\frac{1}{x^2+7x+12} - \frac{1}{x^2+5x+6} \right] dx =$$

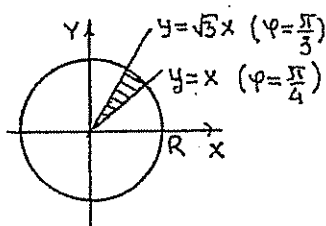
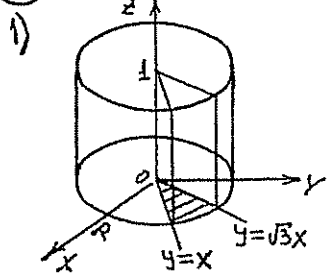
$$= -\frac{1}{2} \int_1^2 \left[\frac{1}{(x+3)(x+4)} - \frac{1}{(x+3)(x+2)} \right] dx = \int_1^2 \frac{dx}{(x+2)(x+3)(x+4)} =$$

$$= \left| \frac{1}{(x+2)(x+3)(x+4)} = \frac{A}{x+2} + \frac{B}{x+3} + \frac{C}{x+4} \right. \\ \left. 1 = A(x+3)(x+4) + B(x+2)(x+4) + C(x+2)(x+3) \Rightarrow A=1/2, B=-1, C=1/2 \right| =$$

$$= \int_1^2 \left[\frac{1}{2(x+2)} - \frac{1}{x+3} + \frac{1}{2(x+4)} \right] dx = \left[\frac{1}{2} \ln(x+2) - \ln(x+3) + \frac{1}{2} \ln(x+4) \right] \Big|_1^2 = \frac{1}{2} \ln \frac{128}{125}$$

האינטגרל המשולש בקואורדינטות גליליות וכדוריות

2)



$$x = \rho \cos \varphi, \quad y = \rho \sin \varphi, \\ z = z, \quad |\vec{r}| = \rho$$

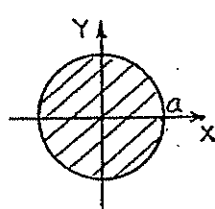
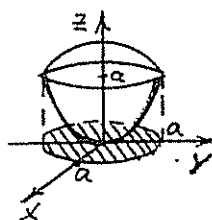
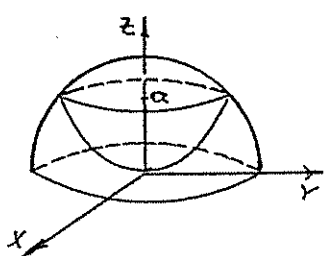
$$x^2 + y^2 = R^2 \Rightarrow \rho = R$$

$$y = x \Rightarrow \varphi = \frac{\pi}{4},$$

$$y = \sqrt{3}x \Rightarrow \varphi = \pi/3$$

$$I = \int_{\pi/4}^{\pi/3} d\varphi \int_0^R \rho d\rho \int_0^1 f(\rho \cos \varphi, \rho \sin \varphi, z) dz$$

2)



משוואת הכדור
: ρ רדיוס הכדור

$$\begin{cases} x^2 + y^2 + z^2 = 2a^2 \\ az = x^2 + y^2 \end{cases}$$

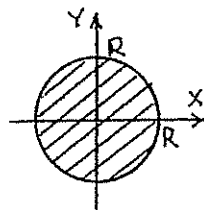
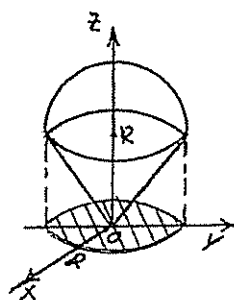
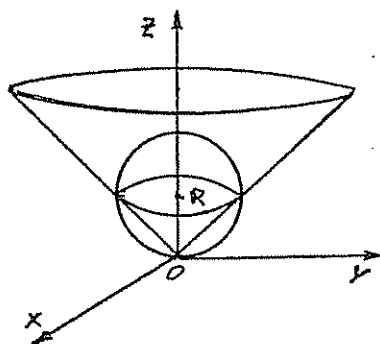
$$z^2 + az - 2a^2 = 0 \Rightarrow z = a$$

$$\begin{cases} az = x^2 + y^2 \\ z = a \end{cases} \Rightarrow x^2 + y^2 = a^2 \Rightarrow (\text{XOY על מישור הסיב}) \Rightarrow \rho = a,$$

$$az = x^2 + y^2 \Rightarrow z = \rho^2/a, \quad x^2 + y^2 + z^2 = 2a^2 \Rightarrow z = \sqrt{2a^2 - \rho^2}$$

$$I = \int_0^{2\pi} d\varphi \int_0^a \rho d\rho \int_{\rho^2/a}^{\sqrt{2a^2 - \rho^2}} f(\rho \cos \varphi, \rho \sin \varphi, z) dz$$

3)



$$x^2 + y^2 + z^2 = 2Rz \Rightarrow x^2 + y^2 + (z-R)^2 = R^2 \Rightarrow (\text{גליל})$$

$$\begin{cases} x^2 + y^2 + z^2 = 2Rz \\ z^2 = x^2 + y^2 \end{cases} \Rightarrow z(z-R) = 0 \Rightarrow z = R \Rightarrow$$

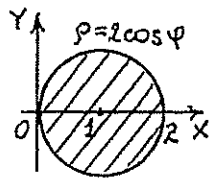
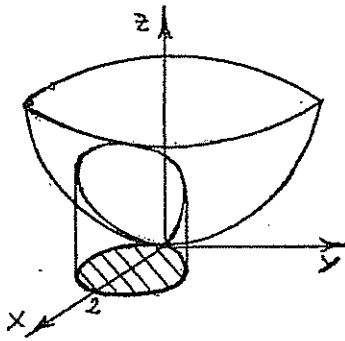
$$\begin{cases} z^2 = x^2 + y^2 \\ z = R \end{cases} \Rightarrow x^2 + y^2 = R^2 \Rightarrow (\text{XOY על מישור הסיב})$$

$$z^2 = x^2 + y^2 \Rightarrow z = \rho, \quad x^2 + y^2 = R^2 \Rightarrow \rho = R$$

$$x^2 + y^2 + (z-R)^2 = R^2 \Rightarrow z = R \pm \sqrt{R^2 - x^2 - y^2} \Rightarrow z = R + \sqrt{R^2 - \rho^2}$$

$$I = \int_0^{2\pi} d\varphi \int_0^R \rho d\rho \int_R^{R + \sqrt{R^2 - \rho^2}} f(\rho \cos \varphi, \rho \sin \varphi, z) dz$$

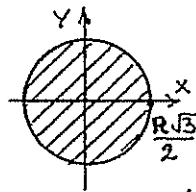
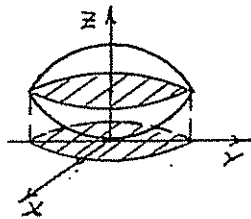
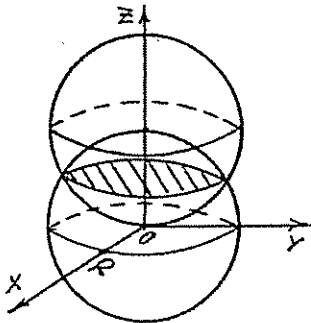
4)



$$\begin{aligned} x^2 + y^2 &= 2x \Rightarrow (x-1)^2 + y^2 = 1^2 \\ x^2 + y^2 &= 2x \Rightarrow \rho^2 = 2\rho \cos \varphi \\ \rho &= 2 \cos \varphi \\ z &= x^2 + y^2 \Rightarrow z = \rho^2 \end{aligned}$$

$$I = \int_{-\pi/2}^{\pi/2} d\varphi \int_0^{2 \cos \varphi} \rho d\rho \int_0^{\rho^2} f(\rho \cos \varphi, \rho \sin \varphi, z) dz$$

5)



$$\begin{cases} x^2 + y^2 + z^2 = R^2 \\ x^2 + y^2 + (z-R)^2 = R^2 \end{cases} \Rightarrow z = R/2$$

$$\begin{cases} x^2 + y^2 + z^2 = R^2 \\ z = R/2 \end{cases}$$

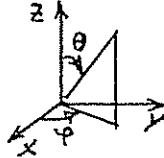
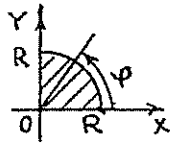
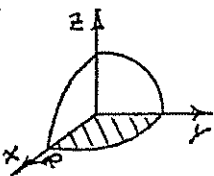
$$(xOy \text{ 平面上 } \delta G' \text{ 的}) \Leftrightarrow x^2 + y^2 = \frac{3}{4} R^2$$

$$x^2 + y^2 = \frac{3}{4} R^2 \Rightarrow \rho = R\sqrt{3}/2, \quad x^2 + y^2 + z^2 = R^2 \Rightarrow z = \pm \sqrt{R^2 - \rho^2}$$

$$x^2 + y^2 + (z-R)^2 = R^2 \Rightarrow z = R \pm \sqrt{R^2 - \rho^2}$$

$$I = \int_0^{2\pi} d\varphi \int_0^{R\sqrt{3}/2} \rho d\rho \int_{R-\sqrt{R^2-\rho^2}}^{\sqrt{R^2-\rho^2}} f(\rho \cos \varphi, \rho \sin \varphi, z) dz$$

6)

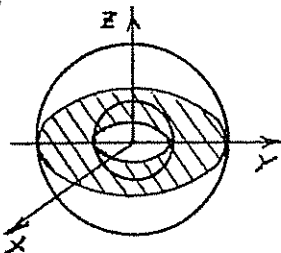


$$\begin{aligned} x &= \rho \sin \theta \cos \varphi, \\ y &= \rho \sin \theta \sin \varphi \\ z &= \rho \cos \theta, \quad |J| = \rho^2 \sin \theta \end{aligned}$$

$$x^2 + y^2 + z^2 = R^2 \Rightarrow \rho = R.$$

$$I = \int_0^{\pi/2} d\varphi \int_0^{\pi/2} \sin \theta d\theta \int_0^R f(\rho \sin \theta \cos \varphi, \rho \sin \theta \sin \varphi, \rho \cos \theta) \rho^2 d\rho$$

7)

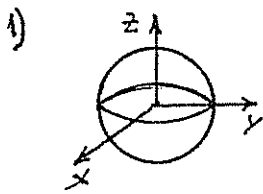


$$x^2 + y^2 + z^2 = R^2 \Rightarrow \rho = R.$$

$$x^2 + y^2 + z^2 = 2R^2 \Rightarrow \rho = \sqrt{2} R$$

$$I = \int_0^{2\pi} d\varphi \int_0^{\pi} \sin \theta d\theta \int_0^{\sqrt{2} R} f(\rho \sin \theta \cos \varphi, \rho \sin \theta \sin \varphi, \rho \cos \theta) \rho^2 d\rho$$

(1)



$$x^2 + y^2 + z^2 = R^2, \quad f(x, y, z) = \sqrt{x^2 + y^2 + z^2}.$$

: מ'ר'ב מ'ר'ב מ'ר'ב

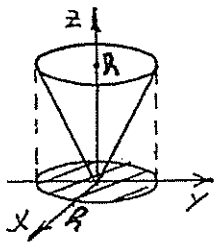
$$f(\varphi, \theta, \rho) = \rho, \quad x^2 + y^2 + z^2 = R^2 \Rightarrow \rho = R$$

$$I = \int_0^{2\pi} d\varphi \int_0^{\pi} \sin\theta d\theta \int_0^R \rho \cdot \rho^2 d\rho = \left| \frac{2\pi \rho^4}{4} \right|_0^R = 4\pi \int_0^{\pi/2} \sin\theta d\theta \int_0^R \rho^3 d\rho =$$

$$= \frac{8\pi}{4} \int_0^{\pi/2} \sin\theta d\theta \left(\rho^4 \Big|_0^R \right) = 2\pi R^4 \int_0^{\pi/2} \sin\theta d\theta \left(\varphi \Big|_0^{\pi/2} \right) = \pi R^4 \int_0^{\pi/2} \sin\theta d\theta =$$

$$= -\pi R^4 \cos\theta \Big|_0^{\pi/2} = \pi R^4$$

2)



$$z^2 = x^2 + y^2 \quad (z \geq 0), \quad z = h \Rightarrow (0, 0, h), \quad f(x, y, z) = z$$

$$\begin{cases} z^2 = x^2 + y^2 \\ z = h \end{cases} \Rightarrow x^2 + y^2 = h^2 \Rightarrow (x, y, z) \text{ is on the circle } z = h$$

: מ'ר'ב מ'ר'ב מ'ר'ב

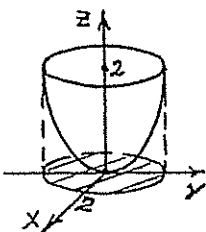
$$f(\varphi, \rho, z) = z, \quad x^2 + y^2 = h^2 \Rightarrow \rho = h$$

: מ'ר'ב מ'ר'ב מ'ר'ב

$$I = 4 \int_0^{\pi/2} d\varphi \int_0^h \rho d\rho \int_0^h z dz = 2 \int_0^{\pi/2} d\varphi \int_0^h \rho d\rho (z^2 \Big|_0^h) = 2 \int_0^{\pi/2} (h^2 \rho - \rho^3) d\rho \int_0^{\pi/2} d\varphi =$$

$$= 2 \int_0^{\pi/2} (h^2 \rho - \rho^3) d\rho \left(\varphi \Big|_0^{\pi/2} \right) = \pi \left(h^2 \frac{\rho^2}{2} - \frac{\rho^4}{4} \right) \Big|_0^h = \frac{\pi R^4}{4}$$

3)



$$2z = x^2 + y^2, \quad z = 2 \Rightarrow (0, 0, 2), \quad f(x, y, z) = x^2 + y^2$$

$$\begin{cases} 2z = x^2 + y^2 \\ z = 2 \end{cases} \Rightarrow x^2 + y^2 = 4 \Rightarrow (x, y, z) \text{ is on the circle } z = 2$$

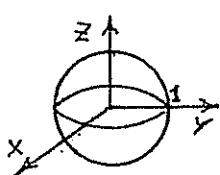
: מ'ר'ב מ'ר'ב מ'ר'ב

$$f(\varphi, \rho, z) = \rho^2, \quad z = \frac{x^2 + y^2}{2} \Rightarrow z = \frac{\rho^2}{2}, \quad x^2 + y^2 = 4 \Rightarrow \rho = 2$$

$$I = 4 \int_0^{\pi/2} d\varphi \int_0^2 \rho d\rho \int_{\rho^2/2}^2 \rho^2 dz = 4 \int_0^{\pi/2} d\varphi \int_0^2 \rho^3 d\rho (z \Big|_{\rho^2/2}^2) = 4 \int_0^{\pi/2} \left(2 - \frac{\rho^2}{2} \right) \rho^3 d\rho \int_0^{\pi/2} d\varphi =$$

$$= 4 \int_0^{\pi/2} \left(2\rho^3 - \frac{1}{2}\rho^5 \right) d\rho \left(\varphi \Big|_0^{\pi/2} \right) = 2\pi \left(\frac{\rho^4}{2} - \frac{\rho^6}{12} \right) \Big|_0^2 = 2\pi \left(8 - \frac{16}{3} \right) = \frac{16}{3}\pi$$

4)



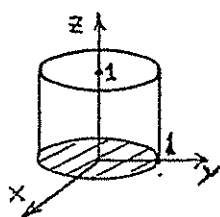
$$x^2 + y^2 + z^2 = 1 \Rightarrow (r, \theta, \phi), \quad f(x, y, z) = \sqrt{1 + (x^2 + y^2 + z^2)^{3/2}}$$

: $\wedge / r, \theta, \phi$ $\wedge / \phi, \theta, r$ $\wedge / \theta, \phi, r$

$$f(\varphi, \theta, \rho) = \sqrt{1 + \rho^3}, \quad x^2 + y^2 + z^2 = 1 \Rightarrow \rho = 1$$

$$\begin{aligned} I &= \int_0^{2\pi} d\varphi \int_0^\pi \sin\theta d\theta \int_0^1 \sqrt{1 + \rho^3} \rho^2 d\rho = \frac{1}{3} \int_0^1 \sqrt{1 + \rho^3} d(1 + \rho^3) \int_0^\pi \sin\theta d\theta (\varphi|_0^{2\pi}) = \\ &= \frac{2\pi}{3} \int_0^1 \sqrt{1 + \rho^3} d(1 + \rho^3) (-\cos\theta|_0^\pi) = \frac{4\pi}{3} \cdot \frac{2}{3} (1 + \rho^3)^{3/2} \Big|_0^1 = \frac{8\pi}{9} (2\sqrt{2} - 1) \end{aligned}$$

5)



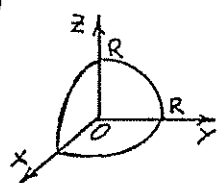
$$x^2 + y^2 = 1, \quad z = 0, \quad z = 1 \Rightarrow (r, \theta, z), \quad f(x, y, z) = (x^2 + y^2 + z)^3$$

: $\wedge / r, \theta, z$ $\wedge / \phi, \theta, r$ $\wedge / \theta, \phi, r$

$$f(\varphi, \rho, z) = (\rho^2 + z)^3, \quad x^2 + y^2 = 1 \Rightarrow \rho = 1$$

$$\begin{aligned} I &= \int_0^{2\pi} d\varphi \int_0^1 \rho d\rho \int_0^1 (\rho^2 + z)^3 dz = \int_0^{2\pi} d\varphi \int_0^1 \rho d\rho \int_0^1 (\rho^2 + z)^3 d(\rho^2 + z) = \\ &= \frac{1}{4} \int_0^{2\pi} d\varphi \int_0^1 \rho d\rho [(\rho^2 + z)^4]_0^1 = \frac{1}{4} \int_0^{2\pi} d\varphi \int_0^1 \rho d\rho [(\rho^2 + 1)^4 - \rho^8] = \\ &= \frac{1}{4} \int_0^{2\pi} [(\rho^2 + 1)^4 \rho - \rho^9] d\rho (\varphi|_0^{2\pi}) = \frac{\pi}{2} \left[\frac{1}{2} \int_0^1 (\rho^2 + 1)^4 d(\rho^2 + 1) - \int_0^1 \rho^9 d\rho \right] = \\ &= \frac{\pi}{2} \left[\frac{1}{10} (\rho^2 + 1)^5 \Big|_0^1 - \frac{1}{10} \rho^{10} \Big|_0^1 \right] = \frac{\pi}{20} [2^5 - 1 - 1] = \frac{\pi}{20} \cdot 30 = \frac{3}{2} \pi \end{aligned}$$

6)



$$x^2 + y^2 + z^2 = R^2, \quad x = 0, \quad y = 0, \quad z = 0, \quad (x \geq 0, y \geq 0, z \geq 0) \} (r, \theta, \phi) \text{ or } (r, \theta, \phi)$$

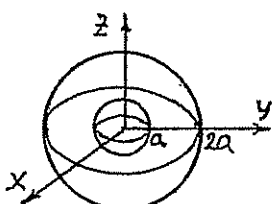
$$f(x, y, z) = z$$

: $\wedge / r, \theta, \phi$ $\wedge / \phi, \theta, r$ $\wedge / \theta, \phi, r$

$$f(\varphi, \theta, \rho) = \rho \cos\theta, \quad x^2 + y^2 + z^2 = R^2 \Rightarrow \rho = R$$

$$\begin{aligned} I &= \int_0^{\pi/2} d\varphi \int_0^{\pi/2} \sin\theta d\theta \int_0^R \rho \cos\theta \cdot \rho^2 d\rho = \int_0^{\pi/2} \sin\theta \cos\theta d\theta \int_0^R \rho^3 d\rho \int_0^{\pi/2} d\varphi = \\ &= \int_0^{\pi/2} \frac{1}{2} \sin 2\theta d\theta \int_0^R \rho^3 d\rho (\varphi|_0^{\pi/2}) = \frac{\pi}{4} \int_0^{\pi/2} \sin 2\theta d\theta \int_0^R \rho^3 d\rho = \frac{\pi}{16} \int_0^{\pi/2} \sin 2\theta d\theta (\rho^4|_0^R) = \\ &= \frac{\pi R^4}{16} \int_0^{\pi/2} \sin 2\theta d\theta = -\frac{\pi R^4}{16} \cdot \frac{1}{2} \cos 2\theta \Big|_0^{\pi/2} = \frac{\pi R^4}{16} \cdot \frac{1}{2} \cdot 2 = \frac{\pi R^4}{16} \end{aligned}$$

7)

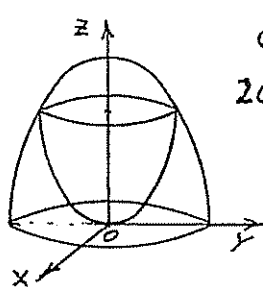
$x^2+y^2+z^2=a^2, x^2+y^2+z^2=4a^2 \Rightarrow$ (שני כדורים)

 $f(x,y,z) = \frac{1}{\sqrt{x^2+y^2+z^2}}$: שני כדורים

$f(\varphi, \theta, \rho) = \frac{1}{\rho}, x^2+y^2+z^2=a^2 \Rightarrow \rho=a, x^2+y^2+z^2=4a^2 \Rightarrow \rho=2a$

$$I = \int_0^{2\pi} d\varphi \int_0^\pi \sin\theta d\theta \int_a^{2a} \frac{1}{\rho} \cdot \rho^2 d\rho = \int_a^{2a} \rho d\rho \int_0^\pi \sin\theta d\theta \int_0^{2\pi} d\varphi = \int_a^{2a} \rho d\rho \int_0^\pi \sin\theta d\theta (\varphi|_0^{2\pi}) =$$

$$= 2\pi \int_a^{2a} \rho d\rho \int_0^\pi \sin\theta d\theta = 2\pi \int_a^{2a} \rho d\rho (-\cos\theta|_0^\pi) = 4\pi \int_a^{2a} \rho d\rho = 2\pi \rho^2|_a^{2a} = 6\pi a^2$$

8)



$az = x^2 + y^2 \rightarrow$ (פרבולה)
 $2az = a^2 - x^2 - y^2 \rightarrow$ (פרבולה)

$$\begin{cases} az = x^2 + y^2 \\ 2az = a^2 - x^2 - y^2 \end{cases} \Rightarrow 3az = a^2 \Rightarrow z = \frac{a}{3}$$

$$\begin{cases} az = x^2 + y^2 \\ z = a/3 \end{cases} \Rightarrow x^2 + y^2 = \frac{a^2}{3}$$

ה'שטח של הפרבולה ושל הפרבולה הפנימית
 :שני כדורים : $a/\sqrt{3}$.

$az = x^2 + y^2 \Rightarrow z = \frac{1}{a} \rho^2, 2az = a^2 - x^2 - y^2 \Rightarrow z = \frac{1}{2a} (a^2 - \rho^2)$

$$V = \int_0^{2\pi} d\varphi \int_0^{a/\sqrt{3}} \rho d\rho \int_{\rho^2/a}^{(a^2-\rho^2)/2a} dz = \int_0^{2\pi} d\varphi \int_0^{a/\sqrt{3}} \rho d\rho (z|_{\rho^2/a}^{(a^2-\rho^2)/2a}) =$$

$$= \int_0^{a/\sqrt{3}} \rho d\rho \left(\frac{a^2 - \rho^2}{2a} - \frac{\rho^2}{a} \right) \int_0^{2\pi} d\varphi = \frac{1}{2a} \int_0^{a/\sqrt{3}} \rho (a^2 - 3\rho^2) d\rho (\varphi|_0^{2\pi}) =$$

$$= \frac{\pi}{a} \int_0^{a/\sqrt{3}} (a^2 \rho - 3\rho^3) d\rho = \frac{\pi}{a} \left(\frac{a^2 \rho^2}{2} - \frac{3}{4} \rho^4 \right) \Big|_0^{a/\sqrt{3}} = \frac{\pi}{a} \left(\frac{a^4}{6} - \frac{3a^4}{4 \cdot 9} \right) = \frac{\pi a^3}{12}$$

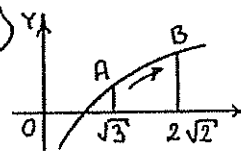
$$= \frac{2\pi a^3}{3} \cdot \frac{\sqrt{2}-1}{\sqrt{2}} = \frac{\pi a^3}{3} (2-\sqrt{2}).$$

האינטגרל הקווי ממין ראשון והאינטגרל הקווי ממין שני

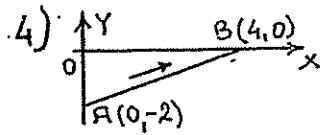
$$\begin{aligned}
 1) I &= \int_L \sqrt{y} \, ds, \quad L: x = a(t - \sin t), \quad y = a(1 - \cos t) \quad (0 \leq t \leq 2\pi) \\
 ds &= \sqrt{x'^2 + y'^2} \, dt = \sqrt{a^2(1 - \cos t)^2 + a^2(\sin t)^2} \, dt = \\
 &= a \sqrt{1 - 2\cos t + \cos^2 t + \sin^2 t} \, dt = a\sqrt{2} \sqrt{1 - \cos t} \, dt \\
 I &= \int_0^{2\pi} \sqrt{a(1 - \cos t)} \cdot a\sqrt{2} \sqrt{1 - \cos t} \, dt = a\sqrt{2a} \int_0^{2\pi} (1 - \cos t) \, dt = \\
 &= a\sqrt{2a} (t - \sin t) \Big|_0^{2\pi} = a\sqrt{2a} (2\pi - 0) = 2\pi a\sqrt{2a}
 \end{aligned}$$

$$\begin{aligned}
 2) \int_L \sqrt{x^2 + y^2} \, ds, \quad L: x &= a(\cos t + t \sin t), \quad y = a(\sin t - t \cos t) \quad (0 \leq t \leq 2\pi) \\
 \sqrt{x^2 + y^2} &= a \sqrt{(\cos t + t \sin t)^2 + (\sin t - t \cos t)^2} = \\
 &= a \sqrt{\cos^2 t + \sin^2 t + t^2 \sin^2 t + t^2 \cos^2 t} = a \sqrt{1 + t^2} \\
 x' &= a(-\sin t + \sin t + t \cos t) = at \cos t, \\
 y' &= a(\cos t - \cos t + t \sin t) = at \sin t \Rightarrow ds = \sqrt{x'^2 + y'^2} \, dt = at \, dt \\
 \int_L \sqrt{x^2 + y^2} \, ds &= \int_0^{2\pi} a \sqrt{1 + t^2} \cdot at \, dt = a^2 \int_0^{2\pi} \sqrt{1 + t^2} \, t \, dt = \\
 &= \frac{a^2}{2} \int_0^{2\pi} \sqrt{1 + t^2} \, d(1 + t^2) = \frac{a^2}{2} \cdot \frac{2}{3} (1 + t^2)^{3/2} \Big|_0^{2\pi} = \frac{a^2}{3} [(1 + 4\pi^2)^{3/2} - 1]
 \end{aligned}$$

3)



$$\begin{aligned}
 I &= \int_{AB} x^2 \, ds, \quad AB: y = \ln x, \quad x_A = \sqrt{3}, \quad x_B = 2\sqrt{2} \\
 ds &= \sqrt{1 + y'^2} \, dx = \sqrt{1 + \left(\frac{1}{x}\right)^2} \, dx = \frac{1}{x} \sqrt{1 + x^2} \, dx \\
 I &= \int_{\sqrt{3}}^{2\sqrt{2}} x^2 \cdot \frac{1}{x} \sqrt{1 + x^2} \, dx = \int_{\sqrt{3}}^{2\sqrt{2}} \sqrt{1 + x^2} \, x \, dx = \frac{1}{2} \int_{\sqrt{3}}^{2\sqrt{2}} (1 + x^2)^{1/2} \, d(1 + x^2) = \\
 &= \frac{1}{2} \cdot \frac{2}{3} (1 + x^2)^{3/2} \Big|_{\sqrt{3}}^{2\sqrt{2}} = \frac{1}{3} (9^{3/2} - 4^{3/2}) = \frac{1}{3} (27 - 8) = 19/3
 \end{aligned}$$

4)  $I = \int_{AB} \frac{ds}{\sqrt{x^2+y^2}}$, $A(0, -2)$, $B(4, 0)$

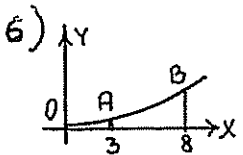
$AB: m = \frac{0+2}{4-0} = \frac{1}{2} \Rightarrow y = \frac{1}{2}x - 2, y' = \frac{1}{2}$

$$\begin{aligned} ds &= \sqrt{1+y'^2} dx = \sqrt{1+\frac{1}{4}} dx = \frac{\sqrt{5}}{2} dx \Rightarrow I = \int_0^4 \frac{\frac{\sqrt{5}}{2} dx}{2\sqrt{x^2+(\frac{1}{2}x-2)^2}} = \\ &= \frac{\sqrt{5}}{2} \int_0^4 \frac{dx}{\sqrt{\frac{5}{4}x^2 - 2x + 4}} = \frac{\sqrt{5}}{2} \int_0^4 \frac{dx}{\sqrt{(\frac{\sqrt{5}}{2}x)^2 - 2x \cdot \frac{\sqrt{5}}{2} + (\frac{2}{\sqrt{5}})^2 - (\frac{2}{\sqrt{5}})^2 + 4}} = \\ &= \frac{\sqrt{5}}{2} \int_0^4 \frac{dx}{\sqrt{(\frac{\sqrt{5}}{2}x - \frac{2}{\sqrt{5}})^2 + \frac{16}{5}}} = \int_0^4 \frac{d(\frac{\sqrt{5}}{2}x - \frac{2}{\sqrt{5}})}{\sqrt{(\frac{\sqrt{5}}{2}x - \frac{2}{\sqrt{5}})^2 + \frac{16}{5}}} = \left| \int \frac{dz}{\sqrt{z^2 + k}} \right| = \\ &= \ln \left| \frac{\sqrt{5}}{2}x - \frac{2}{\sqrt{5}} + \sqrt{\frac{5}{4}x^2 - 2x + 4} \right| \Big|_0^4 = \ln \left| 2\sqrt{5} - \frac{2}{\sqrt{5}} + 4 \right| - \ln \left| 2 - \frac{2}{\sqrt{5}} \right| = \\ &= \ln \frac{8+4\sqrt{5}}{2\sqrt{5}-2} = \ln \frac{2(2+\sqrt{5})}{\sqrt{5}-1} = \ln \frac{2(2+\sqrt{5})(\sqrt{5}+1)}{5-1} = \ln \frac{3\sqrt{5}+7}{2} \end{aligned}$$

5)  $I = \int_L \frac{ds}{x^2+y^2+z^2}$, $L: x = a \cos t, y = a \sin t, z = bt$ ($0 \leq t \leq 2\pi$)

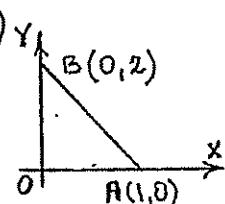
$ds = \sqrt{x'^2 + y'^2 + z'^2} dt = \sqrt{(-a \sin t)^2 + (a \cos t)^2 + b^2} dt$

$$\begin{aligned} ds &= \sqrt{a^2 + b^2} dt \Rightarrow I = \int_0^{2\pi} \frac{\sqrt{a^2 + b^2} dt}{a^2 \cos^2 t + a^2 \sin^2 t + b^2 t^2} = \\ &= \sqrt{a^2 + b^2} \int_0^{2\pi} \frac{dt}{a^2 + b^2 t^2} = \frac{\sqrt{a^2 + b^2}}{b} \int_0^{2\pi} \frac{d(bt)}{a^2 + (bt)^2} = \\ &= \frac{\sqrt{a^2 + b^2}}{b} \cdot \frac{1}{a} \arctg \left(\frac{bt}{a} \right) \Big|_0^{2\pi} = \frac{\sqrt{a^2 + b^2}}{ab} \arctg \frac{2\pi b}{a} \end{aligned}$$

6)  $I = \int_{AB} \frac{y}{\sqrt{x}} ds$, $AB: y^2 = \frac{4}{9}x^3 \Rightarrow y = \frac{2}{3}x^{3/2}, x_A = 3, x_B = 8$

$ds = \sqrt{1+y'^2} dx = \sqrt{1+(\frac{3}{2}\sqrt{x})^2} dx = \sqrt{1+x} dx$

$$\begin{aligned} I &= \int_3^8 \frac{2x\sqrt{x}}{3\sqrt{x}} \sqrt{1+x} dx = \frac{2}{3} \int_3^8 x\sqrt{1+x} dx = \left| \frac{1}{2}x^2 = t^2 \quad dx = 2t dt \right| = \\ &= \frac{4}{3} \int_2^3 (t^2 - 1)t^2 dt = \frac{4}{3} \left(\frac{t^5}{5} - \frac{t^3}{3} \right) \Big|_2^3 = \frac{4}{3} \left[\frac{243}{5} - 9 - \frac{32}{5} + \frac{8}{3} \right] = \frac{2152}{45} \end{aligned}$$

7)  $I = \int_L (2x+y) ds, O(0,0), A(1,0), B(0,2).$

$$\int_L = \int_{OA} + \int_{AB} + \int_{BO}$$

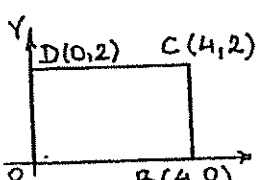
OA: $y=0, ds = \sqrt{1+y'^2} dx = dx \Rightarrow \int_{OA} = \int_0^1 (2x+0) dx = x^2 \Big|_0^1 = 1$

AB: $m = \frac{0-2}{1-0} = -2 \Rightarrow y = -2x+2 \Rightarrow ds = \sqrt{1+(-2)^2} dx = \sqrt{5} dx$

$$\int_{AB} = \int_0^1 [2x + (2-2x)] \sqrt{5} dx = \int_0^1 2\sqrt{5} dx = 2\sqrt{5} x \Big|_0^1 = 2\sqrt{5}$$

BO: $x=0 \Rightarrow ds = \sqrt{1+x'^2} dy = dy \Rightarrow \int_{BO} = \int_0^2 (2 \cdot 0 + y) dy = \frac{y^2}{2} \Big|_0^2 = 2$

$$I = 1 + 2\sqrt{5} + 2 = 3 + 2\sqrt{5}$$

8)  $I = \int_L xy ds, O(0,0), B(4,0), C(4,2), D(0,2)$

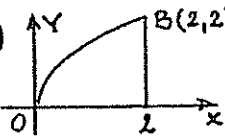
$$\int_L = \int_{OB} + \int_{BC} + \int_{CD} + \int_{DO}$$

OB: $y=0 \Rightarrow \int_{OB} xy ds = 0$

BC: $x=4 \Rightarrow ds = \sqrt{1+x'^2} dy = dy \Rightarrow \int_{BC} = \int_0^2 4y dy = 2y^2 \Big|_0^2 = 8$

CD: $y=2 \Rightarrow ds = \sqrt{1+y'^2} dx = dx \Rightarrow \int_{CD} = \int_0^4 2x dx = x^2 \Big|_0^4 = 16$

DO: $x=0 \Rightarrow \int_{DO} xy ds = 0 \Rightarrow I = 0 + 8 + 16 + 0 = 24$

9)  $I = \int_{OB} y ds, OB: y = \sqrt{2x}, O(0,0), B(2,2)$

$$ds = \sqrt{1+y'^2} dx = \sqrt{1 + \left(\frac{1}{\sqrt{2x}}\right)^2} dx = \sqrt{\frac{2x+1}{2x}} dx$$

$$I = \int_0^2 \sqrt{2x} \cdot \frac{\sqrt{2x+1}}{\sqrt{2x}} dx = \int_0^2 \sqrt{2x+1} dx = \frac{1}{2} \cdot \frac{2}{3} (2x+1)^{3/2} \Big|_0^2 = \frac{1}{3} (5\sqrt{5} - 1)$$

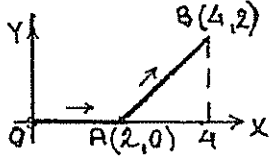
10) $I = \int_L (x^2+y^2)^n ds, L: x = a \cos t, y = a \sin t \quad (0 \leq t \leq 2\pi)$

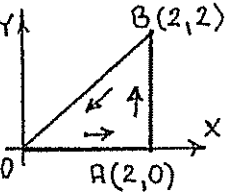
$$ds = \sqrt{x'^2 + y'^2} dt = \sqrt{a^2 \sin^2 t + a^2 \cos^2 t} dt = a dt$$

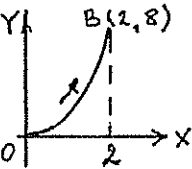
$$I = \int_0^{2\pi} (a^2 \cos^2 t + a^2 \sin^2 t)^n \cdot a dt = \int_0^{2\pi} (a^2)^n \cdot 1^n \cdot a dt = a^{2n+1} t \Big|_0^{2\pi} = 2\pi a^{2n+1}$$

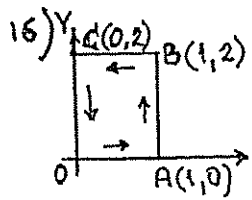
11) $I = \int_L xy dx + y^2 dy$, $L: x=t^2, y=t, 1 \leq t \leq 2$
 $dx=2t dt, dy=dt \Rightarrow I = \int_1^2 (t^2 \cdot t \cdot 2t + t^2) dt = \int_1^2 (2t^4 + t^2) dt =$
 $= \left(\frac{2}{5} t^5 + \frac{t^3}{3} \right) \Big|_1^2 = \frac{64}{5} + \frac{8}{3} - \frac{2}{5} - \frac{1}{3} = \frac{221}{15}$

12) $I = \int_L (x+y) dx + (x-y) dy$, $L: x=R \cos t, y=R \sin t (0 \leq t \leq \frac{\pi}{2})$
 $dx = -R \sin t dt, dy = R \cos t dt \Rightarrow$
 $I = R^2 \int_0^{\pi/2} [(\cos t + \sin t) (-\sin t) + (\cos t - \sin t) \cos t] dt =$
 $= R^2 \int_0^{\pi/2} (\cos^2 t - \sin^2 t - 2 \sin t \cos t) dt = R^2 \int_0^{\pi/2} (\cos 2t - \sin 2t) dt =$
 $= \frac{1}{2} R^2 (\sin 2t + \cos 2t) \Big|_0^{\pi/2} = \frac{1}{2} R^2 (\sin \pi + \cos \pi - \sin 0 - \cos 0) = -R^2$

13)  $I = \int_{OAB} (x+y) dx - x dy$. $\int_{OAB} = \int_{OA} + \int_{AB}$
 $\underline{OA}: y=0, dy=0 \Rightarrow \int_{OA} = \int_0^2 x dx = \frac{x^2}{2} \Big|_0^2 = 2$
 $\underline{AB}: m = \frac{2-0}{4-2} = 1 \Rightarrow y=x-2, dy=1 \cdot dx$
 $\int_{AB} = \int_2^4 [(x+x-2) dx - x \cdot dx] = \int_2^4 (x-2) dx = \left(\frac{x^2}{2} - 2x \right) \Big|_2^4 = 2 \Rightarrow I = 2+2=4$

14)  $I = \oint_L x dy$. $\oint_L = \int_{OA} + \int_{AB} + \int_{BO}$
 $\underline{OA}: y=0, dy=0 \Rightarrow \int_{OA} = \int_{OA} x dy = 0$
 $\underline{AB}: x=2 \Rightarrow \int_{AB} = \int_0^2 x dy = \int_0^2 2 dy = 2y \Big|_0^2 = 4$
 $\underline{BO}: m=1 \Rightarrow y=x, dy=dx \Rightarrow \int_{BO} = \int_2^0 x dx = \frac{x^2}{2} \Big|_2^0 = -2 \Rightarrow I = 4-2=2$

15)  $I = \int_{OB} x dy - y dx$, $OB: y=x^3, O(0,0), B(2,8)$
 $I = \int_0^2 (x \cdot 3x^2 dx - x^3 dx) = \int_0^2 2x^3 dx = \frac{1}{2} x^4 \Big|_0^2 = 8$



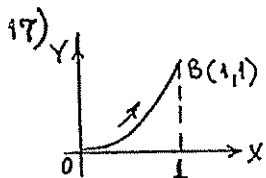
$$I = \oint_L (x^2 - y) dx, \quad \oint = \int_{OA} + \int_{AB} + \int_{BC} + \int_{CO}$$

OA: $y=0 \Rightarrow \int_{OA} = \int_0^1 x^2 dx = \frac{x^3}{3} \Big|_0^1 = \frac{1}{3}$

AB: $x=1 \Rightarrow dx=0 \Rightarrow \int_{AB} = 0$

BC: $y=2 \Rightarrow \int_{BC} = \int_1^0 (x^2 - 2) dx = \left(\frac{x^3}{3} - 2x \right) \Big|_1^0 = 0 - \frac{1}{3} + 2 = 2 - \frac{1}{3}$

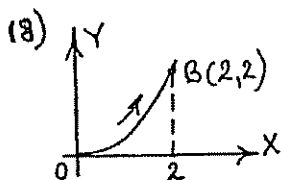
CO: $x=0 \Rightarrow dx=0 \Rightarrow \int_{CO} = 0 \Rightarrow I = \frac{1}{3} + 0 + (2 - \frac{1}{3}) + 0 = 2$



$$I = \int_{OB} \sqrt{x} dx + \sqrt{y} dy, \quad OB: y=x^2, \quad O(0,0), \quad B(1,1)$$

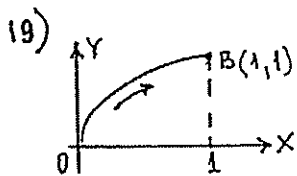
$$I = \int_0^1 (\sqrt{x} dx + \sqrt{x^2} \cdot 2x dx) = \int_0^1 (\sqrt{x} + 2x^2) dx =$$

$$= \left(\frac{2}{3} x^{3/2} + \frac{2}{3} x^3 \right) \Big|_0^1 = 4/3$$



$$I = \int_{OB} (x+y) dx, \quad OB: y=x^2/2, \quad O(0,0), \quad B(2,2)$$

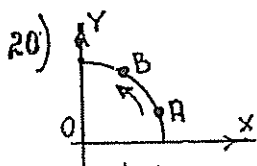
$$I = \int_0^2 \left(x + \frac{x^2}{2} \right) dx = \left(\frac{x^2}{2} + \frac{x^3}{6} \right) \Big|_0^2 = 2 + \frac{4}{3} = 10/3$$



$$I = \int_{OB} x^2 dx + y^2 dy, \quad OB: y=\sqrt{x}, \quad O(0,0), \quad B(1,1)$$

$$I = \int_0^1 \left(x^2 dx + x \cdot \frac{1}{2\sqrt{x}} dx \right) = \int_0^1 \left(x^2 + \frac{1}{2} \sqrt{x} \right) dx =$$

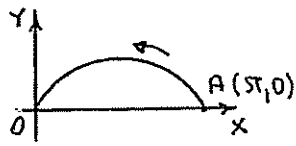
$$= \left(\frac{x^3}{3} + \frac{1}{2} \cdot \frac{2}{3} x^{3/2} \right) \Big|_0^1 = \frac{1}{3} + \frac{1}{3} = 2/3$$



$$I = \int_{AB} \frac{dy}{x} - \frac{dx}{y}, \quad AB (\text{arc}): x=r \cos t, \quad y=r \sin t$$

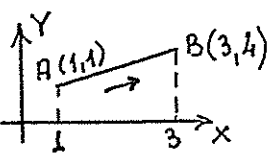
$$t_A = \pi/6, \quad t_B = \pi/3$$

$$I = \int_{\pi/6}^{\pi/3} \left(\frac{r \cos t dt}{r \cos t} - \frac{-r \sin t dt}{r \sin t} \right) = \int_{\pi/6}^{\pi/3} 2 dt = 2t \Big|_{\pi/6}^{\pi/3} = 5\pi/3$$

21)  $I = \int xy dx$, AO: $y = \sin x$, $A(\pi, 0)$, $O(0, 0)$

$$I = \int_{\pi}^0 x \sin x dx = \left| \begin{matrix} x=u & du=dx \\ \sin x dx = dv & v = -\cos x \end{matrix} \right| =$$

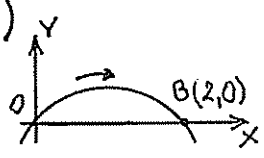
$$= -x \cos x \Big|_{\pi}^0 + \int_{\pi}^0 \cos x dx = \pi \cos \pi + \sin x \Big|_{\pi}^0 = -\pi$$

22)  $I = \int_{AB} (x^2 - y^2) dx + xy dy$; $A(1, 1)$, $B(3, 4)$

AB: $m = \frac{4-1}{3-1} = 1.5 \Rightarrow y-1 = 1.5(x-1) \Rightarrow y = 1.5x - 0.5$

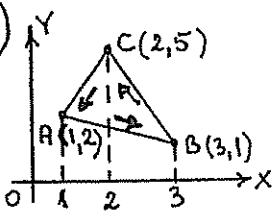
$$I = \int_1^3 [(x^2 - (1.5x - 0.5)^2) dx + x(1.5x - 0.5) \cdot 1.5 dx] = \int_1^3 (x^2 + 0.75x - 0.25) dx$$

$$= \left(\frac{x^3}{3} + 0.75 \cdot \frac{1}{2} x^2 - 0.25x \right) \Big|_1^3 = \left(\frac{27}{3} + \frac{3}{4} \cdot \frac{9}{2} - \frac{3}{4} \right) - \left(\frac{1}{3} + \frac{3}{8} - \frac{1}{4} \right) = 67/6$$

23)  $I = \int_{OB} y dx - (y + x^2) dy$, OB: $y = 2x - x^2$, $O(0, 0)$, $B(2, 0)$

$$I = \int_0^2 [(2x - x^2) dx - (2x - x^2 + x^2) \cdot (2 - 2x) dx] =$$

$$= \int_0^2 (3x^2 - 2x) dx = (x^3 - x^2) \Big|_0^2 = 8 - 4 = 4$$

24)  $I = \oint_L 2x dy - 3y dx$. $\oint = \int + \int + \int$
L AB BC CA

AB: $m = \frac{1-2}{3-1} = -\frac{1}{2}$, $y-2 = -\frac{1}{2}(x-1) \Rightarrow y = -\frac{1}{2}x + \frac{5}{2}$

$$\int_{AB} = \int_1^3 [2x(-\frac{1}{2}) dx - 3(-\frac{1}{2}x + \frac{5}{2}) dx] = \int_1^3 (\frac{1}{2}x - \frac{15}{2}) dx =$$

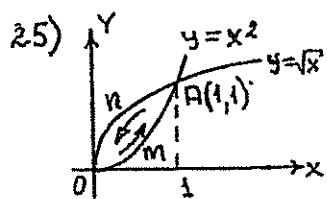
$$= \left(\frac{x^2}{4} - \frac{15}{2}x \right) \Big|_1^3 = \frac{9}{4} - \frac{45}{2} - \frac{1}{4} + \frac{15}{2} = -13$$

BC: $m = -4$, $y-1 = -4(x-3) \Rightarrow y = -4x + 13$

$$\int_{BC} = \int_3^2 [2x(-4) dx - 3(13 - 4x) dx] = \int_3^2 (4x - 39) dx = (2x^2 - 39x) \Big|_3^2 = 8 - 78 - 18 + 117 = 29$$

CA: $m = 3 \Rightarrow y = 3x - 1 \Rightarrow \int_{CA} = \int_2^1 [2x \cdot 3 dx - 3(3x - 1) dx] = \int_2^1 (3 - 3x) dx =$

$$= 3 \left(\frac{x^2}{2} - x \right) \Big|_1^2 = 3 \left(2 - 2 - \frac{1}{2} + 1 \right) = \frac{3}{2} \Rightarrow I = -13 + 29 + \frac{3}{2} = 17.5$$



$$I = \oint_L x^2 y dx + x^3 dy, \quad L: y=x^2, y^2=x \Rightarrow y=\sqrt{x}$$

$$\begin{cases} y=x^2 \\ y=\sqrt{x} \end{cases} \Rightarrow x^2=\sqrt{x} \Rightarrow x^4=x \Rightarrow x=1$$

$$\oint_L = \int_{O_m A} + \int_{A_n O}$$

$$\underline{O_m A}: y=x^2, dy=2x dx \Rightarrow \int_{O_m A} = \int_0^1 (x^2 \cdot x^2 + x^3 \cdot 2x) dx = 3 \int_0^1 x^4 dx = \frac{3}{5} x^5 \Big|_0^1 = \frac{3}{5}$$

$$\underline{A_n O}: y=\sqrt{x}, dy=\frac{dx}{2\sqrt{x}} \Rightarrow \int_{A_n O} = \int_1^0 (x^2 \sqrt{x} + x^3 \frac{1}{2\sqrt{x}}) dx = \frac{3}{2} \int_1^0 x^{5/2} dx = \frac{3}{7} x^{7/2} \Big|_1^0 = -\frac{3}{7}$$

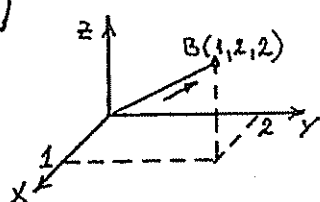
$$I = \frac{3}{5} - \frac{3}{7} = \frac{6}{35}$$

$$26) I = \int_{OB} xy dx + yz dy + zx dz, \quad OB: x=\cos t, y=\sin t, z=1 \quad (0 \leq t \leq \pi/2)$$

$$I = \int_0^{\pi/2} [\cos t \cdot \sin t \cdot (-\sin t) + \sin t \cdot 1 \cdot \cos t + 0] dt = \int_0^{\pi/2} (\sin t - \sin^3 t) \cos t dt =$$

$$= \int_0^{\pi/2} (\sin t - \sin^3 t) d(\sin t) = \left(\frac{\sin^2 t}{2} - \frac{\sin^4 t}{4} \right) \Big|_0^{\pi/2} = \frac{1}{2} - \frac{1}{4} = \frac{1}{4}$$

27)



$$I = \int_{OB} x dx + y dy + z dz, \quad O(0,0,0), \quad B(1,2,2)$$

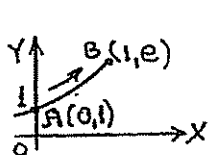
$$\text{כיוון } (1,2,2) : \text{ וקטור } \vec{v} \text{ } \parallel \text{ } \vec{v} = \frac{1}{\sqrt{6}}(1,2,2)$$

$$\text{נורמליזציה } \frac{x}{1} = \frac{y}{2} = \frac{z}{2} : \text{ וקטור } \vec{v} \text{ } \parallel \text{ } \vec{v} = \frac{1}{\sqrt{6}}(1,2,2)$$

$$\frac{x}{1} = \frac{y}{2} = \frac{z}{2} = t \Rightarrow x=t, y=2t, z=2t, t_0=0, t_1=1$$

$$I = \int_0^1 (t + 2t \cdot 2 + 2t \cdot 2) dt = 9 \int_0^1 t dt = 9 \cdot \frac{1}{2} t^2 \Big|_0^1 = 4.5$$

28)



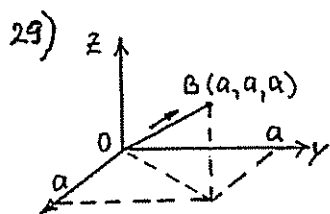
$$I = \int_{AB} (x^2 + y^2) dx + xy dy, \quad AB: y=e^x, \quad A(0,1), \quad B(1,e)$$

$$I = \int_0^1 [(x^2 + e^{2x}) dx + x e^x \cdot e^x dx] = \int_0^1 (x^2 + e^{2x}) dx + \int_0^1 x e^{2x} dx =$$

$$= \left(\frac{x^3}{3} + \frac{1}{2} e^{2x} \right) \Big|_0^1 + \int_0^1 x e^{2x} dx = \left| \frac{x^3}{3} + \frac{1}{2} e^{2x} \right|_0^1 + \int_0^1 x e^{2x} dx = \frac{1}{3} + \frac{1}{2} e^2 - \frac{1}{2} +$$

$$+ \left(\frac{1}{2} x e^{2x} \Big|_0^1 - \frac{1}{2} \int_0^1 e^{2x} dx \right) = \frac{1}{2} e^2 - \frac{1}{6} + \left(\frac{1}{2} e^2 - \frac{1}{4} e^{2x} \Big|_0^1 \right) =$$

$$= \frac{1}{2} e^2 - \frac{1}{6} + \frac{1}{2} e^2 - \frac{1}{4} e^2 + \frac{1}{4} = \frac{3}{4} e^2 + \frac{1}{12}$$



$$I = \int_{OB} ydx + zdy + xdz, \quad O(0,0,0), \quad B(a,a,a)$$

ପଦ, $(1,1,1)$ ଲାଇ OB ରେ $\vec{e}$ // OB ର $1/\sqrt{3}$

: $\therefore$ $\vec{r}(t) = t\vec{e} = \frac{t}{\sqrt{3}}(1,1,1)$: $t=0$ ରେ O ଓ $t=\sqrt{3}$ ରେ B ଥାଏ

$$\frac{x}{1} = \frac{y}{1} = \frac{z}{1} = t \Rightarrow x=t, y=t, z=t, \quad t_0=0, \quad t_B=a \Rightarrow$$

$$I = \int_0^a (t dt + t dt + t dt) = 3 \int_0^a t dt = \frac{3}{2} t^2 \Big|_0^a = \frac{3}{2} a^2$$

$$30) I = \int_C x dy - y dx, \quad C: x = a \cos^3 t, \quad y = a \sin^3 t \quad (0 \leq t \leq 2\pi)$$

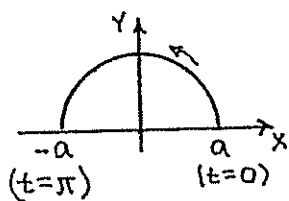
$$dx = 3a \cos^2 t (-\sin t) dt, \quad dy = 3a \sin^2 t \cos t dt \Rightarrow$$

$$I = 3a^2 \int_0^{2\pi} (\sin^2 t \cos^4 t + \sin^4 t \cos^2 t) dt = 3a^2 \int_0^{2\pi} \sin^2 t \cos^2 t \cdot 1 dt =$$

$$= \frac{3}{4} a^2 \int_0^{2\pi} \sin^2 2t dt = \frac{3}{8} a^2 \int_0^{2\pi} 2 \sin^2 2t dt = \frac{3a^2}{8} \int_0^{2\pi} (1 - \cos 4t) dt =$$

$$= \frac{3a^2}{8} \left[t - \frac{1}{4} \sin 4t \right]_0^{2\pi} = \frac{3a^2}{8} \cdot 2\pi = \frac{3}{4} \pi a^2$$

31)



$$I = \int_C \frac{y^2 dx - x^2 dy}{x^2 + y^2}, \quad C: x = a \cos t, \quad y = a \sin t \quad (0 \leq t \leq \pi)$$

$$dx = -a \sin t dt, \quad dy = a \cos t dt$$

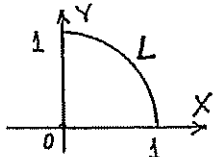
$$I = \int_0^\pi \frac{a^2 \sin^2 t (-a \sin t) dt - a^2 \cos^2 t (a \cos t) dt}{a^2 \cos^2 t + a^2 \sin^2 t} =$$

$$= - \int_0^\pi \frac{a^3 (\sin^3 t + \cos^3 t)}{a^2} dt = -a \int_0^\pi \sin^3 t dt - a \int_0^\pi \cos^3 t dt =$$

$$= a \int_0^\pi (1 - \cos^2 t) d \cos t - a \int_0^\pi (1 - \sin^2 t) d(\sin t) = a \left(\cos t - \frac{1}{3} \cos^3 t \right) \Big|_0^\pi$$

$$- a \left(\sin t - \frac{1}{3} \sin^3 t \right) \Big|_0^\pi = a \left(-1 + \frac{1}{3} - 1 + \frac{1}{3} \right) - a \cdot 0 = -\frac{4}{3} a$$

32)



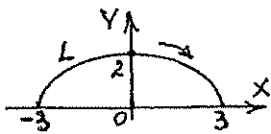
$$\gamma = f(x, y) = x + y, \quad x^2 + y^2 = 1, \quad (x \geq 0, y \geq 0)$$

$$m = \int_L \gamma ds = \int_L (x+y) \sqrt{1+y'^2} dx = \left| y = \sqrt{1-x^2}, y' = -\frac{x}{\sqrt{1-x^2}} \right| =$$

$$= \int_0^1 (x + \sqrt{1-x^2}) \sqrt{1 + \frac{x^2}{1-x^2}} dx = \int_0^1 \frac{x + \sqrt{1-x^2}}{\sqrt{1-x^2}} dx =$$

$$= \int_0^1 \frac{x dx}{\sqrt{1-x^2}} + \int_0^1 dx = -\frac{1}{2} \int_0^1 (1-x^2)^{-1/2} d(1-x^2) + x \Big|_0^1 = -\frac{1}{2} \cdot 2 \sqrt{1-x^2} \Big|_0^1 + 1 = 1 + 1 = 2$$

33)



$$\vec{F} = -y^2 \vec{i} + x^2 \vec{j}, \quad \frac{x^2}{9} + \frac{y^2}{4} = 1 \quad (y \geq 0)$$

$$W = \int_L \vec{F} d\vec{r} = \int_L -y^2 dx + x^2 dy$$

הערה: נבחרת את המסלול כעקמומיות של המעגל.

$$x = 3 \cos t \rightarrow dx = -3 \sin t dt, \quad y = 2 \sin t \rightarrow dy = 2 \cos t dt, \quad 0 \leq t \leq \pi$$

$$W = \int_0^\pi [-4 \sin^2 t (-3 \sin t) + 9 \cos^2 t \cdot 2 \cos t] dt =$$

$$= \int_0^\pi (12 \sin^3 t + 18 \cos^3 t) dt = -12 \int_0^\pi (1 - \cos^2 t) d(\cos t) + 18 \int_0^\pi (1 - \sin^2 t) d(\sin t) =$$

$$= -12 \left(\cos t - \frac{\cos^3 t}{3} \right) \Big|_0^\pi + 18 \left(\sin t - \frac{\sin^3 t}{3} \right) \Big|_0^\pi = -12 \left(\frac{2}{3} + \frac{2}{3} \right) + 0 = -16$$

34)



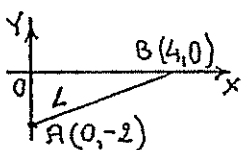
$$\vec{F} = \arctan \frac{x}{y} \vec{j}, \quad y = \sqrt{x}, \quad O(0,0), \quad A(1,1)$$

$$W = \int_L \vec{F} d\vec{r} = \int_L 0 dx + \arctan \frac{x}{y} dy = \int_L \arctan \frac{x}{y} dy =$$

$$= \int_0^1 \arctan \frac{y^2}{y} dy = \int_0^1 \arctan y dy = \left| \arctan y = u, \quad du = \frac{dy}{1+y^2} \right|$$

$$= y \arctan y \Big|_0^1 - \int_0^1 \frac{y dy}{1+y^2} = \frac{\pi}{4} - \frac{1}{2} \int_0^1 \frac{d(1+y^2)}{1+y^2} = \frac{\pi}{4} - \frac{1}{2} \ln(1+y^2) \Big|_0^1 = \frac{\pi}{4} - \frac{1}{2} \ln 2$$

35)



$$\gamma = \frac{1}{x-y}, \quad A(0,-2), \quad B(4,0)$$

$$m = \int_L \gamma ds = \int_L \frac{ds}{x-y} = \int_{AB} \frac{ds}{x-y}$$

$$\text{AB: } m = \frac{0+2}{4-0} = \frac{1}{2} \Rightarrow y = \frac{1}{2}x - 2, \quad ds = \sqrt{1 + \left(\frac{1}{2}\right)^2} dx = \frac{\sqrt{5}}{2} dx$$

$$I = \int_0^4 \frac{\sqrt{5} dx}{2(x - \frac{1}{2}x + 2)} = \sqrt{5} \int_0^4 \frac{dx}{x+4} = \sqrt{5} \ln(x+4) \Big|_0^4 = \sqrt{5} (\ln 8 - \ln 4) = \sqrt{5} \ln 2$$

36) $\vec{F} = yz\vec{i} + xz\vec{j} + xy\vec{k}$, $L: x = a \cos t, y = a \sin t, z = bt$ ($0 \leq t \leq 2\pi$)

$$W = \int_L \vec{F} d\vec{r} = \int_L yz dx + xz dy + xy dz$$

$$W = \int_0^{2\pi} [abt \sin t (-a \sin t dt) + abt \cos t (a \cos t dt) + a^2 \sin t \cos t \cdot b dt] =$$

$$= a^2 b \int_0^{2\pi} t (\cos^2 t - \sin^2 t) dt + a^2 b \int_0^{2\pi} \sin t \cos t dt =$$

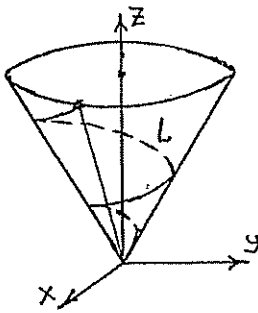
$$= a^2 b \int_0^{2\pi} t \cos 2t dt + a^2 b \int_0^{2\pi} \sin t d(\sin t) = a^2 b \int_0^{2\pi} t \cos 2t dt + a^2 b \frac{\sin^2 t}{2} \Big|_0^{2\pi} =$$

$$= \left| \begin{matrix} t = u & du = dt \\ \cos 2t dt = dv & v = \frac{1}{2} \sin 2t \end{matrix} \right| = a^2 b \left(\frac{1}{2} t \sin 2t \Big|_0^{2\pi} - \frac{1}{2} \int_0^{2\pi} \sin 2t dt \right) + 0 =$$

$$= a^2 b (0 + \frac{1}{4} \cos 2t) \Big|_0^{2\pi} = 0$$

$W = 0$, $\int_0^{2\pi} \sin 2t dt = 0$

37)



$\gamma = f(x, y, z) = z$, $x = t \cos t, y = t \sin t, z = t$
 $0 \leq t \leq 4\pi$

$$m = \int_L \gamma ds = \int_L z \sqrt{x'^2 + y'^2 + z'^2} dt =$$

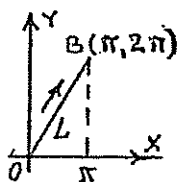
$$= \int_0^{4\pi} t \sqrt{(\cos t - t \sin t)^2 + (\sin t + t \cos t)^2 + 1^2} dt =$$

$$= \int_0^{4\pi} t \sqrt{\cos^2 t - 2t \cos t \sin t + t^2 \sin^2 t + \sin^2 t + 2t \sin t \cos t + t^2 \cos^2 t + 1} dt =$$

$$= \int_0^{4\pi} t \sqrt{(\cos^2 t + \sin^2 t)(1 + t^2) + 1} dt = \int_0^{4\pi} t \sqrt{2 + t^2} dt = \frac{1}{2} \int_0^{4\pi} (2 + t^2)^{1/2} d(2 + t^2)$$

$$= \frac{1}{2} \cdot \frac{2}{3} [(2 + t^2)^{3/2}] \Big|_0^{4\pi} = \frac{1}{3} [(2 + 16\pi^2)^{3/2} - 2^{3/2}] = \frac{2}{3} \sqrt{2} [(1 + 8\pi^2)^{3/2} - 1]$$

38)



$\vec{F} = -x \cos y \vec{i} + y \sin x \vec{j}$, $O(0,0)$, $B(\pi, 2\pi)$

$$W = \int_L \vec{F} d\vec{r} = \int_{OB} -x \cos y dx + y \sin x dy$$

$OB: m = 2, y = 2x \Rightarrow dy = 2dx$

$$W = \int_0^\pi (-x \cos 2x + 2x \sin x \cdot 2) dx = 4 \int_0^\pi x \sin x dx - \int_0^\pi x \cos 2x dx$$

$$\int_0^\pi x \sin x dx = \left| \begin{matrix} x = u & du = dx \\ v = -\cos x \end{matrix} \right| = -x \cos x \Big|_0^\pi + \int_0^\pi \cos x dx = \pi + \sin x \Big|_0^\pi = \pi$$

$$\int_0^\pi x \cos 2x dx = \left| \begin{matrix} x = u & du = dx \\ v = \frac{1}{2} \sin 2x \end{matrix} \right| = \frac{1}{2} x \sin 2x \Big|_0^\pi - \frac{1}{2} \int_0^\pi \sin 2x dx = 0 + \frac{1}{4} \cos 2x \Big|_0^\pi = 0$$

$W = 4\pi - 0 = 4\pi$

39) $\gamma = f(x, y) = e^x \cdot y$, $L: x = \ln(1+t^2)$, $y = 2\arctan t - t$ ($0 \leq t \leq 1$)

$$m = \int_L \gamma ds = \int_L e^x y \cdot ds, \quad ds = \sqrt{x'^2 + y'^2} dt =$$

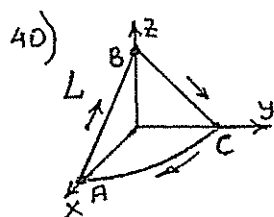
$$= \sqrt{\left(\frac{2t}{1+t^2}\right)^2 + \left(\frac{2}{1+t^2} - 1\right)^2} dt = \sqrt{\frac{4t^2}{(1+t^2)^2} + \frac{(1-t^2)^2}{(1+t^2)^2}} dt =$$

$$= \frac{1}{1+t^2} \sqrt{4t^2 + (1-t^2)^2} dt = \frac{1}{1+t^2} \sqrt{1+2t^2+t^4} dt = \frac{\sqrt{(1+t^2)^2}}{1+t^2} dt = dt$$

$$m = \int_L \frac{y}{e^x} dt = \int_0^1 \frac{2\arctan t - t}{e^{\ln(1+t^2)}} dt = \int_0^1 \frac{2\arctan t - t}{1+t^2} dt =$$

$$= 2 \int_0^1 \frac{\arctan t}{1+t^2} dt - \int_0^1 \frac{t dt}{1+t^2} = 2 \int_0^1 \arctan t d(\arctan t) -$$

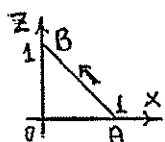
$$- \frac{1}{2} \int_0^1 \frac{d(1+t^2)}{1+t^2} = (\arctan t)^2 \Big|_0^1 - \frac{1}{2} \ln(1+t^2) \Big|_0^1 = \frac{\pi^2}{16} - \frac{1}{2} \ln 2$$



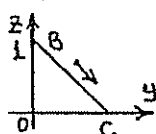
$$\vec{F} = xy \cos xz \vec{i} + \cos(xz) \vec{j} + xy \vec{k}$$

$$W = \int_L \vec{F} d\vec{r} = \int_L xy \cos xz dx + \cos(xz) dy + xy dz$$

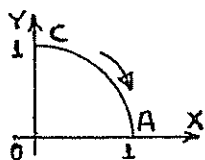
$$\int_L \vec{F} d\vec{r} = \int_{AB} \vec{F} d\vec{r} + \int_{BC} \vec{F} d\vec{r} + \int_{CA} \vec{F} d\vec{r}$$



$$\underline{AB}: y=0, dy=0 \Rightarrow \int_{AB} \vec{F} d\vec{r} = 0$$



$$\underline{BC}: x=0, dx=0, 0 \leq y \leq 1 \Rightarrow \int_{BC} \vec{F} d\vec{r} = \int_0^1 \cos 0 \cdot dy = \int_0^1 dy = y \Big|_0^1 = 1$$



$$\underline{AC}: z=0, dz=0, 0 \leq x \leq 1, 0 \leq y \leq 1, x^2 + y^2 = 1$$

$$\int_{CA} \vec{F} d\vec{r} = \int_{CA} xy dx + dy = \int_{CA} xy dx + \int_{CA} dy =$$

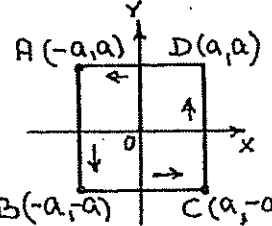
$$= \int_0^1 x \sqrt{1-x^2} dx + \int_1^0 dy = -\frac{1}{2} \cdot \int_0^1 (1-x^2)^{1/2} d(1-x^2) + y \Big|_1^0 =$$

$$= -\frac{1}{2} \cdot \frac{2}{3} (1-x^2)^{3/2} \Big|_0^1 - 1 = -\frac{1}{3} (0-1) - 1 = -\frac{2}{3}$$

$$W = 0 + 1 - \frac{2}{3} = \frac{1}{3}$$

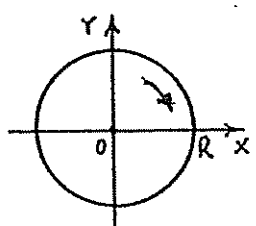
משפט גרין ואי-תלות של אינטגרל קווי במסלול האינטגרציה

(*)

1)  $\oint_L (x-y)dx + xdy$, $L: (ABCD \text{ חל'ה})$

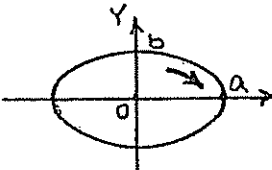
$$\oint_L (x-y)dx + xdy = \iint_D \left(\frac{\partial x}{\partial x} - \frac{\partial (x-y)}{\partial y} \right) dx dy =$$

$$= \iint_D 2 dx dy = 2 \iint_D dx dy = 2 \cdot S_{ABCD} = 2(2a)^2 = 8a^2$$

2)  $\oint_L (x-y)dx + (x+y)dy$, $L: x^2 + y^2 = R^2$

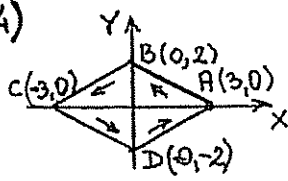
$$\oint_L (x-y)dx + (x+y)dy = - \iint_D \left[\frac{\partial (x+y)}{\partial x} - \frac{\partial (x-y)}{\partial y} \right] dS =$$

$$= - \iint_D 2 dS = -2 \iint_D dS = -2 S_D = -2\pi R^2$$

3)  $\oint_L (x+y)dx - (x-y)dy$, $L: \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

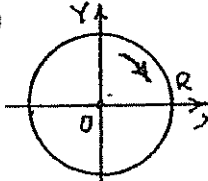
$$\oint_L (x+y)dx - (x-y)dy = - \iint_D \left(\frac{\partial (y-x)}{\partial x} - \frac{\partial (x+y)}{\partial y} \right) dS =$$

$$= - \iint_D (-2) dS = 2 \iint_D dS = 2 S_D = 2 \cdot \pi a \cdot b = 2\pi ab$$

4)  $\oint_L y dx + 2x dy$, $L: (ABCD \text{ חל'ה})$

$$\oint_L y dx + 2x dy = \iint_D \left(\frac{\partial (2x)}{\partial x} - \frac{\partial y}{\partial y} \right) dS = \iint_D (2-1) dS =$$

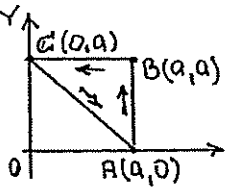
$$= \iint_D dS = S_D = \frac{1}{2} AC \cdot BD = \frac{1}{2} 6 \cdot 4 = 12$$

5)  $I = \oint_L -x^2y dx + xy^2 dy$, $L: x^2 + y^2 = R^2$

$$I = - \iint_D \left(\frac{\partial (xy^2)}{\partial x} - \frac{\partial (-x^2y)}{\partial y} \right) dS = - \iint_D (x^2 + y^2) dx dy =$$

$$= \left| \begin{matrix} x = \rho \cos \varphi \\ y = \rho \sin \varphi \end{matrix} \right| = - \iint_D \rho^2 \rho d\rho d\varphi = - \int_0^{2\pi} d\varphi \int_0^R \rho^3 d\rho = - \frac{1}{4} \int_0^{2\pi} d\varphi (\rho^4 \Big|_0^R) =$$

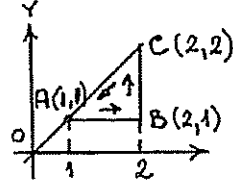
$$= - \frac{1}{4} R^4 \int_0^{2\pi} d\varphi = - \frac{1}{4} R^4 \varphi \Big|_0^{2\pi} = - \frac{1}{2} \pi R^4$$

6)  $\oint_L y^2 dx + (x+y)^2 dy$, $L: (ABC \text{ e } \delta/\text{err})$

$$\oint_L y^2 dx + (x+y)^2 dy = \iint_D \left(\frac{\partial (x+y)^2}{\partial x} - \frac{\partial y^2}{\partial y} \right) dS =$$

$$= \iint_D (2x+2y-2y) dS = 2 \iint_D x dx dy = \left| \begin{array}{l} AC: y=-x+a \\ CB: y=a \end{array} \right| = 2 \int_0^a dx \int_{a-x}^a x dy =$$

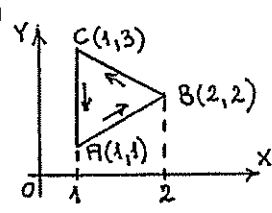
$$= 2 \int_0^a x (y|_{a-x}^a) dx = 2 \int_0^a x (a-a+x) dx = 2 \int_0^a x^2 dx = \frac{2}{3} x^3 \Big|_0^a = \frac{2}{3} a^3$$

7)  $\oint_L \frac{dx}{y} - \frac{dy}{x}$, $L: (ABC \text{ e } \delta/\text{err})$

$$\oint_L \frac{dx}{y} - \frac{dy}{x} = \iint_D \left[\frac{\partial}{\partial x} \left(-\frac{1}{x} \right) - \frac{\partial}{\partial y} \left(\frac{1}{y} \right) \right] dS = \iint_D \left(\frac{1}{x^2} + \frac{1}{y^2} \right) dS =$$

$$= \left| \begin{array}{l} AC: y=x \\ AB: y=1 \end{array} \right| = \int_1^2 dx \int_1^x \left(\frac{1}{x^2} + \frac{1}{y^2} \right) dy = \int_1^2 dx \left[\frac{1}{x^2} y - \frac{1}{y} \right]_1^x =$$

$$= \int_1^2 dx \left[\frac{1}{x} - \frac{1}{x} - \frac{1}{x^2} + 1 \right] = \int_1^2 \left(1 - \frac{1}{x^2} \right) dx = \left(x + \frac{1}{x} \right) \Big|_1^2 = 2 + \frac{1}{2} - 1 - 1 = \frac{1}{2}$$

8)  $I = \oint_L 2(x^2+y^2) dx + (x+y)^2 dy$, $L: (ABC \text{ e } \delta/\text{err})$

$$\oint_L 2(x^2+y^2) dx + (x+y)^2 dy = \iint_D \left[\frac{\partial (x+y)^2}{\partial x} - \frac{\partial (2x^2+2y^2)}{\partial y} \right] dS =$$

$$= \iint_D (2x+2y-4y) dS = 2 \iint_D (x-y) dS$$

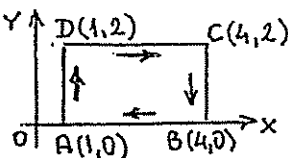
$AB: m = \frac{2-1}{2-1} = 1 \Rightarrow y-1 = 1 \cdot (x-1) \Rightarrow y = x$

$CB: m = \frac{2-3}{2-1} = -1 \Rightarrow y-3 = -1 \cdot (x-1) \Rightarrow y = -x+4$

$$I = 2 \int_1^2 dx \int_x^{4-x} (x-y) dy = 2 \int_1^2 dx \left[xy - \frac{y^2}{2} \right]_x^{4-x} = 2 \int_1^2 dx \left[x(4-x) - \frac{1}{2}(4-x)^2 - x^2 + \frac{x^2}{2} \right] =$$

$$= 2 \int_1^2 (-2x^2 + 8x - 8) dx = -4 \int_1^2 (x^2 - 4x + 4) dx = -4 \int_1^2 (x-2)^2 d(x-2) =$$

$$= -4 \cdot \frac{1}{3} (x-2)^3 \Big|_1^2 = -\frac{4}{3} [0 - (-1)^3] = -\frac{4}{3}$$

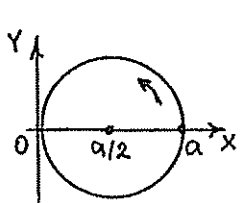
9)  $I = \oint_L \sqrt{x^2+y^2} dx + y[xy + \ln(x+\sqrt{x^2+y^2})] dy,$
 $L: (ABCD) \text{ по } \delta n$

$$\oint_L P dx + Q dy = - \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy$$

$$\frac{\partial Q}{\partial x} = y \left[y + \frac{1}{x + \sqrt{x^2+y^2}} \left(1 + \frac{x}{\sqrt{x^2+y^2}} \right) \right] = y \left[y + \frac{\sqrt{x^2+y^2} + x}{x + \sqrt{x^2+y^2}} \cdot \frac{1}{\sqrt{x^2+y^2}} \right] =$$

$$= y^2 + \frac{y}{\sqrt{x^2+y^2}}, \quad \frac{\partial P}{\partial y} = \frac{y}{\sqrt{x^2+y^2}} \Rightarrow \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = y^2$$

$$I = - \iint_D y^2 dx dy = - \int_1^4 dx \int_0^2 y^2 dy = - \frac{1}{3} \int_1^4 dx (y^3 \Big|_0^2) = - \frac{8}{3} \int_1^4 dx = - \frac{8}{3} x \Big|_1^4 = -8$$

10)  $I = \oint_L (xy + x + y) dx + (xy + x - y) dy, \quad L: x^2 + y^2 = ax$

$$I = \iint_D \left[\frac{\partial}{\partial x} (xy + x - y) - \frac{\partial}{\partial y} (xy + x + y) \right] dS =$$

$$= \iint_D (y + 1 - x - 1) dS = \iint_D (y - x) dS = \left| \begin{array}{l} x = \rho \cos \varphi \\ y = \rho \sin \varphi \end{array} \quad \begin{array}{l} x^2 + y^2 = ax \Rightarrow \rho = a \cos \varphi \\ -\frac{\pi}{2} \leq \varphi \leq \frac{\pi}{2} \end{array} \right| =$$

$$= \int_{-\pi/2}^{\pi/2} d\varphi \int_0^{a \cos \varphi} (\rho \sin \varphi - \rho \cos \varphi) \rho d\rho = \int_{-\pi/2}^{\pi/2} (\sin \varphi - \cos \varphi) d\varphi \int_0^{a \cos \varphi} \rho^2 d\rho =$$

$$= \frac{1}{3} \int_{-\pi/2}^{\pi/2} (\sin \varphi - \cos \varphi) d\varphi \left(\rho^3 \Big|_0^{a \cos \varphi} \right) = \frac{a^3}{3} \int_{-\pi/2}^{\pi/2} (\sin \varphi - \cos \varphi) \cos^3 \varphi d\varphi =$$

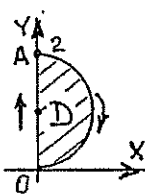
$$= -\frac{a^3}{3} \int_{-\pi/2}^{\pi/2} \cos^3 \varphi d(\cos \varphi) - \frac{a^3}{3} \int_{-\pi/2}^{\pi/2} \cos^4 \varphi d\varphi = -\frac{a^3}{3} \cdot \frac{1}{4} \cos^4 \varphi \Big|_{-\pi/2}^{\pi/2} - \frac{a^3}{12} \int_{-\pi/2}^{\pi/2} (2 \cos^2 \varphi)^2 d\varphi =$$

$$= 0 - \frac{a^3}{12} \int_{-\pi/2}^{\pi/2} (1 + \cos 2\varphi)^2 d\varphi = -\frac{a^3}{12} \left[\int_{-\pi/2}^{\pi/2} d\varphi + 2 \int_{-\pi/2}^{\pi/2} \cos 2\varphi d\varphi + \int_{-\pi/2}^{\pi/2} \cos^2 2\varphi d\varphi \right] =$$

$$= -\frac{a^3}{12} \left[\varphi \Big|_{-\pi/2}^{\pi/2} + \sin 2\varphi \Big|_{-\pi/2}^{\pi/2} + \frac{1}{2} \int_{-\pi/2}^{\pi/2} (1 + \cos 4\varphi) d\varphi \right] =$$

$$= -\frac{a^3}{12} \left[\pi + 0 + \frac{1}{2} \left(\varphi \Big|_{-\pi/2}^{\pi/2} + \frac{1}{4} \sin 4\varphi \Big|_{-\pi/2}^{\pi/2} \right) \right] = -\frac{a^3}{12} \left(\pi + \frac{1}{2} \pi + 0 \right) = -\frac{\pi a^3}{8}$$

(1)

1)  $I = \int_L (y + e^{x^2}) dx + (2x + ye^{y^2}) dy, L: x^2 + y^2 = 2y$
 $A(0,2), O(0,0)$
 . (x=0) OA חגורת 'ג'ל-סח L סלסומח תח "רלסח"

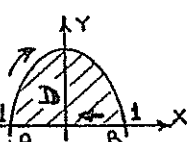
$$\oint = I + \int_{OA} \Rightarrow \oint = \oint_{LUOA} (y + e^{x^2}) dx + (2x + ye^{y^2}) dy =$$

$$= - \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy = - \iint_D (2-1) dx dy = - \iint_D dx dy = - S_D = - \frac{1}{2} \pi \cdot 1^2 = - \frac{\pi}{2}$$

. (1 סלסח סחח סלסח-לסח סח חלס חלסחח חלסחח חלסחח חלסחח חלסחחח)

$$\int_{OA} = \int_{OA} (y + e^{x^2}) dx + (2x + ye^{y^2}) dy = \left| \frac{x^2}{2} \right|_{x=0}^x = \int_0^2 ye^{y^2} dy = \frac{1}{2} \int_0^2 e^{y^2} d(y^2) =$$

$$= \frac{1}{2} e^{y^2} \Big|_0^2 = \frac{1}{2} (e^4 - 1) \Rightarrow -\frac{\pi}{2} = I + \frac{1}{2} (e^4 - 1) \Rightarrow I = \frac{1}{2} (1 - e^4) - \frac{\pi}{2}$$

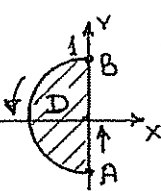
2)  $\vec{F} = (2xy + e^{y^2}) \vec{i} + x(x + 2ye^{y^2} + 1) \vec{j}, L: y = 1 - x^2, A(-1,0)$
 $B(1,0)$
 . (y=0) BA חגורת 'ג'ל-סח L סלסומח תח "רלסח"

$$I = \int_L \vec{F} d\vec{r} \Rightarrow \oint = I + \int_{BA} \Rightarrow \oint = \oint_{LUBA} \vec{F} d\vec{r} = - \iint_D \left(\frac{\partial Q}{\partial x} + \frac{\partial P}{\partial y} \right) dx dy =$$

$$= - \iint_D (2x + 2ye^{y^2} + 1 - 2x - 2ye^{y^2}) dx dy = - \iint_D dx dy = - S_D = - 2 \int_0^1 (1 - x^2) dx =$$

$$= - 2 \left(x - \frac{x^3}{3} \right) \Big|_0^1 = - \frac{4}{3}$$

$$\int_{BA} = \int_{BA} \vec{F} d\vec{r} = \left| \frac{y^2}{2} \right|_{y=0}^y = \int_{-1}^1 e^0 dx = - 2 \Rightarrow - \frac{4}{3} = I - 2 \Rightarrow I = \frac{2}{3}$$

3)  $I = \int_L \vec{F} d\vec{r}, \vec{F} = \frac{x+2}{\sqrt{(x+2)^2 + y^2}} \vec{i} + \left(\frac{y}{\sqrt{(x+2)^2 + y^2}} + 2x \right) \vec{j}, L: x^2 + y^2 = 1$
 $B(0,1), A(0,-1)$
 . (x=0) AB חגורת 'ג'ל-סח L סלסומח תח "רלסח"

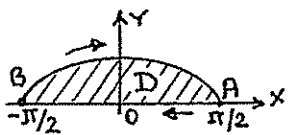
$$\oint = I + \int_{AB} \Rightarrow \oint = \oint_{LUAB} \vec{F} d\vec{r} = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy.$$

$$\frac{\partial Q}{\partial x} = - \frac{1}{2} \cdot \frac{y \cdot 2(x+2)}{[(x+2)^2 + y^2]^{3/2}} + 2, \frac{\partial P}{\partial y} = - \frac{1}{2} \cdot \frac{(x+2) \cdot 2y}{[(x+2)^2 + y^2]^{3/2}} \Rightarrow \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 2$$

$$\oint = \iint_D 2 dx dy = 2 S_D = 2 \cdot \frac{1}{2} \pi \cdot 1^2 = \pi, \int_{AB} = \int_{AB} \vec{F} d\vec{r} = \left| \frac{y^2}{2} \right|_{y=-1}^y = \int_{-1}^1 \frac{y dy}{\sqrt{4+y^2}} = 0$$

. (0-ס חלס חלסח'ס חלסחח חלסחח-חלס חלסחח'ס חלסחח חלסחחח)

$$\pi = I + 0 \Rightarrow I = \pi$$

4)  $I = \int_L (2y \cos^2 x + 2xy e^{x^2 y}) dx + x^2 e^{x^2 y} dy$
 $L: y = \cos x, x \in [-\pi/2, \pi/2]$
 $(y=0) \text{ AB } \text{ וְהָיָה 'ג'־סֵף } L \text{ סִבֵּסוֹנָה בַּת "רֵזֶם"}$

$$\oint = I + \int_{AB} \Rightarrow \oint = \int_{L \cup AB} (2y \cos^2 x + 2xy e^{x^2 y}) dx + x^2 e^{x^2 y} dy = - \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy$$

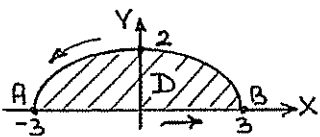
$$\oint = - \iint_D (2x e^{x^2 y} + 2x^3 y e^{x^2 y} - 2 \cos^2 x - 2x e^{x^2 y} - 2x^3 y e^{x^2 y}) dx dy =$$

$$= 2 \iint_D \cos^2 x dx dy = 2 \int_{-\pi/2}^{\pi/2} \cos^2 x dx \int_0^{\cos x} dy = 2 \int_{-\pi/2}^{\pi/2} \cos^2 x (y|_0^{\cos x}) dx =$$

$$= 2 \int_{-\pi/2}^{\pi/2} \cos^3 x dx = 2 \cdot 2 \int_0^{\pi/2} \cos^3 x dx = 4 \int_0^{\pi/2} (1 - \sin^2 x) d(\sin x) =$$

$$= 4 \left(\sin x - \frac{\sin^3 x}{3} \right) \Big|_0^{\pi/2} = 4(1 - 1/3) = 8/3$$

$$\int_{AB} = \int_{AB} (2y \cos^2 x + 2xy e^{x^2 y}) dx + x^2 e^{x^2 y} dy = \Big|_{y=0}^{y=0} = 0 \Rightarrow \frac{8}{3} = I + 0 \Rightarrow I = \frac{8}{3}$$

5)  $I = \int_L \vec{F} d\vec{r}, \vec{F} = y^2 \vec{i} + (x^2 + \sin(y^2)) \vec{j}$
 $L: \frac{x^2}{9} + \frac{y^2}{4} = 1, y \geq 0$
 $(y=0) \text{ AB } \text{ וְהָיָה 'ג'־סֵף } L \text{ סִבֵּסוֹנָה בַּת "רֵזֶם"}$

$$\oint = I + \int_{AB} \Rightarrow \oint = \oint_{L \cup AB} \vec{F} d\vec{r} = \oint_{L \cup AB} y^2 dx + (x^2 + \sin(y^2)) dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy$$

$$\oint = \iint_D (2x - 2y) dx dy$$

$r = ap \cos \varphi, y = bp \sin \varphi, x = ap \cos \varphi$: אִלְטֵסֶה אֶלְבִּילְתִּי אֶלְבִּילְתִּי אֶלְבִּילְתִּי
 $x = 3p \cos \varphi, y = 2p \sin \varphi, r = 6p$

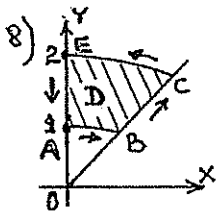
$$\frac{x^2}{9} + \frac{y^2}{4} = 1 \Rightarrow \frac{9p^2 \cos^2 \varphi}{9} + \frac{4p^2 \sin^2 \varphi}{4} = 1 \Rightarrow p = 1$$

$D_{p\varphi} = \{(p, \varphi) | 0 \leq p \leq 1, 0 \leq \varphi < \pi\}$: אִלְטֵסֶה אֶלְבִּילְתִּי אֶלְבִּילְתִּי אֶלְבִּילְתִּי

$$\oint = 2 \iint_{D_{p\varphi}} (3p \cos \varphi - 2p \sin \varphi) 6p d\varphi dp = 12 \int_0^{\pi} (3 \cos \varphi - 2 \sin \varphi) d\varphi \int_0^1 p^2 dp =$$

$$= 12 \int_0^{\pi} (3 \cos \varphi - 2 \sin \varphi) d\varphi \left(\frac{p^3}{3} \Big|_0^1 \right) = 4 \int_0^{\pi} (3 \cos \varphi - 2 \sin \varphi) d\varphi = 4(3 \sin \varphi + 2 \cos \varphi) \Big|_0^{\pi} = -16$$

$$\int_{AB} = \int_{AB} y^2 dx + (x^2 + \sin(y^2)) dy = \Big|_{y=0}^{y=0} = 0 \Rightarrow -16 = I + 0 \Rightarrow I = -16$$



$$I = \int_L \vec{F} d\vec{r}, \quad \vec{F} = (x^2 \sin x - 2y^2) \vec{i} + (x^2 + \cos \frac{\pi y}{2}) \vec{j}$$

$$L: x^2 + 2y^2 = 2, \quad y = x, \quad x^2 + 2y^2 = 8$$

ආකෘතිගත කිරීමේදී, ඛණ්ඩාංක මත පදනම්ව '3' ට

$$\frac{x^2}{2} + \frac{y^2}{1} = 1, \quad \frac{x^2}{8} + \frac{y^2}{4} = 1$$

.(x=0) EA ඡේදන '3'-යෙහි දිශාවට මෙ "නැවතුනු"

$$\oint_{\partial D} \vec{F} \cdot d\vec{r} = \oint_{\partial D} (x^2 \sin x - 2y^2) dx + (x^2 + \cos \frac{\pi y}{2}) dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy$$

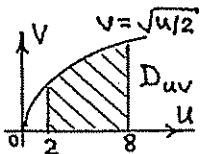
$$\oint_{\partial D} = \iint_D (2x + 4y) dx dy$$

ආකෘතිගත කිරීමේදී, ඛණ්ඩාංක මත පදනම්ව

$$x^2 + 2y^2 = 2, \quad x^2 + 2y^2 = 8 \Rightarrow x^2 + 2y^2 = u, \quad 2 \leq u \leq 8$$

$$y - x = 0 \Rightarrow y - x = v \Rightarrow v = 0$$

$$\begin{cases} x^2 + 2y^2 = u \\ y - x = v \end{cases} \Rightarrow x = 0 \Rightarrow \begin{cases} 2y^2 = u \\ y = v \end{cases} \Rightarrow v = \sqrt{\frac{u}{2}} \quad (y > 0) \rightarrow (v > 0)$$



$$\frac{1}{J} = \begin{vmatrix} u'_x & v'_x \\ u'_y & v'_y \end{vmatrix} = \begin{vmatrix} 2x & -1 \\ 4y & 1 \end{vmatrix} = 2x + 4y \Rightarrow J = \frac{1}{2(x+2y)}$$

$$\oint_{\partial D} = \iint_{D_{uv}} (2x + 4y) \frac{1}{2(x+2y)} du dv = \iint_{D_{uv}} du dv = \int_{D_{uv}} du dv = \int_2^8 \int_0^{\sqrt{u/2}} du dv =$$

$$= \frac{1}{\sqrt{2}} \cdot \frac{2}{3} u^{3/2} \Big|_2^8 = \frac{\sqrt{2}}{3} (8^{3/2} - 2^{3/2}) = \frac{\sqrt{2}}{3} 2^{3/2} (8 - 1) = \frac{4}{3} \cdot 7 = \frac{28}{3}$$

$$\int_{EA} = \int_{EA} (x^2 \sin x - 2y^2) dx + (x^2 + \cos \frac{\pi y}{2}) dy = \Big|_{dx=0}^{x=0} = \int_2^1 \cos \frac{\pi y}{2} dy =$$

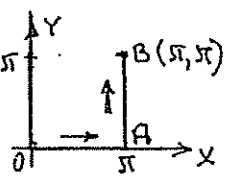
$$= \frac{2}{\pi} \sin \frac{\pi y}{2} \Big|_2^1 = \frac{2}{\pi} (\sin \frac{\pi}{2} - \sin \pi) = \frac{2}{\pi} \Rightarrow \frac{28}{3} = I + \frac{2}{\pi} \Rightarrow I = \frac{28}{3} - \frac{2}{\pi}$$

(λ)

1) $I = \int_{AB} (y \cos(xy) - \cos y) dx + (x \cos(xy) - x \sin y) dy$, $A(-1,0)$, $B(1,0)$

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = [\cos(xy) - xy \sin(xy) - \sin y] - [\cos(xy) - xy \sin xy + \sin y] = -2 \sin y \neq 0$$

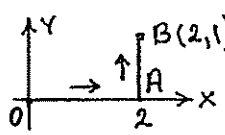
ה'צגות'הן ש'סומא'ל'סנ'סג'הן

2)  $I = \int_{AB} (x+y) dx + (x-y) dy$, $O(0,0)$, $B(\pi, \pi)$

ה'צגות'הן ש'סומא'ל'סנ'סג' $\Leftrightarrow \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 1-1=0$

$$\int_{OB} = \int_{OA} + \int_{AB} \rightarrow \underline{OA}: y=0, dy=0 \Rightarrow \int_{OA} = \int_0^{\pi} x dx = \frac{1}{2} x^2 \Big|_0^{\pi} = \frac{\pi^2}{2}$$

$$\underline{AB}: x=\pi, dx=0 \Rightarrow \int_{AB} = \int_0^{\pi} (\pi-y) dy = \left(\pi y - \frac{y^2}{2} \right) \Big|_0^{\pi} = \frac{\pi^2}{2} \Rightarrow I = \frac{\pi^2}{2} + \frac{\pi^2}{2} = \pi^2$$

3)  $I = \int_{AB} 2xy dx + x^2 dy$, $O(0,0)$, $B(2,1)$

ה'צגות'הן ש'סומא'ל'סנ'סג' $\Leftrightarrow \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 2y-2y=0$

$$\int_{OB} = \int_{OA} + \int_{AB} \rightarrow \underline{OA}: y=0 \Rightarrow dy=0, 2xy=0 \Rightarrow \int_{OA} = 0$$

$$\underline{AB}: x=2 \Rightarrow dx=0 \Rightarrow \int_{AB} = \int_0^1 2^2 dy = 4y \Big|_0^1 = 4 \Rightarrow I = 0+4=4$$

4) $I = \int_{AB} \left(\frac{1}{x+y} + \frac{x}{y} \right) dx + \left(\frac{1}{x+y} - \frac{x^2}{y^2} \right) dy$, $x+y \neq 0, y \neq 0$, $A(2,1)$, $B(3,5)$

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = \left[-\frac{1}{(x+y)^2} - \frac{2x}{y^2} \right] - \left[-\frac{1}{(x+y)^2} - \frac{x}{y^2} \right] = -\frac{2x}{y^2} + \frac{x}{y^2} = -\frac{x}{y^2} \neq 0$$

ה'צגות'הן ש'סומא'ל'סנ'סג'הן

5) $I = \int_{AB} \left(\frac{y}{x^2+y^2} - \frac{y}{x^2} \right) dx - \left(\frac{x}{x^2+y^2} + \frac{1}{x} \right) dy$, $(x,y) \neq (0,0), x \neq 0$, $A(2,1)$, $B(1,0)$

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = \left[-\left(\frac{x^2+y^2-2x^2}{(x^2+y^2)^2} - \frac{1}{x^2} \right) \right] - \left[\frac{x^2+y^2-2y^2}{(x^2+y^2)^2} - \frac{1}{x^2} \right] =$$

$$= \frac{x^2-y^2}{(x^2+y^2)^2} + \frac{1}{x^2} - \frac{x^2-y^2}{(x^2+y^2)^2} + \frac{1}{x^2} = \frac{2}{x^2} \neq 0$$

ה'צגות'הן ש'סומא'ל'סנ'סג'הן

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 0 - 0 = 0$$

$$u = \int_0^x x \cos x dx + \int_0^y dy = \int_0^x x \cos x dx + y \Big|_0^y = \left| \begin{array}{l} x=u \\ \cos x dx = dv \\ v = \sin x \end{array} \right| =$$

$$u = x \sin x + \cos x + y + C$$

$$^A \quad -(2\pi \sin 2\pi + \cos 2\pi + 1) = (-1-1) - (1+1) = -4$$

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = (2x e^{xy} + x^2 y e^{xy}) - (x e^{xy} + (1+x y) \cdot x e^{xy}) = 0$$

$$W = \int_0^y x^2 e^{xy} dy + \int_0^x e^0 dx = x \int_0^y e^{xy} x dy + x \Big|_0^x = x \int_0^y e^{xy} d(xy) + x =$$

$$u = x e^{xy} + C$$

$$\int_A^B F dz = u(B) - u(A) = u(2,0) - u(0,1) = 2e^0 - 0e^0 = 2$$

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = (\ln x - 1 + 1) - \ln x = 0$$

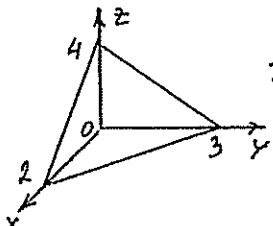
$$U = \int_0^y x(\ln x - 1) dy + \int_1^x 0 dx = x(\ln x - 1) \int_0^y dy = x(\ln x - 1) y \Big|_0^y \Rightarrow$$

$$u = xy(\ln x - 1) + C, \quad x \neq 0$$

$$\int_B F dF = U(B) - U(A) = U(e, 2\pi) - U(1, \pi) = e \cdot 2\pi (\ln e - 1) -$$

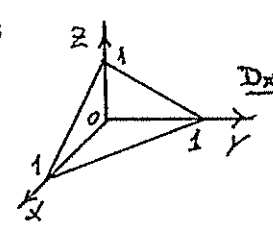
$$-1.5\pi(\ln 1 - 1) = 0 - \pi(-1) = \pi$$

האינטגרל המשטחי

1)  $I = \iint_S (z+2x+\frac{4y}{3}) dS, S: 6x+4y+3z=12$
 $\begin{cases} 6x+4y+3z=12 \\ z=0 \end{cases} \Rightarrow y = -\frac{3}{2}x+3$
 $z = 4-2x-\frac{4}{3}y \Rightarrow z'_x = -2, z'_y = -\frac{4}{3}$

$$dS = \sqrt{1+z'^2_x+z'^2_y} dx dy = \sqrt{1+2^2+(\frac{4}{3})^2} dx dy = \frac{\sqrt{61}}{3} dx dy$$

$$I = \iint_{D_{xy}} [(4-2x-\frac{4}{3}y)+2x+\frac{4}{3}y] \frac{\sqrt{61}}{3} dx dy = \frac{4}{3} \sqrt{61} \iint_{D_{xy}} dx dy = \frac{4}{3} \sqrt{61} S_{\Delta} = \frac{4}{3} \sqrt{61} \cdot \frac{1}{2} \cdot 2 \cdot 3 = 4\sqrt{61}$$

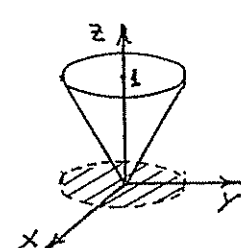
2)  $I = \iint_S xy z dS, S: x+y+z=1$
 $\begin{cases} x+y+z=1 \\ z=0 \end{cases} \Rightarrow y = -x+1$
 $z = 1-x-y, z'_x = -1, z'_y = -1$

$$dS = \sqrt{1+(-1)^2+(-1)^2} dx dy = \sqrt{3} dx dy \Rightarrow I = \iint_{D_{xy}} xy(1-x-y) \sqrt{3} dx dy =$$

$$= \sqrt{3} \int_0^1 x dx \int_0^{1-x} [(1-x)y - y^2] dy = \sqrt{3} \int_0^1 x dx \left[(1-x) \frac{y^2}{2} - \frac{y^3}{3} \right] \Big|_0^{1-x} =$$

$$= \sqrt{3} \int_0^1 x dx \left[\frac{1}{2}(1-x)^3 - \frac{1}{3}(1-x)^3 \right] = \frac{\sqrt{3}}{6} \int_0^1 x(1-x)^3 dx = \frac{\sqrt{3}}{6} \int_0^1 (x-3x^2+3x^3-x^4) dx =$$

$$= \frac{\sqrt{3}}{6} \left(\frac{x^2}{2} - x^3 + \frac{3}{4}x^4 - \frac{x^5}{5} \right) \Big|_0^1 = \frac{\sqrt{3}}{6} \left(\frac{1}{2} - 1 + \frac{3}{4} - \frac{1}{5} \right) = \frac{\sqrt{3}}{120}$$

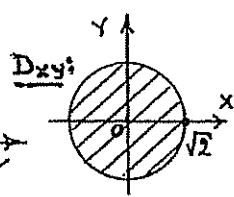
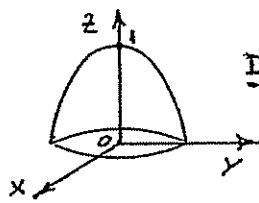
3)  $I = \iint_S (x^2+y^2+3z^2) dS, S: z=\sqrt{x^2+y^2}$
 $\begin{cases} z=\sqrt{x^2+y^2} \\ z=1 \end{cases} \Rightarrow x^2+y^2=1$
 $z'_x = \frac{x}{\sqrt{x^2+y^2}}, z'_y = \frac{y}{\sqrt{x^2+y^2}}$

$$dS = \sqrt{1 + \frac{x^2}{x^2+y^2} + \frac{y^2}{x^2+y^2}} dx dy = \sqrt{\frac{2(x^2+y^2)}{x^2+y^2}} dx dy = \sqrt{2} dx dy$$

$$I = \iint_{D_{xy}} (x^2+y^2+3x^2+3y^2) \sqrt{2} dx dy = 4\sqrt{2} \iint_{D_{xy}} (x^2+y^2) dx dy = \left| \begin{matrix} x = \rho \cos \varphi \\ y = \rho \sin \varphi \end{matrix} \right| =$$

$$= 4\sqrt{2} \int_0^{2\pi} \int_0^1 \rho^2 \rho d\rho d\varphi = 4\sqrt{2} \int_0^{2\pi} d\varphi \int_0^1 \rho^3 d\rho = \sqrt{2} \int_0^{2\pi} d\varphi (\rho^4 \Big|_0^1) = \sqrt{2} \int_0^{2\pi} d\varphi = \sqrt{2} \varphi \Big|_0^{2\pi} = 2\sqrt{2} \pi$$

4)



$$I = \iint_S (x^2 + y^2 + z - \frac{1}{2}) dS, \quad S: z = 2 - x^2 - y^2$$

$$\begin{cases} z = 2 - x^2 - y^2 \\ z = 0 \end{cases} \Rightarrow x^2 + y^2 = 2$$

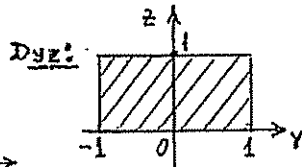
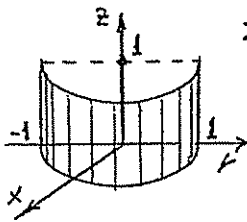
$$z'_x = -x, \quad z'_y = -y \Rightarrow dS = \sqrt{1 + x^2 + y^2} dx dy$$

$$I = \iint_{D_{xy}} \left[x^2 + y^2 + \left(2 - \frac{x^2}{2} - \frac{y^2}{2} \right) - \frac{1}{2} \right] \sqrt{1 + x^2 + y^2} dx dy = \frac{1}{2} \iint_{D_{xy}} (x^2 + y^2 + 1)^{3/2} dx dy =$$

$$= \begin{vmatrix} x = \rho \cos \varphi \\ y = \rho \sin \varphi \end{vmatrix} = \frac{1}{2} \iint_D (1 + \rho^2)^{3/2} \rho d\rho d\varphi = \frac{1}{4} \int_0^{2\pi} d\varphi \int_0^{\sqrt{2}} (1 + \rho^2)^{3/2} d(1 + \rho^2) =$$

$$= \frac{1}{4} \cdot \frac{2}{5} \int_0^{2\pi} d\varphi \left[(1 + \rho^2)^{5/2} \right]_0^{\sqrt{2}} = \frac{1}{10} (3^{5/2} - 1) \int_0^{2\pi} d\varphi = \frac{1}{10} (9\sqrt{3} - 1) \varphi \Big|_0^{2\pi} = \frac{\pi}{5} (9\sqrt{3} - 1)$$

5)



$$I = \iint_S x(y+z) dS, \quad S: x = \sqrt{1-y^2}$$

$$x = \sqrt{1-y^2} \Rightarrow (\delta'_1 \delta'_2 \delta'_3 n)$$

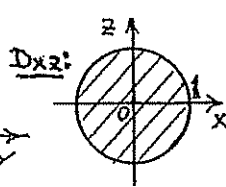
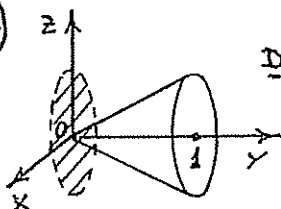
$$x'_y = -\frac{y}{\sqrt{1-y^2}}, \quad x'_z = 0$$

$$dS = \sqrt{1 + (x'_y)^2 + (x'_z)^2} dy dz = \sqrt{1 + \frac{y^2}{1-y^2}} dy dz = \frac{1}{\sqrt{1-y^2}} dy dz$$

$$I = \iint_{D_{yz}} \sqrt{1-y^2} (y+z) \frac{dy dz}{\sqrt{1-y^2}} = \iint_{D_{yz}} (y+z) dy dz = \int_{-1}^1 dy \int_0^1 (y+z) dz = \int_{-1}^1 dy \left[yz + \frac{z^2}{2} \right]_0^1 =$$

$$= \int_{-1}^1 \left(y + \frac{1}{2} \right) dy = \left(\frac{y^2}{2} + \frac{1}{2} y \right) \Big|_{-1}^1 = \frac{1}{2} + \frac{1}{2} - \frac{1}{2} + \frac{1}{2} = 1$$

6)



$$I = \iint_S (3x^2 + 5y^2 + 3z^2 - 2) dS, \quad S: y = \sqrt{x^2 + z^2}$$

$$\begin{cases} y = \sqrt{x^2 + z^2} \\ y = 1 \end{cases} \Rightarrow x^2 + z^2 = 1$$

$$dS = \sqrt{1 + \frac{x^2}{x^2 + z^2} + \frac{z^2}{x^2 + z^2}} dx dz = \sqrt{2} dx dz$$

$$I = \iint_{D_{xz}} (3x^2 + 5x^2 + 5z^2 + 3z^2 - 2) \sqrt{2} dx dz = 2\sqrt{2} \iint_{D_{xz}} (4x^2 + 4z^2 + 1) dx dz =$$

$$= \begin{vmatrix} x = \rho \cos \varphi \\ z = \rho \sin \varphi \end{vmatrix} = 2\sqrt{2} \int_0^{2\pi} d\varphi \int_0^1 (4\rho^2 - 1) \rho d\rho = 2\sqrt{2} \int_0^{2\pi} d\varphi \left[\frac{4}{3} \rho^3 - \frac{\rho^2}{2} \right]_0^1 =$$

$$= \sqrt{2} \int_0^{2\pi} d\varphi = \sqrt{2} \varphi \Big|_0^{2\pi} = 2\sqrt{2} \pi$$

$$z'_x = -\frac{x}{\sqrt{R^2 - x^2 - y^2}}, \quad z'_y = \frac{y}{\sqrt{R^2 - x^2 - y^2}} \Rightarrow dS = \sqrt{1 + z'^2_x + z'^2_y} dx dy \Rightarrow$$

$$ds = \sqrt{1 + \frac{x^2}{R^2 - x^2 + y^2} + \frac{y^2}{R^2 - x^2 - y^2}} \, dx dy = \frac{R \, dx \, dy}{\sqrt{R^2 - x^2 - y^2}} \Rightarrow$$

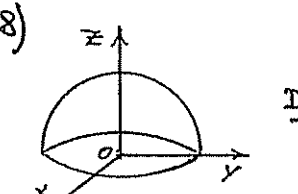
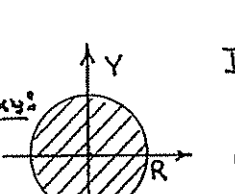
$$I_1 = \iint_D (x^2 + y^2) \frac{R dx dy}{\sqrt{R^2 - x^2 - y^2}} = \int_0^{2\pi} \int_0^R \frac{\rho^2}{\sqrt{R^2 - \rho^2}} d\rho d\varphi =$$

$$= -R \int_0^R \frac{R^2 - \rho^2 - R^2}{\sqrt{R^2 - \rho^2}} \rho d\rho \int_0^{2\pi} d\varphi = -R \int_0^R \left(\sqrt{R^2 - \rho^2} - \frac{R^2}{\sqrt{R^2 - \rho^2}} \right) \rho d\rho \left(\varphi \Big|_0^{2\pi} \right) =$$

$$= \pi R \int_0^R \sqrt{R^2 - \rho^2} d(R^2 - \rho^2) - \pi R^3 \int_0^R \frac{d(R^2 - \rho^2)}{\sqrt{R^2 - \rho^2}} = \pi R \cdot \frac{2}{3} (R^2 - \rho^2)^{\frac{3}{2}} \Big|_0^R -$$

$$-2\pi R^3 \sqrt{R^2 - p^2} \Big|_0^R = -\frac{2}{3}\pi R^4 + 2\pi R^4 = \frac{4}{3}\pi R^4.$$

$I = 2I_1 = \frac{8}{3} \pi R^4$: ଧ୍ରୁବ, ଶାଖିକା ତଥାଓ ଅନ୍ୟାନ୍ୟ ନିର୍ଦ୍ଦିଷ୍ଟ ଦିଗରେ ବ୍ୟବହୃତ।

8)  

$$I = \iint_S (x^2 + y^2 + z) dS, \quad S: z = \sqrt{R^2 - x^2 - y^2}$$

$$dS = \sqrt{1 + z_x'^2 + z_y'^2} dx dy = \frac{R}{\sqrt{R^2 - x^2 - y^2}} dx dy$$

$$I = \iint_{D_{xy}} (x^2 + y^2 + \sqrt{R^2 - x^2 - y^2}) \frac{R}{\sqrt{R^2 - x^2 - y^2}} dx dy =$$

$$= \left| \begin{matrix} x = \rho \cos \varphi \\ y = \rho \sin \varphi \end{matrix} \right| = R \iint_D \left(\frac{\rho^2}{\sqrt{R^2 - \rho^2}} + 1 \right) \rho d\rho d\varphi = R \int_0^R \left(1 + \frac{\rho^2}{\sqrt{R^2 - \rho^2}} \right) \rho d\rho \int_0^{2\pi} d\varphi =$$

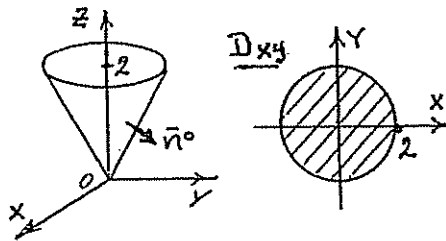
$$= R \int_0^R \left[\rho - \frac{R^2 - \rho^2 - R^2}{\sqrt{R^2 - \rho^2}} \rho \right] d\rho \left(\varphi \Big|_0^{2\pi} \right) = 2\pi R \int_0^R \rho d\rho - 2\pi R \int_0^R \left[\sqrt{R^2 - \rho^2} - \frac{R^2}{\sqrt{R^2 - \rho^2}} \right] \rho d\rho =$$

$$= \pi R \rho^2 \Big|_0^R + \pi R \int_0^R \sqrt{R^2 - \rho^2} d(R^2 - \rho^2) - \pi R^3 \int_0^R \frac{d(R^2 - \rho^2)}{\sqrt{R^2 - \rho^2}} =$$

$$= \pi R^3 + \pi R \cdot \frac{2}{3} (R^2 - \rho^2)^{3/2} \Big|_0^R - 2\pi R^3 \sqrt{R^2 - \rho^2} \Big|_0^R = \pi R^3 - \frac{2}{3} \pi R^4 + 2\pi R^4$$

$$I = \pi R^3 + \frac{4}{3} \pi R^4$$

9)



$$I = \iint_S \vec{F} d\vec{S}, \vec{F} = (x^2 + y^2 + 3z^2) \vec{k}$$

$$S: z = \sqrt{x^2 + y^2}, \cos(\hat{n} \cdot \vec{k}) < 0 \Rightarrow$$

$$I = - \iint_{D_{xy}} 0 dy dz + 0 dx dz + (x^2 + y^2 + 3z^2) dx dy$$

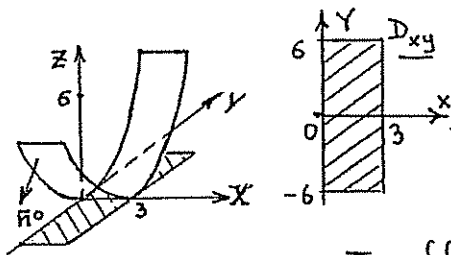
$$D_{xy}: \begin{cases} z = \sqrt{x^2 + y^2} \\ z = 2 \end{cases} \Rightarrow x^2 + y^2 = 4 \Rightarrow$$

$$I = - \iint_{D_{xy}} (x^2 + y^2 + 3z^2) dx dy = - \iint_{D_{xy}} (x^2 + y^2 + 3(x^2 + y^2)) dx dy =$$

$$= -4 \iint_{D_{xy}} (x^2 + y^2) dx dy = \left| \begin{matrix} x = \rho \cos \varphi \\ y = \rho \sin \varphi \end{matrix} \right| = -4 \int_0^{2\pi} d\varphi \int_0^2 \rho^2 \cdot \rho d\rho = - \int_0^{2\pi} d\varphi (\rho^4 \Big|_0^2) =$$

$$= -16 \cdot \varphi \Big|_0^{2\pi} = -32\pi$$

10)



$$I = \iint_S (x^2 - 2y^2 + 6z) dx dy, S: 6z = y^2$$

$$D_{xy}: \begin{cases} 6z = y^2 \\ z = 6 \end{cases} \Rightarrow y = \pm 6$$

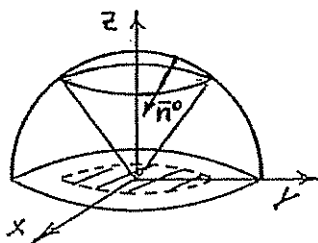
$$\cos(\hat{n} \cdot \vec{k}) < 0$$

$$I = - \iint_{D_{xy}} (x^2 - 2y^2 + y^2) dx dy = \iint_{D_{xy}} (y^2 - x^2) dx dy =$$

$$= \int_0^3 dx \int_{-6}^6 (y^2 - x^2) dy = 2 \int_0^3 dx \int_0^6 (y^2 - x^2) dy = 2 \int_0^3 dx \left[\left(\frac{y^3}{3} - x^2 y \right) \Big|_0^6 \right] =$$

$$= 12 \int_0^3 (12 - x^2) dx = 12 \left(12x - \frac{x^3}{3} \right) \Big|_0^3 = 12 \left(36 - \frac{27}{3} \right) = 324$$

11)



$$I = \iint_S \vec{F} d\vec{S}, \vec{F} = (x^2 + y^2 + z^2) \vec{k} \Rightarrow$$

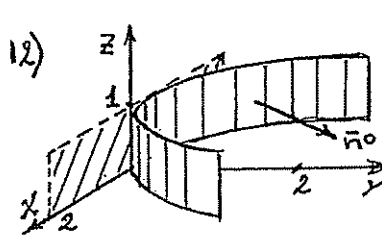
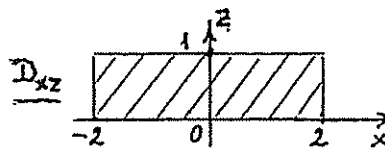
$$S: z = \sqrt{R^2 - x^2 - y^2}, z = \sqrt{x^2 + y^2},$$

$$D_{xy}: \begin{cases} z = \sqrt{R^2 - x^2 - y^2} \\ z = \sqrt{x^2 + y^2} \end{cases} \Rightarrow x^2 + y^2 = \left(\frac{R}{\sqrt{2}} \right)^2$$

$$\cos(\hat{n} \cdot \vec{k}) < 0 \Rightarrow$$

$$I = - \iint_{D_{xy}} (x^2 + y^2 + (R^2 - x^2 - y^2)) dx dy = -R^2 \iint_{D_{xy}} dx dy = -R^2 S_D = -R^2 S_{\odot} =$$

$$= -R^2 \pi \left(\frac{R}{\sqrt{2}} \right)^2 = -\frac{1}{2} \pi R^4$$

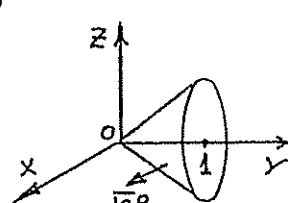
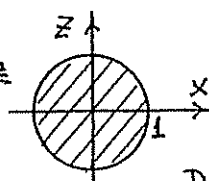
12)  
$$I = \iint_S (x^2 + y + z^2) dx dz$$

$$S: 2y = x^2$$

$$\cos(\hat{n}_0, \hat{j}) > 0 \quad D_{xz}: \begin{cases} 2y = x^2 \\ y = 2 \end{cases} \Rightarrow x = \pm 2$$

$$I = \iint_{D_{xz}} (x^2 + \frac{x^2}{2} + z^2) dx dz = 2 \int_0^2 dx \int_0^1 (\frac{3}{2}x^2 + z^2) dz = 2 \int_0^2 dx \left[\left(\frac{3}{2}x^2 z + \frac{z^3}{3} \right) \Big|_0^1 \right] =$$

$$= 2 \int_0^2 (\frac{3}{2}x^2 + \frac{1}{3}) dx = 2 \left(\frac{1}{2}x^3 + \frac{x}{3} \right) \Big|_0^2 = 8 + \frac{4}{3} = \frac{28}{3}$$

13)  
$$I = \iint_S \vec{F} \cdot d\vec{S} = \iint_S (x^2 + y^2 + z^2) dx dz$$

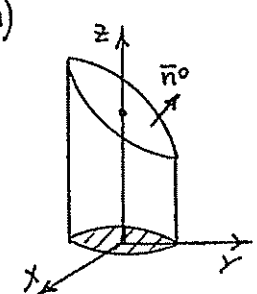
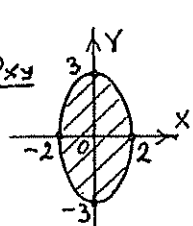
$$S: y = \sqrt{x^2 + z^2}$$

$$D_{xz}: \begin{cases} y = \sqrt{x^2 + z^2} \\ y = 1 \end{cases} \Rightarrow x^2 + z^2 = 1$$

$$\cos(\hat{n}_0, \hat{j}) < 0$$

$$I = - \iint_{D_{xz}} (x^2 + z^2 + (x^2 + z^2)) dx dz = -2 \iint_D (x^2 + y^2) dx dz = \left| \begin{matrix} x = \rho \cos \varphi \\ z = \rho \sin \varphi \end{matrix} \right| =$$

$$= -2 \int_0^{2\pi} d\varphi \int_0^1 \rho^2 \rho d\rho = -\frac{1}{2} \int_0^{2\pi} d\varphi (\rho^4 \Big|_0^1) = -\frac{1}{2} \int_0^{2\pi} d\varphi = -\frac{1}{2} \varphi \Big|_0^{2\pi} = -\pi$$

14)  
$$I = \iint_S (2x + 3y + 4z) dx dy,$$

$$S: x + y + z = 6, \quad \frac{x^2}{4} + \frac{y^2}{9} = 1 \Rightarrow y = \pm \frac{3}{2} \sqrt{4 - x^2}$$

$$\cos(\hat{n}, \hat{k}) > 0$$

$$I = \iint_D (2x + 3y + 4(6 - x - y)) dx dy = \iint_{D_{xy}} (24 - 2x - y) dx dy =$$

$$= \int_{-2}^2 dx \int_{-\frac{3}{2}\sqrt{4-x^2}}^{\frac{3}{2}\sqrt{4-x^2}} (24 - 2x - y) dy = \int_{-2}^2 dx \left[(24 - 2x)y - \frac{1}{2}y^2 \right]_{-\frac{3}{2}\sqrt{4-x^2}}^{\frac{3}{2}\sqrt{4-x^2}} = \int_{-2}^2 dx (24 - 2x) \cdot 2 \cdot \frac{3}{2} \sqrt{4 - x^2} =$$

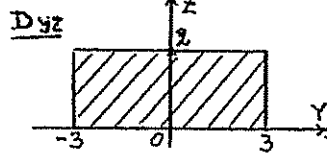
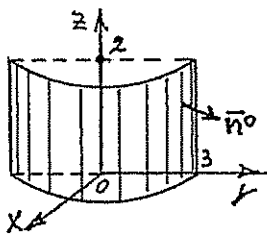
$$= 6 \int_{-2}^2 (12 - x) \sqrt{4 - x^2} dx = 72 \int_{-2}^2 \sqrt{4 - x^2} dx - 6 \int_{-2}^2 x \sqrt{4 - x^2} dx.$$

$$72 \int_{-2}^2 \sqrt{4 - x^2} dx = \left| \begin{matrix} x = 2 \sin t \\ dx = 2 \cos t dt \\ -\frac{\pi}{2} \leq t \leq \frac{\pi}{2} \end{matrix} \right| = 144 \int_{-\pi/2}^{\pi/2} 2 \cos^2 t dt = 144 \cdot 2 \int_0^{\pi/2} (1 + \cos 2t) dt =$$

$$= 144 \cdot 2 \left(t + \frac{1}{2} \sin 2t \right) \Big|_0^{\pi/2} = 144 \pi.$$

$$\therefore \text{by } \int_{-2}^2 \sqrt{4 - x^2} x dx = 0 \quad \therefore I = 144 \pi + 0 = 144 \pi$$

15)



$$I = \iint_S \vec{F} d\vec{S} = \iint_S (x^2 + z^2) dy dz$$

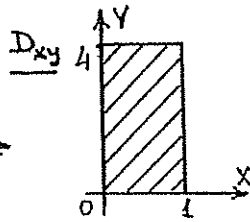
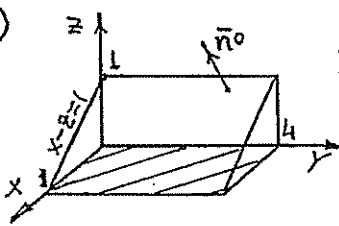
$$S: x = \sqrt{9 - y^2}$$

$$\cos(\vec{n}_0, \vec{e}) > 0$$

$$I = \iint_{D_{yz}} (9 - y^2 + z^2) dy dz =$$

$$= 2 \int_0^3 dy \int_0^2 (9 - y^2 + z^2) dz = 2 \int_0^3 dy \left[(9 - y^2)z + \frac{z^3}{3} \right]_0^2 = \frac{4}{3} \int_0^3 (31 - 3y^2) dy = \frac{4}{3} (31y - y^3) \Big|_0^3 = 88$$

16)



$$I = \iint_S \vec{F} d\vec{S}, \quad \vec{F} = x\vec{i} + y\vec{j} + z\vec{k}$$

$$I = \iint_S P dy dz + Q dx dz + R dx dy =$$

$$= \iint_S x dy dz + y dx dz + z dx dy$$

$$S: f(x, y, z) = x + z - 1 = 0$$

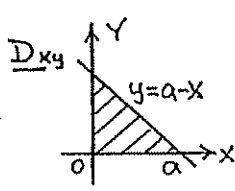
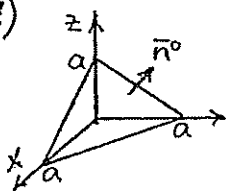
הנורמל הנכנס

$$\iint_S P dy dz + Q dx dz + R dx dy = \pm \iint_{D_{xy}} \left(P \frac{\partial f}{\partial x} + Q \frac{\partial f}{\partial y} + R \frac{\partial f}{\partial z} \right) \frac{dx dy}{\partial f / \partial z}$$

$$\cos(\vec{n}_0, \vec{e}) > 0 \Rightarrow I = \iint_{D_{xy}} (x \cdot 1 + y \cdot 0 + z \cdot 1) \frac{dx dy}{1} = \iint_{D_{xy}} (x + z) dx dy = |z = 1 - x| =$$

$$= \iint_{D_{xy}} (x + 1 - x) dx dy = \iint_{D_{xy}} dx dy = S_D = 1 \cdot 4 = 4$$

17)



$$I = \iint_S x dy dz + y dx dz + z dx dy, \quad S: x + y + z = a$$

האנליזה של פונקציה I היא פונקציה של x ו-y

הפונקציה הזו היא פונקציה של x ו-y.

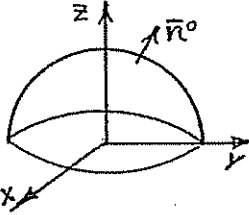
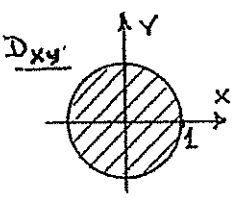
$$I_1 = \iint_{D_{xy}} z dx dy = \int_0^a dx \int_0^{a-x} (a - x - y) dy = \int_0^a dx \left[(a - x)y - \frac{y^2}{2} \right]_0^{a-x} = \int_0^a \frac{(a - x)^2}{2} dx =$$

$$= -\frac{1}{2} \int_0^a (a - x)^2 d(a - x) = -\frac{1}{6} (a - x)^3 \Big|_0^a = a^3/6$$

התוצאה היא $a^3/6$. זהו תוצאה של אינטגרל של פונקציה של x ו-y. הפונקציה הזו היא פונקציה של x ו-y.

$$I = \iint_{D_{xy}} \left(P \frac{\partial f}{\partial x} + Q \frac{\partial f}{\partial y} + R \frac{\partial f}{\partial z} \right) \frac{dx dy}{\partial f / \partial z} = \iint_{D_{xy}} (x \cdot 1 + y \cdot 1 + z \cdot 1) \frac{dx dy}{1} = \iint_{D_{xy}} (x + y + a - x - y) dx dy =$$

$$= a \int_0^a dx \int_0^{a-x} dy = a \int_0^a (a - x) dx = a \left(ax - \frac{x^2}{2} \right) \Big|_0^a = a \cdot \frac{a^2}{2} = a^3/2$$

18)  

$$I = \iint_S x^2 y^2 z \, dx \, dy, \quad S: z = \sqrt{1-x^2-y^2}$$

$$\cos(\hat{n}, \hat{k}) > 0$$

$$I = \iint_{D_{xy}} (x^2 y^2 \sqrt{1-x^2-y^2}) \, dx \, dy \Big|_{\substack{x = \rho \cos \varphi \\ y = \rho \sin \varphi}} =$$

$$= \iint_{D_{xy}} \rho^2 \cos^2 \varphi \cdot \rho^2 \sin^2 \varphi \sqrt{1-\rho^2} \rho \, d\rho \, d\varphi = \frac{1}{4} \int_0^{2\pi} (2 \sin^2 \varphi \cos^2 \varphi) \, d\varphi \int_0^1 \rho^4 \sqrt{1-\rho^2} \cdot 2\rho \, d\rho =$$

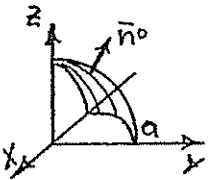
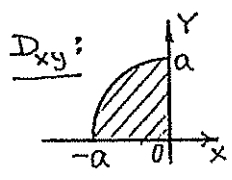
$$= \frac{1}{8} \int_0^{2\pi} \sin^2 2\varphi \, d\varphi \int_0^1 (\rho^2)^2 \sqrt{1-\rho^2} \, d(\rho^2) \Big|_{\substack{\rho^2 = t \\ 0 \leq t \leq 1}} = \frac{1}{8} \int_0^{2\pi} \sin^2 2\varphi \, d\varphi \int_0^1 t^2 \sqrt{1-t} \, dt =$$

$$= \Big|_{\substack{1-t = \rho^2 \quad t=0 \rightarrow \rho=1 \\ dt = -2\rho \, d\rho \quad t=1 \rightarrow \rho=0}} \frac{1}{8} \int_0^{2\pi} \sin^2 2\varphi \, d\varphi \int_1^0 (1-\rho^2)^2 \rho (-2\rho) \, d\rho =$$

$$= \frac{1}{4} \int_0^{2\pi} \sin^2 2\varphi \, d\varphi \int_0^1 (\rho^2 - 2\rho^4 + \rho^6) \, d\rho = \frac{1}{4} \int_0^{2\pi} \sin^2 2\varphi \, d\varphi \left[\frac{\rho^3}{3} - \frac{2}{5} \rho^5 + \frac{\rho^7}{7} \right]_0^1 =$$

$$= \frac{1}{4} \cdot \frac{8}{105} \int_0^{2\pi} \sin^2 2\varphi \, d\varphi = \frac{1}{105} \int_0^{2\pi} 2 \sin^2 2\varphi \, d\varphi = \frac{1}{105} \int_0^{2\pi} (1 - \cos 4\varphi) \, d\varphi =$$

$$= \frac{1}{105} \left(\varphi - \frac{1}{4} \sin 4\varphi \right) \Big|_0^{2\pi} = \frac{2\pi}{105}$$

19)  

$$I = \iint_S x^2 \, dy \, dz + y^2 \, dx \, dz + z^2 \, dx \, dy$$

$$S: f(x, y, z) = x^2 + y^2 + 2az - a^2 = 0$$

$$\frac{\partial f}{\partial x} = 2x, \quad \frac{\partial f}{\partial y} = 2y, \quad \frac{\partial f}{\partial z} = 2a$$

$$I = \iint_S x^2 \, dy \, dz + y^2 \, dx \, dz + z^2 \, dx \, dy = \iint_{D_{xy}} (x^2 \cdot 2x + y^2 \cdot 2y + z^2 \cdot 2a) \frac{dx \, dy}{2a} =$$

$$= \frac{1}{a} \iint_{D_{xy}} (x^3 + y^3 + az^2) \, dx \, dy = \frac{1}{a} \iint_{D_{xy}} \left[x^3 + y^3 + a \left(\frac{a^2 - x^2 - y^2}{2a} \right)^2 \right] \, dx \, dy \Big|_{\substack{x = \rho \cos \varphi \\ y = \rho \sin \varphi}} =$$

$$= \frac{1}{a} \iint_{D_{xy}} \left[\rho^3 (\cos^3 \varphi + \sin^3 \varphi) + \frac{(a^2 - \rho^2)^2}{4a} \right] \rho \, d\rho \, d\varphi = \frac{1}{a} \int_{\pi/2}^{\pi} (\cos^3 \varphi + \sin^3 \varphi) \, d\varphi \int_0^a \rho^4 \, d\rho +$$

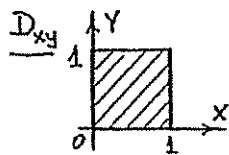
$$+ \frac{1}{4a^2} \int_{\pi/2}^{\pi} d\varphi \int_0^a (a^2 - \rho^2)^2 \rho \, d\rho = \frac{1}{a} \cdot \frac{a^5}{5} \int_{\pi/2}^{\pi} (\cos^3 \varphi + \sin^3 \varphi) \, d\varphi - \frac{1}{8a^2} \int_{\pi/2}^{\pi} d\varphi \int_0^a (a^2 - \rho^2)^2 \, d(a^2 - \rho^2)$$

$$= \frac{a^4}{5} \left[\int_{\pi/2}^{\pi} (1 - \sin^2 \varphi) \, d\sin \varphi - \int_{\pi/2}^{\pi} (1 - \cos^2 \varphi) \, d\cos \varphi \right] - \frac{1}{8a^2} \int_{\pi/2}^{\pi} d\varphi \left[\frac{(a^2 - \rho^2)^3}{3} \right]_0^a =$$

$$= \frac{a^4}{5} \left[\left(\sin \varphi - \frac{\sin^3 \varphi}{3} \right) \Big|_{\pi/2}^{\pi} - \left(\cos \varphi - \frac{\cos^3 \varphi}{3} \right) \Big|_{\pi/2}^{\pi} \right] + \frac{1}{8a^2} \cdot \frac{a^6}{3} \int_{\pi/2}^{\pi} d\varphi =$$

$$= \frac{a^4}{5} \left[-1 + \frac{1}{3} - \left(-1 + \frac{1}{3} \right) \right] + \frac{a^4}{24} \varphi \Big|_{\pi/2}^{\pi} = 0 + \frac{a^4}{24} \cdot \left(\pi - \frac{\pi}{2} \right) = \frac{\pi a^4}{48}$$

20)



$$I = \oint_S x dy dz + y dx dz + z dx dy, S: (x^2 + y^2 + z^2 = 1)$$

[illegible]

הא'טעגראל פ'ן $\iint_D z \, dx \, dy$ איז

על אף שה'הפאקט' - היציאה לארץ - נמצאת במסגרת ה'החלטות' של ה'הנהלה'.

∴ $\vec{r}$ နှင့် $\vec{r}_1$ ၏ $\angle$ $= 1$, $\cos(\vec{r}, \vec{r}_1) > 0$ ∴ $\vec{r}$ နှင့် $\vec{r}_1$ ၏ $\angle$ $< 90^\circ$

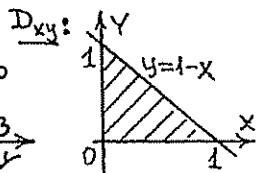
$$I_1 = \int \int_{D_{xy}} 1 \cdot dx dy = S_D = S_{\square} = 1 \cdot 1 = 1.$$

• 0-8 නිමි 872671103 128, dz=0, z=0, $\cos(\vec{n} \cdot \hat{z}) < 0$: නිමි 128 නිමි 128

7. $\iint_R z \, dx \, dy = 1$: z ከ xy -የተፈጠረውን የፍጥነት ስኬት z ለማግኘት ስራውን ይፈቅድ።

[illegible]

21)



$$I = \oint_S xy \, dy \, dz + yz \, dx \, dz + xz \, dx \, dy$$

$$S: x+y+z=1, \quad x=0, \quad y=0, \quad z=0.$$

תעודת זהו פסוק I תעודת

$$\iiint xz \, dx \, dy \quad ; \text{nie } 2 \text{ en } y \cdot p' \delta' z \delta' y' \text{ica}$$
[illegible]

הבא'ן געזיגט צו טאן D_x . ער האט דאס פראגראם AOB און ער האט דאס פראגראם p^n און $d_2=0, z=0$, דאס האט געזיגט

∴ $\cos(\vec{n} \cdot \vec{r}) > 0$ ∴ $\vec{r}$ is in the region ACB of the circle. O is the center of the circle AOB is the arc.

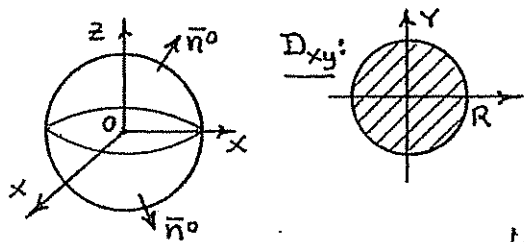
$$I_1 = \iint_{D \times y} xz \, dx \, dy = \iint_{D \times y} x(1-x-y) \, dx \, dy = \int_0^1 x \, dx \int_0^{1-x} ((1-x)-y) \, dy =$$

$$= \int_0^1 x dx \left[(1-x)y - \frac{y^2}{2} \right] \Big|_0^{1-x} = \frac{1}{2} \int_0^1 x(1-x)^2 dx = \frac{1}{2} \int_0^1 (x - 2x^2 + x^3) dx =$$

$$= \frac{1}{2} \left(\frac{x^2}{2} - \frac{2}{3}x^3 + \frac{1}{4}x^4 \right) \Big|_0^1 = \frac{1}{2} \left(\frac{1}{2} - \frac{2}{3} + \frac{1}{4} \right) = \frac{1}{24}$$

: p''₁ ၁၂၆၈၆၁ ၁၆၆၆

$$\iint_S xy \, dy \, dz = \iint_S yz \, dx \, dz = I_1 = \frac{1}{24} \Rightarrow I = 3 \cdot \frac{1}{24} = \frac{1}{8}$$



$$I = \oint_S x dy dz + y dz dx + z dx dy$$

$$S: x^2 + y^2 + z^2 = R^2$$

הנחיה זאת באמצעות: $\iint_S z \, dx \, dy$

1. 1883 : 1917 ገደብ ስለሆነ በጥራት ይገኛል

$$z = -\sqrt{R^2 - x^2 - y^2} : \text{Innen} \quad , \quad z = \sqrt{R^2 - x^2 - y^2} : \text{Außen}$$

עבור $\vec{r}$ ו $\vec{r}'$ תחלק $\vec{r}$ ל $\vec{r}_1$ ו $\vec{r}_2$ כך ש $\vec{r} = \vec{r}_1 + \vec{r}_2$ ו $\vec{r}_1$ ו $\vec{r}_2$ יהיו
מקבילים ל $\vec{r}'$ ו $\vec{r}'$ יהיו מקבילים ל $\vec{r}$.
אם $\vec{r}$ ו $\vec{r}'$ הם וקטורים במישור $\vec{r}$ ו $\vec{r}'$ יהיו מקבילים ל $\vec{r}$ ו $\vec{r}'$ יהיו מקבילים ל $\vec{r}$.
אם $\vec{r}$ ו $\vec{r}'$ הם וקטורים במישור $\vec{r}$ ו $\vec{r}'$ יהיו מקבילים ל $\vec{r}$ ו $\vec{r}'$ יהיו מקבילים ל $\vec{r}$.

$$\iint_S z \, dx \, dy = \iint_{D_{xy}} \sqrt{R^2 - x^2 - y^2} \, dx \, dy - \iint_{D_{xy}} [-\sqrt{R^2 - x^2 - y^2}] \, dx \, dy =$$

$$= 2 \iint_{D_{xy}} \sqrt{R^2 - x^2 - y^2} \, dx \, dy \Big|_{\substack{x = \rho \cos \varphi \\ y = \rho \sin \varphi}} = 2 \iint_{D_{xy}} \sqrt{R^2 - \rho^2} \, \rho \, d\rho \, d\varphi =$$

$$= - \int_0^{2\pi} d\psi \int_0^R \sqrt{R^2 - \rho^2} d(R^2 - \rho^2) = - \frac{2}{3} \int_0^{2\pi} d\psi \left[(R^2 - \rho^2)^{3/2} \right]_0^R = \frac{2}{3} R^3 \int_0^{2\pi} d\psi = \frac{4}{3} \pi R^3$$

∴ $\iiint_S x \, dy \, dz = \iiint_S y \, dx \, dz = \frac{4}{3} \pi R^3$: ρ δ γ η ν δ δ

$$I = 3 \cdot \frac{4}{3} \pi R^3 = 4\pi R^3$$

על שם המלך המשיח:

$$I = \iiint_S (x dy dz + y dx dz + z dx dy),$$

$$f(x, y, z) = x^2 + y^2 + z^2 - R^2 = 0$$

ע/כר ח' ככז/ר ה'תש"ח | מתק"פ:

$$z = \sqrt{R^2 - x^2 - y^2}, \quad \cos(\vec{n}^0, \vec{r}) > 0 \Rightarrow I_1 = \iint_{D_{xy}} \left(x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} + z \frac{\partial f}{\partial z} \right) \frac{dx dy}{\partial f / \partial z} =$$

$$= \iint_{D_{xy}} (x \cdot 2x + y \cdot 2y + z \cdot 2z) \frac{dx dy}{2z} = \iint_{D_{xy}} (x^2 + y^2 + z^2) \frac{dx dy}{z} =$$

$$= \iint_{D_{xy}} (\cancel{x^2 + y^2} + R^2 - \cancel{x^2 - y^2}) \frac{dx dy}{\sqrt{R^2 - x^2 - y^2}} = R^2 \iint_{D_{xy}} \frac{dx dy}{\sqrt{R^2 - x^2 - y^2}}$$

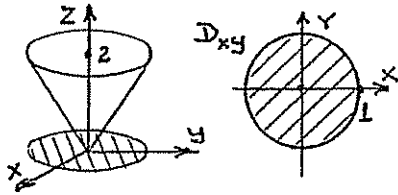
צד 1: $\cos(\vec{n} \cdot \vec{R}) < 0$: פתח 3' $z = -\sqrt{R^2 - x^2 - y^2}$, צד 2:

$$I_2 = - \iint_{D_{xy}} (x^2 + y^2 + z^2) \frac{dx dy}{z} = - \iint_{D_{xy}} (\cancel{x^2 + y^2} + R^2 - \cancel{x^2 - y^2}) \frac{dx dy}{\sqrt{R^2 - x^2 - y^2}} = I_1$$

$$I = 2R^2 \iint_{D_{xy}} \frac{dx dy}{\sqrt{R^2 - x^2 - y^2}} = 2R^2 \int_0^{2\pi} d\varphi \int_0^R \frac{\rho d\rho}{\sqrt{R^2 - \rho^2}} = -2R^2 \int_0^{2\pi} d\varphi (\sqrt{R^2 - \rho^2}) \Big|_0^R =$$

$$= 2R^3 \int_0^{2\pi} d\varphi = 2R^3 \varphi \Big|_0^{2\pi} = 4\pi R^3$$

23)



$$S: \frac{z^2}{4} = x^2 + y^2, \quad 0 \leq z \leq 2, \quad r = \sqrt{x^2 + y^2}$$

$$m = \iint_S r \, dS \Rightarrow D_{xy}: \begin{cases} \frac{z^2}{4} = x^2 + y^2 \\ z = 2 \end{cases} \Rightarrow x^2 + y^2 = 1$$

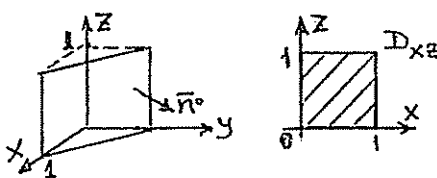
$$z = 2\sqrt{x^2 + y^2} \Rightarrow z'_x = \frac{2x}{\sqrt{x^2 + y^2}}, \quad z'_y = \frac{2y}{\sqrt{x^2 + y^2}} \Rightarrow dS = \sqrt{1 + z'^2_x + z'^2_y} \, dx \, dy$$

$$dS = \sqrt{1 + \frac{4x^2}{x^2 + y^2} + \frac{4y^2}{x^2 + y^2}} \, dx \, dy = \sqrt{1 + \frac{4(x^2 + y^2)}{x^2 + y^2}} \, dx \, dy = \sqrt{5} \, dx \, dy$$

$$m = \iint_{D_{xy}} \sqrt{x^2 + y^2} \sqrt{5} \, dx \, dy = \left| \begin{matrix} x = \rho \cos \varphi \\ y = \rho \sin \varphi \end{matrix} \right| = \sqrt{5} \iint_{D_{xy}} \rho \cdot \rho \, d\varphi \, d\rho = \sqrt{5} \int_0^{2\pi} d\varphi \int_0^1 \rho^2 \, d\rho =$$

$$= \sqrt{5} \cdot 2\pi \cdot \frac{1}{3} \rho^3 \Big|_0^1 = \frac{2}{3} \pi \sqrt{5}$$

24)



$$\vec{F} = z\vec{i} + (1-z)\vec{j} + xy\vec{k}, \quad S: x+y=1, \quad 0 \leq z \leq 1, \quad x, y \geq 0$$

$$I = \iint_S \vec{F} \cdot d\vec{S} = \iint_S P \, dy \, dz + Q \, dx \, dz + R \, dx \, dy =$$

$$= \iint_S z \, dy \, dz + (1-z) \, dx \, dz + xy \, dx \, dy, \quad \cos(\vec{n} \cdot \vec{j}) > 0$$

$$S: f(x, y, z) = x + y - 1 = 0 \Rightarrow I = \iint_{D_{xz}} (P f'_x + Q f'_y + R f'_z) \frac{dx \, dz}{f'_y} =$$

$$= \iint_{D_{xz}} (z \cdot 1 + (1-z) \cdot 1 + xy \cdot 0) \frac{dx \, dz}{1} = \iint_{D_{xz}} dx \, dz = S_D = 1 \cdot 1 = 1.$$

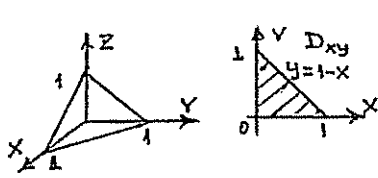
הנורמל $\vec{n}$ הוא $(1, 1, 0)$

$$I = \iint_S \vec{F} \cdot \vec{n} \, dS \Rightarrow f(x, y, z) = x + y - 1 = 0 \Rightarrow \vec{n} = (1, 1, 0) \Rightarrow \vec{n}^0 = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0 \right)$$

$$y = 1 - x \Rightarrow y'_x = -1, \quad y'_z = 0 \Rightarrow dS = \sqrt{1 + y'^2_x + y'^2_z} \, dx \, dz = \sqrt{2} \, dx \, dz$$

$$I = \iint_{D_{xz}} \vec{F} \cdot \vec{n}^0 \, dS = \iint_{D_{xz}} (z, 1-z, xy) \cdot \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0 \right) \cdot \sqrt{2} \, dx \, dz = \iint_{D_{xz}} dx \, dz = S_D = 1$$

25)



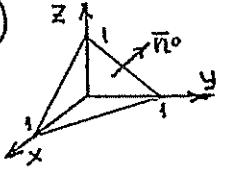
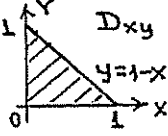
$$S: x + y + z = 1, \quad x \geq 0, \quad y \geq 0, \quad z \geq 0, \quad r = \frac{1}{(1+x+z)^2}$$

$$m = \iint_S r \, dS \Rightarrow dS = \sqrt{1 + z'^2_x + z'^2_y} \, dx \, dy =$$

$$= |z = 1 - x - y| = \sqrt{1 + 1^2 + 1^2} \, dx \, dy = \sqrt{3} \, dx \, dy$$

$$m = \iint_{D_{xy}} \frac{\sqrt{3} \, dx \, dy}{(1+x+z)^2} = \iint_{D_{xy}} \frac{\sqrt{3} \, dx \, dy}{[1+x+(1-x-y)]^2} = \sqrt{3} \int_0^1 dx \int_0^{1-x} \frac{dy}{(2-y)^2} = \sqrt{3} \int_0^1 dx \int_0^{1-x} \frac{d(y-2)}{(y-2)^2} =$$

$$= \sqrt{3} \int_0^1 dx \left(\frac{-1}{y-2} \Big|_0^{1-x} \right) = \sqrt{3} \int_0^1 \left(\frac{1}{x+1} - \frac{1}{2} \right) dx = \sqrt{3} \left[\ln(x+1) - \frac{x}{2} \right] \Big|_0^1 = \sqrt{3} \left(\ln 2 - \frac{1}{2} \right)$$

26)   $\vec{F} = x^2\vec{i} + y^2\vec{j} + z^2\vec{k}$, $S: f(x,y,z) = x+y+z-1=0$

$$I = \iint_S \vec{F} d\vec{S} = \iint_S P dy dz + Q dx dz + R dx dy =$$

$$= \iint_S x^2 dy dz + y^2 dx dz + z^2 dx dy = \left| \cos \hat{n} \cdot \hat{k} > 0 \right| = \iint_{D_{xy}} (P f'_x + Q f'_y + R f'_z) \frac{dx dy}{f'_z} =$$

$$= \iint_{D_{xy}} (x^2 + y^2 + z^2) \frac{dx dy}{1} = \left| z = 1-x-y \right| = \iint_{D_{xy}} [x^2 + y^2 + (x+y-1)^2] dx dy =$$

$$= \int_0^1 dx \int_0^{1-x} [x^2 + y^2 + (x+y-1)^2] dy = \int_0^1 \left[x^2 y + \frac{1}{3} y^3 + \frac{1}{3} (x+y-1)^3 \right] \Big|_0^{1-x} dx =$$

$$= \int_0^1 \left[x^2(1-x) + \frac{1}{3}(1-x)^3 + 0 - \frac{1}{3}(x-1)^3 \right] dx = \int_0^1 \left[x^2 - x^3 - \frac{2}{3}(x-1)^3 \right] dx =$$

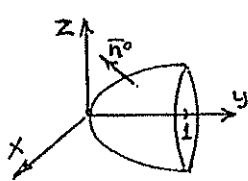
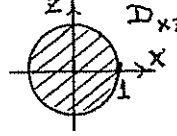
$$= \left[\frac{x^3}{3} - \frac{x^4}{4} - \frac{2}{3} \cdot \frac{1}{4} (x-1)^4 \right] \Big|_0^1 = \frac{1}{3} - \frac{1}{4} - 0 + \frac{1}{6} = \frac{1}{4}$$

$I = \iint_S \vec{F} \cdot \vec{n} dS$: $\vec{n} = (1, 1, 1)$, $|\vec{n}| = \sqrt{3}$, $\vec{n}^0 = (\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}})$

$S: x+y+z-1=0 \Rightarrow \vec{n} = (1, 1, 1)$, $|\vec{n}| = \sqrt{3}$, $\vec{n}^0 = (\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}})$

$dS = \sqrt{1+z_x'^2 + z_y'^2} dx dy = |z = 1-x-y| = \sqrt{1+1+1} dx dy = \sqrt{3} dx dy$

$I = \iint_{D_{xy}} (x^2, y^2, z^2) \cdot (\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}) \sqrt{3} dx dy = \iint_{D_{xy}} (x^2 + y^2 + z^2) dx dy \Rightarrow$

27)   $\vec{F} = y\vec{j} + z^2\vec{k}$, $S: y = x^2 + z^2$, $0 \leq y \leq 1$

$$I = \iint_S \vec{F} d\vec{S} = \iint_S 0 dy dz + y dx dz + z^2 dx dy$$

$f(x,y,z) = y - x^2 - z^2 = 0$, $\cos(\hat{n}^0 \cdot \hat{j}) < 0 \Rightarrow$

$I = - \iint_{D_{xz}} (P f'_x + Q f'_y + R f'_z) \frac{dx dz}{f'_y} = - \iint_{D_{xz}} (y \cdot 1 + z^2 (-2z)) \frac{dx dz}{1} =$

$= - \iint_{D_{xz}} (y - 2z^3) dx dz = \left| y = x^2 + z^2 \right| = - \iint_{D_{xz}} (x^2 + z^2 - 2z^3) dx dz = \iint_{D_{xz}} (2z^3 - x^2 - z^2) dx dz$

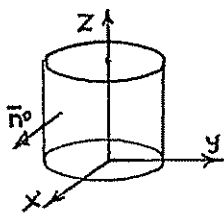
$D_{xz}: \begin{cases} y = x^2 + z^2 \\ y = 1 \end{cases} \Rightarrow x^2 + z^2 = 1 \Rightarrow x = \rho \cos \varphi, z = \rho \sin \varphi, 0 \leq \varphi \leq 2\pi, 0 \leq \rho \leq 1$

$I = \iint_{D_{xz}} (2\rho^3 \sin^3 \varphi - \rho^2) \rho d\varphi d\rho = \int_0^{2\pi} d\varphi \int_0^1 (2\rho^4 \sin^3 \varphi - \rho^3) d\rho =$

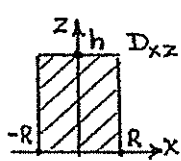
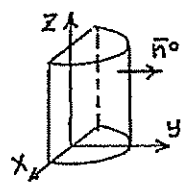
$= \int_0^{2\pi} d\varphi \left(\frac{2}{5} \rho^5 \sin^3 \varphi - \frac{\rho^4}{4} \right) \Big|_0^1 = \int_0^{2\pi} \left(\frac{2}{5} \sin^3 \varphi - \frac{1}{4} \right) d\varphi = -\frac{2}{5} \int_0^{2\pi} (1 - \cos^2 \varphi) d(\cos \varphi) -$

$-\frac{1}{4} \varphi \Big|_0^{2\pi} = -\frac{2}{5} (\cos \varphi - \frac{\cos^3 \varphi}{3}) \Big|_0^{2\pi} - \frac{1}{4} \cdot 2\pi = 0 - \frac{\pi}{2} = -\frac{\pi}{2}$

28)



$$\vec{F} = x\vec{i} + y\vec{j} + z\vec{k}, \quad S: x^2 + y^2 = R^2, \quad 0 \leq z \leq h$$



יש נגזרות חלקיות
y=0 וזוהי גבולות הפונקציה
: y ≥ 0 וזוהי

$$I_1 = \iint_{S_1} \vec{F} d\vec{S}, \quad \cos(\vec{n}^0, \vec{j}) > 0$$

$$f(x, y, z) = x^2 + y^2 - R^2 = 0 \Rightarrow I_1 = \iint_{S_1} x dy dz + y dx dz + z dx dy =$$

$$= \iint_{D_{xz}} (P'_x + Q'_y + R'_z) \frac{dx dz}{f'_y} = \iint_{D_{xz}} (x \cdot 2x + y \cdot 2y + z \cdot 0) \frac{dx dz}{2y} = \iint_{D_{xz}} (x^2 + y^2) \frac{dx dz}{y} =$$

$$= |x^2 + y^2 = R^2| = \iint_{D_{xz}} R^2 \frac{dx dz}{y} = R^2 \iint_{D_{xz}} \frac{dx dz}{\sqrt{R^2 - x^2}} = R^2 \int_{-R}^R \frac{dx}{\sqrt{R^2 - x^2}} \int_0^h dz =$$

$$= R^2 \left(z \Big|_0^h \right) 2 \int_0^R \frac{dx}{\sqrt{R^2 - x^2}} = 2R^2 h \arcsin \frac{x}{R} \Big|_0^R = 2R^2 h \arcsin 1 = 2R^2 h \cdot \frac{\pi}{2} = \pi R^2 h$$

: פונקציה y ≤ 0 וזוהי נגזרת של הפונקציה וזוהי

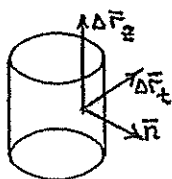
$$y = -\sqrt{R^2 - x^2}, \quad \cos(\vec{n}^0, \vec{j}) < 0 \Rightarrow I_2 = - \iint_{D_{xz}} x dy dz + y dx dz + z dx dy =$$

$$= - \iint_{D_{xz}} (P'_x + Q'_y + R'_z) \frac{dx dz}{f'_y} = - \iint_{D_{xz}} (x^2 + y^2) \frac{dx dz}{y} = -R^2 \iint_{D_{xz}} \frac{dx dz}{\sqrt{R^2 - x^2}} = -\pi R^2 h$$

$$I = I_1 + I_2 = 2\pi R^2 h : \text{נגזרת של הפונקציה וזוהי}$$

הפונקציה יחידה וזוהי נגזרת של הפונקציה וזוהי

$$S: \vec{r}(t, z) = R \cos t \vec{i} + R \sin t \vec{j} + z \vec{k}, \quad 0 \leq t < 2\pi, \quad 0 \leq z \leq h$$



$$I = \iint_S \vec{F} \cdot (\vec{r}'_t \times \vec{r}'_z) dt dz$$

$$\vec{r}'_t = -R \sin t \vec{i} + R \cos t \vec{j} + 0 \vec{k}, \quad \vec{r}'_z = 0 \vec{i} + 0 \vec{j} + \vec{k} \Rightarrow$$

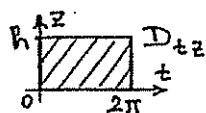
$$x = R \cos t, \quad y = R \sin t, \quad z = z$$

: פונקציה נגזרת של הפונקציה וזוהי

: פונקציה נגזרת של הפונקציה וזוהי

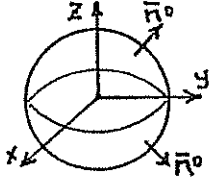
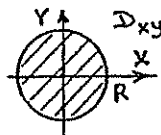
$$\vec{F} = x\vec{i} + y\vec{j} + z\vec{k} = R \cos t \vec{i} + R \sin t \vec{j} + z\vec{k} \Rightarrow$$

$$\vec{F} \cdot (\vec{r}'_t \times \vec{r}'_z) = \begin{vmatrix} R \cos t & R \sin t & z \\ -R \sin t & R \cos t & 0 \\ 0 & 0 & 1 \end{vmatrix} = R^2 \cos^2 t + R^2 \sin^2 t = R^2 \Rightarrow$$



$$I = \iint_{D_{tz}} \vec{F} \cdot (\vec{r}'_t \times \vec{r}'_z) dt dz = \iint_{D_{tz}} R^2 dt dz = R^2 \iint_{D_{tz}} dt dz =$$

$$= R^2 S_D = R^2 \cdot 2\pi \cdot h = 2\pi R^2 h$$

29)  $\vec{F} = |\vec{r}| \cdot \vec{r} = \sqrt{x^2+y^2+z^2}(\vec{i} + \vec{j} + \vec{k})$, $S: x^2+y^2+z^2=R^2$
 $\vec{F} = |\vec{r}| \cdot \vec{r} = \sqrt{x^2+y^2+z^2}(\vec{i} + \vec{j} + \vec{k})$, $S: x^2+y^2+z^2=R^2$
 $I_1 = \iint_{S_1} \vec{F} \cdot d\vec{S}$, $\cos(\vec{n}_0 \cdot \vec{k}) > 0$, $z = \sqrt{R^2 - x^2 - y^2}$

$$f(x,y,z) = x^2+y^2+z^2-R^2=0 \Rightarrow I_1 = \iint_{S_1} |\vec{F}| x dy dz + |\vec{F}| y dx dz + |\vec{F}| z dx dy =$$

$$= \iint_{D_{xy}} (P f'_x + Q f'_y + R f'_z) \frac{dx dy}{f'_z} = \iint_{D_{xy}} |\vec{F}| (x \cdot 2x + y \cdot 2y + z \cdot 2z) \frac{dx dy}{2z} =$$

$$= \iint_{D_{xy}} \sqrt{x^2+y^2+z^2} (x^2+y^2+z^2) \frac{dx dy}{z} = \iint_{D_{xy}} \frac{(x^2+y^2+z^2)^{3/2}}{\sqrt{R^2-x^2-y^2}} dx dy = |x^2+y^2+z^2=R^2| =$$

$$= R^3 \iint_{D_{xy}} \frac{dx dy}{\sqrt{R^2-x^2-y^2}} = \left| \begin{matrix} x = \rho \cos \varphi \\ y = \rho \sin \varphi \end{matrix} \right| = R^3 \iint_{D_{xy}} \frac{\rho d\varphi d\rho}{\sqrt{R^2-\rho^2}} = -\frac{R^3}{2} \int_0^{2\pi} d\varphi \int_0^R \frac{d(R^2-\rho^2)}{\sqrt{R^2-\rho^2}} =$$

$$= -\frac{R^3}{2} (\varphi|_0^{2\pi}) 2\sqrt{R^2-\rho^2}|_0^R = -2\pi R^3 (0-R) = 2\pi R^4$$

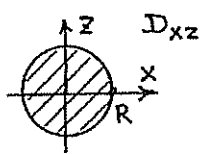
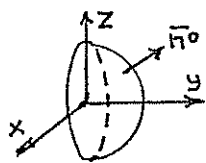
התוצאה היא $2\pi R^4$ עבור החלק העליון של הכדור.

$$z = -\sqrt{R^2-x^2-y^2}, \cos(\vec{n}_0 \cdot \vec{k}) < 0 \Rightarrow I_2 = -\iint_{S_2} |\vec{F}| x dy dz + |\vec{F}| y dx dz + |\vec{F}| z dx dy =$$

$$= -\iint_{D_{xy}} (P f'_x + Q f'_y + R f'_z) \frac{dx dy}{f'_z} = -\iint_{D_{xy}} \frac{(x^2+y^2+z^2)^{3/2}}{-\sqrt{R^2-x^2-y^2}} dx dy = I_1 = 2\pi R^4$$

$$I = I_1 + I_2 = 4\pi R^4 \quad \text{התוצאה היא } 4\pi R^4 \text{ עבור כל הכדור.}$$

30)



$$\vec{F} = yz\vec{i} + x^2\vec{j} + yz\vec{k}$$

$$S: f(x,y,z) = x^2+y^2+z^2-R^2=0, y \geq 0$$

$$I = \iint_S \vec{F} \cdot d\vec{S}, \cos(\vec{n}_0 \cdot \vec{j}) > 0$$

$$I = \iint_S yz dy dz + x^2 dx dz + yz dx dy = \iint_{D_{xz}} (P f'_x + Q f'_y + R f'_z) \frac{dx dz}{f'_y} =$$

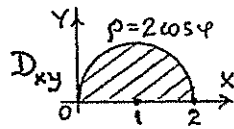
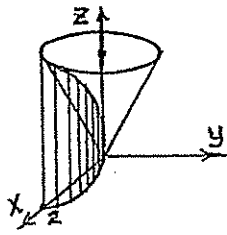
$$= \iint_{D_{xz}} (yz \cdot 2x + x^2 \cdot 2y + yz \cdot 2z) \frac{dx dz}{2y} = \iint_{D_{xz}} (xz + x^2 + z^2) dx dz = \left| \begin{matrix} x = \rho \cos \varphi \\ z = \rho \sin \varphi \end{matrix} \right| =$$

$$= \iint_{D_{xz}} (\rho^2 \sin \varphi \cos \varphi + \rho^2 (\cos^2 \varphi + \sin^2 \varphi)) \rho d\varphi d\rho = \iint_{D_{xz}} (\sin \varphi \cos \varphi + 1) \rho^3 d\varphi d\rho =$$

$$= \int_0^{2\pi} (\sin \varphi \cos \varphi + 1) d\varphi \int_0^R \rho^3 d\rho = \frac{1}{4} (\rho^4|_0^R) \int_0^{2\pi} (\sin \varphi \cos \varphi + 1) d\varphi =$$

$$= \frac{R^4}{4} \left[\int_0^{2\pi} \sin \varphi d(\sin \varphi) + \int_0^{2\pi} d\varphi \right] = \frac{R^4}{4} \left(\frac{1}{2} \sin^2 \varphi|_0^{2\pi} + 2\pi \right) = \frac{R^4}{4} (0 + 2\pi) = \frac{\pi R^4}{2}$$

31)



$$S: z = \sqrt{x^2 + y^2}, \quad x^2 + y^2 = 2x, \quad y \geq 0$$

$$\sigma = xy + yz + xz$$

$$m = \iint_S \sigma \, dS, \quad D_{xy}: (x-1)^2 + y^2 = 1, \quad y \geq 0$$

$$z = \sqrt{x^2 + y^2} \Rightarrow z'_x = \frac{x}{\sqrt{x^2 + y^2}}, \quad z'_y = \frac{y}{\sqrt{x^2 + y^2}} \Rightarrow dS = \sqrt{1 + z'^2_x + z'^2_y} \, dx \, dy \Rightarrow$$

$$dS = \sqrt{1 + \frac{x^2}{x^2 + y^2} + \frac{y^2}{x^2 + y^2}} \, dx \, dy = \sqrt{1 + \frac{x^2 + y^2}{x^2 + y^2}} \, dx \, dy = \sqrt{2} \, dx \, dy$$

$$m = \iint_{D_{xy}} (xy + yz + xz) \sqrt{2} \, dx \, dy = \left| z = \sqrt{x^2 + y^2} \right| = \iint_{D_{xy}} [xy + (y+x)\sqrt{x^2 + y^2}] \sqrt{2} \, dx \, dy =$$

$$= \left| \begin{matrix} x = \rho \cos \varphi \\ y = \rho \sin \varphi \end{matrix} \right| = \sqrt{2} \iint_{D_{xy}} [\rho^2 \cos \varphi \sin \varphi + (\sin \varphi + \cos \varphi) \rho^2] \rho \, d\varphi \, d\rho =$$

$$= \sqrt{2} \iint_{D_{xy}} (\sin \varphi \cos \varphi + \sin \varphi + \cos \varphi) \rho^3 \, d\varphi \, d\rho = \left| \begin{matrix} D_{xy}: x^2 + y^2 = 2x \Rightarrow \\ \rho = 2 \cos \varphi \end{matrix} \right| =$$

$$= \sqrt{2} \int_0^{\pi/2} (\sin \varphi + \cos \varphi + \sin \varphi + \cos \varphi) \, d\varphi \int_0^{2 \cos \varphi} \rho^3 \, d\rho =$$

$$= \frac{\sqrt{2}}{4} \int_0^{\pi/2} (\sin \varphi \cos \varphi + \sin \varphi + \cos \varphi) (\rho^4 \Big|_0^{2 \cos \varphi}) \, d\varphi =$$

$$= 4 \cdot \sqrt{2} \int_0^{\pi/2} (\cos^5 \varphi \sin \varphi + \cos^4 \sin \varphi + \cos^5 \varphi) \, d\varphi = 4 \sqrt{2} (I_1 + I_2 + I_3)$$

$$I_1 = \int_0^{\pi/2} \cos^5 \varphi \sin \varphi \, d\varphi = - \int_0^{\pi/2} \cos^5 \varphi \, d(\cos \varphi) = - \frac{\cos^6 \varphi}{6} \Big|_0^{\pi/2} = - (0 - \frac{1}{6}) = \frac{1}{6}$$

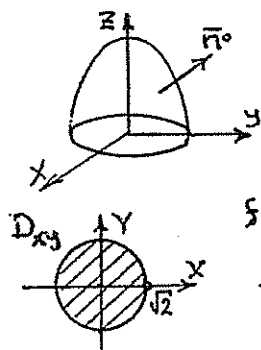
$$I_2 = \int_0^{\pi/2} \cos^4 \varphi \sin \varphi \, d\varphi = - \int_0^{\pi/2} \cos^4 \varphi \, d(\cos \varphi) = - \frac{\cos^5 \varphi}{5} \Big|_0^{\pi/2} = \frac{1}{5}$$

$$I_3 = \int_0^{\pi/2} \cos^5 \varphi \, d\varphi = \int_0^{\pi/2} \cos^4 \varphi \cdot \cos \varphi \, d\varphi = \int_0^{\pi/2} (1 - \sin^2 \varphi)^2 \, d(\sin \varphi) =$$

$$= \int_0^{\pi/2} (1 - 2 \sin^2 \varphi + \sin^4 \varphi) \, d(\sin \varphi) = \left(\sin \varphi - \frac{2}{3} \sin^3 \varphi + \frac{1}{5} \sin^5 \varphi \right) \Big|_0^{\pi/2} = \frac{8}{15}$$

$$m = 4 \sqrt{2} \left(\frac{1}{6} + \frac{1}{5} + \frac{8}{15} \right) = \frac{18 \sqrt{2}}{5}$$

32)



$$\vec{F} = x^3 \vec{i} + y^3 \vec{j} + z^3 \vec{k}, \quad S: z = 2 - x^2 - y^2, \quad z \geq 0$$

$$I = \iint_S \vec{F} \cdot d\vec{S}, \quad \cos(\hat{n}, \hat{k}) > 0 \quad z = 2 - x^2 - y^2 \Rightarrow$$

$$f(x, y, z) = x^2 + y^2 + z - 2 \Rightarrow I = \iint_S x^3 dy dz + y^3 dx dz + z^3 dx dy =$$

$$= \iint_{D_{xy}} (P f'_x + Q f'_y + R f'_z) \frac{dx dy}{f'_z} =$$

$$= \iint_{D_{xy}} (x^3 \cdot 2x + y^3 \cdot 2y + z^3 \cdot 1) \frac{dx dy}{1} = \left| z = 2 - x^2 - y^2 \right| = \iint_{D_{xy}} (2x^4 + 2y^4 + (2 - x^2 - y^2)^3) dx dy =$$

$$= \iint_{D_{xy}} [2(x^4 + 2x^2y^2 + y^4 - 2x^2y^2) + (2 - x^2 - y^2)^3] dx dy =$$

$$= \iint_{D_{xy}} [2(x^2 + y^2)^2 - 4x^2y^2 + (2 - (x^2 + y^2))^3] dx dy = \begin{vmatrix} x = \rho \cos \varphi \\ y = \rho \sin \varphi \end{vmatrix} =$$

$$= \iint_{D_{xy}} [2\rho^4 - 4\rho^4 \cos^2 \varphi \sin^2 \varphi + (2 - \rho^2)^3] \rho d\varphi d\rho =$$

$$= \iint_{D_{xy}} [2\rho^4 - \rho^4 (2 \sin \varphi \cos \varphi)^2 + 8 - 12\rho^2 + 6\rho^4 - \rho^6] \rho d\varphi d\rho =$$

$$= \iint_{D_{xy}} [-\rho^6 + 8\rho^4 - 12\rho^2 + 8 - \rho^4 \sin^2 2\varphi] \rho d\varphi d\rho =$$

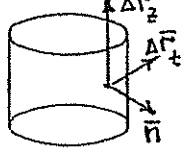
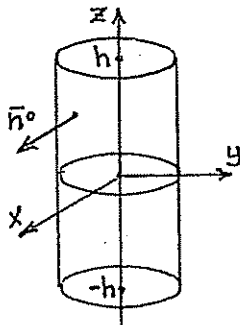
$$= \int_0^{2\pi} d\varphi \int_0^{\sqrt{2}} (-\rho^7 + 8\rho^5 - 12\rho^3 + 8\rho - \rho^5 \sin^2 2\varphi) d\rho = \int_0^{2\pi} \left(-\frac{\rho^8}{8} + \frac{4}{3}\rho^6 - 3\rho^4 + 4\rho^2 - \frac{\rho^6}{6} \sin^2 2\varphi \right) \Big|_0^{\sqrt{2}} d\varphi =$$

$$= \int_0^{2\pi} \left(-2 + \frac{32}{3} - 12 + 8 - \frac{4}{3} \sin^2 2\varphi \right) d\varphi = \int_0^{2\pi} \left(\frac{14}{3} - \frac{4}{3} \sin^2 \varphi \right) d\varphi =$$

$$= \frac{14}{3} \varphi \Big|_0^{2\pi} - \frac{2}{3} \int_0^{2\pi} 2 \sin^2 2\varphi d\varphi = \frac{28}{3} \pi - \frac{2}{3} \int_0^{2\pi} (1 - \cos 4\varphi) d\varphi =$$

$$= \frac{28}{3} \pi - \frac{2}{3} \left(\varphi - \frac{1}{4} \sin 4\varphi \right) \Big|_0^{2\pi} = \frac{28}{3} \pi - \frac{4}{3} \pi = 8\pi$$

33)



$$\vec{F} = x\vec{i} + y^2\vec{j} + z^2\vec{k}, \quad S_1: x^2 + y^2 = R^2, \quad z = h, \quad z = -h$$

: ס'כנה של הצורה הנלמדת וקצת וקטור שדה ונעלה את

$$S_1: x^2 + y^2 = R^2, \quad -h \leq z \leq h$$

: וקטור הנורמה הנלמדת את הצורה הנלמדת

$$\vec{r} = R\cos t\vec{i} + R\sin t\vec{j} + z\vec{k}, \quad 0 \leq t < 2\pi, \quad -h \leq z \leq h$$

$$I_1 = \iint_{S_1} \vec{F} \cdot (\vec{r}_t' \times \vec{r}_z') dt dz : \text{נלמדת וקצת } \vec{F} \text{ וקטור שדה ונעלה את}$$

: וס, $z = z$, $y = R\sin t$, $x = R\cos t$: נלמדת של וקטור הנורמה

$$\vec{F} = R\cos t\vec{i} + R^2\sin^2 t\vec{j} + z^2\vec{k}$$

: נלמדת וקצת ונעלה את

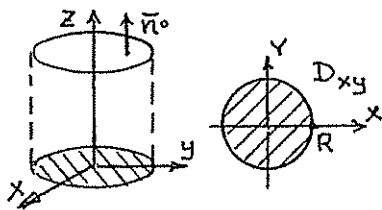
$$\vec{F} \cdot (\vec{r}_t' \times \vec{r}_z') = \begin{vmatrix} R\cos t & R^2\sin^2 t & z^2 \\ -R\sin t & R\cos t & 0 \\ 0 & 0 & 1 \end{vmatrix} = R^2\cos^2 t + R^3\sin^3 t$$

$$I_1 = \iint_{D_{tz}} (R^2\cos^2 t + R^3\sin^3 t) dt dz = \int_0^{2\pi} (R^2\cos^2 t + R^3\sin^3 t) dt \int_{-h}^h dz =$$

$$= \left(\int_{-h}^h dz \right) \left[\int_0^{2\pi} R^2\cos^2 t dt + \int_0^{2\pi} R^3\sin^3 t dt \right] = 2hR^2 \cdot \frac{1}{2} \int_0^{2\pi} (1 + \cos 2t) dt -$$

$$- 2hR^3 \int_0^{2\pi} (1 - \cos^2 t) d(\cos t) = hR^2 \left(t + \frac{1}{2}\sin 2t \right) \Big|_0^{2\pi} - 2hR^3 \left(\cos t - \frac{\cos^3 t}{3} \right) \Big|_0^{2\pi} =$$

$$= hR^2 \cdot 2\pi - 0 = 2\pi R^2 h.$$



: וסכנה סכנה וקצת וקטור שדה ונעלה את

$$S_2: x^2 + y^2 \leq R^2, \quad z = h, \quad \cos(\vec{n} \cdot \vec{k}) > 0$$

$$I_2 = \iint_{S_2} \vec{F} d\vec{S} = \iint_{S_2} x dy dz + y^2 dx dz + z^2 dx dy =$$

$$= \left| \frac{dz}{dz=0} \right| = \iint_{D_{xy}} h^2 dx dy = h^2 \iint_{D_{xy}} dx dy = h^2 S_D = h^2 \pi R^2 = \pi R^2 h$$

: וקטור הנורמה הנלמדת וקצת וקטור שדה ונעלה את

$$S_3: x^2 + y^2 \leq R^2, \quad z = -h, \quad \cos(\vec{n} \cdot \vec{k}) < 0$$

$$I_3 = \iint_{S_3} \vec{F} d\vec{S} = - \iint_{S_3} x dy dz + y^2 dx dz + z^2 dx dy = \left| \frac{dz}{dz=0} \right| =$$

$$= - \iint_{D_{xy}} h^2 dx dy = -\pi R^2 h \Rightarrow \iint_S \vec{F} d\vec{S} = I_1 + I_2 + I_3 = 2\pi R^2 h$$

משפט גאוס

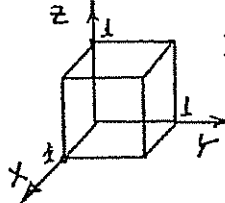
$$1) \oint_S (xy^2 dy dz + yz^2 dx dz + x^2 z dx dy) = \\ = \iiint_V \left[\frac{\partial}{\partial x}(xy^2) + \frac{\partial}{\partial y}(yz^2) + \frac{\partial}{\partial z}(x^2 z) \right] dV = \iiint_V (y^2 + z^2 + x^2) dx dy dz$$

$$2) I = \oint_S \frac{x dy dz}{\sqrt{x^2 + y^2 + z^2}} + \frac{y dx dz}{\sqrt{x^2 + y^2 + z^2}} + \frac{z dx dy}{\sqrt{x^2 + y^2 + z^2}} = \\ = \iiint_V \left[\frac{\partial}{\partial x} \left(\frac{x}{\sqrt{x^2 + y^2 + z^2}} \right) + \frac{\partial}{\partial y} \left(\frac{y}{\sqrt{x^2 + y^2 + z^2}} \right) + \frac{\partial}{\partial z} \left(\frac{z}{\sqrt{x^2 + y^2 + z^2}} \right) \right] dV \\ \frac{\partial}{\partial x} \left(\frac{x}{\sqrt{x^2 + y^2 + z^2}} \right) = \frac{\sqrt{x^2 + y^2 + z^2} - \frac{x^2}{\sqrt{x^2 + y^2 + z^2}}}{x^2 + y^2 + z^2} = \frac{y^2 + z^2}{(x^2 + y^2 + z^2)^{3/2}}, \\ \frac{\partial}{\partial y} \left(\frac{y}{\sqrt{x^2 + y^2 + z^2}} \right) = \frac{x^2 + z^2}{(x^2 + y^2 + z^2)^{3/2}}, \quad \frac{\partial}{\partial z} \left(\frac{z}{\sqrt{x^2 + y^2 + z^2}} \right) = \frac{x^2 + y^2}{(x^2 + y^2 + z^2)^{3/2}}$$

$$I = \iiint_V \frac{(y^2 + z^2) + (x^2 + z^2) + (x^2 + y^2)}{(x^2 + y^2 + z^2)^{3/2}} dV = 2 \iiint_V \frac{dx dy dz}{\sqrt{x^2 + y^2 + z^2}}$$

$$3) \oint_S \frac{\partial f}{\partial x} dy dz + \frac{\partial f}{\partial y} dx dz + \frac{\partial f}{\partial z} dx dy = \iiint_V \left[\frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right) + \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial y} \right) + \frac{\partial}{\partial z} \left(\frac{\partial f}{\partial z} \right) \right] dV = \\ = \iiint_V \left(\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} \right) dx dy dz$$

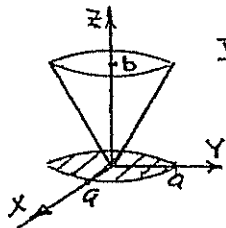
$$4) I = \oint_S x dy dz + y dx dz + z dx dy, \quad V: x^2 + y^2 + z^2 = R^2 \\ I = \iiint_V \left(\frac{\partial x}{\partial x} + \frac{\partial y}{\partial y} + \frac{\partial z}{\partial z} \right) dV = \iiint_V (1+1+1) dV = 3 \iiint_V dV = 3V = 3 \cdot \frac{4}{3} \pi R^3 = 4\pi R^3$$

$$5) \text{  } I = \oint_S x^2 dy dz + y^2 dx dz + z^2 dx dy, \quad V: (0,1|1). \\ I = \iiint_V \left(\frac{\partial}{\partial x}(x^2) + \frac{\partial}{\partial y}(y^2) + \frac{\partial}{\partial z}(z^2) \right) dV = 2 \iiint_V (x+y+z) dx dy dz = \\ = 2 \int_0^1 dx \int_0^1 dy \int_0^1 (x+y+z) dz = 2 \int_0^1 dx \int_0^1 dy \left[(x+y)z + \frac{z^2}{2} \right] \Big|_0^1 = \\ = 2 \int_0^1 dx \int_0^1 \left(x+y+\frac{1}{2} \right) dy = 2 \int_0^1 dx \left[\left(x+\frac{1}{2} \right)y + \frac{y^2}{2} \right] \Big|_0^1 = 2 \int_0^1 (x+1) dx = \\ = 2 \left(\frac{1}{2}x^2 + x \right) \Big|_0^1 = 2 \cdot \frac{3}{2} = 3$$

6) $I = \iiint_S x^3 dy dz + y^3 dx dz + z^3 dx dy$, $V: x^2 + y^2 + z^2 = R^2$ (> 0)

$$\begin{aligned} I &= \iiint_V \left(\frac{\partial(x^3)}{\partial x} + \frac{\partial(y^3)}{\partial y} + \frac{\partial(z^3)}{\partial z} \right) dV = 3 \iiint_V (x^2 + y^2 + z^2) dx dy dz = \begin{vmatrix} x = \rho \sin \theta \cos \varphi \\ y = \rho \sin \theta \sin \varphi \\ z = \rho \cos \theta \end{vmatrix} \\ &= 3 \iiint_V \rho^2 \cdot \rho^2 \sin \theta d\rho d\varphi d\theta = 3 \int_0^{2\pi} d\varphi \int_0^\pi \sin \theta d\theta \int_0^R \rho^4 d\rho = \\ &= \frac{3}{5} \int_0^{2\pi} d\varphi \int_0^\pi \sin \theta d\theta (\rho^5|_0^R) = \frac{3}{5} R^5 \int_0^{2\pi} d\varphi \int_0^\pi \sin \theta d\theta = -\frac{3}{5} R^5 \int_0^{2\pi} d\varphi (\cos \theta|_0^\pi) = \\ &= \frac{6}{5} R^5 \int_0^{2\pi} d\varphi = \frac{6}{5} R^5 \varphi|_0^{2\pi} = \frac{12}{5} \pi R^5 \end{aligned}$$

7) $I = \iiint_S x^2 dy dz + y^2 dx dz + z^2 dx dy$, $\frac{z^2}{b^2} = \frac{x^2}{a^2} + \frac{y^2}{a^2}$, $z = b$



$$\begin{aligned} I &= \iiint_V \left(\frac{\partial(x^2)}{\partial x} + \frac{\partial(y^2)}{\partial y} + \frac{\partial(z^2)}{\partial z} \right) dV = 2 \iiint_V (x + y + z) dx dy dz = \\ &= \begin{vmatrix} x = \rho \cos \varphi \\ y = \rho \sin \varphi \\ z = z \end{vmatrix} \quad z = \frac{b}{a} \sqrt{x^2 + y^2} \Rightarrow z = \frac{b}{a} \rho \end{vmatrix} = \end{aligned}$$

$$= 2 \iiint_V (\rho \cos \varphi + \rho \sin \varphi + z) \rho d\rho d\varphi dz = 2 \int_0^{2\pi} d\varphi \int_0^a \rho d\rho \int_{\frac{b}{a}\rho}^b (\rho \cos \varphi + \rho \sin \varphi + z) dz$$

$$I = \underbrace{2 \int_0^{2\pi} d\varphi \int_0^a \rho d\rho \int_{\frac{b}{a}\rho}^b \rho \cos \varphi dz}_{I_1} + \underbrace{2 \int_0^{2\pi} d\varphi \int_0^a \rho d\rho \int_{\frac{b}{a}\rho}^b \rho \sin \varphi dz}_{I_2} + \underbrace{2 \int_0^{2\pi} d\varphi \int_0^a \rho d\rho \int_{\frac{b}{a}\rho}^b z dz}_{I_3}$$

$$I_1 = 2 \int_0^{2\pi} \cos \varphi d\varphi \int_0^a \rho^2 d\rho (z|_{\frac{b}{a}\rho}^b) = 2 \int_0^{2\pi} \cos \varphi d\varphi \int_0^a \rho^2 (b - \frac{b}{a}\rho) d\rho =$$

$$= 2 \int_0^{2\pi} \cos \varphi d\varphi \left[\frac{\rho^3}{3} b - \frac{b}{a} \frac{\rho^4}{4} \right]_0^a = \frac{a^3 b}{6} \int_0^{2\pi} \cos \varphi d\varphi = \frac{a^3 b}{6} \sin \varphi|_0^{2\pi} = 0$$

$$-I_2 = 0 \quad \text{for } \varphi \text{ from } 0 \text{ to } 2\pi$$

$$I_3 = \int_0^{2\pi} d\varphi \int_0^a \rho d\rho (z^2|_{\frac{b}{a}\rho}^b) = \int_0^{2\pi} d\varphi \int_0^a \rho d\rho (b^2 - \frac{b^2}{a^2} \rho^2) = \frac{b^2}{a^2} \int_0^{2\pi} d\varphi \int_0^a (a^2 - \rho^2) \rho d\rho =$$

$$= \frac{b^2}{a^2} \int_0^{2\pi} d\varphi \left(\frac{a^2 \rho^2}{2} - \frac{\rho^4}{4} \right) \Big|_0^a = \frac{b^2}{a^2} \cdot \frac{a^4}{4} \int_0^{2\pi} d\varphi = \frac{a^2 b^2}{4} \varphi|_0^{2\pi} = \frac{\pi a^2 b^2}{2}. \quad I = \frac{\pi a^2 b^2}{2}$$

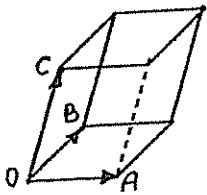
8) $\vec{F} = \vec{m} \times \vec{p}$, $\vec{p} = x\vec{i} + y^2\vec{j} + z^3\vec{k}$, $\vec{m} = a\vec{i} + b\vec{j} + c\vec{k}$, S: $x^2 + y^2 + z^2 = R^2$

$$I = \oint_S \vec{F} d\vec{S} = \iiint_V \text{div } \vec{F} dv$$

$$\vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ a & b & c \\ x & y^2 & z^3 \end{vmatrix} = (bz^3 - cy^2)\vec{i} + (cx - az^3)\vec{j} + (ay^2 - bx)\vec{k},$$

$$\operatorname{div} \vec{F} = \frac{\partial}{\partial x}(bx^3 - cy^2) + \frac{\partial}{\partial y}(cx - az^3) + \frac{\partial}{\partial z}(ay^2 - bx) = 0 \Rightarrow I = 0$$

g)



$$\vec{F} = yz^2\vec{i} - 6yz^2\vec{j} - (z - 2z^3)\vec{k}, \quad 0(0,0,0)$$

$O(0,0,0), A(1,-2,1), B(3,2,1), C(1,0,-1)$

$$\begin{aligned} I &= \oint_S \vec{F} \cdot d\vec{s} = \iiint_V \operatorname{div} \vec{F} \, dV = \iiint_V \left(\frac{\partial}{\partial x}(yz^2) - \frac{\partial}{\partial y}(xyz^2) - \frac{\partial}{\partial z}(z-2z^3) \right) dV = \\ &= \iiint_V (0 - 6xz^2 - 1 + 6z^3) dV = - \iiint_V dV = -V \end{aligned}$$

מנצח: מר. יצחק גולדברג, מנצח: מר. יצחק גולדברג, מנצח: מר. יצחק גולדברג.

$$\overrightarrow{OA} = (1, -2, 1), \quad \overrightarrow{OB} = (3, 2, 1), \quad \overrightarrow{OC} = (1, 0, -1) \Rightarrow V = \pm \overrightarrow{OC} \cdot (\overrightarrow{OA} \times \overrightarrow{OB}) \Rightarrow$$

$$V = \pm \begin{vmatrix} 1 & 0 & -1 \\ 1 & -2 & 1 \\ 3 & 2 & 1 \end{vmatrix} = \pm(-2+0-2-6-2-0) = \pm(-12) = 12 \Rightarrow I = -12$$

10) $\vec{F} = \frac{x^3}{a^2} \vec{i} + \frac{y^3}{b^2} \vec{j} + \frac{z^3}{c^2} \vec{k}$, $S: \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$, (внутри эллипсоида)

$$I = \oint_S \vec{F} \cdot d\vec{S} = \iiint_V \text{div } \vec{F} \, dv = - \iiint_V \left(\frac{\partial}{\partial x} \left(\frac{x^3}{a^2} \right) + \frac{\partial}{\partial y} \left(\frac{y^3}{b^2} \right) + \frac{\partial}{\partial z} \left(\frac{z^3}{c^2} \right) \right) dv =$$

$$= -3 \iiint_V \left(\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} \right) dx dy dz$$

נח'ס חת ה'אל'צ'ג'ל'מ:

$$x = a\rho \sin\theta \cos\varphi, \quad y = b\rho \sin\theta \sin\varphi, \quad z = c\rho \cos\theta, \quad 0 \leq \rho < \infty, \quad 0 \leq \varphi < 2\pi, \quad 0 \leq \theta \leq \pi$$

[illegible]

$$\frac{a^2 p^2 \sin^2 \theta \cos^2 \varphi}{a^2} + \frac{b^2 p^2 \sin^2 \theta \sin^2 \varphi}{b^2} + \frac{c^2 p^2 \cos^2 \theta}{c^2} = 1 \Rightarrow p^2 = 1 \Rightarrow p = 1$$

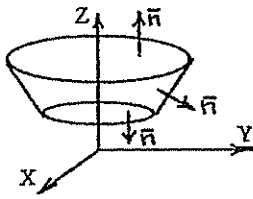
$$J = \frac{D(x, y, z)}{D(\rho, \theta, \phi)} = \begin{vmatrix} a \sin \theta \cos \phi & b \sin \theta \sin \phi & c \cos \theta \\ \rho a \cos \theta \cos \phi & \rho b \cos \theta \sin \phi & -\rho c \sin \theta \\ -\rho a \sin \theta \cos \phi & -\rho b \sin \theta \sin \phi & 0 \end{vmatrix} = abc \rho^2 \sin \theta$$

$$I = -3 \iiint_V \rho^2 |J| d\rho d\varphi d\theta = -3 \iiint_V \rho^2 \cdot abc \rho^2 \sin\theta d\rho d\varphi d\theta =$$

$$= -3abc \int_0^{2\pi} d\varphi \int_0^{\pi} \sin\theta d\theta \int_0^1 \rho^4 d\rho = -3abc (\varphi|_0^{2\pi}) (\frac{1}{5}\rho^5|_0^1) (-\cos\theta|_0^{\pi}) =$$

$$= -3abc \cdot \frac{1}{5} \cdot 2 = -\frac{12}{5}\pi abc$$

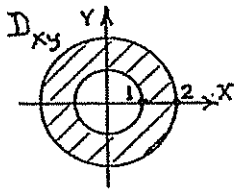
11)



$$\vec{F} = x\vec{i} + y\vec{j} + z\vec{k}, \quad S_1: z = \sqrt{x^2 + y^2}, \quad S_2: z = 1, \quad S_3: z = 2$$

$$I = \oint_S \vec{F} \cdot d\vec{S} \quad \text{ע"פ חוק גאוס} \quad \text{הנורמל הוא } \vec{n} \text{ כלפי מעלה}$$

$$I = I_1 + I_2 + I_3 \quad \text{הנורמל הוא } \vec{n} \text{ כלפי מעלה}$$

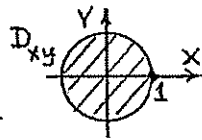
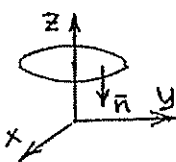


$$I_1 = \iint_{S_1} \vec{F} \cdot d\vec{S}, \quad S_1: z = \sqrt{x^2 + y^2}, \quad 1 \leq z \leq 2, \quad \cos(\hat{n}, \vec{k}) < 0$$

$$I_1 = \iint_{S_1} x dy dz + y dx dz + z dx dy, \quad f(x, y, z) = z^2 - x^2 - y^2 = 0$$

$$I_1 = - \iint_{D_{xy}} (P f'_x + Q f'_y + R f'_z) \frac{dx dy}{f'_z} =$$

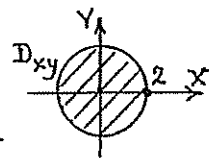
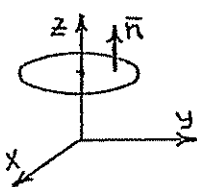
$$= - \iint_{D_{xy}} (-2x^2 - 2y^2 + 2z^2) \frac{dx dy}{2} = \iint_{D_{xy}} (x^2 + y^2 - z^2) \frac{dx dy}{2} = \left| z^2 = x^2 + y^2 \right| = 0$$



$$I_2 = \iint_{S_2} \vec{F} \cdot d\vec{S}, \quad S_2: z = 1, \quad \cos(\hat{n}, \vec{k}) < 0$$

$$I_2 = \iint_{S_2} x dy dz + y dx dz + z dx dy = \left| \frac{z=1}{dz=0} \right| =$$

$$= - \iint_{D_{xy}} dx dy = -S_D = -\pi \cdot 1^2 = -\pi$$



$$I_3 = \iint_{S_3} \vec{F} \cdot d\vec{S}, \quad S_3: z = 2, \quad \cos(\hat{n}, \vec{k}) > 0$$

$$I_3 = \iint_{S_3} x dy dz + y dx dz + z dx dy = \left| \frac{z=2}{dz=0} \right| =$$

$$= \iint_{D_{xy}} 2 dx dy = 2 \iint_{D_{xy}} dx dy = 2S_D = 2 \cdot \pi \cdot 2^2 = 8\pi \Rightarrow$$

$$I = 0 + (-\pi) + 8\pi \Rightarrow I = 7\pi.$$

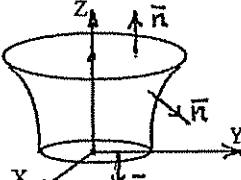
הנורמל הוא כלפי מעלה, $\vec{F} = x\vec{i} + y\vec{j} + z\vec{k}$, V - הנפח הכלוא בתחתית.

$$I = \oint_S \vec{F} \cdot d\vec{S} = \iiint_V \text{div } \vec{F} dv = \iiint_V (1+1+1) dv = 3 \iiint_V dv = 3V$$

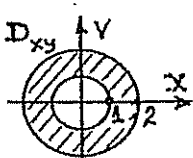
הנפח של התחתית הוא $V = \frac{\pi}{3} h(R^2 + r^2 + Rr)$: הנורמל הוא כלפי מעלה

הנורמל הוא כלפי מעלה, $R=2, r=1, h=1$: הנורמל הוא כלפי מעלה

$$V = \frac{\pi}{3} \cdot 1 \cdot (2^2 + 1^2 + 2 \cdot 1) = \frac{7\pi}{3} \Rightarrow I = 3V = 7\pi$$

12)  $\vec{F} = -xz\vec{i} - yz\vec{j} + (x^2 + y^2)\vec{k}$, $S_1: z^2 + 1 = x^2 + y^2$, $S_2: z=0$, $S_3: z=\sqrt{3}$

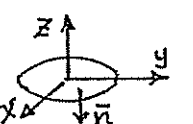
(N) $I = \oint_S \vec{F} d\vec{S} = I_1 + I_2 + I_3$

 $I_1 = \iint_{S_1} \vec{F} d\vec{S}$, $S_1: z^2 + 1 = x^2 + y^2$, $0 \leq z \leq \sqrt{3}$, $\cos(\vec{n}, \vec{k}) < 0$

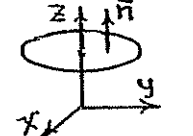
$I_1 = \iint_{S_1} (-xz)dydz - yzdx dz + (x^2 + y^2)dx dy$, $f(x, y, z) = 1 + z^2 - x^2 - y^2 = 0$

$I_1 = - \iint_{D_{xy}} (Pf'_x + Qf'_y + Rf'_z) \frac{dx dy}{f'_z} = - \iint_{D_{xy}} [2x^2z + 2y^2z + (x^2 + y^2)2z] \frac{dx dy}{2z} =$

$= -2 \iint_{D_{xy}} (x^2 + y^2) dx dy = \left| \begin{matrix} x = \rho \cos \varphi \\ y = \rho \sin \varphi \end{matrix} \right| = -2 \iint_{D_{xy}} \rho^2 \rho d\varphi d\rho = -2 \int_0^{2\pi} d\varphi \int_1^2 \rho^3 d\rho = -15\pi$

 $I_2 = \iint_{S_2} \vec{F} d\vec{S}$, $S_2: z=0$, $\cos(\vec{n}, \vec{k}) < 0$

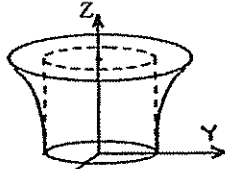
$I_2 = \left| \frac{dz}{dz=0} \right| = - \iint_{D_{xy}} (x^2 + y^2) dx dy = - \int_0^{2\pi} d\varphi \int_0^1 \rho^2 \rho d\rho = -\frac{\pi}{2}$

 $I_3 = \iint_{S_3} \vec{F} d\vec{S}$, $S_3: z=\sqrt{3}$, $\cos(\vec{n}, \vec{k}) > 0$

$I_3 = \left| \frac{dz}{dz=0} \right| = \iint_{D_{xy}} (x^2 + y^2) dx dy = \int_0^{2\pi} d\varphi \int_0^2 \rho^2 \rho d\rho = 8\pi$

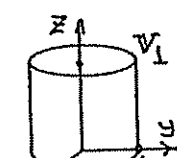
$I = -15\pi - \frac{\pi}{2} + 8\pi = -\frac{15}{2}\pi$

(I)

 $I = \oint_S \vec{F} d\vec{S} = \iiint_V \text{div } \vec{F} dv = \iiint_V (-z - z + 0) dv = -2 \iiint_V z dv$

שם $\delta' \delta z$ כלל $V_1 \cdot V_2 = V_1: \rho' \leq \rho \leq \rho''$ ו V הוא $\rho \leq \rho''$

$I = I_1 + I_2$

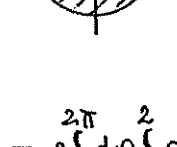
 $I_1 = -2 \iiint_{V_1} z dx dy dz = \left| \begin{matrix} x = \rho \cos \varphi \\ y = \rho \sin \varphi \\ z = z \end{matrix} \right| = -2 \iiint_{V_1} z \rho d\varphi d\rho dz =$

$= -2 \int_0^{2\pi} d\varphi \int_0^1 \rho d\rho \int_0^{\sqrt{3}} z dz = -2 \cdot 2\pi \cdot \frac{1}{2} \cdot \frac{1}{2} (\sqrt{3})^2 = -3\pi$

W פלחחחחח V_1 הוא "ר' $\rho' \leq \rho \leq \rho''$ " ו V_2 הוא "ר' $\rho \leq \rho''$ "

שם $\delta' \delta z$ כלל $V_1 \cdot V_2 = V_1: \rho' \leq \rho \leq \rho''$ ו V הוא $\rho \leq \rho''$

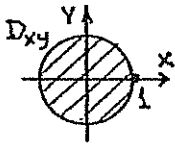
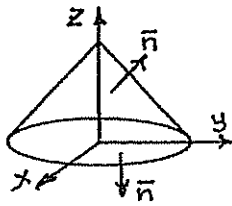
$z^2 + 1 = x^2 + y^2 \Rightarrow z^2 + 1 = \rho^2 \Rightarrow z = \sqrt{\rho^2 - 1} \Rightarrow$

 $I_2 = -2 \iiint_{V_2} z dx dy dz = -2 \iiint_{V_2} z \rho d\varphi d\rho dz =$

$= -2 \int_0^{2\pi} d\varphi \int_1^2 \rho d\rho \int_{\sqrt{\rho^2 - 1}}^{\sqrt{3}} z dz = -2 \cdot 2\pi \int_1^2 \rho d\rho \left(\frac{1}{2} z^2 \Big|_{\sqrt{\rho^2 - 1}}^{\sqrt{3}} \right) = -2\pi \int_1^2 (4\rho - \rho^3) d\rho = -\frac{9}{2}\pi$

$I = -3\pi - \frac{9}{2}\pi = -\frac{15}{2}\pi$

13)



$$\vec{F} = (xz^2 + y^2)\vec{i} + (yx^2 + z^2)\vec{j} + (zy^2 + x^2)\vec{k}; S: z = 1 - \sqrt{x^2 + y^2}$$

'3'-סח נלענח תיח "רלח" $I = \iint_S \vec{F} d\vec{S}$ תיח לענח '3'

:פ"ר תיח S_0 נלענח תיח /נו" $z=0$ רלע'ת

$$\oint_S = I + \iint_{(z=0)} \Rightarrow \oint_S = \iint_S \vec{F} d\vec{S} = \iiint_V \text{div } \vec{F} dV = \iiint_V (z^2 + x^2 + y^2) dV$$

:לרנח סע תללע'ת רל'ס'ס תל'3'רל'ר

$$x = \rho \cos \varphi, y = \rho \sin \varphi, z = z \Rightarrow z = 1 - \sqrt{x^2 + y^2} \Rightarrow z = 1 - \rho$$

$$\oint_S = \iiint_V (z^2 + \rho^2) \rho d\varphi d\rho dz = \int_0^{2\pi} d\varphi \int_0^1 \rho d\rho \int_0^{1-\rho} (z^2 + \rho^2) dz =$$

$$= 2\pi \int_0^1 \rho d\rho \left(\frac{z^3}{3} + \rho^2 z \right) \Big|_0^{1-\rho} = \frac{2\pi}{3} \int_0^1 \rho d\rho [(1-\rho)^3 + 3\rho^2(1-\rho)] =$$

$$= \frac{2\pi}{3} \int_0^1 (1-3\rho+3\rho^2-\rho^3+3\rho^2-3\rho^3) \rho d\rho = \frac{2\pi}{3} \int_0^1 (\rho-3\rho^2+6\rho^3-4\rho^4) d\rho =$$

$$= \frac{2\pi}{3} \left(\frac{\rho^2}{2} - \rho^3 + \frac{3}{2}\rho^4 - \frac{4}{5}\rho^5 \right) \Big|_0^1 = \frac{2\pi}{3} \left(\frac{1}{2} - 1 + \frac{3}{2} - \frac{4}{5} \right) = \frac{2\pi}{15}$$

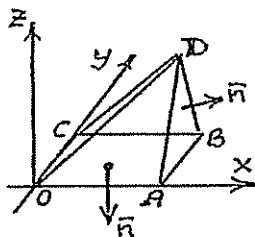
$$\iint_{(z=0)} \vec{F} d\vec{S} = \iint_{S_0} (xz^2 + y^2) dy dz + (yx^2 + z^2) dx dz + (zy^2 + x^2) dx dy \Big|_{dz=0} =$$

$$= - \iint_{D_{xy}} x^2 dx dy = - \iint_{D_{xy}} \rho^2 \cos^2 \varphi \cdot \rho d\varphi d\rho = - \int_0^{2\pi} \cos^2 \varphi d\varphi \int_0^1 \rho^3 d\rho =$$

$$= - \left(\frac{\rho^4}{4} \Big|_0^1 \right) \cdot \frac{1}{2} \int_0^{2\pi} \cos^2 \varphi d\varphi = - \frac{1}{8} \int_0^{2\pi} (1 + \cos 2\varphi) d\varphi = - \frac{1}{8} \left(\varphi + \frac{1}{2} \sin 2\varphi \right) \Big|_0^{2\pi} = - \frac{1}{8} \cdot 2\pi = - \frac{\pi}{4}$$

$$\frac{2\pi}{15} = I - \frac{\pi}{4} \Rightarrow I = \frac{23\pi}{60}$$

14)



$$\vec{F} = (x + y^2 z)\vec{i} + (y + x z^2)\vec{j} + (1 - z)\vec{k}, O(0,0,0), A(4,0,0), B(4,3,0), D(3,2,5)$$

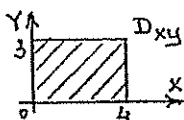
'3'-סח נלענח תיח "רלח" $I = \iint_S \vec{F} d\vec{S}$ תיח לענח '3'

$\oint_S = I + \iint_{(z=0)}$ $\Leftarrow$ (S_0 -נו" $z=0$ רלע'ת)

$$\oint_S = \iint_S \vec{F} d\vec{S} = \iiint_V \text{div } \vec{F} dV = \iiint_V (1 + 1 - 1) dV = \iiint_V dV = V = \frac{1}{3} S_0 h = \frac{1}{3} \cdot 4 \cdot 3 \cdot 5 = 20$$

($D(3,2,5)$: תיח '3', $h=5$ תל'ת, תל'ת, תל'ת סע תל'ת)

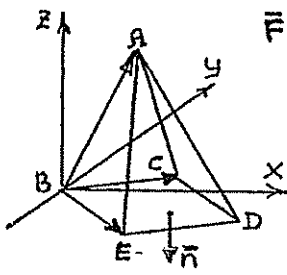
$$\iint_{(z=0)} \vec{F} d\vec{S} = \iint_{S_0} (x + y^2 z) dy dz + (y + x z^2) dx dz + (1 - z) dx dy \Big|_{dz=0} = - \iint_{D_{xy}} dx dy =$$



$$= -S_0 = -4 \cdot 3 = -12$$

$$20 = I - 12 \Rightarrow I = 32$$

15)



$$\vec{F} = (x+z)\vec{i} + (y+x)\vec{j} + z\vec{k}, \quad A(4,5,3), B(0,0,0), C(1,1,0), D(3,-1,0) \\ E(2,-2,0)$$

כך נחשב את $I = \iint_S \vec{F} \cdot d\vec{S}$ את "הזרם" דרך S_0

המשטח S_0 הוא המישור $z=0$ (הבסיס של הטטרהדרון).

$$\Phi = I + \iint_{(z=0)} \Rightarrow \Phi = \iint_{S \cup S_0} \vec{F} \cdot d\vec{S} = \iiint_V \text{div } \vec{F} \, dv =$$

$$= \iiint_V (1+1+1) \, dv = 3 \iiint_V \, dv = 3V$$

$$\vec{BC} = (1,1,0), \vec{BE} = (2,-2,0), \vec{CD} = (3-1, -1-1, 0) = (2,-2,0) \Rightarrow \vec{BE} = \vec{CD}$$

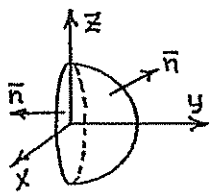
הכיוון הנורמלי של המישור $z=0$ הוא $\vec{n} = (0,0,1)$. לכן, $\vec{BE} \parallel \vec{CD}$ ו- $\vec{BC} \perp \vec{n}$.
 $\vec{BA}, \vec{BE}, \vec{BC}$ הם וקטורים המגדירים את המישור $z=0$.

$$V = \pm \frac{1}{3} \vec{BA} \cdot (\vec{BE} \times \vec{BC}) = \pm \frac{1}{3} \begin{vmatrix} 4 & 5 & 3 \\ 2 & -2 & 0 \\ 1 & 1 & 0 \end{vmatrix} = \pm \frac{1}{3} (12) = 4 \Rightarrow \Phi = 3 \cdot 4 = 12$$

$$\iint_{(z=0)} \vec{F} \cdot d\vec{S} = \iint_{S_0} (x+z) \, dy \, dz + (y+x) \, dx \, dz + z \, dx \, dy \Big|_{z=0} = 0$$

$$12 = I + 0 \Rightarrow I = 12$$

16)



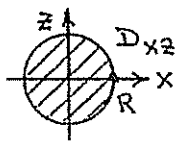
$$u = x^2 + yz, \quad \vec{F} = \text{grad } u, \quad S: x^2 + y^2 + z^2 = R^2, \quad y \geq 0$$

$$\vec{F} = \text{grad } u = \frac{\partial u}{\partial x} \vec{i} + \frac{\partial u}{\partial y} \vec{j} + \frac{\partial u}{\partial z} \vec{k} = 2x\vec{i} + z\vec{j} + y\vec{k}$$

כך נחשב את $I = \iint_S \vec{F} \cdot d\vec{S}$ את "הזרם" דרך S_0

המשטח S_0 הוא המישור $y=0$ (הבסיס של המישור).

$$\Phi = \iint_{S \cup S_0} \vec{F} \cdot d\vec{S} = \iiint_V \text{div } \vec{F} \, dv = \iiint_V (2+0+0) \, dv = 2 \iiint_V \, dv = 2V = \frac{4}{3} \pi R^3$$



$$\iint_{(y=0)} \vec{F} \cdot d\vec{S} = \iint_{S_0} 2x \, dy \, dz + z \, dx \, dz + y \, dx \, dy \Big|_{y=0} = 0$$

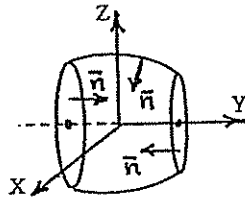
$$= - \iint_{D_{xz}} z \, dx \, dz = \int_0^R \int_0^{2\pi} \rho \sin \varphi \cdot \rho \, d\varphi \, d\rho =$$

$$= - \int_0^{2\pi} \sin \varphi \, d\varphi \int_0^R \rho^2 \, d\rho = (\cos \varphi \Big|_0^{2\pi}) \left(\frac{1}{3} \rho^3 \Big|_0^R \right) = 0$$

האפשרות האחרת היא $\iint_{D_{xz}} z \, dx \, dz$ - עולה ו-0, כי z הוא פונקציה של x ו- z בלבד.
 אי-אפשרות אחרת (התחלה) - $\iint_{D_{xz}} z \, dx \, dz = 0$.

$$\frac{4}{3} \pi R^3 = I + 0 \Rightarrow I = \frac{4}{3} \pi R^3$$

17)



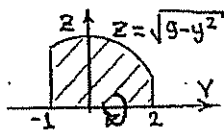
$$\vec{F} = 2xy\vec{i} - y^2\vec{j} + 2z\vec{k}, S: x^2 + y^2 + z^2 = 9, -1 \leq y \leq 2$$

נבחר את "הצורה", $I = \iint_S \vec{F} d\vec{S}$ את פניו של

ה'צורה' $(S_2 / \text{מסגרת})$ $y=2$ ו- $(S_1 / \text{מסגרת})$ $y=-1$ ו- $(S_3 / \text{מסגרת})$ $y=0$

$$I = I_1 + I_2 + I_3 \Rightarrow I = \iint_{S_1 \cup S_2 \cup S_3} \vec{F} d\vec{S} = - \iiint_V \text{div } \vec{F} dv = - \iiint_V (2y - 2y + 2) dv =$$

$$= - \iiint_V 2 dv = -2 \iiint_V dv = -2V$$

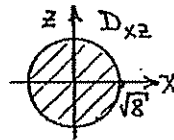
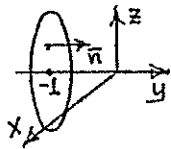


נחשב את V כנפח של חצי כדור.

את המישור $x=0$, נקח $z = \sqrt{9-y^2}$, $-1 \leq y \leq 2$

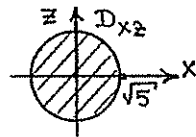
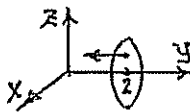
כאשר השטח מתחת לזוהי מסתובב סביב ציר ה- y , מתקבל הגוף המבוקש.

$$V = \pi \int_{-1}^2 z^2 dy = \pi \int_{-1}^2 (9-y^2) dy = \pi \left(9y - \frac{y^3}{3} \right) \Big|_{-1}^2 = 24\pi \Rightarrow I = -2 \cdot 24\pi = -48\pi$$



$$I_1 = \iint_{S_1} \vec{F} d\vec{S} = \iint_{S_1} 2xy dy dz - y^2 dx dz + 2z dx dy =$$

$$= \left| \frac{dy}{dy} = 0 \right| = - \iint_{D_{xz}} dx dz = -S_D = -\pi(\sqrt{8})^2 = -8\pi$$



$$I_2 = \iint_{S_2} 2xy dy dz - y^2 dx dz + 2z dx dy =$$

$$= \left| \frac{dy}{dy} = 0 \right| = - \iint_{D_{xz}} (-2^2) dx dz = 4 \iint_{D_{xz}} dx dy = 4S_D = 4 \cdot \pi(\sqrt{5})^2 = 20\pi$$

$$-48\pi = I_1 - 8\pi + 20\pi \Rightarrow I = -60\pi$$

רוטור של וקטור ומשפט סטוקס

(X)

$$1a) \vec{F} = z^2 \vec{i} + y^2 \vec{j} + x^2 \vec{k}$$

$$\text{rot } \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ z^2 & y^2 & x^2 \end{vmatrix} = \vec{i}(0-0) + \vec{j}(2z-2x) + \vec{k}(0-0) = 2(z-x)\vec{j}$$

$$1b) \vec{F} = (x+z)\vec{i} + (y+z)\vec{j} + (x^2+z)\vec{k}$$

$$\text{rot } \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x+z & y+z & x^2+z \end{vmatrix} = \vec{i}(0-1) + \vec{j}(1-2x) + \vec{k}(0-0) = -\vec{i} + (1-2x)\vec{j}$$

$$1c) \vec{F} = (x^2+y^2)\vec{i} + (y^2+z^2)\vec{j} + (z^2+x^2)\vec{k}$$

$$\text{rot } \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2+y^2 & y^2+z^2 & z^2+x^2 \end{vmatrix} = \vec{i}(0-2z) + \vec{j}(0-2x) + \vec{k}(0-2y) = -2z\vec{i} - 2x\vec{j} - 2y\vec{k}$$

$$2) \vec{F} = |\vec{r}| \vec{r}, \vec{r} = x\vec{i} + y\vec{j} + z\vec{k} \Rightarrow |\vec{r}| = \sqrt{x^2+y^2+z^2}, \vec{F} = x|\vec{r}|\vec{i} + y|\vec{r}|\vec{j} + z|\vec{r}|\vec{k}$$

$$\text{rot } \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x|\vec{r}| & y|\vec{r}| & z|\vec{r}| \end{vmatrix} =$$

$$= \vec{i}(z \frac{\partial}{\partial y} |\vec{r}| - y \frac{\partial}{\partial z} |\vec{r}|) + \vec{j}(x \frac{\partial}{\partial z} |\vec{r}| - z \frac{\partial}{\partial x} |\vec{r}|) + \vec{k}(y \frac{\partial}{\partial x} |\vec{r}| - x \frac{\partial}{\partial y} |\vec{r}|)$$

הנכונות של הנוסחה עבור $\vec{r}$ היא $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$ וכן $|\vec{r}| = \sqrt{x^2+y^2+z^2}$

$$z \frac{\partial}{\partial y} |\vec{r}| - y \frac{\partial}{\partial z} |\vec{r}| = z \frac{\partial}{\partial y} (\sqrt{x^2+y^2+z^2}) - y \frac{\partial}{\partial z} (\sqrt{x^2+y^2+z^2}) =$$

$$= z \cdot \frac{y}{\sqrt{x^2+y^2+z^2}} - y \cdot \frac{z}{\sqrt{x^2+y^2+z^2}} = 0 \Rightarrow \text{rot } \vec{F} = 0\vec{i} + 0\vec{j} + 0\vec{k} = 0$$

$$3) \vec{F} = P(x,y,z)\vec{i} + Q(x,y,z)\vec{j} + R(x,y,z)\vec{k} \Rightarrow \lambda \vec{F} = \lambda P\vec{i} + \lambda Q\vec{j} + \lambda R\vec{k}$$

$$\text{rot}(\lambda \vec{F}) = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \lambda P & \lambda Q & \lambda R \end{vmatrix} = \lambda \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & R \end{vmatrix} = \lambda \cdot \text{rot } \vec{F}$$

4) $\vec{F} = \vec{m} |\vec{r}|$, $\vec{m} = a\vec{i} + b\vec{j} + c\vec{k}$, $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k} \Rightarrow |\vec{r}| = \sqrt{x^2 + y^2 + z^2}$

$$\vec{F} = m \vec{r} = a|\vec{r}|\vec{i} + b|\vec{r}|\vec{j} + c|\vec{r}|\vec{k}$$

$$\text{rot } \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ a|\vec{r}| & b|\vec{r}| & c|\vec{r}| \end{vmatrix} =$$

$$= (c \frac{\partial}{\partial y} |\vec{r}| - b \frac{\partial}{\partial z} |\vec{r}|) \vec{i} + (a \frac{\partial}{\partial z} |\vec{r}| - c \frac{\partial}{\partial x} |\vec{r}|) \vec{j} + (b \frac{\partial}{\partial x} |\vec{r}| - a \frac{\partial}{\partial y} |\vec{r}|) \vec{k}$$

הנהגת מוסדות המערכת תהיה כפי שצוין לעיל.

$$c \frac{\partial}{\partial y} |\vec{r}| - b \frac{\partial}{\partial z} |\vec{r}| = c \frac{\partial}{\partial y} (\sqrt{x^2 + y^2 + z^2}) - b \frac{\partial}{\partial z} (\sqrt{x^2 + y^2 + z^2}) =$$

$$= c \frac{y}{\sqrt{x^2+y^2+z^2}} - b \frac{z}{\sqrt{x^2+y^2+z^2}} = \frac{cy}{|F|} - \frac{bz}{|F|} = \frac{cy-bz}{|F|}$$

$(\bar{x}, \bar{y})$ של P' נמצא במרחק $\frac{bx-ay}{|F|}$, $\frac{ax-cx}{|F|}$: קראו δ ו δ' ו δ'' ו δ'''

$$\text{rot } \vec{F} = \frac{1}{|\vec{r}|^3} [(cy - bz)\vec{i} + (az - cx)\vec{j} + (bx - ay)\vec{k}]$$

את ה'ט"ו בסיון תר"ם חתמו/ה'ר' אפרם רבין כצט"ר/ה'ה:

$$(cy - bz)\vec{i} + (az - cx)\vec{j} + (bx - ay)\vec{k} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ x & y & z \\ a & b & c \end{vmatrix} = \vec{r} \times \vec{m} \Rightarrow \text{rot } \vec{F} = \frac{\vec{r} \times \vec{m}}{|\vec{r}|}$$

5) $\vec{u} = u_x \vec{i} + u_y \vec{j} + u_z \vec{k}$, $\vec{v} = v_x \vec{i} + v_y \vec{j} + v_z \vec{k}$

$$\vec{u} + \vec{v} = (u_x + v_x)\vec{i} + (u_y + v_y)\vec{j} + (u_z + v_z)\vec{k}$$

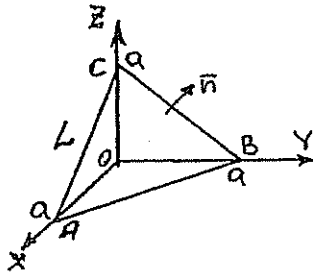
$$\text{rot}(\bar{u} + \bar{v}) = \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ u_x + v_x & u_y + v_y & u_z + v_z \end{vmatrix} =$$

$$= \begin{vmatrix} \bar{c} & \bar{j} & \bar{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ u_x & u_y & u_z \end{vmatrix} + \begin{vmatrix} \bar{c} & \bar{j} & \bar{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ v_x & v_y & v_z \end{vmatrix} = \text{rot } \bar{u} + \text{rot } \bar{v}$$

• ବ୍ୟାପ୍ୟ ବ୍ୟବହାର

(ג)

1)



$$I = \oint_L (z-y)dx + (x-z)dy + (y-x)dz, L: \Delta ABC$$

בתור n נבחר L המקביל לוקטור $\vec{n}$, כלומר L יהיה $\vec{n}$ וכל $\vec{n}$ יהיה $\vec{n}$ וכל $\vec{n}$ יהיה $\vec{n}$. ABC עשוי מן ABC וכל $\vec{n}$ יהיה $\vec{n}$.

$$I = \iint_S \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ z-y & x-z & y-x \end{vmatrix} d\vec{S} =$$

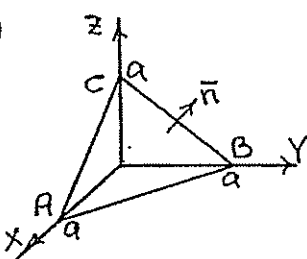
$$= \iint_S \left[\frac{\partial}{\partial y}(y-x) - \frac{\partial}{\partial z}(x-z) \right] dydz - \left[\frac{\partial}{\partial x}(y-x) - \frac{\partial}{\partial z}(z-y) \right] dx dz + \left[\frac{\partial}{\partial x}(x-z) - \frac{\partial}{\partial y}(z-y) \right] dx dy =$$

$$= \iint_S (1+1) dydz - (-1-1) dx dz + (1+1) dx dy = 2 \iint_S dydz + dx dz + dx dy =$$

$$= 2 \left[\iint_{D_{yz}} dydz + \iint_{D_{xz}} dx dz + \iint_{D_{xy}} dx dy \right] = 2 \left(\iint_{\Delta BOC} + \iint_{\Delta AOC} + \iint_{\Delta AOB} \right) =$$

$$= 2 \cdot (3 \cdot \iint_{\Delta AOB}) = 2 \cdot 3 \cdot \frac{1}{2} a \cdot a = 3a^2$$

2)



$$I = \oint_L x(z-y)dx + y(x-z)dy + z(y-x)dz, L: \Delta ABC$$

בתור n נבחר L המקביל לוקטור $\vec{n}$, כלומר L יהיה $\vec{n}$ וכל $\vec{n}$ יהיה $\vec{n}$ וכל $\vec{n}$ יהיה $\vec{n}$. ABC עשוי מן ABC וכל $\vec{n}$ יהיה $\vec{n}$.

$$I = \iint_S \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x(z-y) & y(x-z) & z(y-x) \end{vmatrix} d\vec{S} =$$

$$= \iint_S \left[\frac{\partial}{\partial y}(zy-zx) - \frac{\partial}{\partial z}(yx-yz) \right] dydz - \left[\frac{\partial}{\partial x}(zy-zx) - \frac{\partial}{\partial z}(xz-xy) \right] dx dz +$$

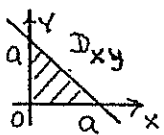
$$+ \left[\frac{\partial}{\partial x}(yx-yz) - \frac{\partial}{\partial y}(xz-xy) \right] dx dy = \iint_S (z+y) dydz + (z+x) dx dz + (x+y) dx dy$$

הנורמל $\vec{n}$ הוא $\vec{n} = (1, 1, 1)$ וכל $\vec{n}$ יהיה $\vec{n}$ וכל $\vec{n}$ יהיה $\vec{n}$. ABC עשוי מן ABC וכל $\vec{n}$ יהיה $\vec{n}$.

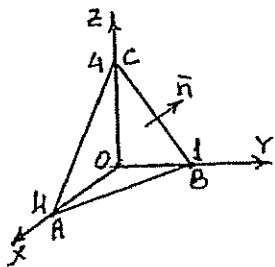
$$1 \cdot (x-a) + 1 \cdot (y-0) + 1 \cdot (z-0) = 0 \Rightarrow x+y+z=a \Rightarrow f(x,y,z) = x+y+z-a=0$$

$$I = \iint_{D_{xy}} (P f'_x + Q f'_y + R f'_z) \frac{dx dy}{f'_z} = \iint_{D_{xy}} [(z+y) + (z+x) + (x+y)] \frac{dx dy}{1} =$$

$$= 2 \iint_{D_{xy}} (x+y+z) dx dy = 2 \iint_{D_{xy}} (x+y+a) dx dy = 2a \iint_{D_{xy}} dx dy = 2a \cdot \frac{a^2}{2} = a^3.$$



3)



$$I = \oint_L (2z-x)dx + (x+2z)dy + 3zdz, \quad L: \Delta ABC$$

בתור המסלול, נבחר את L כעקב השעון, ΔABC עליונה וסביבה
:סביבה סגורה

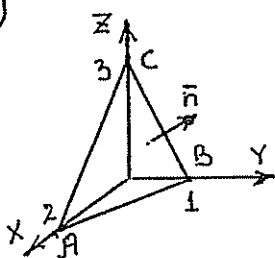
$$I = \iint_S \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2z-x & x+2z & 3z \end{vmatrix} d\bar{S} =$$

$$= \iint_S \left[\frac{\partial(3z)}{\partial y} - \frac{\partial(x+2z)}{\partial z} \right] dydz - \left[\frac{\partial(3z)}{\partial x} - \frac{\partial(2z-x)}{\partial z} \right] dx dz + \left[\frac{\partial(x+2z)}{\partial x} - \frac{\partial(2z-x)}{\partial y} \right] dx dy =$$

$$= \iint_S (-2) dydz - (-2) dx dz + 1 dx dy = -2 \iint_{D_{yz}} dydz + 2 \iint_{D_{xz}} dx dz + \iint_{D_{xy}} dx dy =$$

$$= -2S_{\Delta BOC} + 2S_{\Delta AOC} + S_{\Delta AOB} = -2 \cdot \frac{1}{2} \cdot 1 \cdot 4 + 2 \cdot \frac{1}{2} \cdot 4 \cdot 4 + \frac{1}{2} \cdot 4 \cdot 1 = 14$$

4)



$$I = \oint_L y^3 dx + 3xy^2 dy + x^2 dz, \quad L: \Delta ABC$$

בתור המסלול, נבחר את L כעקב השעון, ΔABC עליונה וסביבה
:סביבה סגורה

$$I = \iint_S \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y^3 & 3xy^2 & x^2 \end{vmatrix} \bar{n} \cdot d\bar{S} =$$

$$= \iint_S \left[\left(\frac{\partial(x^2)}{\partial y} - \frac{\partial(3xy^2)}{\partial z} \right) \bar{i} + \left(\frac{\partial(y^3)}{\partial z} - \frac{\partial(x^2)}{\partial x} \right) \bar{j} + \left(\frac{\partial(3xy^2)}{\partial x} - \frac{\partial(y^3)}{\partial y} \right) \bar{k} \right] \bar{n} \cdot d\bar{S}$$

$$= \iint_S [0\bar{i} + (-2x)\bar{j} + (3y^2 - 3y^2)\bar{k}] \bar{n} \cdot d\bar{S} = \iint_S (0, -2x, 0) \bar{n} \cdot d\bar{S}$$

$$\bar{CB} = (0, 1, -3), \quad \bar{CA} = (2, 0, -3), \quad \text{נורמלית סביבה}$$

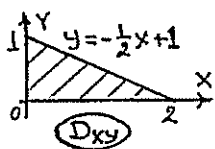
$$\bar{n} = \bar{CA} \times \bar{CB} = \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ 2 & 0 & -3 \\ 0 & 1 & -3 \end{vmatrix} = 3\bar{i} + 6\bar{j} + 2\bar{k} \Rightarrow B(0, 1, 0) \Rightarrow 3(x-0) + 6(y-1) + 2(z-0) = 0$$

$$\bar{n} \cdot \bar{n} = 14 \Rightarrow \bar{n} = \left(\frac{3}{\sqrt{14}}, \frac{6}{\sqrt{14}}, \frac{2}{\sqrt{14}} \right) \Rightarrow \bar{n} \cdot d\bar{S} = \left(\frac{3}{2}, 3, 1 \right) dx dy$$

$$z = 3 - \frac{3}{2}x - 3y \Rightarrow dS = \sqrt{1 + z_x'^2 + z_y'^2} dx dy = \sqrt{1 + \frac{9}{4} + 9} dx dy = \frac{7}{2} dx dy$$

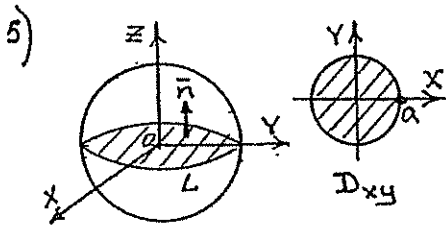
$$|\bar{n}| = \sqrt{9 + 36 + 4} = 7 \Rightarrow \bar{n} = \left(\frac{3}{7}, \frac{6}{7}, \frac{2}{7} \right) \Rightarrow \bar{n} \cdot d\bar{S} = \left(\frac{3}{2}, 3, 1 \right) dx dy$$

$$I = \iint_{D_{xy}} (0, -2x, 0) \cdot \left(\frac{3}{2}, 3, 1 \right) dx dy = \iint_{D_{xy}} (-3x) dx dy = -6 \iint_{D_{xy}} x dx dy =$$



$$= -6 \int_0^2 x dx \int_0^{-\frac{1}{2}x+1} dy = -6 \int_0^2 x dx \left(y \Big|_0^{-\frac{1}{2}x+1} \right) = -6 \int_0^2 x \left(1 - \frac{1}{2}x \right) dx =$$

$$= -6 \int_0^2 \left(x - \frac{x^2}{2} \right) dx = -6 \left(\frac{x^2}{2} - \frac{x^3}{6} \right) \Big|_0^2 = -6 \left(2 - \frac{4}{3} \right) = -4$$



$$I = \oint_L x^2 y^3 dx + dy + z dz, \quad L: \begin{cases} x^2 + y^2 + z^2 = a^2 \\ z = 0 \end{cases}$$

התור המעגלי של התור הוא $z=0$ והוא
הוא $z=0$ והוא $z=0$.

$$I = \iint_S \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \end{vmatrix} \cdot \vec{n}^0 ds =$$

$$= \iint_S \left[\frac{\partial z}{\partial y} - 0 \right] dy dz - \left[\frac{\partial z}{\partial x} - \frac{\partial(x^2 y^3)}{\partial z} \right] dx dz + \left[0 - \frac{\partial(x^2 y^3)}{\partial y} \right] dx dy =$$

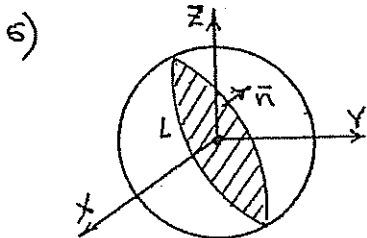
$$= -3 \iint_{D_{xy}} x^2 y^2 dx dy = \left| \begin{matrix} x = \rho \cos \varphi \\ y = \rho \sin \varphi \end{matrix} \right| = -3 \iint_{D_{xy}} \rho^4 \cos^2 \varphi \sin^2 \varphi \rho d\rho d\varphi =$$

$$= -3 \int_0^{2\pi} \cos^2 \varphi \sin^2 \varphi d\varphi \int_0^a \rho^5 d\rho = -\frac{1}{2} \int_0^{2\pi} \cos^2 \varphi \sin^2 \varphi (\rho^6|_0^a) =$$

$$= -a^6 \cdot \frac{1}{2} \cdot \frac{1}{4} \int_0^{2\pi} (\sin^2 2\varphi) d\varphi = -a^6 \cdot \frac{1}{8} \cdot 4 \int_0^{\pi/2} \sin^2 2\varphi d\varphi =$$

$$= -a^6 \cdot \frac{1}{2} \cdot \frac{1}{2} \int_0^{\pi/2} 2 \sin^2 2\varphi d\varphi = -\frac{a^6}{4} \int_0^{\pi/2} (1 - \cos 4\varphi) d\varphi = -\frac{a^6}{4} \left(\varphi - \frac{1}{4} \sin 4\varphi \right) \Big|_0^{\pi/2} \Rightarrow$$

$$I = -\frac{\pi a^6}{8}$$



$$I = \oint_L y dx + z dy + x dz, \quad L: \begin{cases} x^2 + y^2 + z^2 = a^2 \\ x + y + z = 0 \end{cases}$$

התור המעגלי של התור הוא $x+y+z=0$ והוא
הוא $x+y+z=0$ והוא $x+y+z=0$.

$$I = \iint_S \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \end{vmatrix} \cdot \vec{n}^0 ds = \iint_S \left[\left(\frac{\partial(z)}{\partial y} - \frac{\partial(y)}{\partial z} \right) \bar{i} + \left(\frac{\partial(x)}{\partial z} - \frac{\partial(z)}{\partial x} \right) \bar{j} + \left(\frac{\partial(y)}{\partial x} - \frac{\partial(x)}{\partial y} \right) \bar{k} \right] \cdot \vec{n}^0 ds =$$

$$= \iint_S (-\bar{i} - \bar{j} - \bar{k}) \cdot \vec{n}^0 ds = -\iint_S (1, 1, 1) \cdot \vec{n}^0 ds$$

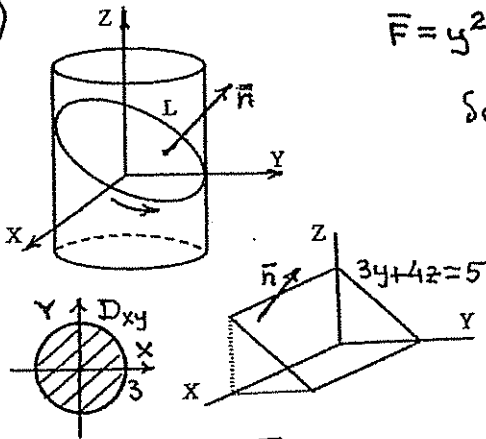
$$\cdot \vec{n}^0 = \frac{1}{\sqrt{3}} (1, 1, 1) : \text{כיוון } n = (1, 1, 1) : \text{הנורמל של } x+y+z=0$$

$$I = -\iint_S (1, 1, 1) \cdot \frac{1}{\sqrt{3}} (1, 1, 1) ds = -\frac{1}{\sqrt{3}} \iint_S (1+1+1) ds = -\frac{3}{\sqrt{3}} \iint_S ds = -\sqrt{3} \cdot S.$$

כאן S הוא שטח המעגל של התור L והוא $S = \pi a^2$ $\Leftarrow$ a הוא רדיוס המעגל.

$$I = -\sqrt{3} \cdot \pi a^2 \Rightarrow I = -\pi a^2 \sqrt{3}.$$

9)



$$\vec{F} = y^2 \vec{i} + z^2 \vec{j}, \quad L: \begin{cases} x^2 + y^2 = 9 \\ 3y + 4z = 5 \end{cases}$$

מסמך של האינטגרל הזה הוא אוליפס (קו ההתאמה) של המישור.

התור המעט S נבחר את הצורה האליפטית של המישור $3y + 4z = 5$ ו- L הוא השדה של 5 . כלומר $3y + 4z = 5$ הוא:

$$I = \oint_L \vec{F} d\vec{r} = \iint_S \text{rot } \vec{F} \vec{n} \cdot d\vec{S}$$

$$\text{rot } \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y^2 & z^2 & 0 \end{vmatrix} = \vec{i}(0 - 2z) + \vec{j}(0 - 0) + \vec{k}(0 - 2y) = -2z\vec{i} + 0\vec{j} - 2y\vec{k}$$

$$|\vec{n}| = \sqrt{9 + 16} = 5 \quad \vec{n} = (0, 3, 4) \quad \text{כדי } 3y + 4z - 5 = 0 \quad \text{המשוואה}$$

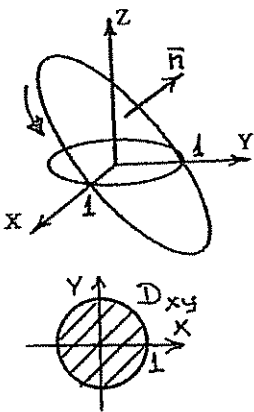
$$\vec{n} = (0, \frac{3}{5}, \frac{4}{5}) \Rightarrow \text{rot } \vec{F} \vec{n} = (-2z, 0, -2y) \cdot (\frac{3}{5}, \frac{4}{5}) = -\frac{8}{5}y$$

$$z = \frac{5}{4} - \frac{3}{4}y \Rightarrow dS = \sqrt{1 + z_x'^2 + z_y'^2} dx dy = \sqrt{1 + 0 + \frac{9}{16}} dx dy = \frac{5}{4} dx dy$$

$$I = \iint_S \text{rot } \vec{F} \vec{n} \cdot d\vec{S} = -\frac{8}{5} \cdot \frac{5}{4} \iint_{D_{xy}} y dx dy = -2 \iint_{D_{xy}} y dx dy = 0$$

(האינטגרל הזה הוא אוליפס של המישור).

10)



$$\vec{F} = (y-x)\vec{i} + (y+x)\vec{j} + (y+z)\vec{k}, \quad L: \begin{cases} x^2 + y^2 = 1 \\ x + y + z = 1 \end{cases}$$

מסמך של האינטגרל הזה הוא אוליפס (קו ההתאמה) של המישור $x + y + z = 1$. התור המעט S נבחר את הצורה האליפטית של המישור $x + y + z = 1$ ו- L הוא השדה של 1 .

$$I = \oint_L \vec{F} d\vec{r} = \iint_S \text{rot } \vec{F} \vec{n} \cdot d\vec{S}$$

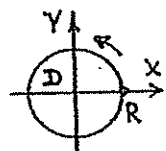
$$\text{rot } \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y-x & y+x & y+z \end{vmatrix} = 1\vec{i} + 0\vec{j} + (1-1)\vec{k} = \vec{i} + 0\vec{j} + 0\vec{k}$$

המשוואה של המישור $x + y + z = 1$ היא $\vec{n} = (1, 1, 1)$ כפי שניתן לראות.

$$\vec{n} = (1, 1, 1) \Rightarrow |\vec{n}| = \sqrt{3} \Rightarrow \vec{n} = \frac{1}{\sqrt{3}} (1, 1, 1)$$

$$z = 1 - x - y \Rightarrow dS = \sqrt{1 + z_x'^2 + z_y'^2} dx dy = \sqrt{1 + 1 + 1} dx dy = \sqrt{3} dx dy$$

$$I = \iint_S \text{rot } \vec{F} \vec{n} \cdot d\vec{S} = \iint_{D_{xy}} (1, 0, 0) \cdot \frac{1}{\sqrt{3}} (1, 1, 1) \cdot \sqrt{3} dx dy = \iint_{D_{xy}} dx dy = \pi \cdot 1^2 = \pi$$



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$$L: x^2 + y^2 = R^2, \quad z = 0$$

• ጸ-፩ ስለሚፈጸምበት ዓ/ታ
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$$I = \iint_S \text{rot } \vec{F} d\vec{S} = \oint_L \vec{F} d\vec{r}$$

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: $\rho \delta$, $z=0$ ላይ የሚገኝ ሲሆን ለ $\delta/\delta_{\text{መጨረሻ}} \ll 1$ ይታወቃል።

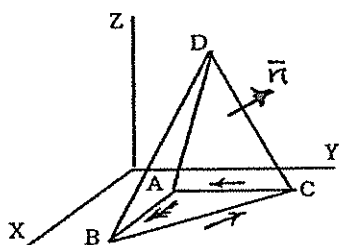
$$\vec{F}|_L = -y^3 \vec{i} + x^3 \vec{j} + 0 \vec{k} \Rightarrow \int_L \vec{F} \cdot d\vec{r} = \int_L -y^3 dx + x^3 dy$$

[illegible]

$$\begin{aligned} I &= \oint_C -y^3 dx + x^3 dy = \iint_D \left[\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right] dx dy = \iint_D \left(\frac{\partial}{\partial x} (x^3) - \frac{\partial}{\partial y} (-y^3) \right) dx dy = \\ &= 3 \iint_D (x^2 + y^2) dx dy = \left| \begin{matrix} x = \rho \cos \varphi \\ y = \rho \sin \varphi \end{matrix} \right| = 3 \iint_D \rho^2 \cdot \rho d\varphi d\rho = 3 \int_0^{2\pi} d\varphi \int_0^R \rho^3 d\rho = \end{aligned}$$

$$= 3 \cdot 2\pi \cdot \frac{1}{4} \rho^4 \Big|_0^R = \frac{3}{2} \pi R^4$$

14)



$$\vec{F} = (2y + xz + \sqrt{x^2 + y^2})\vec{i} + y[z + \ln(x + \sqrt{x^2 + y^2})]\vec{j} + (z^2 + z)\vec{k}$$

$$A(1,1,0), B(5,1,0), C(1,4,0), D(2,2,6)$$

[illegible]

$$I = \oint_S \text{rot } \vec{F} d\vec{S} = \oint_L \vec{F} d\vec{r}$$

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[illegible]

$$\vec{F} = (2y + \sqrt{x^2 + y^2})\vec{i} + y \ln(x + \sqrt{x^2 + y^2})\vec{j}$$

$$I = \oint_C (2y + \sqrt{x^2 + y^2}) dx + y \ln(x + \sqrt{x^2 + y^2}) dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy$$

$$\frac{\partial Q}{\partial x} = y \cdot \frac{1}{x + \sqrt{x^2 + y^2}} \left(1 + \frac{x}{\sqrt{x^2 + y^2}} \right) = y \cdot \frac{x + \sqrt{x^2 + y^2}}{(x + \sqrt{x^2 + y^2}) \sqrt{x^2 + y^2}} = \frac{y}{\sqrt{x^2 + y^2}}$$

$$\frac{\partial P}{\partial y} = 2 + \frac{y}{\sqrt{x^2 + y^2}} \Rightarrow \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = \frac{y}{\cancel{\sqrt{x^2 + y^2}}} - 2 - \frac{y}{\cancel{\sqrt{x^2 + y^2}}} = -2$$

$$I = \iint_D (-2) dx dy = -2 \iint_D dx dy = -2 S_D = -2 \cdot \frac{1}{2} AB \cdot AC = -4 \cdot 3 = -12$$

$\text{rot } \vec{F} = 0$: $\rho''_{\vec{r}}$ מן ρ לא נכנס אצלו $\vec{F}$

$\bar{F}$ הוא עצה משמרת. נבחר $\gamma/7$ כעולה, $\delta \in M(0,0,0)$ סמל,

הנה נא להעביר את המעורבות הזו

$\vec{r}$ הוא NeN . נבחר $N(0,0,0)$ כשדה:

$$u = xy + yz + zx + C$$

2. μ_{eff} ከ μ_{free} የሚለየው

$$5) \vec{F} = yz \cos(xy) \vec{i} + xz \cos(xy) \vec{j} + \sin(xy) \vec{k}$$

$$\text{rot } \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ yz \cos(xy) & xz \cos(xy) & \sin(xy) \end{vmatrix} = \vec{i}(x \cos(xy) - x \cos(xy)) +$$

$$+ \vec{j}(y \cos(xy) - y \cos(xy)) + \vec{k}(z \cos(xy) - xy z \sin(xy) - z \cos(xy) + xy z \sin(xy)) = 0$$

.M(0,0,0) נקודת המנוחה של $\vec{F}$

$$U = \int_0^x yz \cos(xy) dx + \int_0^y 0 dy + \int_0^z 0 dz = z \int_0^x \cos(xy) d(xy) = z \sin(xy) \Big|_0^x = z \sin(xy)$$

$$U = z \sin(xy) + C$$

$$6) \vec{F} = f(x, y, z)(\vec{i} + \vec{j} + \vec{k}), \quad f(x, y, z) = \frac{1}{x+y+z}$$

$$\text{rot } \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{f}{1} & \frac{f}{1} & \frac{f}{1} \end{vmatrix} = \vec{i}(f'_y - f'_z) + \vec{j}(f'_z - f'_x) + \vec{k}(f'_x - f'_y)$$

:מכיוון ש $x+y+z=t$ נקודת המנוחה של $\vec{F}$

$$f'_x = f'_t \cdot t'_x = f'_t(x+y+z)'_x = f'_t \cdot 1 = f'_t \Rightarrow f'_y = f'_t \cdot t'_y = f'_t, \quad f'_z = f'_t \cdot t'_z = f'_t$$

$$\text{rot } \vec{F} = \vec{i}(f'_t - f'_t) + \vec{j}(f'_t - f'_t) + \vec{k}(f'_t - f'_t) = 0$$

.M(0,0,1) נקודת המנוחה של $\vec{F}$

$$U = \int_0^x \frac{dx}{x+y+z} + \int_0^y \frac{dy}{y+z} + \int_1^z \frac{dz}{z} = \int_0^x \frac{d(x+y+z)}{x+y+z} + \int_0^y \frac{d(y+z)}{y+z} + \ln|z| \Big|_1^z =$$

$$= \ln|x+y+z| \Big|_0^x + \ln|y+z| \Big|_0^y + \ln|z| = \ln|x+y+z| - \ln|y+z| + \ln|y+z| - \ln|z| +$$

$$+ \ln|z| = \ln|x+y+z| \Rightarrow U = \ln|x+y+z| + C$$

$$7) \vec{F} = \left(\sqrt{z} + \frac{y}{2\sqrt{x}}\right) \vec{i} + \left(\sqrt{x} + \frac{z}{2\sqrt{y}}\right) \vec{j} + \left(\sqrt{y} + \frac{x}{2\sqrt{z}}\right) \vec{k}$$

$$\text{rot } \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \sqrt{z} + \frac{y}{2\sqrt{x}} & \sqrt{x} + \frac{z}{2\sqrt{y}} & \sqrt{y} + \frac{x}{2\sqrt{z}} \end{vmatrix} = \vec{i}\left(\frac{1}{2\sqrt{y}} - \frac{1}{2\sqrt{y}}\right) + \vec{j}\left(\frac{1}{2\sqrt{z}} - \frac{1}{2\sqrt{z}}\right) + \vec{k}\left(\frac{1}{2\sqrt{x}} - \frac{1}{2\sqrt{x}}\right) = 0$$

.M(1,1,1) נקודת המנוחה של $\vec{F}$

$$U = \int_1^x \left(\sqrt{z} + \frac{y}{2\sqrt{x}}\right) dx + \int_1^y \left(1 + \frac{z}{2\sqrt{y}}\right) dy + \int_1^z \left(1 + \frac{1}{2\sqrt{z}}\right) dz = (x\sqrt{z} + y\sqrt{x}) \Big|_1^x + (y + z\sqrt{y}) \Big|_1^y +$$

$$+ (z + \sqrt{z}) \Big|_1^z = x\sqrt{z} + y\sqrt{x} - \sqrt{z} - y + y + z\sqrt{y} - 1 - z + z + \sqrt{z} - 1 - 1 = x\sqrt{z} + y\sqrt{x} + z\sqrt{y} - 3$$

$$U = x\sqrt{z} + y\sqrt{x} + z\sqrt{y} + C$$

8) $\vec{F} = x\vec{i} - y^2\vec{j} + z\vec{k}$, $A(1, -1, 2)$, $B(2, 1, 3)$

$$\text{rot } \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x & -y^2 & z \end{vmatrix} = (0-0)\vec{i} + (0-0)\vec{j} + (0-0)\vec{k} = 0$$

$\vec{F}$ הוא שדה משמר, ניתן לבדוק את הצורה של $\vec{F}$ בעזרת $M(0,0,0)$ נבחר

$$U = \int_0^x x dx - \int_0^y y^2 dy + \int_0^z z dz = \frac{x^2}{2} \Big|_0^x - \frac{y^3}{3} \Big|_0^y + \frac{z^2}{2} \Big|_0^z = \frac{x^2}{2} - \frac{y^3}{3} + \frac{z^2}{2}$$

$$W = \int_A^B \vec{F} d\vec{r} = U(B) - U(A) = U(2, 1, 3) - U(1, -1, 2) = (2 - \frac{1}{3} + \frac{9}{2}) - (\frac{1}{2} + \frac{1}{3} + 2) = \frac{10}{3}$$

9) $\vec{F} = yz\vec{i} + zx\vec{j} + xy\vec{k}$, $A(0, 2, 3)$, $B(3, 2, 1)$

$$\text{rot } \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ yz & zx & xy \end{vmatrix} = \vec{i}(x-x) + \vec{j}(y-y) + \vec{k}(z-z) = 0$$

$\vec{F}$ הוא שדה משמר, ניתן לבדוק את הצורה של $\vec{F}$ בעזרת $M(0,0,0)$ נבחר

$$U = \int_0^x yz dx + \int_0^y 0 dy + \int_0^z 0 dz = xyz \Big|_0^x = xyz$$

$$W = \int_A^B \vec{F} d\vec{r} = U(B) - U(A) = U(3, 2, 1) - U(0, 2, 3) = 3 \cdot 2 \cdot 1 - 0 = 6$$

10) $\vec{F} = x(y^2 + z^2)\vec{i} + y(x^2 + z^2)\vec{j} + z(x^2 + y^2)\vec{k}$, $A(-1, 1, -1)$, $B(1, 1, 2)$

$$\text{rot } \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x(y^2 + z^2) & y(x^2 + z^2) & z(x^2 + y^2) \end{vmatrix} = \vec{i}(2yz - 2yz) + \vec{j}(2xz - 2xz) + \vec{k}(2xy - 2xy) = 0$$

$\vec{F}$ הוא שדה משמר, ניתן לבדוק את הצורה של $\vec{F}$ בעזרת $M(0,0,0)$ נבחר

$$U = \int_0^x x(y^2 + z^2) dx + \int_0^y yz^2 dy + \int_0^z 0 dz = \frac{1}{2} x^2 (y^2 + z^2) \Big|_0^x + \frac{1}{2} y^2 z^2 \Big|_0^y \Rightarrow$$

$$U = \frac{1}{2} (x^2 y^2 + x^2 z^2 + y^2 z^2)$$

$$W = \int_A^B \vec{F} d\vec{r} = U(B) - U(A) = U(1, 1, 2) - U(-1, 1, -1) = \frac{1}{2} [(1+4+4) - (1+1+1)] = 3$$

$$11) \vec{F} = \frac{yz}{1+x^2y^2z^2} \vec{i} + \frac{xz}{1+x^2y^2z^2} \vec{j} + \frac{xy}{1+x^2y^2z^2} \vec{k}, \quad A(\sqrt{2}, 0, 0), B(1, 1, 1)$$

$$\text{rot } \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{yz}{1+x^2y^2z^2} & \frac{xz}{1+x^2y^2z^2} & \frac{xy}{1+x^2y^2z^2} \end{vmatrix}$$

$\vec{i}$ של $\text{rot } \vec{F}$ תלך

$$\frac{\partial}{\partial y} \left(\frac{xy}{1+x^2y^2z^2} \right) - \frac{\partial}{\partial z} \left(\frac{xz}{1+x^2y^2z^2} \right) = x \cdot \frac{1+x^2y^2z^2 - 2x^2y^2z^2}{(1+x^2y^2z^2)^2} - x \cdot \frac{1+x^2y^2z^2 - 2x^2y^2z^2}{(1+x^2y^2z^2)^2} = 0$$

$\text{rot } \vec{F} = 0 \Leftrightarrow 0$ - δ פ'לל $\vec{k}$ - $\vec{j}$ של $\text{rot } \vec{F}$ תלך

תלך $\vec{i}$ של $\text{rot } \vec{F}$ תלך $M(0,0,0)$ תלך $\vec{k}$ של $\text{rot } \vec{F}$ תלך $\vec{j}$ של $\text{rot } \vec{F}$ תלך $\vec{i}$ של $\text{rot } \vec{F}$ תלך

$$u = \int_0^x \frac{yz dx}{1+x^2y^2z^2} + \int_0^y 0 dy + \int_0^z 0 dz = \int_0^x \frac{d(xy z)}{1+(xy z)^2} = \arctan(xy z) \Big|_0^x \Rightarrow$$

$$u = \arctan(xy z)$$

$$W = \int_A^B \vec{F} d\vec{F} = u(B) - u(A) = u(1,1,1) - u(\sqrt{2}, 0, 0) = \arctan 1 - 0 = \frac{\pi}{4}$$

טורים מספריים

1) $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}} = \frac{1}{1} + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \dots$ $u_n = \frac{1}{\sqrt{n}}$
 נעלה שם בטור $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$ $\leftarrow u_n = \frac{1}{\sqrt{n}} \leftarrow \frac{1}{\sqrt{n}} > \frac{1}{n} \leftarrow n > \sqrt{n} \leftarrow v_n = \frac{1}{n} \leftarrow \sum_{n=1}^{\infty} \frac{1}{n}$
 הטור הנעלה מתבצר בטור $\sum_{n=1}^{\infty} \frac{1}{n}$ (הטור הפשוט)

2) $\sum_{n=1}^{\infty} \frac{1}{n \cdot 5^{n-1}} = \frac{1}{1} + \frac{1}{2 \cdot 5} + \frac{1}{3 \cdot 5^2} + \dots$ $\Rightarrow u_n = \frac{1}{n \cdot 5^{n-1}}$
 נעלה בטור $\sum_{n=1}^{\infty} \frac{1}{5^{n-1}}$ $\leftarrow v_n = \frac{1}{5^{n-1}} \leftarrow \frac{1}{n \cdot 5^{n-1}} \leq \frac{1}{5^{n-1}} \leftarrow \sum_{n=1}^{\infty} \frac{1}{5^{n-1}}$
 הטור הנעלה מתבצר כי הטור $\sum_{n=1}^{\infty} \frac{1}{5^{n-1}}$ מתבצר (סדרה הנעשית ויזנית)

3) $\sum_{n=2}^{\infty} \frac{1}{e_n n} = \frac{1}{e_n 2} + \frac{1}{e_n 3} + \frac{1}{e_n 4} + \dots$ $u_n = \frac{1}{e_n n}$
 כ'ע'ר: $\frac{1}{e_n n} > \frac{1}{n} \leftarrow e_n n < n$ הטור הנעלה מתבצר בטור $\sum_{n=2}^{\infty} \frac{1}{n}$ (הטור הפשוט)

4) $\sum_{n=2}^{\infty} \left(\frac{n}{n-1}\right)^n = 2^2 + 1.5^3 + \left(\frac{4}{3}\right)^4 + \dots$ $u_n = \left(\frac{n}{n-1}\right)^n$
 $\lim_{n \rightarrow \infty} u_n = \lim_{n \rightarrow \infty} \left(\frac{n-1+1}{n-1}\right)^{n-1+1} = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n-1}\right)^{n-1} \left(1 + \frac{1}{n-1}\right) = e \cdot 1 = e \neq 0$
 הטור מתבצר כי הפיתוי ההכרחי לא מתק"ק

5) $\sum_{n=1}^{\infty} \frac{1}{1000n-2} = \frac{1}{998} + \frac{1}{1998} + \frac{1}{2998} + \dots$ $u_n = \frac{1}{1000n-2}$
 $v_n = \frac{1}{n} \cdot \sum_{n=1}^{\infty} \frac{1}{n}$ נעלה בטור
 $\lim_{n \rightarrow \infty} \frac{u_n}{v_n} = \lim_{n \rightarrow \infty} \frac{n}{1000n-2} = \lim_{n \rightarrow \infty} \frac{1}{1000-2/n} = 0.001$
 הטור הנעלה מתבצר כי הטור $\sum_{n=1}^{\infty} \frac{1}{n}$ מתבצר

6) $\sum_{n=2}^{\infty} \frac{1}{n \sqrt{10}} = \frac{1}{\sqrt{10}} + \frac{1}{2\sqrt{10}} + \frac{1}{3\sqrt{10}} + \dots$ $u_n = \frac{1}{n \sqrt{10}}$
 $\lim_{n \rightarrow \infty} u_n = \lim_{n \rightarrow \infty} \frac{1}{n \sqrt{10}} = 0 \neq 0 \rightarrow$ (מתבצר)
 הפיתוי ההכרחי לא מתק"ק.

7) $\sum_{n=2}^{\infty} \frac{1}{\sqrt{n(n-1)}} = \frac{1}{\sqrt{1 \cdot 2}} + \frac{1}{\sqrt{2 \cdot 3}} + \frac{1}{\sqrt{3 \cdot 4}} + \dots$ $u_n = \frac{1}{\sqrt{n(n-1)}}$
 $v_n = \frac{1}{n} \cdot \sum_{n=2}^{\infty} \frac{1}{n}$ נעלה בטור
 $\lim_{n \rightarrow \infty} \frac{u_n}{v_n} = \lim_{n \rightarrow \infty} \frac{n}{\sqrt{n(n-1)}} = \lim_{n \rightarrow \infty} \frac{1}{\sqrt{1-1/n}} = 1.$
 הטור הנעלה מתבצר, כי הטור ההכרחי מתבצר

$$8) \sum_{n=1}^{\infty} n \ln\left(1 + \frac{1}{n}\right) = \ln 2 + 2 \ln 1.5 + 3 \ln \frac{4}{3} + \dots \quad u_n = n \cdot \ln\left(1 + \frac{1}{n}\right)$$

$$\lim_{n \rightarrow \infty} u_n = \lim_{n \rightarrow \infty} \ln\left(1 + \frac{1}{n}\right)^n = \ln e = 1 \neq 0$$

(הגורם n נכנס ל-1) (ההכרחי ל-1) (ההכרחי ל-1)

$$9) 1 + \frac{1}{3} + \frac{1}{5} + \dots \Rightarrow u_n = \frac{1}{2n-1}, \quad f(x) = \frac{1}{2x-1} \quad (n=1, 2, \dots)$$

$$\int_1^{\infty} \frac{dx}{2x-1} = \frac{1}{2} \int_1^{\infty} \frac{d(2x-1)}{2x-1} = \frac{1}{2} \ln(2x-1) \Big|_1^{\infty} = \frac{1}{2} [\lim_{x \rightarrow \infty} (2x-1) - \ln 1] = \infty$$

(הגורם n נכנס ל-1)

$$10) 1 + \frac{1}{\sqrt{4}} + \frac{1}{\sqrt{9}} + \dots \Rightarrow 1, 4, 9, 16, \dots \Rightarrow a_1 = 1, d = 3, a_n = 1 + (n-1)3 = 3n-2$$

$$u_n = \frac{1}{\sqrt{3n-2}}, \quad f(x) = \frac{1}{\sqrt{3x-2}}, \quad (n=1, 2, \dots)$$

$$\int_1^{\infty} \frac{dx}{\sqrt{3x-2}} = \frac{2}{3} \sqrt{3x-2} \Big|_1^{\infty} = \frac{2}{3} [\lim_{x \rightarrow \infty} (\sqrt{3x-2}) - 1] = \infty \rightarrow (הגורם n נכנס ל-1)$$

$$11) \frac{1}{2^3} + \frac{2}{3^3} + \frac{3}{4^3} + \dots \Rightarrow u_n = \frac{n}{(n+1)^3}, \quad f(x) = \frac{x}{(x+1)^3}, \quad (n=1, 2, \dots)$$

$$\int_1^{\infty} \frac{x dx}{(x+1)^3} = \int_1^{\infty} \frac{x+1-1}{(x+1)^3} dx = \int_1^{\infty} \frac{dx}{(x+1)^2} - \int_1^{\infty} \frac{dx}{(x+1)^3} = \left(-\frac{1}{x+1} + \frac{1}{2} \cdot \frac{1}{(x+1)^2} \right) \Big|_1^{\infty} =$$

$$= \lim_{x \rightarrow \infty} \left(\frac{1}{2(x+1)^2} - \frac{1}{x+1} \right) - \left[\frac{1}{2 \cdot 2^2} - \frac{1}{1+1} \right] = 0 - \frac{1}{8} + \frac{1}{2} = \frac{3}{8} \rightarrow (הגורם n נכנס ל-1)$$

$$12) \frac{1}{1+1^2} + \frac{1}{1+2^2} + \frac{1}{1+3^2} + \dots \Rightarrow u_n = \frac{1}{1+n^2}, \quad f(x) = \frac{1}{1+x^2}, \quad (n=1, 2, \dots)$$

$$\int_1^{\infty} \frac{dx}{1+x^2} = \arctg x \Big|_1^{\infty} = \lim_{x \rightarrow \infty} \arctg x - \frac{\pi}{4} = \frac{\pi}{2} - \frac{\pi}{4} = \frac{\pi}{4} \rightarrow (הגורם n נכנס ל-1)$$

$$13) \frac{1}{1+1^2} + \frac{2}{1+2^2} + \frac{3}{1+3^2} + \dots \quad u_n = \frac{n}{1+n^2}, \quad f(x) = \frac{x}{1+x^2}, \quad (n=1, 2, \dots)$$

$$\int_1^{\infty} \frac{x dx}{1+x^2} = \frac{1}{2} \int_1^{\infty} \frac{d(x^2+1)}{x^2+1} = \frac{1}{2} \ln(x^2+1) \Big|_1^{\infty} = \frac{1}{2} [\lim_{x \rightarrow \infty} \ln(x^2+1) - \ln 2] = \infty$$

(הגורם n נכנס ל-1)

$$14) \frac{1}{2^2-1} + \frac{1}{5^2-1} + \frac{1}{7^2-1} + \dots \quad u_n = \frac{1}{(2n+1)^2-1}, \quad f(x) = \frac{1}{(2x+1)^2-1}, \quad n=1, 2, \dots$$

$$\int_1^{\infty} \frac{dx}{(2x+1)^2-1} = \frac{1}{2} \int_1^{\infty} \frac{d(2x+1)}{(2x+1)^2-1} = \frac{1}{2} \cdot \frac{1}{2} \ln \frac{(2x+1)-1}{(2x+1)+1} \Big|_1^{\infty} = \frac{1}{4} \ln \frac{x}{x+1} \Big|_1^{\infty} =$$

$$= \frac{1}{4} [\lim_{x \rightarrow \infty} \ln \frac{x}{x+1} - \ln \frac{1}{2}] = \frac{1}{4} [\lim_{x \rightarrow \infty} \ln \frac{1}{1+1/x} + \ln 2] = \frac{1}{4} \ln 2 \rightarrow (הגורם n נכנס ל-1)$$

$$15) \frac{1}{2\ln 2} + \frac{1}{3\ln 3} + \frac{1}{4\ln 4} + \dots \quad u_n = \frac{1}{n \ln n}, \quad f(x) = \frac{1}{x \ln x} \quad (n=2, 3, \dots)$$

$$\int_2^{\infty} \frac{dx}{x \ln x} = \int_2^{\infty} \frac{d(\ln x)}{\ln x} = \ln \ln x \Big|_2^{\infty} = \lim_{x \rightarrow \infty} \ln(\ln x) - \ln(\ln 2) = \infty \rightarrow (\text{Divergenz})$$

$$16) \frac{1}{2\ln^2 2} + \frac{1}{3\ln^2 3} + \dots \quad u_n = \frac{1}{n \ln^2 n}, \quad f(x) = \frac{1}{x \ln^2 x}, \quad (n=2, 3, \dots)$$

$$\int_2^{\infty} \frac{dx}{x \ln^2 x} = \int_2^{\infty} \frac{d(\ln x)}{\ln^2 x} = -\frac{1}{\ln x} \Big|_2^{\infty} = -\left[\lim_{x \rightarrow \infty} \frac{1}{\ln x} - \frac{1}{\ln 2} \right] = \frac{1}{\ln 2} \rightarrow (0, \text{Konvergenz})$$

$$17) \sum_{n=1}^{\infty} \frac{2n}{3^n} = \frac{2}{3} + \frac{4}{9} + \frac{6}{27} + \dots \quad u_n = \frac{2n}{3^n}, \quad u_{n+1} = \frac{2(n+1)}{3^{n+1}}$$

$$\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = \lim_{n \rightarrow \infty} \frac{2(n+1) \cdot 3^n}{3^{n+1} \cdot 2n} = \frac{1}{3} \lim_{n \rightarrow \infty} \left(\frac{n+1}{n} \right) = \frac{1}{3} \cdot 1 = \frac{1}{3} < 1 \rightarrow (0, \text{Konvergenz})$$

$$18) \sum_{n=1}^{\infty} \frac{2^{n-1}}{n!} = 1 + \frac{2}{2!} + \frac{4}{3!} + \dots \quad u_n = \frac{2^{n-1}}{n!}, \quad u_{n+1} = \frac{2^n}{(n+1)!}$$

$$\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = \lim_{n \rightarrow \infty} \frac{2^n \cdot n!}{(n+1)! \cdot 2^{n-1}} = 2 \lim_{n \rightarrow \infty} \frac{n!}{n! \cdot (n+1)} = 2 \cdot 0 = 0 < 1 \rightarrow (0, \text{Konvergenz})$$

$$19) \sum_{n=1}^{\infty} \frac{n!}{1 \cdot 3 \cdot 5 \cdot \dots \cdot (2n-1)} = \frac{1}{1} + \frac{1 \cdot 2}{1 \cdot 3} + \frac{1 \cdot 2 \cdot 3}{1 \cdot 3 \cdot 5} + \dots \quad u_n = \frac{n!}{1 \cdot 3 \cdot 5 \cdot \dots \cdot (2n-1)}$$

$$u_{n+1} = \frac{(n+1)!}{1 \cdot 3 \cdot 5 \cdot \dots \cdot (2n-1) \cdot (2n+1)}, \quad \lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = \lim_{n \rightarrow \infty} \frac{(n+1)! \cdot 1 \cdot 3 \cdot \dots \cdot (2n-1)}{1 \cdot 3 \cdot \dots \cdot (2n-1) \cdot (2n+1) \cdot n!} =$$

$$= \lim_{n \rightarrow \infty} \frac{n+1}{2n+1} = \lim_{n \rightarrow \infty} \frac{1+1/n}{2+1/n} = \frac{1}{2} < 1 \rightarrow (0, \text{Konvergenz})$$

$$20) \sum_{n=1}^{\infty} \frac{3^{n-1}}{2^{n-1} (2n-1)} = 1 + \frac{3}{2 \cdot 3} + \frac{3^2}{2^2 \cdot 5} + \dots \quad u_n = \frac{3^{n-1}}{2^{n-1} (2n-1)}, \quad u_{n+1} = \frac{3^n}{2^n (2n+1)}$$

$$\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = \lim_{n \rightarrow \infty} \frac{3^n \cdot 2^{n-1} (2n-1)}{2^n (2n+1) \cdot 3^{n-1}} = \frac{3}{2} \lim_{n \rightarrow \infty} \frac{2n-1}{2n+1} = \frac{3}{2} \cdot 1 > 1 \rightarrow (\text{Divergenz})$$

$$21) \sum_{n=1}^{\infty} \frac{(2n-1)!}{2 \cdot 4 \cdot \dots \cdot 2n} = \frac{1}{2} + \frac{3!}{2 \cdot 4} + \frac{5!}{2 \cdot 4 \cdot 6} + \dots \quad u_n = \frac{(2n-1)!}{2 \cdot 4 \cdot \dots \cdot 2n}, \quad u_{n+1} = \frac{(2n+1)!}{2 \cdot 4 \cdot \dots \cdot 2n \cdot (2n+2)}$$

$$\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = \lim_{n \rightarrow \infty} \frac{(2n+1)! \cdot 2 \cdot 4 \cdot \dots \cdot 2n}{2 \cdot 4 \cdot \dots \cdot (2n+2) \cdot (2n-1)!} = \lim_{n \rightarrow \infty} \frac{(2n+1) \cdot 2n \cdot (2n+1) \cdot 2 \cdot 4 \cdot \dots \cdot 2n}{2 \cdot 4 \cdot \dots \cdot 2n \cdot (2n+2) \cdot (2n-1)!} =$$

$$= \lim_{n \rightarrow \infty} \frac{2n(2n+1)}{2n+2} = \lim_{n \rightarrow \infty} \frac{n(2+1/n)}{1+1/n} = \left| \frac{\infty \cdot 2}{1} \right| = \infty > 1 \quad (\text{Divergenz})$$

$$22) \sum_{n=1}^{\infty} \frac{4n-3}{\sqrt{n \cdot 3^n}} = \frac{1}{\sqrt{3}} + \frac{5}{\sqrt{2 \cdot 3^2}} + \frac{9}{\sqrt{3 \cdot 3^3}} + \dots \quad u_n = \frac{4n-3}{\sqrt{n \cdot 3^n}}, \quad u_{n+1} = \frac{4n+1}{\sqrt{(n+1)3^{n+1}}}$$

$$\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = \lim_{n \rightarrow \infty} \frac{(4n+1)\sqrt{n \cdot 3^n}}{\sqrt{(n+1) \cdot 3^{n+1}}(4n-3)} = \lim_{n \rightarrow \infty} \sqrt{\frac{3^n}{3^{n+1}}} \cdot \frac{n(4+1/n)}{n(4-3/n)} \cdot \sqrt{\frac{n}{n+1}} =$$

$$= \frac{1}{\sqrt{3}} \cdot 1 \cdot 1 = \frac{1}{\sqrt{3}} < 1 \rightarrow (0, 1, 0, 0, \dots)$$

$$23) \sum_{n=1}^{\infty} \left(\frac{n+1}{2n-1} \right)^n, \quad a_n = \left(\frac{n+1}{2n-1} \right)^n, \quad \lim_{n \rightarrow \infty} \sqrt[n]{a_n} = \lim_{n \rightarrow \infty} \frac{n+1}{2n-1} = \frac{1}{2} < 1$$

(0, 1, 0, 0, \dots)

$$24) \sum_{n=1}^{\infty} \left(\frac{n}{3n-1} \right)^{2n-1}, \quad a_n = \left(\frac{n}{3n-1} \right)^{2n-1}, \quad \lim_{n \rightarrow \infty} \sqrt[n]{a_n} = \lim_{n \rightarrow \infty} \left(\frac{n}{3n-1} \right)^{\frac{2n-1}{n}} =$$

$$= \lim_{n \rightarrow \infty} \left(\frac{1}{3-1/n} \right)^{2-\frac{1}{n}} = \lim_{n \rightarrow \infty} \left(\frac{1}{3-1/n} \right)^2 \cdot \sqrt[n]{3-1/n} = \left(\frac{1}{3} \right)^2 \cdot 1 = \frac{1}{9} < 1 \rightarrow (0, 1, 0, 0, \dots)$$

$$25) \sum_{n=1}^{\infty} \frac{1}{e n^n (n+1)}, \quad a_n = \left[\frac{1}{e n (n+1)} \right]^n, \quad \lim_{n \rightarrow \infty} \sqrt[n]{a_n} = \lim_{n \rightarrow \infty} \frac{1}{e n (n+1)} = 0 < 1$$

(0, 1, 0, 0, \dots)

$$26) \sum_{n=1}^{\infty} \arcsin^n \frac{1}{n}, \quad a_n = \left(\arcsin \frac{1}{n} \right)^n,$$

$$\lim_{n \rightarrow \infty} \sqrt[n]{a_n} = \lim_{n \rightarrow \infty} \arcsin \frac{1}{n} = \arcsin 0 = 0 < 1 \rightarrow (0, 1, 0, 0, \dots)$$

$$27) \sum_{n=1}^{\infty} \frac{(n+1)n^2}{n^2 \cdot 3^n}, \quad a_n = \left(\frac{n+1}{n} \right)^n \cdot \left(\frac{1}{3} \right)^n$$

$$\lim_{n \rightarrow \infty} \sqrt[n]{a_n} = \lim_{n \rightarrow \infty} \left(\frac{n+1}{n} \right)^n \cdot \frac{1}{3} = \frac{1}{3} \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n} \right)^n = \frac{1}{3} e = \frac{e}{3} < 1 \rightarrow (0, 1, 0, 0, \dots)$$

$$28) \sum_{n=1}^{\infty} \frac{n^3}{e^n}, \quad a_n = \frac{n^3}{e^n}, \quad \lim_{n \rightarrow \infty} \sqrt[n]{\frac{n^3}{e^n}} = \lim_{n \rightarrow \infty} \frac{(\sqrt[n]{n})^3}{e} = \frac{1^3}{e} = \frac{1}{e} < 1 \rightarrow (0, 1, 0, 0, \dots)$$

$$29) \sum_{n=1}^{\infty} \left(\frac{n-2}{n+1} \right)^{n(n-2)}, \quad a_n = \left(\frac{n-2}{n+1} \right)^{n(n-2)}$$

$$\lim_{n \rightarrow \infty} \sqrt[n]{a_n} = \lim_{n \rightarrow \infty} \left(\frac{n-2}{n+1} \right)^{\frac{n(n-2)}{n}} = \lim_{n \rightarrow \infty} \left(\frac{n-2}{n+1} \right)^{n-2} = \lim_{n \rightarrow \infty} \left(1 + \frac{-3}{n+1} \right)^{n+1-3} =$$

$$= \lim_{n \rightarrow \infty} \left(1 + \frac{-3}{n+1} \right)^{n+1} \cdot \left(1 - \frac{3}{n+1} \right)^{-3} = e^{-3} \cdot 1 = \frac{1}{e^3} < 1 \rightarrow (0, 1, 0, 0, \dots).$$

30) $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2+1} \Rightarrow |u_n| = \frac{1}{n^2+1} < \frac{1}{n^2} \Rightarrow \sum_{n=1}^{\infty} \frac{1}{n^2+1} \rightarrow (\text{מתכנס})$
 הסדר הפתוח מתכנס בהכרח.

31) $\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt[3]{n^2}} \Rightarrow |u_n| = \frac{1}{n^{2/3}} \Rightarrow \sum_{n=1}^{\infty} \frac{1}{n^{2/3}} \rightarrow (\text{מתבדר})$

a) $\frac{1}{1^{2/3}} > \frac{1}{2^{2/3}} > \frac{1}{3^{2/3}} > \dots$ b) $\lim_{n \rightarrow \infty} u_n = \lim_{n \rightarrow \infty} \frac{1}{n^{2/3}} = 0$
 הסדר הפתוח מתכנס בתנאי

32) $\sum_{n=1}^{\infty} \frac{3^n \sin n}{5^n} \Rightarrow |u_n| = \frac{3^n |\sin n|}{5^n} \leq \left(\frac{3}{5}\right)^n \Rightarrow \sum_{n=1}^{\infty} \left(\frac{3}{5}\right)^n \rightarrow (\text{מתכנס})$
 הסדר הפתוח מתכנס בהכרח

33) $\sum_{n=2}^{\infty} \frac{(-1)^n n}{n\sqrt{n}-1} = \sum_{n=2}^{\infty} \frac{(-1)^n}{\sqrt{n}-1/n}$, $|u_n| = \frac{1}{\sqrt{n}-1/n} > \frac{1}{\sqrt{n}}$
 הסדר: $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$ מתבדר

a) נתבאר בהוכחה: $|u_n| = \left\{ \frac{n}{n\sqrt{n}-1} \right\}_2^n$ נבחר, באופן שלא ממש/לית' ו' ארצות.

$$|u_{n+1}| - |u_n| = \frac{n+1}{(n+1)\sqrt{n+1}-1} - \frac{n}{n\sqrt{n}-1} = \frac{(n+1)(n\sqrt{n}-1) - n[(n+1)(\sqrt{n+1}-1)]}{[(n+1)\sqrt{n+1}-1](n\sqrt{n}-1)} =$$

$$= \frac{n(n+1)[\sqrt{n}-\sqrt{n+1}] - n-1+n}{[(n+1)\sqrt{n+1}-1](n\sqrt{n}-1)} = -\frac{(\sqrt{n+1}-\sqrt{n})n(n+1)+1}{[(n+1)\sqrt{n+1}-1](n\sqrt{n}-1)} < 0 \Rightarrow$$

$$|u_2| > |u_3| > |u_4| > \dots$$

על אפס/לית' אלה' הוכחה הנ"ל בקצרה של הסדרות:

$$f(x) = \frac{x}{x\sqrt{x}-1}, \quad x=2,3,4,\dots$$

נבאר, באופן פתוח/לית' ו' ארצות (כל ס' מ' פתוח/לית' ו' ארצות):

$$f'(x) = \frac{x\sqrt{x}-1-x(\sqrt{x}+\frac{x}{2\sqrt{x}})}{(x\sqrt{x}-1)^2} = \frac{x\sqrt{x}-1-x\sqrt{x}-\frac{1}{2}x\sqrt{x}}{(x\sqrt{x}-1)^2} = -\frac{2+x\sqrt{x}}{2(x\sqrt{x}-1)^2} < 0.$$

הפתוח/לית' ו' ארצות, נכון: $|u_2| > |u_3| > |u_4| > \dots \Leftrightarrow f(2) > f(3) > f(4) > \dots$

b) $\lim_{n \rightarrow \infty} \frac{1}{\sqrt{n}-1/n} = 0 \Rightarrow$ (הסדר הפתוח מתכנס בתנאי)

34) $\sum_{n=2}^{\infty} (-1)^n \frac{e_n n}{n} \Rightarrow |u_n| = \frac{e_n n}{n} > \frac{1}{n} \Rightarrow \sum_{n=2}^{\infty} \frac{1}{n} \rightarrow (\text{מתבדר})$

$\frac{e_n n}{n}$ - הקצרה של הפתוח/לית' ו' ארצות, $f(x) = \frac{e_n x}{x}$, נבאר/לית' ו' ארצות.

a) $f'(x) = \frac{1-e_n x}{x^2} < 0 \quad (x > e) \rightarrow$ (הסדר הפתוח מתכנס בתנאי).

b) $\lim_{x \rightarrow \infty} \frac{e_n x}{x} = \lim_{x \rightarrow \infty} \frac{1}{x} = 0 \Rightarrow$ (הסדר הפתוח מתכנס בתנאי)

טורים מספריים (תרגילים שונים)

$$1) \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{\sqrt{n}} \cdot |u_n| = \frac{1}{\sqrt{n}} \rightarrow \sum_{n=1}^{\infty} \frac{1}{\sqrt{n}} - (\text{מתבצר})$$

$$a) \frac{1}{\sqrt{1}} > \frac{1}{\sqrt{2}} > \frac{1}{\sqrt{3}} > \dots \quad b) \lim_{n \rightarrow \infty} \frac{1}{\sqrt{n}} = 0 \rightarrow (\text{בטל היתרון בתנאי'})$$

$$2) \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{(2n-1)^2} \cdot |u_n| = \frac{1}{(2n-1)^2} \Rightarrow v_n = \frac{1}{n^2} \rightarrow$$

$$\lim_{n \rightarrow \infty} \frac{|u_n|}{v_n} = \lim_{n \rightarrow \infty} \frac{n^2}{(2n-1)^2} = \lim_{n \rightarrow \infty} \frac{1}{(2-1/n)^2} = \frac{1}{4} \rightarrow \sum_{n=1}^{\infty} \frac{1}{n^2} - (\text{מתבצר})$$

בטל היתרון מתבצר בהחלט

$$3) \sum_{n=1}^{\infty} \frac{1}{(2n-1)\sqrt{2n-1}} \cdot u_n = \frac{1}{(2n-1)^{3/2}} \rightarrow v_n = \frac{1}{n^{3/2}} \rightarrow$$

$$\lim_{n \rightarrow \infty} \frac{u_n}{v_n} = \lim_{n \rightarrow \infty} \frac{n^{3/2}}{(2n-1)^{3/2}} = \lim_{n \rightarrow \infty} \frac{1}{(2-1/n)^{3/2}} = \frac{1}{2\sqrt{2}} \rightarrow \sum_{n=1}^{\infty} \frac{1}{n^{3/2}} - (\text{מתבצר})$$

בטל היתרון מתבצר

$$4) \sum_{n=1}^{\infty} \frac{1}{100n+1} \cdot u_n = \frac{1}{100n+1} \rightarrow v_n = \frac{1}{n} \rightarrow$$

$$\lim_{n \rightarrow \infty} \frac{u_n}{v_n} = \lim_{n \rightarrow \infty} \frac{n}{100n+1} = \frac{1}{100} \Rightarrow \sum_{n=1}^{\infty} \frac{1}{n} \rightarrow (\text{מתבצר})$$

בטל היתרון מתבצר

$$5) \sum_{n=2}^{\infty} \frac{(-1)^n}{n \ln n} \cdot |u_n| = \frac{1}{n \ln n} \Rightarrow f(x) = \frac{1}{x \ln x}$$

$$\int_2^{\infty} \frac{dx}{x \ln x} = \int_2^{\infty} \frac{d(\ln x)}{\ln x} = \ln \ln x \Big|_2^{\infty} = \lim_{x \rightarrow \infty} \ln \ln x - \ln \ln 2 = \infty - (\text{מתבצר})$$

$$a) \frac{1}{2 \ln 2} > \frac{1}{3 \ln 3} > \frac{1}{4 \ln 4} > \dots \quad b) \lim_{n \rightarrow \infty} \frac{1}{n \ln n} = 0$$

בטל היתרון מתבצר בתנאי'

$$6) \sum_{n=1}^{\infty} \frac{n}{1+n^4} \cdot u_n = \frac{n}{1+n^4} < \frac{n}{n^4} = \frac{1}{n^3} \Rightarrow \sum_{n=1}^{\infty} \frac{1}{n^3} \rightarrow (\text{מתבצר})$$

בטל היתרון מתבצר

$$7) \sum_{n=1}^{\infty} \frac{3^n - 1}{(2^n - 1)n} \cdot u_n = \frac{3^n - 1}{(2^n - 1)n} = \frac{3^n(1 - 1/3^n)}{2^n(1 - 1/2^n)n}$$

$$\lim_{n \rightarrow \infty} \sqrt[n]{u_n} = \lim_{n \rightarrow \infty} \frac{3}{2} \sqrt[n]{\frac{1 - 1/3^n}{1 - 1/2^n}} \cdot \frac{1}{\sqrt[n]{n}} = \frac{3}{2} \cdot 1 \cdot 1 > 1 \rightarrow (\text{מתבצר})$$

$$8) \sum_{n=1}^{\infty} \frac{1}{(3n-2)^2} \cdot u_n = \frac{1}{(3n-2)^2} \Rightarrow v_n = \frac{1}{n^2}$$

$$\lim_{n \rightarrow \infty} \frac{u_n}{v_n} = \lim_{n \rightarrow \infty} \frac{n^2}{(3n-2)^2} = \lim_{n \rightarrow \infty} \frac{1}{(3-2/n)^2} = \frac{1}{9} \Rightarrow \sum_{n=1}^{\infty} \frac{1}{n^2} \rightarrow (\text{מתבצר})$$

בטל היתרון מתבצר

$$9) \sum_{n=1}^{\infty} \frac{2^{n-1}}{2^n}. \quad u_n = \frac{2^{n-1}}{2^n}, \quad u_{n+1} = \frac{2^n}{2^{n+1}}. \quad \lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} =$$

$$= \lim_{n \rightarrow \infty} \frac{(2^{n+1}) \cdot 2^n}{2^{n+1} (2^n)} = \frac{1}{2} \lim_{n \rightarrow \infty} \frac{2^{n+1}}{2^n} = \frac{1}{2} \cdot 1 < 1 \rightarrow (\text{סדרת גאומטרית})$$

$$10) \sum_{n=1}^{\infty} \frac{\sin(\pi n/5)}{(\ln 10)^n}. \quad |u_n| = \frac{|\sin(\pi n/5)|}{(\ln 10)^n} \leq \frac{1}{(\ln 10)^n} = \frac{1}{(2.3)^n}$$

הסדרה $\sum_{n=1}^{\infty} \frac{1}{(2.3)^n}$ מתכנסת, לפי בדיקת הסדרה הגאומטרית.

$$11) \sum_{n=1}^{\infty} \frac{2^n}{(2n-1)!}. \quad u_n = \frac{2^n}{(2n-1)!}, \quad u_{n+1} = \frac{2^{n+1}}{(2n+1)!} = \frac{2(n+1)}{(2n-1)! \cdot 2n(2n+1)}$$

$$\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = \lim_{n \rightarrow \infty} \frac{(n+1)(2n+1)!}{(2n+1)! \cdot 2n(2n+1)} = \lim_{n \rightarrow \infty} \frac{n+1}{2n} = \frac{1}{2} < 1 \rightarrow (\text{סדרת גאומטרית})$$

$$12) \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{\sqrt{2n+1}}. \quad |u_n| = \frac{1}{\sqrt{2n+1}} \sim \frac{1}{\sqrt{n}} \rightarrow \sum_{n=1}^{\infty} \frac{1}{\sqrt{n}} \rightarrow (\text{סדרת גאומטרית})$$

$$a) \frac{1}{\sqrt{2 \cdot 1 + 1}} > \frac{1}{\sqrt{2 \cdot 2 + 1}} > \frac{1}{\sqrt{2 \cdot 3 + 1}} > \dots, \quad b) \lim_{n \rightarrow \infty} \frac{1}{\sqrt{2n+1}} = 0$$

הסדרה $\sum_{n=1}^{\infty} \frac{1}{\sqrt{2n+1}}$ מתכנסת, לפי בדיקת הסדרה הגאומטרית.

$$13) \sum_{n=1}^{\infty} \frac{(-1)^n}{n^3}. \quad |u_n| = \frac{1}{n^3} \Rightarrow \sum_{n=1}^{\infty} \frac{1}{n^3} \rightarrow (\text{סדרת גאומטרית})$$

הסדרה $\sum_{n=1}^{\infty} \frac{1}{n^3}$ מתכנסת, לפי בדיקת הסדרה הגאומטרית.

$$14) \sum_{n=1}^{\infty} \frac{\cos \pi n}{n} = \sum_{n=1}^{\infty} \frac{(-1)^n}{n} \rightarrow (\text{סדרת גאומטרית})$$

$$15) \sum_{n=2}^{\infty} \frac{1}{n \ln^3 n}. \quad u_n = \frac{1}{n \ln^3 n}, \quad f(x) = \frac{1}{x \ln^3 x} \quad (n=2, 3, \dots)$$

$$\int_2^{\infty} \frac{dx}{x \ln^3 x} = \int_2^{\infty} \frac{d(\ln x)}{\ln^3 x} = -\frac{1}{2 \ln^2 x} \Big|_2^{\infty} = -\frac{1}{2} \left[\lim_{x \rightarrow \infty} \frac{1}{\ln^2 x} - \frac{1}{\ln^2 2} \right] = \frac{1}{2 \ln^2 2}$$

(סדרת גאומטרית)

$$16) \sum_{n=10}^{\infty} \frac{1}{n \cdot \ln n \cdot \ln \ln n}. \quad u_n = \frac{1}{n \cdot \ln n \cdot \ln \ln n}, \quad f(x) = \frac{1}{x \ln x \cdot \ln \ln x}$$

$$\int_{10}^{\infty} \frac{dx}{x \ln x \cdot \ln \ln x} = \int_{10}^{\infty} \frac{d(\ln \ln x)}{\ln \ln x} = \ln(\ln \ln x) \Big|_{10}^{\infty} =$$

$$= \lim_{x \rightarrow \infty} \ln(\ln \ln x) - \ln(\ln \ln 10) = \infty \rightarrow (\text{סדרת גאומטרית})$$

$$17) \sum_{n=2}^{\infty} \frac{1}{n^2 - n}. \quad u_n = \frac{1}{n^2 - n} \rightarrow v_n = \frac{1}{n^2} \rightarrow \lim_{n \rightarrow \infty} \frac{u_n}{v_n} = \lim_{n \rightarrow \infty} \frac{n^2}{n^2 - n} = 1$$

הסדרה $\sum_{n=2}^{\infty} \frac{1}{n^2}$ מתכנסת, לפי בדיקת הסדרה הגאומטרית.

$$18) \sum_{n=2}^{\infty} \frac{1}{n^2 \ln n}. \quad u_n = \frac{1}{n^2 \ln n} < \frac{1}{n^2} \rightarrow \sum_{n=2}^{\infty} \frac{1}{n^2} \rightarrow (\text{סדרת } n)$$

סדרת $\frac{1}{n^2}$ מתכנסת

$$19) \sum_{n=1}^{\infty} \frac{(n^2+1)^{n^3+1}}{n^2 n^3+2}, \quad u_n = \frac{(n^2+1)^{n^3+1}}{n^2 n^3+2} = \frac{(n^2+1)^{n^3} (n^2+1)}{(n^2)^{n^3} \cdot n^2} = \left(1 + \frac{1}{n^2}\right)^{n^3} \cdot \left(1 + \frac{1}{n^2}\right)$$

$$\lim_{n \rightarrow \infty} \sqrt[n]{u_n} = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n^2}\right)^{n^2} \sqrt[n]{1 + \frac{1}{n^2}} = e \cdot 1 > 1 \rightarrow (\text{סדרת } n)$$

$$20) \sum_{n=1}^{\infty} \frac{1}{\sqrt{n} e^{\sqrt{n}}}: \quad u_n = \frac{e^{-\sqrt{n}}}{\sqrt{n}}, \quad f(x) = \frac{e^{-\sqrt{x}}}{\sqrt{x}}, \quad (n=1, 2, \dots)$$

$$\int_1^{\infty} \frac{e^{-\sqrt{x}}}{\sqrt{x}} dx \Rightarrow 2 \int_1^{\infty} e^{-\sqrt{x}} d(-\sqrt{x}) = -2 e^{-\sqrt{x}} \Big|_1^{\infty} = -2 \left[\lim_{x \rightarrow \infty} \frac{1}{e^{\sqrt{x}}} - \frac{1}{e} \right] = \frac{2}{e} (\text{סדרת } n)$$

$$21) \sum_{n=2}^{\infty} n \cdot \ln\left(\frac{n+1}{n-1}\right). \quad \lim_{n \rightarrow \infty} u_n = \lim_{n \rightarrow \infty} \ln\left(1 + \frac{2}{n-1}\right)^{n-1+1} =$$

$$= \lim_{n \rightarrow \infty} \ln\left(1 + \frac{2}{n-1}\right)^{n-1} \cdot \left(1 + \frac{2}{n-1}\right) = \ln e^2 \cdot 1 = 2 \neq 0 \rightarrow (\text{סדרת } n)$$

$$22) \sum_{n=1}^{\infty} \frac{3^n}{(2n)!}. \quad u_n = \frac{3^n}{(2n)!}, \quad u_{n+1} = \frac{3^{n+1}}{(2n+2)!} = \frac{3^n \cdot 3}{(2n)! (2n+1)(2n+2)}$$

$$\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = \lim_{n \rightarrow \infty} \frac{3^{n+1} \cdot (2n)!}{(2n+1)(2n+2) \cdot 3^n} = \lim_{n \rightarrow \infty} \frac{3}{(2n+1)(2n+2)} = 0 < 1 \rightarrow (\text{סדרת } n)$$

$$23) \sum_{n=1}^{\infty} \frac{n^3}{e^n}. \quad u_n = \frac{n^3}{e^n}. \quad \lim_{n \rightarrow \infty} \sqrt[n]{u_n} = \lim_{n \rightarrow \infty} \frac{(\sqrt[n]{n})^3}{e} = \frac{1}{e} < 1 \rightarrow (\text{סדרת } n)$$

$$24) \sum_{n=2}^{\infty} \frac{1}{\sqrt[n]{e^2}}. \quad \lim_{n \rightarrow \infty} u_n = \lim_{n \rightarrow \infty} \frac{1}{\sqrt[n]{e^2}} = 1 \neq 0 \rightarrow (\text{סדרת } n)$$

$$25) \sum_{n=1}^{\infty} \frac{e^n}{\sqrt{n} 2^n}. \quad u_n = \frac{e^n}{\sqrt{n} 2^n} = \frac{e^n}{(\sqrt{2})^n} \cdot \frac{1}{\sqrt{n}}.$$

$$\lim_{n \rightarrow \infty} \sqrt[n]{u_n} = \lim_{n \rightarrow \infty} \frac{e}{\sqrt{2}} \left(\frac{1}{\sqrt{n}}\right)^{1/n} = \frac{e}{\sqrt{2}} > 1 \rightarrow (\text{סדרת } n)$$

$$26) \sum_{n=1}^{\infty} \frac{n^2}{(3n)!}. \quad u_n = \frac{n^2}{(3n)!}, \quad u_{n+1} = \frac{(n+1)^2}{(3n+3)!}$$

$$\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = \lim_{n \rightarrow \infty} \frac{(n+1)^2 (3n)!}{(3n+3)! n^2} = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^2 \frac{1}{(3n+1)(3n+2)(3n+3)} = 0 < 1$$

(סדרת n)

$$27) \sum_{n=1}^{\infty} \frac{n^n}{n!}. \quad u_n = \frac{n^n}{n!}, \quad u_{n+1} = \frac{(n+1)^{n+1}}{(n+1)!}$$

$$\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = \lim_{n \rightarrow \infty} \frac{(n+1)^{n+1} \cdot n!}{(n+1)! \cdot n^n} = \lim_{n \rightarrow \infty} \frac{(n+1)^{n+1}}{(n+1) \cdot n^n} = \lim_{n \rightarrow \infty} \frac{(n+1)^n}{n^n} = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = e > 1$$

(סדרת n)

תחום ההתכנסות של טור חזקות

2)

$$1) \sum_{n=1}^{\infty} \frac{x^{n-1}}{3^{n-1} \cdot n} \quad u_n = \frac{x^{n-1}}{3^{n-1} \cdot n}, \quad u_{n+1} = \frac{x^n}{3^n(n+1)}$$

$$\lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right| = \lim_{n \rightarrow \infty} \frac{|x|^n \cdot 3^{n-1} \cdot n}{3^n(n+1) |x|^{n-1}} = \frac{1}{3} \lim_{n \rightarrow \infty} \frac{n}{n+1} |x| = \frac{1}{3} \cdot 1 \cdot |x|$$

$$\frac{1}{3} |x| < 1 \Rightarrow |x| < 3$$

$$x=3 \Rightarrow u_n = \frac{3^{n-1}}{3^{n-1} \cdot n} = \frac{1}{n} \rightarrow \sum_{n=1}^{\infty} \frac{1}{n} \rightarrow (\text{ג'ורג'י ס' קר'ס})$$

$$x=-3 \Rightarrow u_n = \frac{(-3)^{n-1}}{3^{n-1} \cdot n} = \frac{(-1)^{n-1} (3)^{n-1}}{3^{n-1} \cdot n} = \frac{(-1)^{n-1}}{n}$$

$$-3 \leq x < 3 \Leftrightarrow (\text{ג'ורג'י ס' קר'ס}) \text{ מתכנס בנקודה } \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} \text{ וכן}$$

$$2) \sum_{n=1}^{\infty} \frac{(-1)^n x^{2n-1}}{5^n \sqrt{n+1}} \quad u_n = \frac{(-1)^n x^{2n-1}}{5^n \sqrt{n+1}}, \quad u_{n+1} = \frac{(-1)^{n+1} x^{2n+1}}{5^{n+1} \sqrt{n+2}}$$

$$\lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right| = \lim_{n \rightarrow \infty} \frac{|x|^{2n+1} \cdot 5^n \sqrt{n+1}}{5^{n+1} \sqrt{n+2} |x|^{2n-1}} = \frac{1}{5} \lim_{n \rightarrow \infty} \frac{\sqrt{n+1}}{\sqrt{n+2}} |x|^2 = \frac{1}{5} |x|^2$$

$$\frac{1}{5} |x|^2 < 1 \Rightarrow |x| < \sqrt{5}$$

$$x=\sqrt{5} \Rightarrow u_n = \frac{(-1)^n 5^{n-1/2}}{5^n \sqrt{n+1}} = \frac{(-1)^n}{\sqrt{5n+5}}$$

$$(\text{ג'ורג'י ס' קר'ס}) \text{ מתכנס בנקודה } \sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{5n+5}} \text{ וכן}$$

$$x=-\sqrt{5} \Rightarrow u_n = \frac{(-1)^n (-1)^{2n-1}}{\sqrt{5n+5}} = \frac{(-1)^{3n-1}}{\sqrt{5n+5}} = \frac{[-(-1)^3]^n (-1)}{\sqrt{5n+5}} = \frac{(-1)^{n+1}}{\sqrt{5n+5}}$$

$$-\sqrt{5} \leq x \leq \sqrt{5} \Leftrightarrow (\text{ג'ורג'י ס' קר'ס}) \text{ מתכנס בנקודה } \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{\sqrt{5n+5}} \text{ וכן}$$

$$3) \sum_{n=1}^{\infty} \frac{2^n x^n}{(2n+1)^2 \sqrt{3}^n} \quad u_n = \frac{2^n x^n}{(2n+1)^2 \sqrt{3}^n}, \quad u_{n+1} = \frac{2^{n+1} x^{n+1}}{(2n+3)^2 \sqrt{3}^{n+1}}$$

$$\lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right| = \lim_{n \rightarrow \infty} \frac{2^{n+1} |x|^{n+1} (2n+1)^2 \sqrt{3}^n}{(2n+3)^2 \sqrt{3}^{n+1} \cdot 2^n |x|^n} = \frac{2}{\sqrt{3}} \lim_{n \rightarrow \infty} \left(\frac{2n+1}{2n+3} \right)^2 |x| = \frac{2}{\sqrt{3}} \cdot 1 \cdot |x|$$

$$\frac{2}{\sqrt{3}} |x| < 1 \Rightarrow |x| < \frac{\sqrt{3}}{2}$$

$$x=\frac{\sqrt{3}}{2} \Rightarrow u_n = \frac{2^n \cdot \left(\frac{\sqrt{3}}{2}\right)^n}{(2n+1)^2 \sqrt{3}^n \cdot 2^n} = \frac{1}{(2n+1)^2} \sim \frac{1}{n^2} \Rightarrow \sum_{n=1}^{\infty} \frac{1}{n^2} \rightarrow (\text{ג'ורג'י ס' קר'ס})$$

$$x=-\frac{\sqrt{3}}{2} \Rightarrow u_n = \frac{(-1)^n}{(2n+1)^2} \Rightarrow \sum_{n=1}^{\infty} \frac{(-1)^n}{(2n+1)^2} \rightarrow (\text{ג'ורג'י ס' קר'ס})$$

$$-\frac{\sqrt{3}}{2} \leq x \leq \frac{\sqrt{3}}{2}$$

$$4) \sum_{n=1}^{\infty} \frac{x^n}{n!} \quad u_n = \frac{x^n}{n!}, \quad u_{n+1} = \frac{x^{n+1}}{(n+1)!}$$

$$\lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right| = \lim_{n \rightarrow \infty} \frac{|x|^{n+1} n!}{(n+1)! |x|^n} = \lim_{n \rightarrow \infty} \frac{|x|}{n+1} = 0. \quad 0 < 1 \rightarrow (x \text{ 32 222})$$

$-\infty < x < \infty$

$$5) \sum_{n=1}^{\infty} \frac{(-x)^{n-1}}{n} = \sum_{n=1}^{\infty} \frac{(-1)^{n-1} x^{n-1}}{n} \quad u_n = \frac{(-1)^{n-1} x^{n-1}}{n}, \quad u_{n+1} = \frac{(-1)^n x^n}{n+1}$$

$$\lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right| = \lim_{n \rightarrow \infty} \frac{|x|^n n}{(n+1) |x|^{n-1}} = \lim_{n \rightarrow \infty} \frac{n}{n+1} |x| = 1 \cdot |x|. \Rightarrow |x| < 1.$$

$$x=1 \quad u_n = \frac{(-1)^{n-1}}{n} \Rightarrow \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} \rightarrow (\text{123222 22222})$$

$$x=-1 \quad u_n = \frac{(-1)^{n-1}}{n} = \frac{1}{n} \Rightarrow \sum_{n=1}^{\infty} \frac{1}{n} \Rightarrow (\text{232222}) \Rightarrow \underline{-1 < x \leq 1}$$

$$6) \sum_{n=1}^{\infty} \frac{3^n x^n}{\sqrt{(3n-2)2^n}}. \quad u_n = \frac{3^n x^n}{\sqrt{(3n-2)2^n}}, \quad u_{n+1} = \frac{3^{n+1} x^{n+1}}{\sqrt{(3n+1)2^{n+1}}}$$

$$\lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right| = \lim_{n \rightarrow \infty} \frac{3^{n+1} |x|^{n+1} \sqrt{(3n-2)2^n}}{\sqrt{(3n+1)2^{n+1}} 3^n |x|^n} = \frac{3}{\sqrt{2}} \lim_{n \rightarrow \infty} \sqrt{\frac{3n-2}{3n+1}} |x| = \frac{3}{\sqrt{2}} \sqrt{1} |x|$$

$$\frac{3}{\sqrt{2}} |x| < 1 \Rightarrow x < \sqrt{2}/3$$

$$x = \frac{\sqrt{2}}{3} \Rightarrow u_n = \frac{3^n \cdot \sqrt{2}^n}{3^n \sqrt{(3n-2)2^n}} = \frac{1}{\sqrt{3n-2}} \sim \frac{1}{\sqrt{n}} \Rightarrow \sum_{n=1}^{\infty} \frac{1}{\sqrt{n}} \rightarrow (\text{232222})$$

$$x = -\frac{\sqrt{2}}{3} \Rightarrow u_n = \frac{(-1)^n}{\sqrt{3n-2}} \Rightarrow \sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{3n-2}} \rightarrow (\text{123222 22222})$$

$$\underline{-\frac{\sqrt{2}}{3} \leq x < \frac{\sqrt{2}}{3}}$$

$$7) \sum_{n=1}^{\infty} \frac{(-1)^{n-1} x^{2n-2}}{3^{n-1} n \sqrt{n}}. \quad u_n = \frac{(-1)^{n-1} x^{2n-2}}{3^{n-1} n \sqrt{n}}, \quad u_{n+1} = \frac{(-1)^n x^{2n}}{3^n (n+1) \sqrt{n+1}}$$

$$\lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right| = \lim_{n \rightarrow \infty} \frac{|x|^{2n} 3^{n-1} n \sqrt{n}}{3^n (n+1) \sqrt{n+1} |x|^{2n-2}} = \frac{1}{3} \lim_{n \rightarrow \infty} \left(\frac{n}{n+1} \right)^{3/2} |x|^2 = \frac{1}{3} \cdot 1 \cdot |x|^2$$

$$\frac{1}{3} |x|^2 < 1 \Rightarrow |x| < \sqrt{3}$$

$$x = \sqrt{3} \Rightarrow u_n = \frac{(-1)^{n-1} (\sqrt{3})^{2n-2}}{3^{n-1} n \sqrt{n}} = \frac{(-1)^{n-1}}{n^{3/2}} \Rightarrow \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^{3/2}} \rightarrow (\text{222222})$$

$$x = -\sqrt{3} \Rightarrow u_n = \frac{(-1)^{n-1} (-1)^{2n-2}}{n^{3/2}} = \frac{(-1)^{3n-3}}{n^{3/2}} = \frac{(-1)^{3n-1}}{n^{3/2}} = \frac{(-1)^{n-1}}{n^{3/2}}$$

$$\underline{-\sqrt{3} \leq x \leq \sqrt{3}} \quad \Leftarrow \text{662222 22222} \quad \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^{3/2}} \quad \text{2222}$$

$$8) \sum_{n=1}^{\infty} \frac{(10x)^n}{\sqrt{n}} \quad u_n = \frac{10^n x^n}{\sqrt{n}}, \quad u_{n+1} = \frac{10^{n+1} x^{n+1}}{\sqrt{n+1}}$$

$$\lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right| = \lim_{n \rightarrow \infty} \frac{10^{n+1} |x|^{n+1} \sqrt{n}}{\sqrt{n+1} 10^n |x|^n} = \lim_{n \rightarrow \infty} \frac{10}{1} \sqrt{\frac{n}{n+1}} |x| = 10 \cdot \sqrt{1} |x|$$

$$10|x| < 1 \Rightarrow |x| < 0.1$$

$$x = 1/10 \Rightarrow u_n = \frac{1}{\sqrt{n}} \Rightarrow \sum_{n=1}^{\infty} \frac{1}{\sqrt{n}} \rightarrow (\text{רצבטר})$$

$$x = -1/10 \Rightarrow u_n = \frac{(-1)^n}{\sqrt{n}} \Rightarrow \sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n}} \rightarrow (\text{לסדר רצבטר})$$

$$\underline{-1/10 \leq x < 1/10}$$

$$9) \sum_{n=1}^{\infty} x^{n-1} \quad u_n = x^{n-1}, \quad u_{n+1} = x^n \quad \lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right| = \lim_{n \rightarrow \infty} \frac{|x|^n}{|x|^{n-1}} = |x|$$

$$|x| < 1 \Rightarrow x = 1 \Rightarrow \sum_{n=1}^{\infty} 1^{n-1} = 1+1+\dots \Rightarrow (\text{רצבטר})$$

$$x = -1 \Rightarrow \sum_{n=1}^{\infty} (-1)^{n-1} = 1-1+1-1+\dots \Rightarrow (\text{רצבטר})$$

$$\underline{-1 < x \leq 1}$$

$$10) \sum_{n=1}^{\infty} \frac{x^{2n-1}}{2n-1} \quad u_n = \frac{x^{2n-1}}{2n-1}, \quad u_{n+1} = \frac{x^{2n+1}}{2n+1}$$

$$\lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right| = \lim_{n \rightarrow \infty} \frac{|x|^{2n+1} (2n-1)}{(2n+1) |x|^{2n-1}} = \lim_{n \rightarrow \infty} \frac{2n-1}{2n+1} |x|^2 = 1 \cdot |x|^2$$

$$|x|^2 < 1 \Rightarrow |x| < 1$$

$$x = 1 \rightarrow u_n = \frac{1}{2n-1} \sim \frac{1}{n} \Rightarrow \sum_{n=1}^{\infty} \frac{1}{n} \rightarrow (\text{רצבטר})$$

$$x = -1 \rightarrow u_n = \frac{(-1)^{2n-1}}{2n-1} = -\frac{1}{2n-1} \Rightarrow \sum_{n=1}^{\infty} \left(-\frac{1}{2n-1} \right) = -\sum_{n=1}^{\infty} \frac{1}{2n-1} \rightarrow (\text{רצבטר})$$

$$\underline{-1 < x < 1}$$

$$11) \sum_{n=1}^{\infty} \frac{n^5 x^{2n}}{2n-1} \quad u_n = \frac{n^5 x^{2n}}{2n-1}, \quad u_{n+1} = \frac{(n+1)^5 x^{2n+2}}{2n+1}$$

$$\lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right| = \lim_{n \rightarrow \infty} \frac{(n+1)^5 |x|^{2n+2} (2n-1)}{(2n+1) n^5 |x|^{2n}} = \lim_{n \rightarrow \infty} \left(\frac{n+1}{n} \right)^5 \frac{2n-1}{2n+1} |x|^2 = 1^5 \cdot 1 \cdot |x|^2$$

$$|x| < 1$$

$$x = 1 \quad u_n = \frac{n^5}{2n-1} \quad \lim_{n \rightarrow \infty} \frac{n^5}{2n-1} = \infty \neq 0 \Rightarrow (\text{לסדר רצבטר})$$

$$x = -1 \quad u_n = \frac{n^5}{2n-1} \rightarrow (\text{רצבטר})$$

$$\underline{-1 < x < 1}$$

$$12) \sum_{n=1}^{\infty} n! x^n \quad u_n = n! x^n, \quad u_{n+1} = (n+1)! x^{n+1}$$

$$\lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right| = \lim_{n \rightarrow \infty} \frac{(n+1)! |x|^{n+1}}{n! |x|^n} = \lim_{n \rightarrow \infty} (n+1) |x| = \begin{cases} 0, & x=0 \\ \infty, & x \neq 0 \end{cases}$$

$x=0$ נעבור רק שגורר ו/או ∞ , $x \neq 0$ נעבור רק $1-n$ וגרר ∞

$$13) \sum_{n=2}^{\infty} \frac{x^{n-1}}{n \cdot 3^n \ln n}. \quad u_n = \frac{x^{n-1}}{n \cdot 3^n \ln n}, \quad u_{n+1} = \frac{x^n}{(n+1) 3^{n+1} \ln(n+1)}$$

$$\lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right| = \lim_{n \rightarrow \infty} \frac{|x|^n n \cdot 3^n \ln n}{(n+1) 3^{n+1} \ln(n+1) |x|^{n-1}} = \frac{1}{3} \lim_{n \rightarrow \infty} \frac{n}{n+1} \frac{\ln n}{\ln(n+1)} |x| =$$

$$= \frac{1}{3} |x| \lim_{n \rightarrow \infty} \frac{\ln n}{\ln(n+1)} = \frac{1}{3} |x| = \frac{1}{3} |x| \lim_{n \rightarrow \infty} \frac{1/n}{1/(n+1)} = \frac{1}{3} |x| \cdot 1. \quad \frac{1}{3} |x| < 1 \Rightarrow |x| < 3$$

$$x=3 \Rightarrow u_n = \frac{3^{n-1}}{n \cdot 3^n \ln n} = \frac{1}{3 \cdot n \ln n} \Rightarrow \int_2^{\infty} \frac{1}{x \ln x} \Rightarrow$$

$$\int_2^{\infty} \frac{dx}{x \ln x} = \ln \ln x \Big|_2^{\infty} = \lim_{x \rightarrow \infty} \ln \ln x - \ln \ln 2 = \infty. \Rightarrow \sum_{n=2}^{\infty} \frac{1}{3n \ln n} \rightarrow (\text{נפרט})$$

$$x=-3 \Rightarrow u_n = \frac{(-1)^{n-1}}{3n \ln n} \Rightarrow \sum_{n=2}^{\infty} \frac{(-1)^{n-1}}{3n \ln n} \rightarrow (\text{נפרט})$$

$$\underline{-3 \leq x < 3}$$

$$14) \sum_{n=1}^{\infty} \frac{(x-3)^n}{n \cdot 5^n}. \quad u_n = \frac{(x-3)^n}{n \cdot 5^n}, \quad u_{n+1} = \frac{(x-3)^{n+1}}{(n+1) 5^{n+1}}$$

$$\lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right| = \lim_{n \rightarrow \infty} \frac{|x-3|^{n+1} \cdot n \cdot 5^n}{(n+1) 5^{n+1} |x-3|^n} = \frac{1}{5} \lim_{n \rightarrow \infty} \frac{n}{n+1} |x-3| = \frac{1}{5} |x-3|$$

$$\frac{1}{5} |x-3| < 1 \Rightarrow |x-3| < 5 \Rightarrow -5 < x-3 < 5 \Rightarrow -2 < x < 8$$

$$x=8 \quad u_n = \frac{5^n}{n \cdot 5^n} = \frac{1}{n} \Rightarrow \sum_{n=1}^{\infty} \frac{1}{n} \rightarrow (\text{נפרט})$$

$$x=-2 \quad u_n = \frac{(-5)^n}{n \cdot 5^n} = \frac{(-1)^n \cdot 5^n}{n \cdot 5^n} = \frac{(-1)^n}{n} \Rightarrow \sum_{n=1}^{\infty} \frac{(-1)^n}{n} \rightarrow (\text{נפרט})$$

$$\underline{-2 \leq x < 8}$$

$$15) \sum_{n=1}^{\infty} \frac{(x-1)^{2n}}{n \cdot 9^n}. \quad u_n = \frac{(x-1)^{2n}}{n \cdot 9^n}, \quad u_{n+1} = \frac{(x-1)^{2n+2}}{(n+1) 9^{n+1}}$$

$$\lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right| = \lim_{n \rightarrow \infty} \frac{|x-1|^{2n+2} \cdot n \cdot 9^n}{(n+1) 9^{n+1} |x-1|^{2n}} = \frac{1}{9} \lim_{n \rightarrow \infty} \frac{n}{n+1} |x-1|^2 = \frac{1}{9} |x-1|^2$$

$$\frac{1}{9} |x-1|^2 < 1 \Rightarrow |x-1| < 3 \Rightarrow -3 < x-1 < 3 \Rightarrow -2 < x < 4$$

$$x=4 \quad u_n = \frac{3^{2n}}{n \cdot 9^n} = \frac{1}{n} \Rightarrow \sum_{n=1}^{\infty} \frac{1}{n} \rightarrow (\text{נפרט})$$

$$x=-2 \quad u_n = \frac{(-1)^{2n} \cdot 3^{2n}}{n \cdot 9^n} = \frac{((-1)^2)^n}{n} = \frac{1}{n} \Rightarrow \sum_{n=1}^{\infty} \frac{1}{n} \rightarrow (\text{נפרט})$$

$$\underline{-2 < x < 4}$$

$$6) \sum_{n=1}^{\infty} \frac{(x+3)^n}{n^2}, \quad u_n = \frac{(x+3)^n}{n^2}, \quad u_{n+1} = \frac{(x+3)^{n+1}}{(n+1)^2}$$

$$\lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right| = \lim_{n \rightarrow \infty} \frac{|x+3|^{n+1} \cdot n^2}{(n+1)^2 |x+3|^n} = \lim_{n \rightarrow \infty} \left(\frac{n}{n+1} \right)^2 |x+3| = |x+3|$$

$$|x+3| < 1 \Rightarrow -1 < x+3 < 1 \Rightarrow -4 < x < -2$$

$$x = -2 \Rightarrow u_n = \frac{1}{n^2} \Rightarrow \sum_{n=1}^{\infty} \frac{1}{n^2} \rightarrow (\text{סדרת})$$

$$x = -4 \Rightarrow u_n = \frac{(-1)^n}{n^2} \Rightarrow \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} \rightarrow (\text{סדרת סדרת})$$

$$\underline{-4 \leq x \leq -2}$$

$$7) \sum_{n=1}^{\infty} \frac{(x+5)^{2n-1}}{2n \cdot 4^n}, \quad u_n = \frac{(x+5)^{2n-1}}{2n \cdot 4^n}, \quad u_{n+1} = \frac{(x+5)^{2n+1}}{(2n+2) \cdot 4^{n+1}}$$

$$\lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right| = \lim_{n \rightarrow \infty} \frac{|x+5|^{2n+1} \cdot 2n \cdot 4^n}{(2n+2) 4^{n+1} |x+5|^{2n-1}} = \frac{1}{4} \lim_{n \rightarrow \infty} \frac{2n}{2n+2} |x+5|^2 = \frac{1}{4} |x+5|^2$$

$$\frac{1}{4} |x+5|^2 < 1 \Rightarrow |x+5| < 2 \Rightarrow -2 < x+5 < 2 \Rightarrow -7 < x < -3$$

$$x = -3 \rightarrow u_n = \frac{2^{2n-1}}{2n \cdot 4^n} = \frac{4^n \cdot 2^{-1}}{2n \cdot 4^n} = \frac{1}{4n} \sim \frac{1}{n} \Rightarrow \sum_{n=1}^{\infty} \frac{1}{n} \rightarrow (\text{סדרת})$$

$$x = -7 \rightarrow u_n = \frac{(-2)^{2n-1}}{2n \cdot 4^n} = \frac{(-1)^{2n-1} \cdot 2^{2n-1}}{2n \cdot 4^n} = \frac{(-1)^{2n-1} \cdot (-1) \cdot 4^n \cdot 2^{-1}}{2n \cdot 4^n} = -\frac{1}{4n}$$

$$\underline{-7 < x < -3} \quad \Leftarrow \text{סדרת } -\frac{1}{4} \sum_{n=1}^{\infty} \frac{1}{n} \rightarrow (\text{סדרת})$$

$$8) \sum_{n=2}^{\infty} \frac{(x+1)^{n-1}}{n \ln^2 n}, \quad u_n = \frac{(x+1)^{n-1}}{n \ln^2 n}, \quad u_{n+1} = \frac{(x+1)^n}{(n+1) \ln^2(n+1)}$$

$$\lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right| = \lim_{n \rightarrow \infty} \frac{|x+1|^n \cdot n \cdot \ln^2 n}{(n+1) \ln^2(n+1) |x+1|^{n-1}} = \lim_{n \rightarrow \infty} \frac{n}{n+1} \left(\frac{\ln n}{\ln(n+1)} \right)^2 |x+1| =$$

$$= \left| \lim_{n \rightarrow \infty} \frac{\ln n}{\ln(n+1)} = 1 \right| = 1 \cdot 1^2 |x+1| \Rightarrow |x+1| < 1 \Rightarrow -2 < x < 0$$

$$\left| \lim_{n \rightarrow \infty} \frac{1/n}{1/(n+1)} = 1 \right|$$

$$x = 0 \Rightarrow u_n = \frac{1}{n \ln^2 n} \Rightarrow f(x) = \frac{1}{x \ln^2 x} \rightarrow \int_2^{\infty} \frac{dx}{x \ln^2 x} = -\frac{1}{\ln x} \Big|_2^{\infty} = \frac{1}{\ln 2}$$

(סדרת)

$$x = -2 \Rightarrow u_n = \frac{(-1)^{n-1}}{n \ln^2 n} \Rightarrow \sum_{n=2}^{\infty} \frac{(-1)^{n-1}}{n \ln^2 n} \rightarrow (\text{סדרת סדרת})$$

$$\underline{-2 \leq x \leq 0}$$

$$9) \sum_{n=1}^{\infty} n^n (x+3)^n, \quad u_n = n^n (x+3)^n, \quad u_{n+1} = (n+1)^{n+1} (x+3)^{n+1}$$

$$\lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right| = \lim_{n \rightarrow \infty} \frac{(n+1)^{n+1} |x+3|^{n+1}}{n^n |x+3|^n} = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n} \right)^n |x+3| = |x+3| \cdot e \cdot \lim_{n \rightarrow \infty} (n+1)$$

$x = -3$ סדרת סדרת סדרת סדרת, $x = -3$ סדרת סדרת סדרת סדרת

טור טיילור

(א)

1) $f(x) = e^{-x^2}$, $x_0 = 0$

ערשטע און צווייטע

$$e^x = 1 + x + \frac{x^2}{2!} + \dots + \frac{x^n}{n!} + \dots \quad -\infty < x < \infty$$

$$e^{-x^2} = 1 + (-x^2) + \frac{(-x^2)^2}{2!} + \dots + \frac{(-x^2)^n}{n!} + \dots = 1 - x^2 + \frac{x^4}{2!} - \frac{x^6}{3!} + \frac{x^8}{4!} - \dots$$

$$e^{-x^2} = 1 - x^2 + \frac{x^4}{2!} - \frac{x^6}{3!} + \dots + (-1)^n \frac{x^{2n}}{n!} + \dots \quad -\infty < x < \infty$$

2) $f(x) = \sinh x = \frac{1}{2}(e^x - e^{-x})$, $x_0 = 0$

$$\sinh x = \frac{1}{2}(e^x - e^{-x}) = \frac{1}{2} \left[\left(1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots \right) - \left(1 + (-x) + \frac{(-x)^2}{2!} + \frac{(-x)^3}{3!} + \dots \right) \right] =$$

$$= \frac{1}{2} \left[\left(1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots \right) - \left(1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \dots \right) \right] =$$

$$= \frac{1}{2} \left(2x + 2 \cdot \frac{x^3}{3!} + 2 \cdot \frac{x^5}{5!} + \dots \right) \Rightarrow$$

$$\sinh x = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \dots + \frac{x^{2n+1}}{(2n+1)!} + \dots \quad -\infty < x < \infty$$

3) $f(x) = \cosh x = \frac{1}{2}(e^x + e^{-x})$, $x_0 = 0$

$$\cosh x = \frac{1}{2}(e^x + e^{-x}) = \frac{1}{2} \left[\left(1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots \right) + \left(1 + (-x) + \frac{(-x)^2}{2!} + \frac{(-x)^3}{3!} + \dots \right) \right] =$$

$$= \frac{1}{2} \left[\left(1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots \right) + \left(1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \dots \right) \right] =$$

$$= \frac{1}{2} \left(2 + 2 \cdot \frac{x^2}{2!} + 2 \cdot \frac{x^4}{4!} + 2 \cdot \frac{x^6}{6!} + \dots \right) \Rightarrow$$

$$\cosh x = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \frac{x^6}{6!} + \dots + \frac{x^{2n}}{(2n)!} + \dots \quad -\infty < x < \infty$$

4) $f(x) = 2^x$, $x_0 = 0$

ערשטע און צווייטע

$$e^x = 1 + x + \frac{x^2}{2!} + \dots + \frac{x^n}{n!} + \dots \quad -\infty < x < \infty$$

$$2^x = (e^{\ln 2})^x = e^{x \ln 2} = 1 + (x \ln 2) + \frac{(x \ln 2)^2}{2!} + \dots + \frac{(x \ln 2)^n}{n!} + \dots$$

$$2^x = 1 + \ln 2 \cdot x + \frac{(\ln 2)^2}{2!} x^2 + \frac{(\ln 2)^3}{3!} x^3 + \dots + \frac{(\ln 2)^n}{n!} x^n + \dots \quad -\infty < x < \infty$$

5) $f(x) = \cos^2 x, \quad x_0 = 0$

$$\cos^2 x = \frac{1}{2} \cdot 2 \cos^2 x = \frac{1}{2} (1 + \cos 2x)$$

: 7/3'а 1/6а еннеу,

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots + (-1)^n \frac{x^{2n}}{(2n)!} + \dots \quad -\infty < x < \infty$$

$$\cos^2 x = \frac{1}{2} (1 + \cos 2x) = \frac{1}{2} \left[1 + \left(1 - \frac{(2x)^2}{2!} + \frac{(2x)^4}{4!} - \frac{(2x)^6}{6!} + \dots \right) \right] =$$

$$= 1 + \frac{1}{2} \left[-\frac{2^2}{2!} x^2 + \frac{2^4}{4!} x^4 - \frac{2^6}{6!} x^6 + \dots \right] \Rightarrow$$

$$\cos^2 x = 1 + \frac{1}{2} \left(-\frac{2^2}{2!} x^2 + \frac{2^4}{4!} x^4 - \frac{2^6}{6!} x^6 + \dots + (-1)^{n+1} \frac{2^{2n+2}}{(2n+2)!} x^{2n+2} + \dots \right) \quad -\infty < x < \infty$$

6) $f(x) = \sin^2 x, \quad x_0 = 0 \quad \Rightarrow \quad \sin^2 x = \frac{1}{2} (1 - \cos 2x)$

$$\sin^2 x = \frac{1}{2} (1 - \cos 2x) = \frac{1}{2} \left[1 - \left(1 - \frac{(2x)^2}{2!} + \frac{(2x)^4}{4!} - \frac{(2x)^6}{6!} + \dots \right) \right] =$$

$$= \frac{1}{2} \left(\frac{2^2}{2!} x^2 - \frac{2^4}{4!} x^4 + \frac{2^6}{6!} x^6 - \dots \right) = \frac{2^2}{2!} x^2 - \frac{2^4}{4!} x^4 + \frac{2^6}{6!} x^6 - \dots$$

$$\sin^2 x = \frac{2^2}{2!} x^2 - \frac{2^4}{4!} x^4 + \frac{2^6}{6!} x^6 - \dots + (-1)^n \frac{2^{2n+2}}{(2n+2)!} x^{2n+2} + \dots \quad -\infty < x < \infty$$

7) $f(x) = \frac{1}{2-3x}, \quad x_0 = 0$

: 7/6а еннеу

$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + \dots + x^n + \dots \quad -1 < x < 1$$

$$\frac{1}{2-3x} = \frac{1}{2(1-3x/2)} = \frac{1}{2} \cdot \frac{1}{1-3x/2} = \frac{1}{2} \left[1 + \frac{3x}{2} + \left(\frac{3x}{2} \right)^2 + \dots + \left(\frac{3x}{2} \right)^n + \dots \right]$$

$$-1 < \frac{3x}{2} < 1 \Rightarrow -\frac{2}{3} < x < \frac{2}{3} \quad \Rightarrow$$

$$\frac{1}{2-3x} = \frac{1}{2} \left[1 + \frac{3}{2} x + \left(\frac{3}{2} \right)^2 x^2 + \dots + \left(\frac{3}{2} \right)^n x^n + \dots \right], \quad -\frac{2}{3} < x < \frac{2}{3}$$

8) $f(x) = \frac{x}{9+x^2} = x \cdot \frac{1}{9+x^2}, \quad x_0 = 0$

: 7/6а еннеу

$$\frac{1}{1+x} = 1 - x + x^2 - x^3 + \dots + (-1)^n x^n + \dots \quad -1 < x < 1$$

$$\frac{x}{9+x^2} = \frac{1}{9} x \cdot \frac{1}{1+\left(\frac{x}{3}\right)^2} = \frac{1}{9} x \cdot \left[1 - \left(\frac{x}{3} \right)^2 + \left(\frac{x}{3} \right)^4 - \left(\frac{x}{3} \right)^6 + \dots + (-1)^n \left(\frac{x}{3} \right)^{2n} + \dots \right] =$$

$$= \frac{x}{3^2} - \frac{x^3}{3^4} + \frac{x^5}{3^6} - \frac{x^7}{3^8} + \dots + (-1)^n \frac{x^{2n+1}}{3^{2n+2}}$$

$$-1 < \frac{x}{3} < 1 \Rightarrow -3 < x < 3$$

$$\frac{x}{9+x^2} = \frac{x}{3^2} - \frac{x^3}{3^4} + \frac{x^5}{3^6} - \frac{x^7}{3^8} + \dots + (-1)^n \frac{x^{2n+1}}{3^{2n+2}} + \dots \quad -3 < x < 3$$

9) $f(x) = x e^{-2x}$, $x_0 = 0$

$$x e^{-2x} = x \left[1 + (-2x) + \frac{(-2x)^2}{2!} + \frac{(-2x)^3}{3!} + \dots + \frac{(-2x)^n}{n!} + \dots \right] =$$

$$= x - 2x^2 + \frac{2^2 x^3}{2!} - \frac{2^3 x^4}{3!} + \dots + (-1)^n \frac{2^n}{n!} x^{n+1} + \dots, \quad -\infty < x < \infty$$

10) $f(x) = x \cos 3x^2$, $x_0 = 0$

$$x \cos 3x^2 = x \left[1 - \frac{(3x^2)^2}{2!} + \frac{(3x^2)^4}{4!} - \frac{(3x^2)^6}{6!} + \dots + (-1)^n \frac{(3x^2)^{2n}}{(2n)!} \right] =$$

$$= x - \frac{3^2 x^5}{2!} + \frac{3^4 x^9}{4!} - \frac{3^6 x^{13}}{6!} + \dots + (-1)^n \frac{3^{2n} x^{4n+1}}{(2n)!}$$

$$x \cos 3x^2 = x - \frac{3^2}{2!} x^5 + \frac{3^4}{4!} x^9 - \dots + (-1)^n \frac{3^{2n}}{(2n)!} x^{4n+1}, \quad -\infty < x < \infty$$

11) $f(x) = \ln x$, $x_0 = 1$.

: x | 3 | 5 | 7 | 6 n n e j

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots + (-1)^n \frac{x^{n+1}}{n+1} + \dots \quad -1 < x \leq 1$$

$$\ln x = \ln(1+x-1) = \ln[1+(x-1)] = (x-1) - \frac{(x-1)^2}{2} + \frac{(x-1)^3}{3} - \frac{(x-1)^4}{4} + \dots$$

$$-1 < x-1 \leq 1 \Rightarrow 0 < x \leq 2$$

$$\ln x = (x-1) - \frac{(x-1)^2}{2} + \frac{(x-1)^3}{3} - \frac{(x-1)^4}{4} + \dots + (-1)^n \frac{(x-1)^{n+1}}{n+1} + \dots, \quad 0 < x \leq 2$$

12) $f(x) = \frac{1}{x}$, $x_0 = 2$

: x | 4 n e n n e j

$$\frac{1}{1+x} = 1 - x + x^2 - x^3 + \dots + (-1)^n x^n + \dots \quad -1 < x < 1$$

$$\frac{1}{x} = \frac{1}{2+x-2} = \frac{1}{2} \cdot \frac{1}{1+\frac{x-2}{2}} = \frac{1}{2} \left[1 - \frac{x-2}{2} + \left(\frac{x-2}{2}\right)^2 - \left(\frac{x-2}{2}\right)^3 + \dots \right]$$

$$-1 < \frac{x-2}{2} < 1 \Rightarrow 0 < x < 4$$

$$\frac{1}{x} = \frac{1}{2} \left[1 - \frac{x-2}{2} + \frac{(x-2)^2}{2^2} - \frac{(x-2)^3}{2^3} + \dots + (-1)^n \frac{(x-2)^n}{2^n} + \dots \right], \quad 0 < x < 4$$

13) $f(x) = \frac{1}{\sqrt{1+x}}$, $x_0 = 0$

г/бд еннеј

$$(1+x)^m = 1 + mx + \frac{m(m-1)}{2!} x^2 + \frac{m(m-1)(m-2)}{3!} x^3 + \dots + \frac{m(m-1)\dots(m-n+1)}{n!} x^n, |x| < 1$$

$$\frac{1}{\sqrt{1+x}} = (1+x)^{-1/2} = 1 - \frac{1}{2}x + \frac{-\frac{1}{2}(-\frac{1}{2}-1)}{2!} x^2 + \frac{-\frac{1}{2}(-\frac{1}{2}-1)(-\frac{1}{2}-2)}{3!} x^3 + \dots =$$

$$= 1 - \frac{1}{2}x + \frac{\frac{1}{2} \cdot \frac{3}{2}}{2!} x^2 - \frac{\frac{1}{2} \cdot \frac{3}{2} \cdot \frac{5}{2}}{3!} x^3 + \frac{\frac{1}{2} \cdot \frac{3}{2} \cdot \frac{5}{2} \cdot \frac{7}{2}}{4!} x^4 - \dots$$

$$\frac{1}{\sqrt{1+x}} = 1 - \frac{1}{2}x + \frac{1 \cdot 3}{2! \cdot 2^2} x^2 - \frac{1 \cdot 3 \cdot 5}{3! \cdot 2^3} x^3 + \frac{1 \cdot 3 \cdot 5 \cdot 7}{4! \cdot 2^4} x^4 - \dots \quad -1 < x < 1$$

14) $\ln(2-3x+x^2)$, $x_0 = 0$

г/бд еннеј

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots + (-1)^n \frac{x^{n+1}}{n+1} + \dots, \quad -1 < x \leq 1.$$

$$\ln(2-3x+x^2) = \ln(x^2-3x+2) = \ln[(x-1)(x-2)] = \ln[(1-x)(2-x)] =$$

$$= \ln[2(1-x)(1-\frac{x}{2})] = \ln 2 + \ln(1-x) + \ln(1-\frac{x}{2}) =$$

$$= \ln 2 + \left[-x - \frac{(-x)^2}{2} + \frac{(-x)^3}{3} - \frac{(-x)^4}{4} + \dots \right] + \left[-\frac{x}{2} - \frac{(-\frac{x}{2})^2}{2} + \frac{(-\frac{x}{2})^3}{3} - \frac{(-\frac{x}{2})^4}{4} + \dots \right] =$$

$$= \ln 2 - \left(x + \frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{4} + \dots \right) - \left(\frac{x}{2} + \frac{x^2}{2 \cdot 2^2} + \frac{x^3}{3 \cdot 2^3} + \frac{x^4}{4 \cdot 2^4} + \dots \right) =$$

$$= \ln 2 - \left[\left(1 + \frac{1}{2}\right)x + \left(\frac{1}{2} + \frac{1}{2 \cdot 2^2}\right)x^2 + \left(\frac{1}{3} + \frac{1}{3 \cdot 2^3}\right)x^3 + \left(\frac{1}{4} + \frac{1}{4 \cdot 2^4}\right)x^4 + \dots \right]$$

$$p \left\{ \begin{array}{l} -1 < x \leq 1 \\ -1 < \frac{x}{2} \leq 1 \end{array} \right\} \Rightarrow p \left\{ \begin{array}{l} -1 < x \leq 1 \\ -2 < x \leq 2 \end{array} \right\} \Rightarrow -1 < x < 1$$

$$\ln(2-3x+x^2) = \ln 2 - \left[\left(1 + \frac{1}{2}\right)x + \left(1 + \frac{1}{2^2}\right)\frac{x^2}{2} + \dots + \left(1 + \frac{1}{2^{n+1}}\right)\frac{x^{n+1}}{n+1} + \dots \right] \quad -1 < x < 1$$

15) $f(x) = \frac{2x+3}{x^2+3x+2}$, $x_0 = 0$.

$$\frac{2x+3}{x^2+3x+2} = \frac{2x+3}{(x+1)(x+2)} = \frac{A}{x+1} + \frac{B}{x+2} \Rightarrow 2x+3 = A(x+2) + B(x+1)$$

$$\left. \begin{array}{l} x=-1 \Rightarrow 1=A \\ x=-2 \Rightarrow -1=-B \Rightarrow B=1 \end{array} \right\} \Rightarrow f(x) = \frac{1}{x+1} + \frac{1}{x+2} = \frac{1}{1+x} + \frac{1}{2} \cdot \frac{1}{1+x/2} =$$

$$= (1-x+x^2-x^3+\dots) + \frac{1}{2} \left(1 - \frac{x}{2} + \frac{x^2}{4} - \frac{x^3}{8} + \dots \right) =$$

$$= \frac{3}{2} - \left(1 + \frac{1}{4}\right)x + \left(1 + \frac{1}{8}\right)x^2 - \left(1 + \frac{1}{16}\right)x^3 + \left(1 + \frac{1}{32}\right)x^4 - \dots$$

$$p \left\{ \begin{array}{l} -1 < x < 1 \\ -1 < \frac{x}{2} < 1 \end{array} \right\} \Rightarrow p \left\{ \begin{array}{l} -1 < x < 1 \\ -2 < x < 2 \end{array} \right\} \Rightarrow -1 < x < 1$$

$$\frac{2x+3}{x^2+3x+2} = \frac{3}{2} - \left[\left(1 + \frac{1}{2^2}\right)x - \left(1 + \frac{1}{2^3}\right)x^2 + \left(1 + \frac{1}{2^4}\right)x^3 - \dots + (-1)^n \left(1 + \frac{1}{2^{n+2}}\right)x^{n+1} + \dots \right] \quad -1 < x < 1$$

(ב)

$$1) \sqrt[5]{1.1} = (1+0.1)^{1/5} \Rightarrow (1+x)^m = 1 + mx + \frac{m(m-1)}{2!}x^2 + \frac{m(m-1)(m-2)}{3!}x^3 + \dots$$

$$\sqrt[5]{1.1} = \left| \begin{matrix} x=0.1 \\ m=1/5 \end{matrix} \right| = 1 + \frac{1}{5} \cdot 0.1 + \frac{\frac{1}{5} \cdot (-\frac{4}{5})}{2} \cdot 0.1^2 + \frac{\frac{1}{5} \cdot (-\frac{4}{5}) \cdot (-\frac{9}{5})}{3!} \cdot 0.1^3 + \frac{\frac{1}{5} \cdot (-\frac{4}{5}) \cdot (-\frac{9}{5}) \cdot (-\frac{14}{5})}{4!} \cdot (0.1)^4 + \dots$$

נחשב את פאטור האחרון בפיתוח:

$$\left| -\frac{1 \cdot 4 \cdot 9 \cdot 14}{4! \cdot 5^4 \cdot 10^4} \right| = \frac{36 \cdot 14}{24 \cdot 625 \cdot 10^4} \approx \frac{1}{300,000} < \frac{1}{100,000}$$

ע"פ משפט טיורינג שאנר, פטור עם סמנים מתחלפים יאחר צדדים
מפסק פאטור של פאטור הראשון של פאטור. מכאן:

$$\sqrt[5]{1.1} \approx 1 + \frac{1}{50} - \frac{4}{5000} + \frac{36}{750,000} = 1.01925$$

$$2) \ln 1.04 \Rightarrow \ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$$

$$\ln(1+0.04) = 0.04 - \frac{(0.04)^2}{2} + \frac{(0.04)^3}{3} - \frac{(0.04)^4}{4} + \dots$$

$$= 0.04 - \frac{16}{2 \cdot 10^4} + \frac{64}{3 \cdot 10^6} - \frac{256}{4 \cdot 10^8} + \dots \Rightarrow (\text{פ'ס מתחלפים סמנים וצדדים})$$

$$\frac{256}{4 \cdot 10^8} < \frac{1}{100,000} \Rightarrow \ln 1.04 \approx 0.04 - \frac{8}{10^4} + \frac{64}{3 \cdot 10^6} = 0.03922$$

$$3) \frac{1}{\sqrt[5]{e}} = e^{-1/5} \Rightarrow e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

$$e^{-1/5} = 1 + (-\frac{1}{5}) + \frac{1}{2!}(-\frac{1}{5})^2 + \frac{1}{3!}(-\frac{1}{5})^3 + \frac{1}{4!}(-\frac{1}{5})^4 + \frac{1}{5!}(-\frac{1}{5})^5 + \dots =$$

$$= 1 - \frac{1}{5} + \frac{1}{50} - \frac{1}{750} + \frac{1}{15,000} - \frac{1}{5! \cdot 5^5} \Rightarrow (\text{פ'ס מתחלפים סמנים וצדדים})$$

$$\frac{1}{5! \cdot 5^5} = \frac{1}{375000} < \frac{1}{100,000} \Rightarrow e^{-1/5} \approx 1 - \frac{1}{5} + \frac{1}{50} - \frac{1}{750} + \frac{1}{15,000} = 0.81873$$

$$4) \cos 18^\circ = \cos \frac{\pi}{10} \Rightarrow \cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$$

$$\cos 18^\circ = \cos \frac{\pi}{10} = 1 - \frac{\pi^2}{10^2 \cdot 2!} + \frac{\pi^4}{10^4 \cdot 4!} - \frac{\pi^6}{10^6 \cdot 6!} + \dots =$$

$$= 1 - \frac{\pi^2}{200} + \frac{\pi^4}{240,000} - \frac{\pi^6}{720 \cdot 10^6} + \dots \Rightarrow (\text{פ'ס מתחלפים סמנים וצדדים})$$

$$\left| -\frac{\pi^6}{720 \cdot 10^6} \right| \approx \frac{1.33}{10^6} < \frac{1}{100,000} \Rightarrow$$

$$\cos 18^\circ \approx 1 - \frac{\pi^2}{200} + \frac{\pi^4}{240,000} = 0.95106$$

(λ)

∴ 1/δ777 7/68 $f(x) = \frac{1}{1+x}$ 1/377777 7/68 7/68 7/68 7/68 7/68 (a.1)

$$\frac{1}{1+x} = 1 - x + x^2 - x^3 + \dots \quad -1 < x < 1$$

$$\int_0^x \frac{dx}{1+x} = \int_0^x (1 - x + x^2 - x^3 + \dots) dx \Rightarrow \ln(1+x) \Big|_0^x = \left(x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots \right) \Big|_0^x$$

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots + (-1)^n \frac{x^{n+1}}{n+1} + \dots \quad -1 < x < 1$$

∴ 1/δ777 7/68 $f(x) = \frac{1}{1+x^2}$ 1/377777 7/68 7/68 7/68 7/68 (b)

$$\frac{1}{1+x^2} = 1 - x^2 + x^4 - x^6 + \dots \quad -1 < x < 1$$

$$\int_0^x \frac{dx}{1+x^2} = \int_0^x (1 - x^2 + x^4 - x^6 + \dots) dx \Rightarrow \arctan x \Big|_0^x = \left(x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \dots \right) \Big|_0^x$$

$$\arctan x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \dots + (-1)^n \frac{x^{2n+1}}{2n+1} + \dots \quad -1 < x < 1$$

∴ 1/δ777 7/68 $f(x) = \frac{1}{1-x^2}$ 1/377777 7/68 7/68 7/68 7/68 (c)

$$\frac{1}{1-x^2} = 1 + x^2 + x^4 + x^6 + \dots \quad -1 < x < 1$$

$$\int_0^x \frac{dx}{1-x^2} = \int_0^x (1 + x^2 + x^4 + x^6 + \dots) dx \Rightarrow \frac{1}{2} \ln \frac{1+x}{1-x} \Big|_0^x = \left(x + \frac{x^3}{3} + \frac{x^5}{5} + \frac{x^7}{7} + \dots \right) \Big|_0^x$$

$$\ln \frac{1+x}{1-x} = 2 \left(x + \frac{x^3}{3} + \frac{x^5}{5} + \dots + \frac{x^{2n+1}}{2n+1} + \dots \right) \quad -1 < x < 1$$

∴ $f(x) = \frac{1}{\sqrt{1-x^2}}$ 1/377777 7/68 7/68 7/68 7/68 (d)

$$\frac{1}{\sqrt{1-x^2}} = (1-x^2)^{-\frac{1}{2}} = 1 + \left(-\frac{1}{2}\right)(-x^2) + \frac{-\frac{1}{2}(-\frac{1}{2}-1)}{2!}(-x^2)^2 + \frac{-\frac{1}{2}(-\frac{3}{2})(-\frac{1}{2}-2)}{3!}(-x^2)^3 + \dots =$$

$$= 1 + \frac{1}{2}x^2 + \frac{1 \cdot 3}{2 \cdot 4}x^4 + \frac{1 \cdot 3 \cdot 5}{2 \cdot 4 \cdot 6}x^6 + \frac{1 \cdot 3 \cdot 5 \cdot 7}{2 \cdot 4 \cdot 6 \cdot 8}x^8 + \dots \quad -1 < x < 1$$

$$\int_0^x \frac{dx}{\sqrt{1-x^2}} = \int_0^x \left(1 + \frac{1}{2}x^2 + \frac{1 \cdot 3}{2 \cdot 4}x^4 + \dots \right) dx \Rightarrow$$

$$\arcsin x \Big|_0^x = \left(x + \frac{1}{2} \cdot \frac{x^3}{3} + \frac{1 \cdot 3}{2 \cdot 4} \cdot \frac{x^5}{5} + \dots \right) \Big|_0^x \Rightarrow$$

$$\arcsin x = x + \frac{1}{2} \cdot \frac{x^3}{3} + \frac{1 \cdot 3}{2 \cdot 4} \cdot \frac{x^5}{5} + \frac{1 \cdot 3 \cdot 5}{2 \cdot 4 \cdot 6} \cdot \frac{x^7}{7} + \frac{1 \cdot 3 \cdot 5 \cdot 7}{2 \cdot 4 \cdot 6 \cdot 8} \cdot \frac{x^9}{9} + \dots$$

$$-1 < x < 1$$

2a) $1 + 2x + 3x^2 + \dots + nx^{n-1} + \dots \quad -1 < x < 1$

$nx^{n-1} = (x^n)'$: פונקציה רגילה $\Rightarrow$ $S(x)$ - פונקציה רגילה
 $(-1 < x < 1)$, $\frac{1}{1-x} = 1 + x + x^2 + \dots + x^n + \dots$: פונקציה רגילה

$$\left(\frac{1}{1-x}\right)' = (1 + x + x^2 + \dots)' \Rightarrow \frac{1}{(1-x)^2} = 1 + 2x + 3x^2 + \dots \Rightarrow$$

$$1 + 2x + 3x^2 + \dots + nx^{n-1} = \frac{1}{(1-x)^2} \quad , \quad S(x) = \frac{1}{(1-x)^2}$$

b) $x + \frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{4} + \dots + \frac{x^n}{n} + \dots \quad -1 < x < 1$

: פונקציה רגילה $\Rightarrow$ $S(x)$ - פונקציה רגילה

$$S(x) = x + \frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{4} + \dots \Rightarrow S'(x) = (x + \frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{4} + \dots)' \Rightarrow$$

$$S'(x) = 1 + x + x^2 + x^3 + \dots \Rightarrow S'(x) = \frac{1}{1-x} \quad , \quad -1 < x < 1 \Rightarrow$$

$$S(x) = \int_0^x \frac{dx}{1-x} = -\ln(1-x) \Big|_0^x = -\ln(1-x) \Rightarrow S(x) = \ln \frac{1}{1-x}$$

c) $-2x + 4x^3 - 6x^5 + \dots + (-1)^n \cdot 2n \cdot x^{2n-1} \quad , \quad -1 < x < 1$

: פונקציה רגילה $\Rightarrow$ $S(x)$ - פונקציה רגילה

$$S(x) = -2x + 4x^3 - 6x^5 + \dots \Rightarrow \int_0^x S(x) dx = -x^2 + x^4 - x^6 + \dots + (-1)^n x^{2n+1} + \dots$$

$$\int_0^x S(x) dx = -1 + 1 - x^2 + x^4 - \dots = -1 + [1 - x^2 + (x^2)^2 - (x^2)^3 + \dots]$$

$$\int_0^x S(x) dx = -1 + \frac{1}{1+x^2} \quad , \quad -1 < x < 1 \Rightarrow S(x) = \left(\int_0^x S(x) dx\right)' = \left(-1 + \frac{1}{1+x^2}\right)'$$

$$S(x) = -\frac{2x}{(1+x^2)^2}$$

d) $2x + \frac{3}{1!}x^2 + \frac{4}{2!}x^3 + \frac{5}{3!}x^4 + \dots \quad , \quad -\infty < x < \infty$

: פונקציה רגילה $\Rightarrow$ $S(x)$ - פונקציה רגילה

$$S(x) = 2x + \frac{3}{1!}x^2 + \frac{4}{2!}x^3 + \dots \Rightarrow \int_0^x S(x) dx = \int_0^x (2x + 3x^2 + \frac{4}{2!}x^3 + \dots) dx$$

$$\int_0^x S(x) dx = x^2 + x^3 + \frac{x^4}{2!} + \frac{x^5}{3!} + \dots = x^2(1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots)$$

$$\int_0^x S(x) dx = x^2 e^x \quad S(x) = \left(\int_0^x S(x) dx\right)' = (x^2 e^x)' = 2xe^x + x^2 e^x$$

$$S(x) = x(2+x)e^x$$

(T)

$$1) I = \int_0^{0.2} \frac{\sin x}{x} dx, \quad \varepsilon = 0,0001$$

$$I = \int_0^{0.2} \frac{1}{x} \sin x dx = \int_0^{0.2} \frac{1}{x} \left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots \right) dx = \int_0^{0.2} \left(1 - \frac{x^2}{3!} + \frac{x^4}{4!} - \dots \right) dx =$$

$$= \left(x - \frac{x^3}{3 \cdot 3} + \frac{x^5}{5 \cdot 5} - \dots \right) \Big|_0^{1/5} = \frac{1}{5} - \frac{1}{3! \cdot 3 \cdot 5^3} + \frac{1}{5! \cdot 5 \cdot 5^5} - \dots$$

, p'adnann p'jw'o p'x rlg d'atn

$$\frac{1}{5! \cdot 5^6} = \frac{1}{120 \cdot 15,625} < \frac{1}{10,000} \Rightarrow I \approx \frac{1}{5} - \frac{1}{3! \cdot 3 \cdot 5^3} = \frac{1}{5} - \frac{1}{2250}$$

$$I \approx \frac{449}{2250} = 0,1996$$

$$2) I = \int_0^{0.1} \frac{\ln(1+x)}{x} dx, \quad \varepsilon = 0,001$$

$$I = \int_0^{0.1} \frac{1}{x} \ln(1+x) dx = \int_0^{0.1} \frac{1}{x} \left(x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots \right) dx =$$

$$= \int_0^{0.1} \left(1 - \frac{x}{2} + \frac{x^2}{3} - \frac{x^3}{4} + \dots \right) dx = \left(x - \frac{x^2}{4} + \frac{x^3}{9} - \frac{x^4}{16} + \dots \right) \Big|_0^{0.1} =$$

$$= \frac{1}{10} - \frac{1}{400} + \frac{1}{9000} - \frac{1}{160,000} + \dots \Rightarrow \quad (p'adnann p'jw'o p'x rlg)$$

$$\frac{1}{9000} < \frac{1}{1000} \Rightarrow I \approx \frac{1}{10} - \frac{1}{400} = \frac{39}{400} = 0,098$$

$$3) I = \int_0^{0.5} \frac{1 - \cos x}{x^2} dx, \quad \varepsilon = 0,0001$$

$$I = \int_0^{0.5} \frac{1}{x^2} (1 - \cos x) dx = \int_0^{0.5} \frac{1}{x^2} \left[1 - \left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots \right) \right] dx =$$

$$= \int_0^{1/2} \frac{1}{x^2} \left(\frac{x^2}{2!} - \frac{x^4}{4!} + \frac{x^6}{6!} - \dots \right) dx = \int_0^{1/2} \left(\frac{1}{2!} - \frac{x^2}{4!} + \frac{x^4}{6!} - \dots \right) dx =$$

$$= \left(\frac{1}{2!} x - \frac{x^3}{4! \cdot 3} + \frac{x^5}{6! \cdot 5} - \dots \right) \Big|_0^{1/2} = \frac{1}{2! \cdot 2} - \frac{1}{4! \cdot 3 \cdot 2^3} + \frac{1}{6! \cdot 5 \cdot 2^5} - \dots$$

, p'adnann p'jw'o p'x rlg d'atn

$$\frac{1}{6! \cdot 5 \cdot 2^5} = \frac{1}{720 \cdot 5 \cdot 32} < \frac{1}{10,000} \Rightarrow I \approx \frac{1}{2! \cdot 2} - \frac{1}{4! \cdot 3 \cdot 2^3} =$$

$$= \frac{1}{4} - \frac{1}{576} = \frac{143}{576} = 0,2483$$

(7)

1) $f(x) = \frac{x}{1-x}$, $n=5$

פונקציע $f(x)$ איז א'צייגן פונעם אלגעמיינעם פארמולע
: $\frac{1}{n!} f^{(n)}(0) =$ דעם קoefficient פון x^n אין

$$f(x) = x \cdot \frac{1}{1-x} = x(1+x+x^2+x^3+x^4+\dots) = x+x^2+x^3+x^4+x^5+\dots$$

: פונקציע פאר x^5 איז דעם קoefficient

$$\frac{1}{5!} f^{(5)}(0) = 1 \Rightarrow f^{(5)}(0) = 5! = 120$$

2) $f(x) = \frac{1}{1+2x^2}$, a) $n=6$, b) $n=9$

$$f(x) = \frac{1}{1+2x^2} = 1 - (2x^2) + (2x^2)^2 - (2x^2)^3 + \dots = 1 - 2x^2 + 4x^4 - 8x^6 + \dots$$

: פונקציע פאר x^6 איז דעם קoefficient

$$\frac{f^{(6)}(0)}{6!} = -8 \Rightarrow f^{(6)}(0) = -8 \cdot 6! = -5760$$

(b) פאר $n=9$ איז דעם קoefficient פון x^9 אין
: $f^{(9)}(0) = 0$ פונקציע פאר $x=0$ איז דעם קoefficient

3) $f(x) = \frac{x}{1-x^2}$, $n=5$

$$f(x) = x \cdot \frac{1}{1-x^2} = x(1+x^2+x^4+x^6+\dots) = x+x^3+x^5+x^7+\dots$$

$$\frac{f^{(5)}(0)}{5!} = 1 \Rightarrow f^{(5)}(0) = 5! = 120$$

4) $f(x) = x \sin x$, $n=8$

$$f(x) = x \sin x = x \left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots \right) = x^2 - \frac{x^4}{3!} + \frac{x^6}{5!} - \frac{x^8}{7!} + \dots$$

$$\frac{f^{(8)}(0)}{8!} = -\frac{1}{7!} \Rightarrow f^{(8)}(0) = -\frac{8!}{7!} \Rightarrow f^{(8)}(0) = -8$$